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(54) **BIOMARKERS FOR THE DIAGNOSIS AND TREATMENT OF FIBROTIC LUNG DISEASE**

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(51) **Int. Cl.**

C12Q 1/68	(2018.01)
A61K 9/00	(2006.01)
A61K 31/197	(2006.01)
A61K 31/4412	(2006.01)
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A61P 11/00	(2006.01)
C12P 19/34	(2006.01)
C12Q 1/6883	(2018.01)
A61K 31/198	(2006.01)
A61K 31/4418	(2006.01)

(52) **U.S. Cl.**

CPC ***A61K 31/496*** (2013.01); ***A61K 9/0056*** (2013.01); ***A61K 31/197*** (2013.01); ***A61K 31/4412*** (2013.01); ***A61P 11/00*** (2018.01); ***C12Q 1/6883*** (2013.01); *A61K 31/198* (2013.01); *A61K 31/4418* (2013.01); *C12Q 2600/106* (2013.01); *C12Q 2600/156* (2013.01)

(58) **Field of Classification Search**

CPC C12Q 2600/106; C12Q 2600/156; C12Q 1/6883

See application file for complete search history.

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DbSNP short genetic variations—SNP linked to Gene (geneID:727897) Via Contig Annotation, SNPs in the MUC5B gene region, printed from ncbi.nlm.nih.gov, Mar. 3, 2022, pp. 1-30 provided (Year: 2022).

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(74) *Attorney, Agent, or Firm* — Cooley LLP

(57) **ABSTRACT**

The present disclosure provides a method of treating a fibrotic lung disease in a subject comprising administering to the subject an effective amount of a therapeutic agent, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease.

7 Claims, 33 Drawing Sheets

Specification includes a Sequence Listing.

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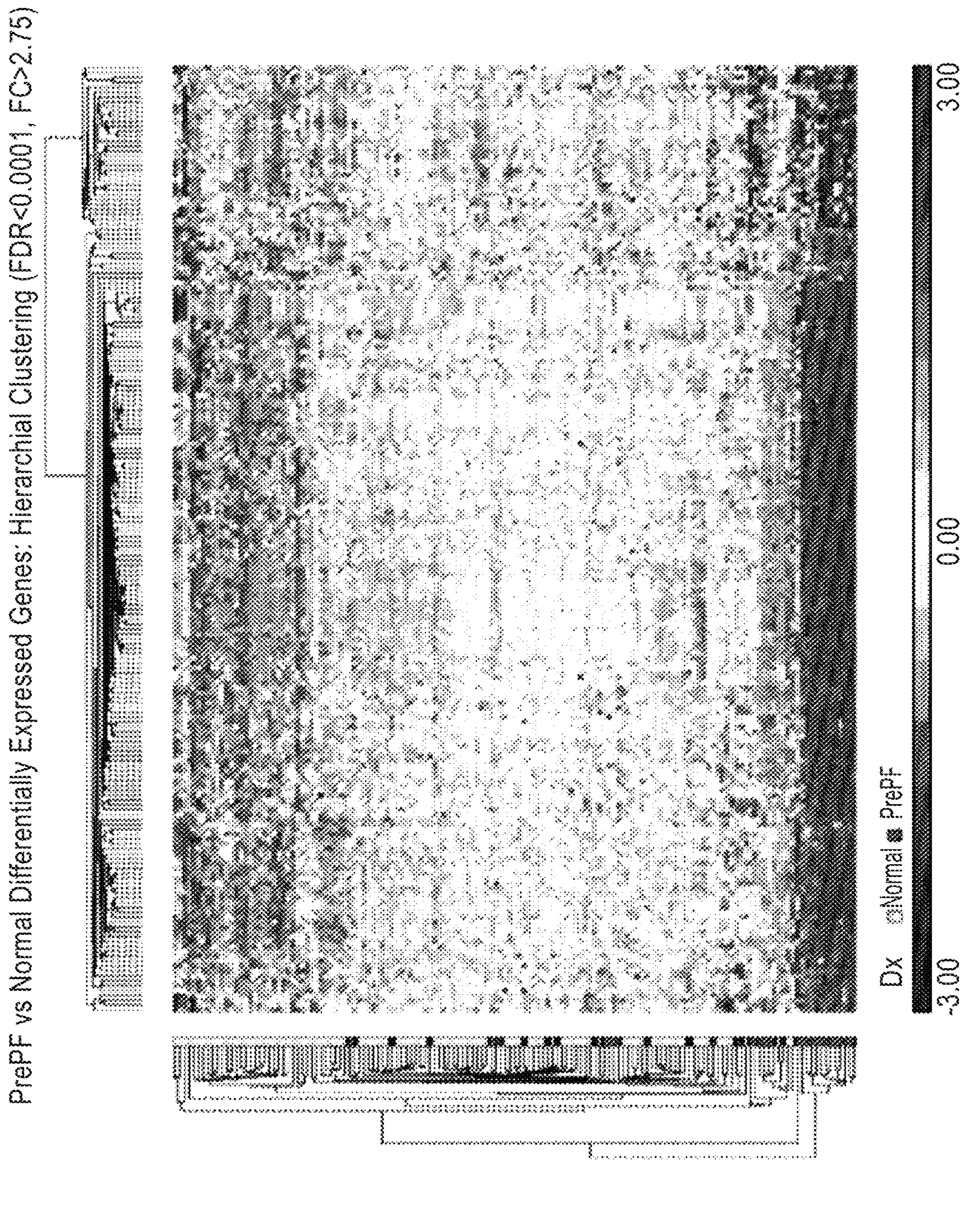

PrePF vs Normal Differentially Expressed Genes: Hierarchial Clustering (FDR<0.0001, FC>2.75)
Dx
Normal
PrePF
-3.00
0.00
3.00

FIG. 1

FIG. 2A

Before outlier exclusion

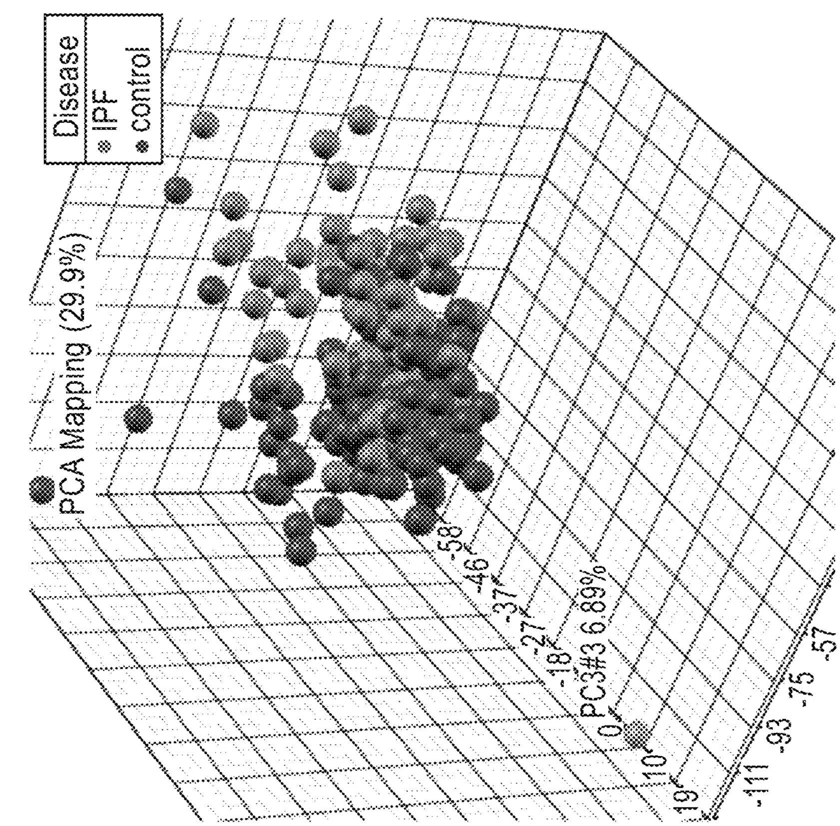


FIG. 2B

After outlier exclusion

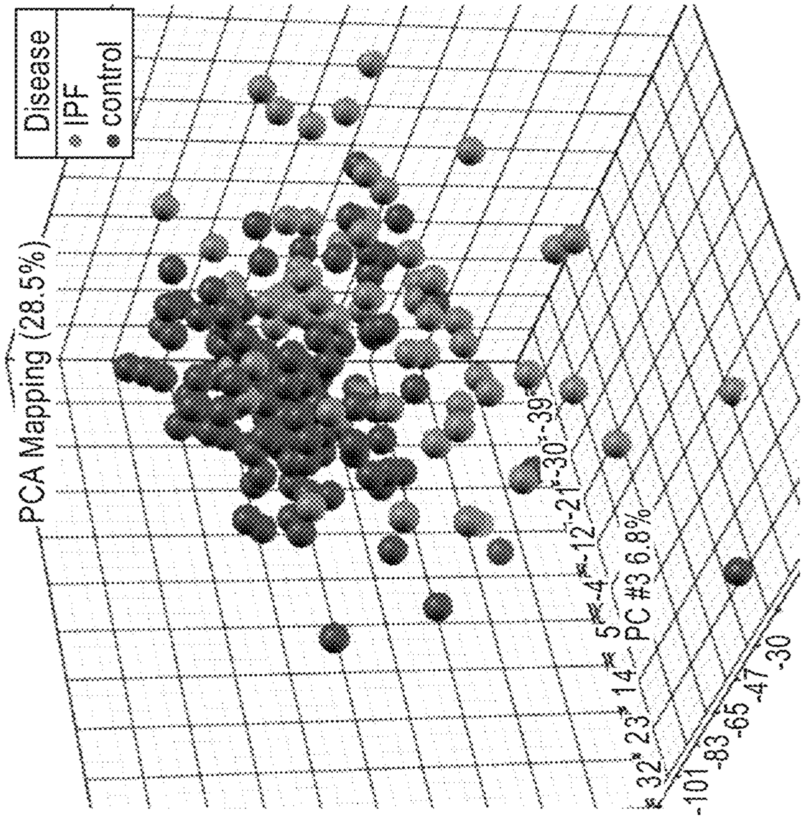


FIG. 3

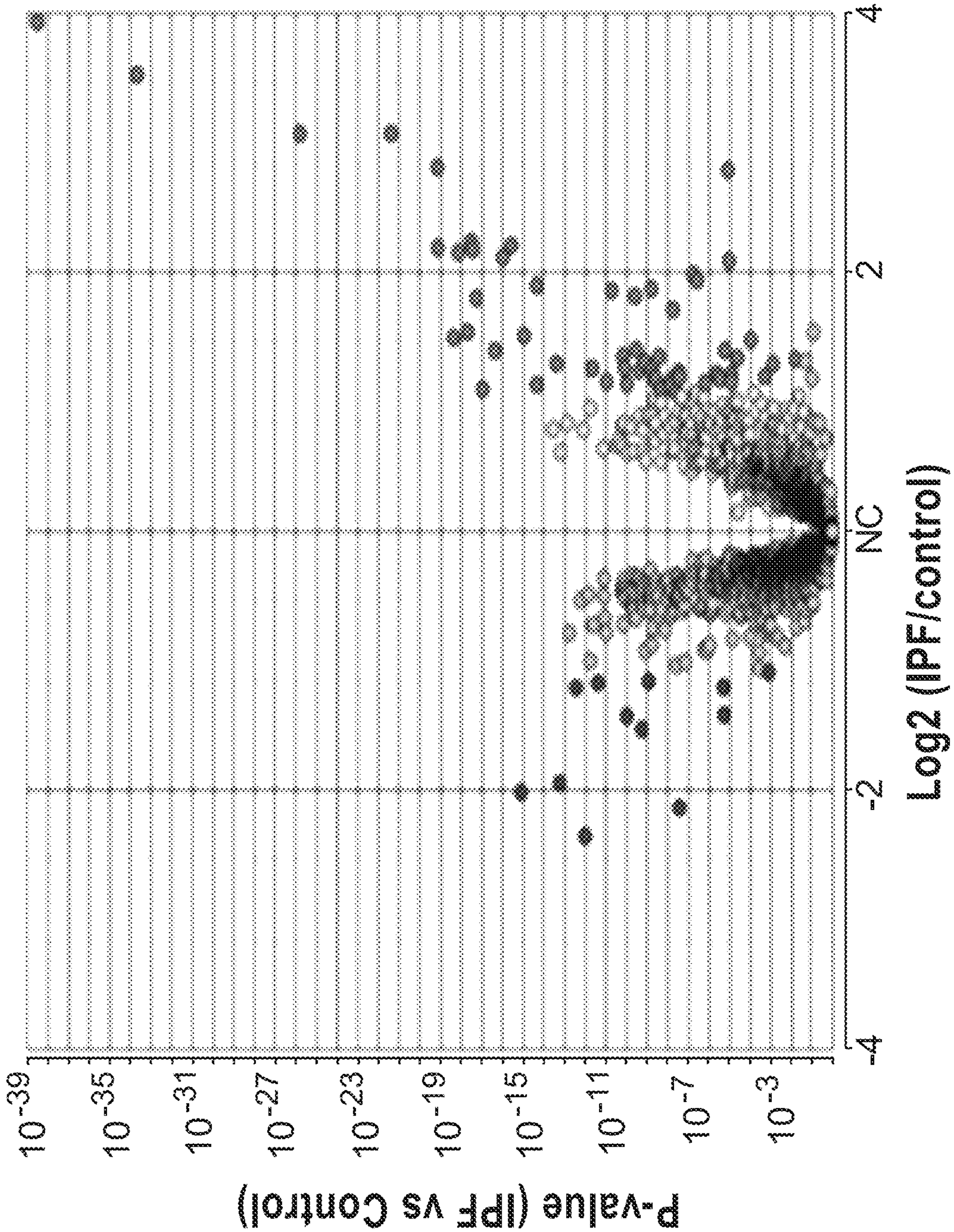

P-value (IPF vs Control)
10^{-39}
10^{-35}
10^{-31}
10^{-27}
10^{-23}
10^{-19}
10^{-15}
10^{-11}
10^{-7}
10^{-3}
-4
-2
NC
2
4
Log2 (IPF/control)

FIG. 4

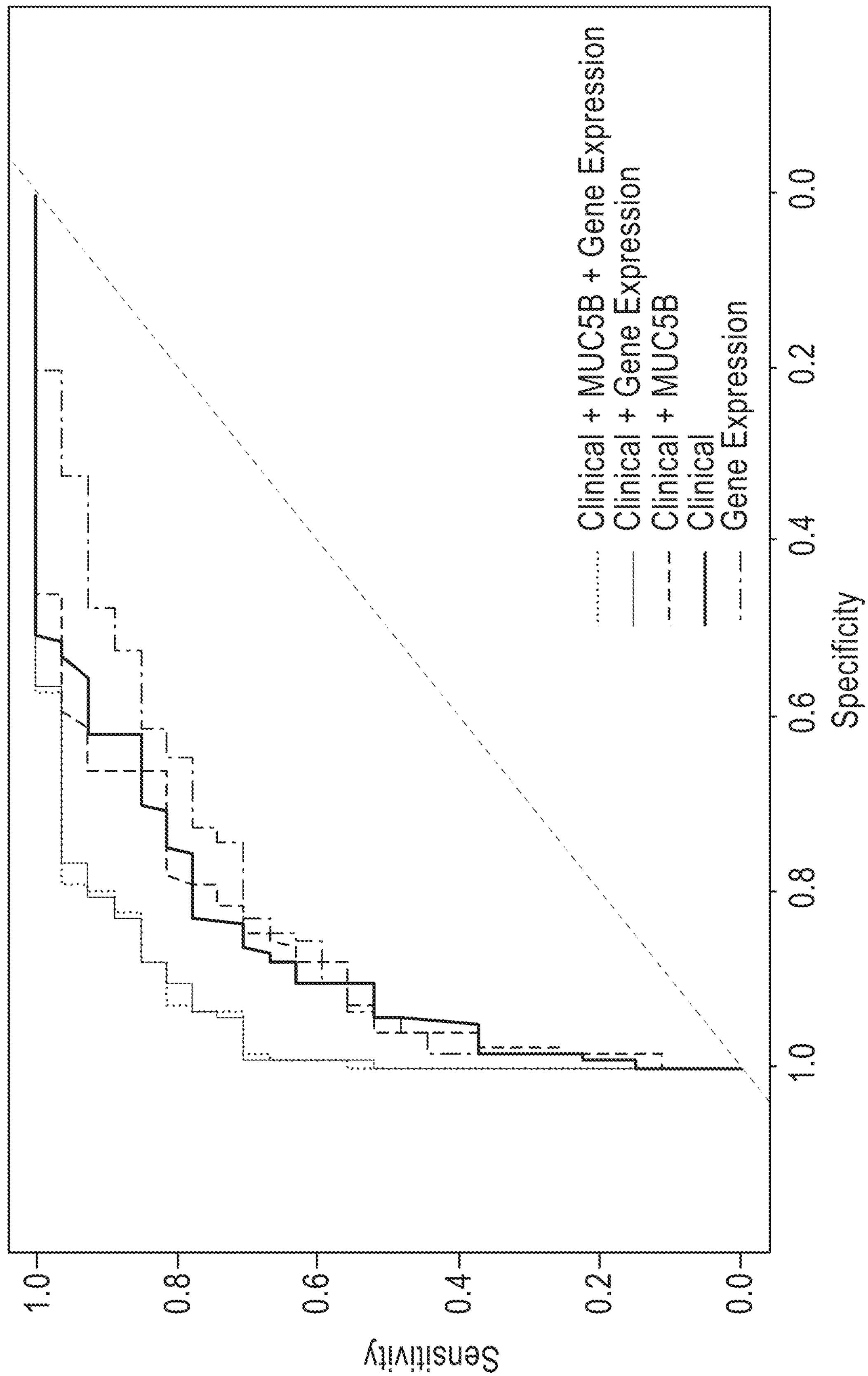

Clinical + MUC5B + Gene Expression
Clinical + Gene Expression
Clinical + MUC5B
Clinical
Gene Expression
Specificity
Sensitivity
1.0
0.8
0.6
0.4
0.2
0.0

FIG. 5

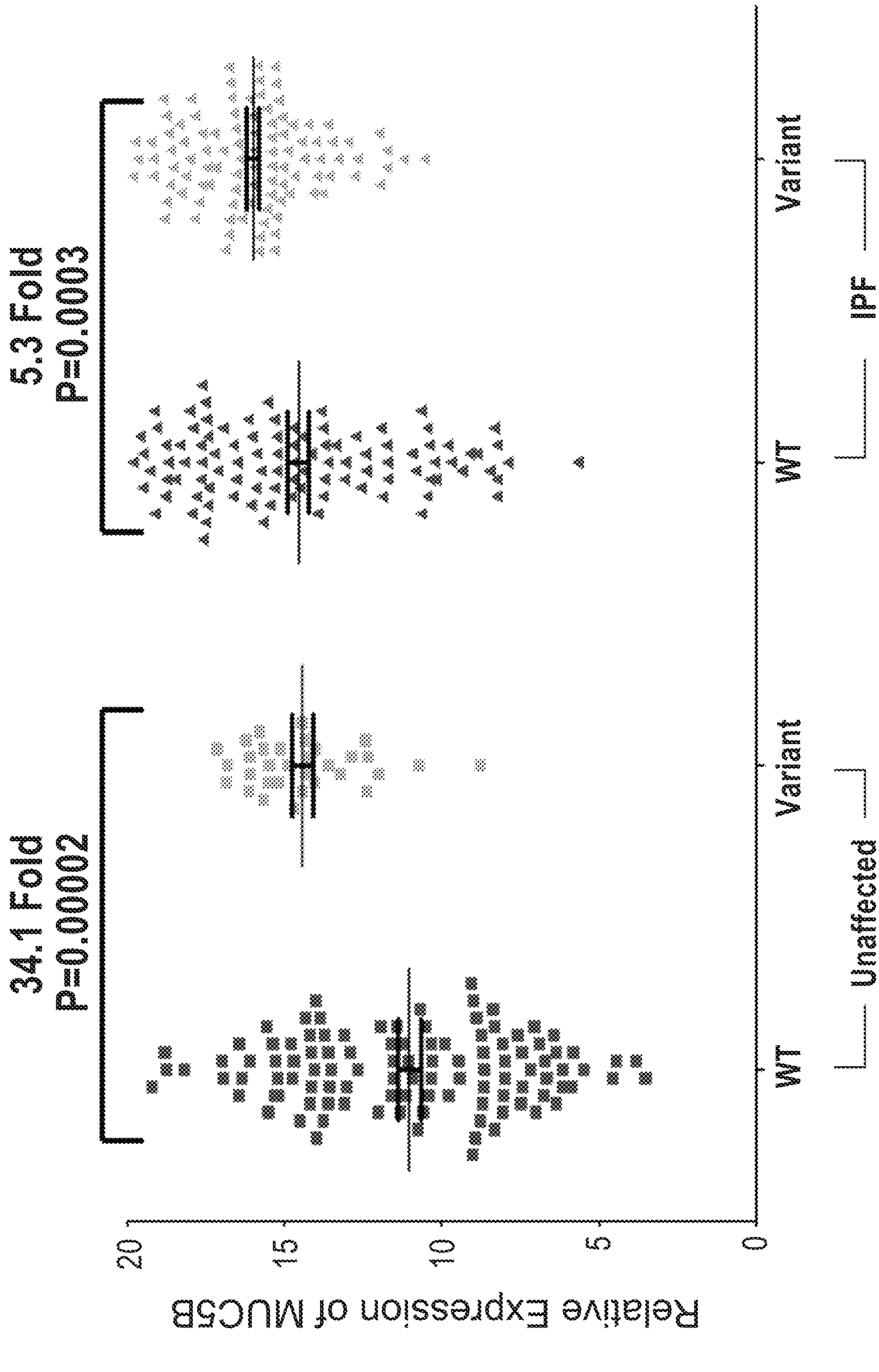

34.1 Fold
P=0.00002
5.3 Fold
P=0.0003
Relative Expression of MUC5B
20
15
10
5
0
WT
Variant
WT
Variant
Unaffected
IPF

FIG. 6B

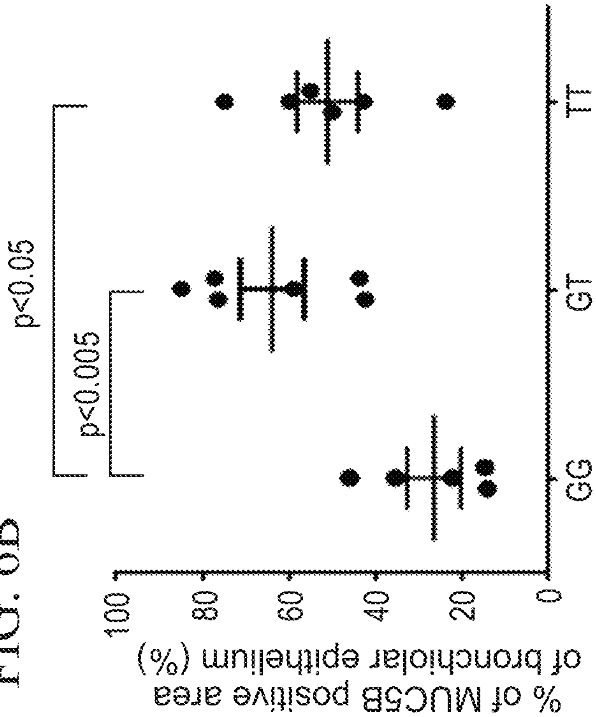
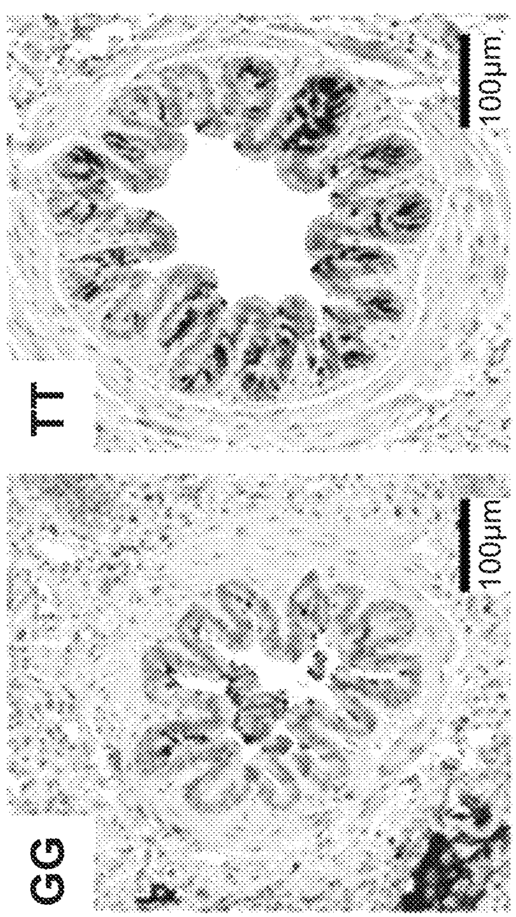
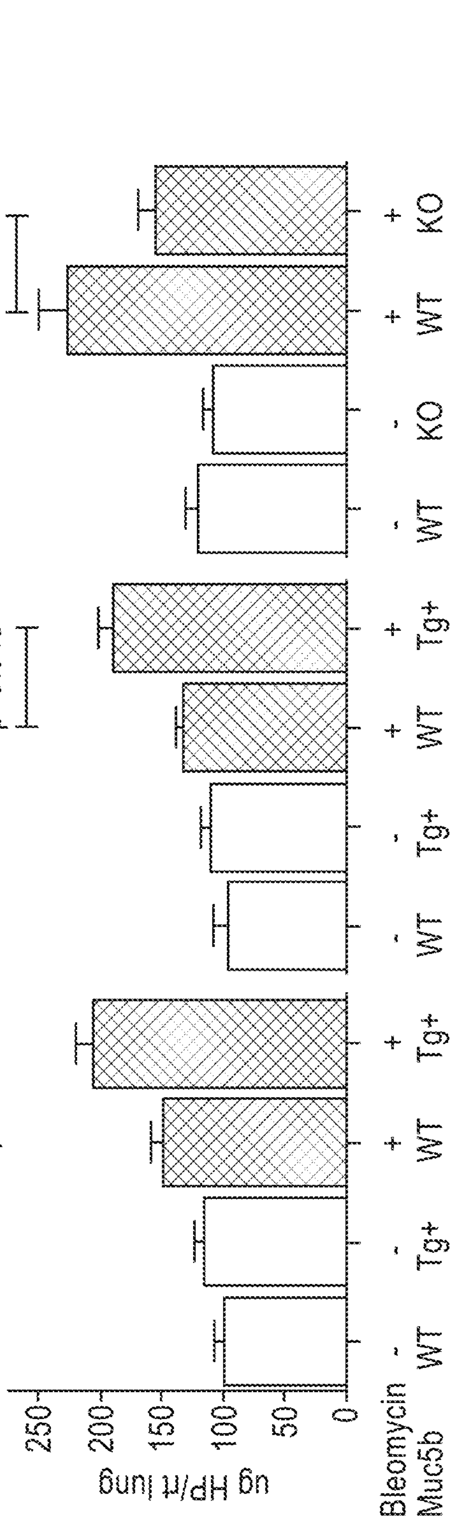

p<0.05
p<0.005
% of MUC5B positive area
of bronchiolar epithelium (%)
100
80
60
40
20
0
GG
GT
TT

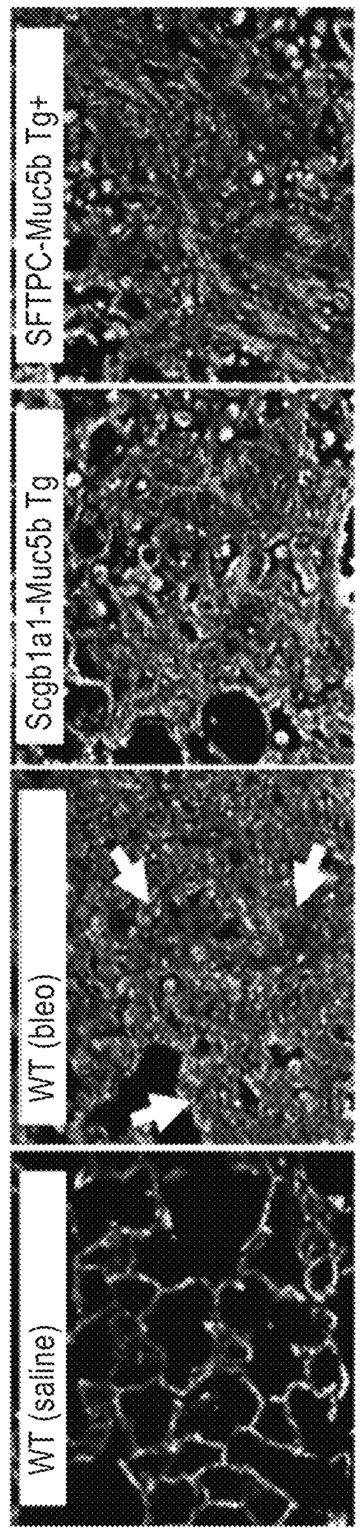
FIG. 6A


GG
TT
100μm
100μm



Muc5b-Scgb1a1
p=0.002
Muc5b-SPC
p=0.015
Muc5b-/-
p=0.015
ug HP/rt lung
250
200
150
100
50
0
Bleomycin
Muc5b
- WT
- Tg+
+ WT
+ Tg+
- WT
- Tg+
+ WT
+ Tg+
- WT
- KO
+ WT
+ KO

FIG. 7A



WT (saline)
WT (bleo)
Scgb1a1-Muc5b Tg
SFTPC-Muc5b Tg+

FIG. 7B

FIG. 8

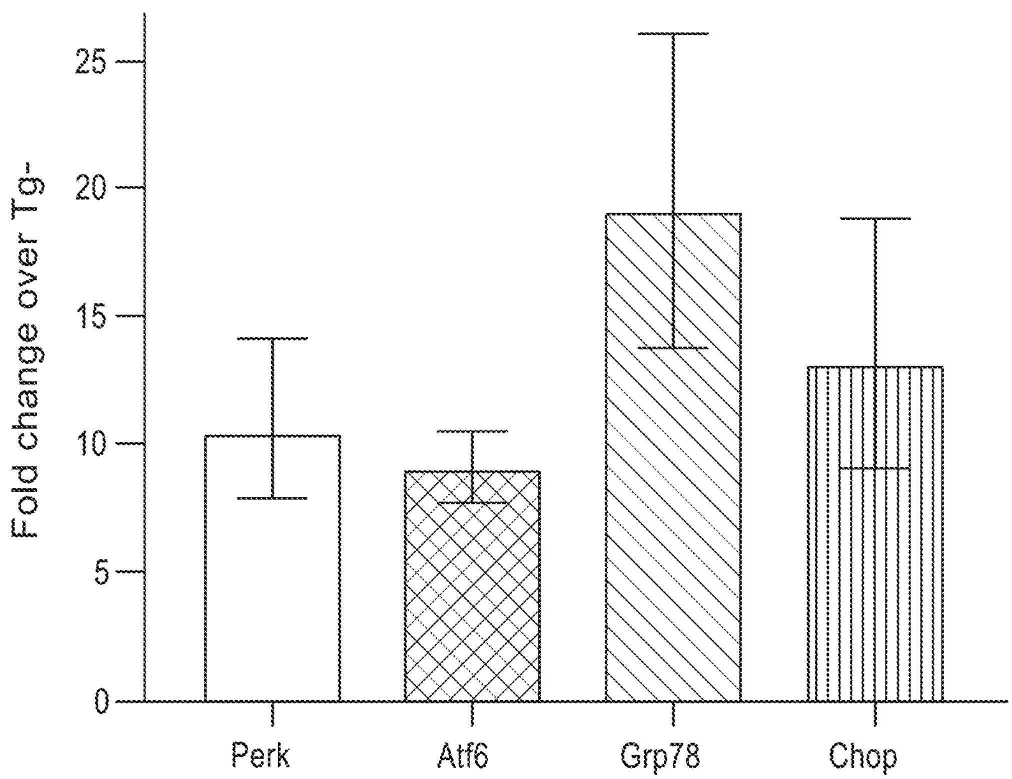
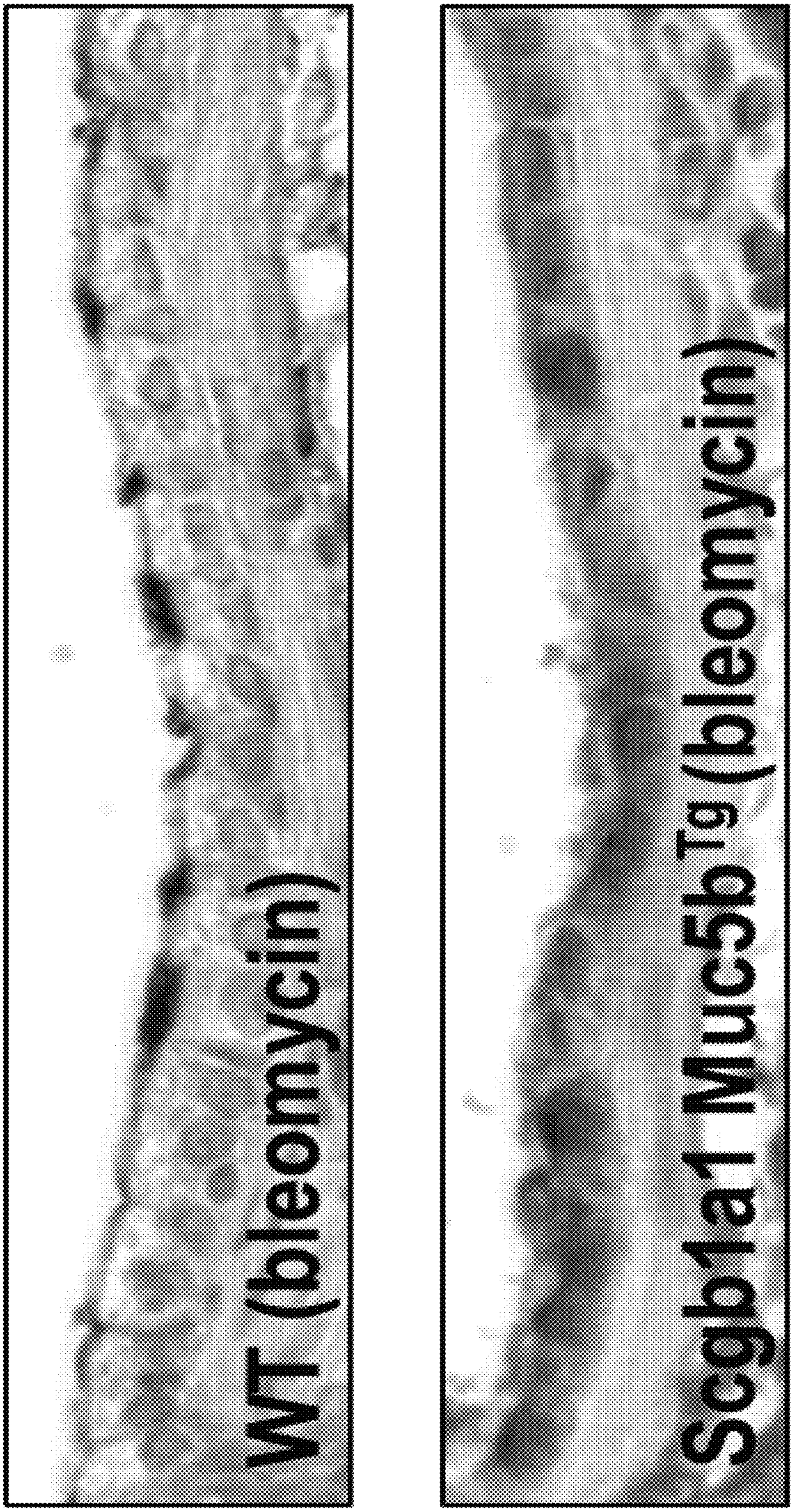


FIG. 9



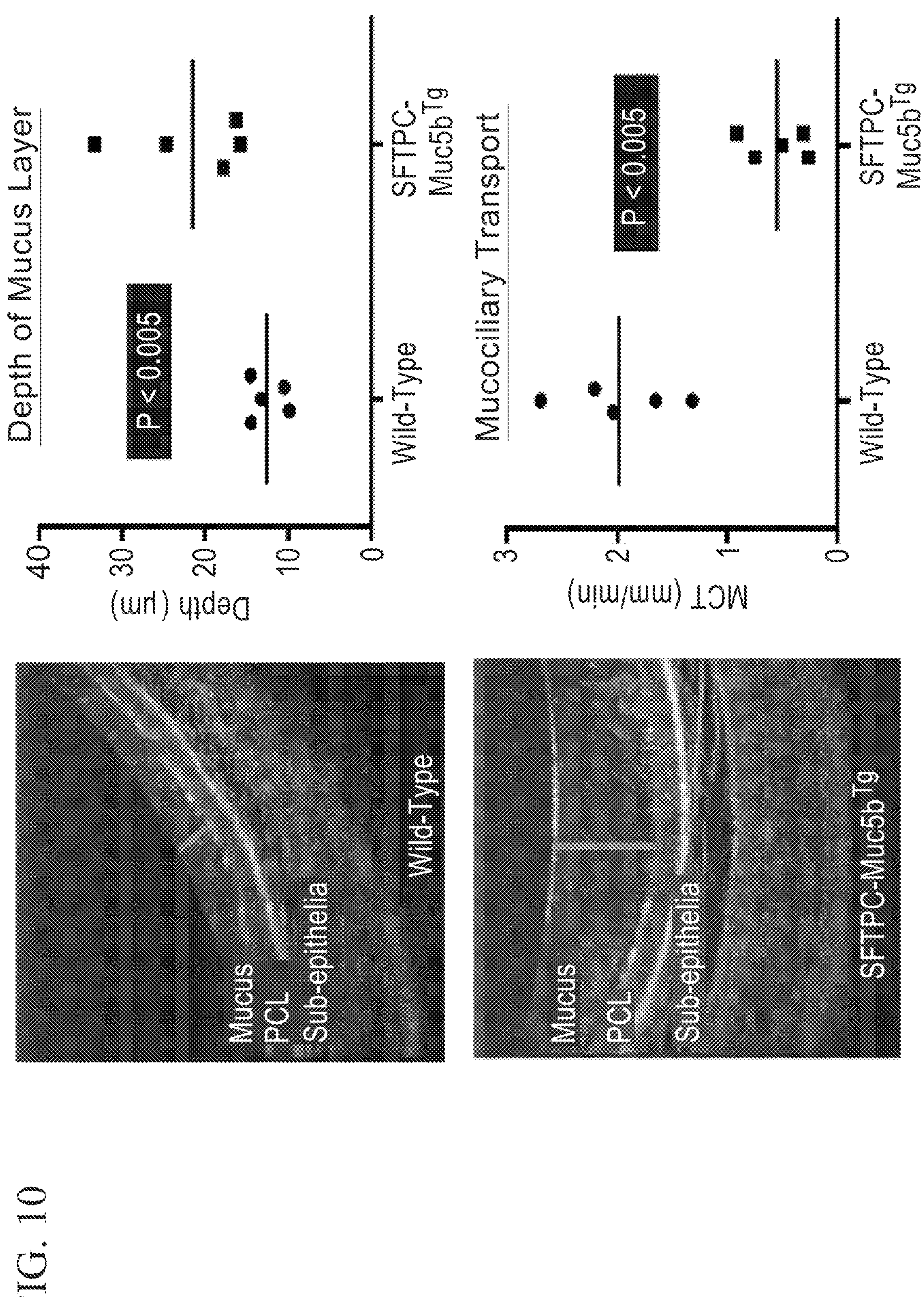

Depth of Mucus Layer
P < 0.005
Depth (μm)
40
30
20
10
0
Wild-Type
SFTPC-Muc5b Tg
Mucociliary Transport
P < 0.005
MCT (mm/min)
3
2
1
0
Wild-Type
SFTPC-Muc5b Tg
Mucus
PCL
Sub-epithelia
Wild-Type
Mucus
PCL
Sub-epithelia
SFTPC-Muc5b Tg

FIG. 10

FIG. 11

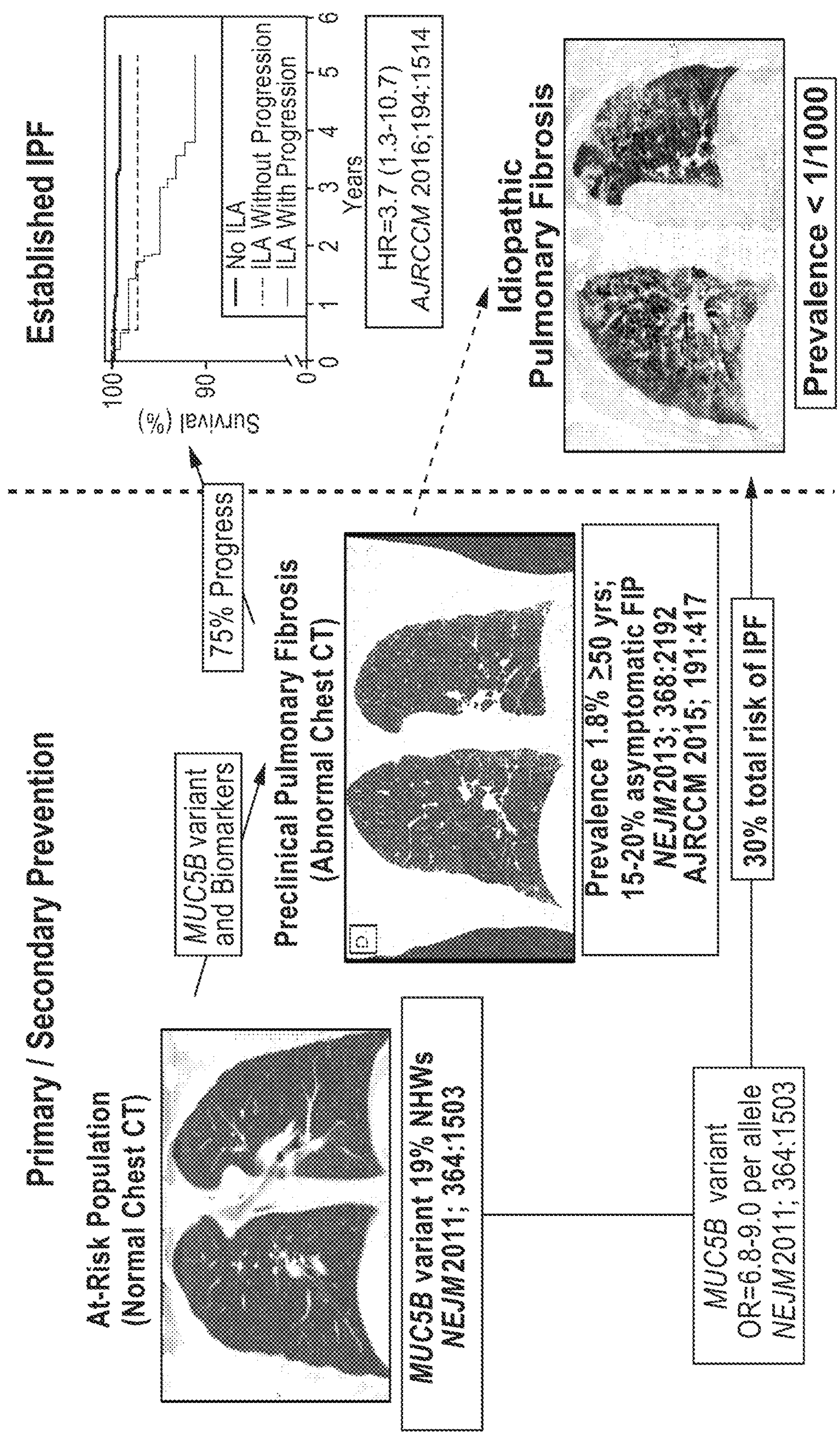

Primary / Secondary Prevention
Established IPF
At-Risk Population
(Normal Chest CT)
MUC5B variant 19% NHWs
NEJM 2011; 364:1503
MUC5B variant
and Biomarkers
Preclinical Pulmonary Fibrosis
(Abnormal Chest CT)
Prevalence 1.8% ≥50 yrs;
15-20% asymptomatic FIP
NEJM 2013; 368:2192
AJRCCM 2015; 191:417
75% Progress
Survival (%)
100
90
0
0
1
2
3
4
5
6
Years
No ILA
ILA Without Progression
ILA With Progression
HR=3.7 (1.3-10.7)
AJRCCM 2016;194:1514
Idiopathic
Pulmonary Fibrosis
Prevalence < 1/1000
30% total risk of IPF
MUC5B variant
OR=6.8-9.0 per allele
NEJM 2011; 364:1503

FIG. 12

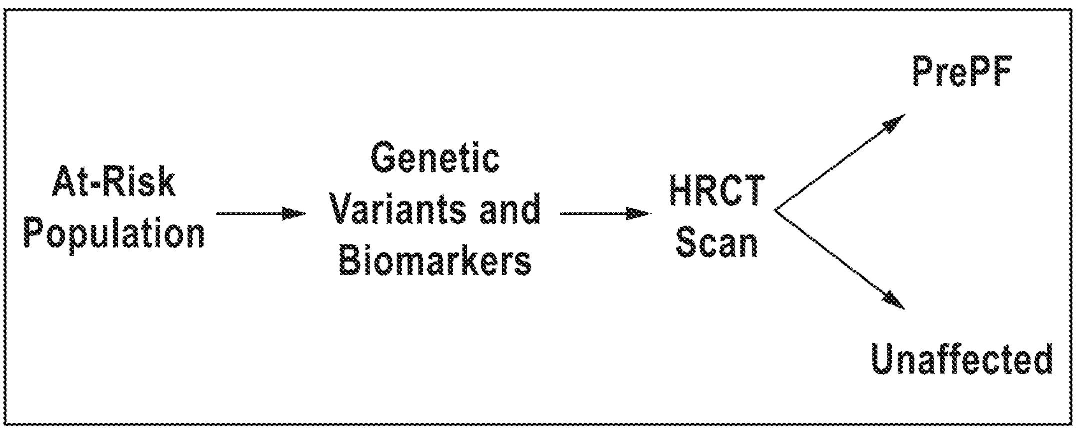

At-Risk Population
Genetic Variants and Biomarkers
HRCT Scan
PrePF
Unaffected

FIG. 13

Characteristic	RA-ILD (N=620)	RA-noILD (N=614)	Crude P Value	Adjusted P Value*
Female-no. (%)	345/565(61.1)	446/540 (82.6)	1.5×10^{-3}	3.7×10^{-12}
Age at inclusion	69.0±10.8	60.4±12.6	1.20×10^{-24}	1.3×10^{-21}
Age at RA onset-yr	55.7±14.6	45.7±13.5	7.0×10^{-23}	5.6×10^{-14}
RA duration-yr	13.3±11.5	14.8±10.2	0.034	0.38
Age at ILD onset-yr	62.7±11.8			
ILD duration-yr	4.3±4.0			
Ever smoker-no. (%)	282/516 (54.7)	168/465 (36.1)	4.6×10^{-4}	0.53
Methotrexate use ever[t]-no. (%)	260/318 (81.8)	142/153 (92.8)	0.42	0.69
RA manifestations				
ACPA and/or RF-positive-no. (%)	449/506 (88.7)	446/468 (95.3)	0.47	0.72
Erosive disease-no. (%)	224/482 (46.5)	274/469 (58.4)	0.045	0.30
Chest CT scan pattern				
UIP and possible UIP-no. (%)	207/505 (41.0)			
Inconsistent UIP-no. (%)	298/505 (59.0)			
Pulmonary function testing				
FVC% of predicted value	78.2±25.0			
DL_{CO}% of predicted value	57.6±23.4			
TLC% of predicted value	81.3±20.3			

RA:rheumatoid arthritis, ILD:interstitial lung disease, ACPA: anti-citrullinated protein antibody, RF: rheumatoid factor, CT computed tomography, UIP: usual interstitial pneumonia, FVC: forced vital capacity, DLco: diffusion capacity of carbon monoxide, TLC: total lung capacity

* Adjusted *P* value for sex and cohort.

FIG. 14

Cohorts	RA-ILD	RA-noILD	Controls	MUC5B rs35705950 Minor Allele Frequency			Genotypic Association Test RA-ILD vs Controls					Genotypic Association Test RA-noILD vs Controls					Genotypic Association Test RA-ILD vs RA-noILD		
	N of individuals			RA-ILD	RA-noILD	Controls	P value	Crude Odds Ratio for RA-ILD (95% CI)	P value	Adjusted* Odds Ratio for RA-ILD (95% CI)	P value	Crude Odds Ratio for RA-ILD (95% CI)	P value	Adjusted* Odds Ratio for RA-ILD (95% CI)	P value	Crude Odds Ratio for RA-ILD (95% CI)	P value	Adjusted** Odds Ratio for RA-ILD (95% CI)
				Percent														
France (Discovery)	118	105	1229	32.6	12.9	10.9	$3.8x10^{-17}$	3.8(2.8-5.2)	$9.7x10^{-17}$	3.8(2.8-5.2)	0.40	1.2(0.8-1.8)	0.28	1.3(0.8-1.9)	$5.9x10^{-6}$	3.8(2.2-6.8)	$9.4x10^{-4}$	3.1(1.6-6.3)
Greece	56	-	1795	26.8	-	3.8	$2.2x10^{-20}$	13.2(7.6-22.9)	$6.2x10^{-20}$	13.2(7.6-23.0)	-	-	-	-	-	-	-	-
The Netherlands	40	-	249	30.0	-	9.0	$5.0x10^{-7}$	5.6(2.9-11.2)	$1.2x10^{-4}$	4.9(2.2-11.5)	-	-	-	-	-	-	-	-
USA-1	99	68	500	28.8	11.0	1 0.7	$5.8x10^{-11}$	4.1(2.7-6.3)	$5.6x10^{-11}$	4.1(2.7-6.3)	0.91	1.0(0.6-1.8)	0.99	1.0(0.5-1.7)	$7.9x10^{-6}$	5.4(2.6-11.7)	NA	NA
USA-2	48	75	-	13.5	12.5	-	-	-	-	-	-	-	-	-	0.80	1.1(0.5-2.5)	NA	NA
Mexico	55	69	347	16.4	3.6	5.3	$1.1x10^{-4}$	3.4(1.8-6.2)	$2.2x10^{-4}$	3.6(1.8-7.3)	0.42	0.7(0.2-1.6)	0.42	0.7(0.2-1.7)	$1.5x10^{-3}$	5.7(2.1-18.6)	0.03	3.8(1.2-13.3)
Japan	182	300	315	1.1	0.5	0.2	0.08	7.1(1.0-138.6)	0.16	5.5(0.6-119.1)	0.32	3.2(0.4-64.3)	0.26	3.7(0.5-75.1)	0.30	2.2(0.5-11.4)	0.30	3.1(0.3-28.0)
China	22	-	1013	2.3	-	0.8	0.30	3.0(0.2-15.6)	0.14	4.9(0.3-27.5)	-	-	-	-	-	-	-	-
Multi Ethnic Replication Sample	454	437	4219	-	-	-	$3.9x10^{-35}$	5.5(4.2-7.2)	$4.7x10^{-35}$	5.5(4.2-7.3)	0.90	1.0(0.6-1.5)	0.83	1.0(0.6-1.5)	$5.3x10^{-7}$	3.1(2.0-5.0)	0.04	2.9(1.1-8.4)
Combined Analysis	572	542	5448	-	-	-	$1.3x10^{-49}$	4.7(3.8-5.8)	$1.3x10^{-49}$	4.7(3.9-5.8)	0.60	1.1(0.8-1.5)	0.54	1.1(0.8-1.5)	$1.6x10^{-11}$	3.4(2.4-4.8)	$7.4x10^{-5}$	3.1(1.8-5.4)

RA: rheumatoid arthritis, ILD: interstitial lung disease, RA-ILD: patients with rheumatoid arthritis associated interstitual lung disease, RA-noILD: rheumatoid arthritis patients without interstitial lung disease.

*P values and odds ratios in the category were adjusted for sex and cohort.

** P values and odds ratios in this category were adjusted for sex, age at inclusion, ever smoking status and cohort.

NA: All above-cited covariates not available for adjusting

FIG. 15

Cohorts	**RA-ILD Cases**		***MUC5B* rs35705950 MinorAllele Frequency**		***Genotypic Association Test***			
	RA-UIP	RA-nonUIP	RA-UIP	RA-nonUIP	*P* value	Crude Odds Ratio for RA-UIP (95%Cl)	*P* value	Adjusted* Odds Ratio for RA-UIP (95%Cl)
	N of individuals		*percent*					
France (Discovery)	50	31	34.0	12.9	3.9×10^{-4}	6.1(2.3-17.5)	2.7×10^{-3}	4.9(1.8-14.6)
Greece	18	38	36.1	21.1	0.04	3.6(1.1-13.1)	0.12	2.9(0.8-12.1)
The Netherlands	18	22	33.3	25.0	0.29	2.0(0.6-7.6)	0.51	1.6(0.4-6.7)
USA-1	34	42	33.8	23.8	0.08	2.3(0.9-6.0)	0.18	2.1(0.7-6.3)
Mexico	19	36	28.9	8.3	2.8×10^{-3}	6.9(2.0-26.0)	0.07	3.8(0.9-16.8)
Japan	60	122	1.7	0.8	0.47	2.1(0.2-17.6)	0.99	NA[t]
China	8	7	0	7.1	1.0	NA[t]	1.0	NA[t]
Multi-Ethnic Replication Sample	157	267	-	-	1.5×10^{-4}	2.9(1.7-5.0)	6.3×10^{-3}	2.3(1.3-4.1)
Combined Analysis	207	298	-	-	3.6×10^{-7}	3.5(2.2-5.6)	5.1×10^{-5}	2.9(1.7-4.8)

RA: rheumatoid arthritis, UIP: usual interstitial pneumonia, ILD: interstitial lung disease
The RA-UIP subset includes RA-ILD patients with the following HRCT scan patterns: usual interstitial pneumonia and possible usual interstitial pneumonia.

FIG. 16A

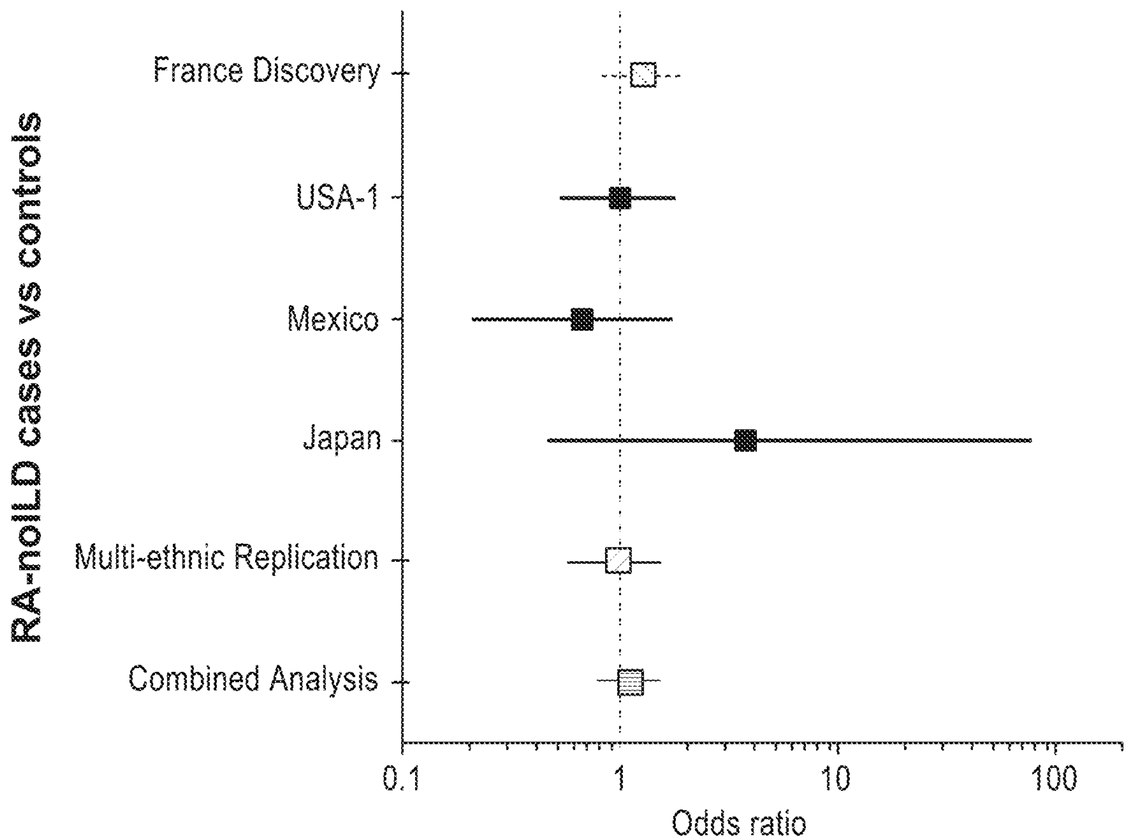

RA-noILD cases vs controls
France Discovery
USA-1
Mexico
Japan
Multi-ethnic Replication
Combined Analysis
0.1
1
10
100
Odds ratio

FIG. 16B

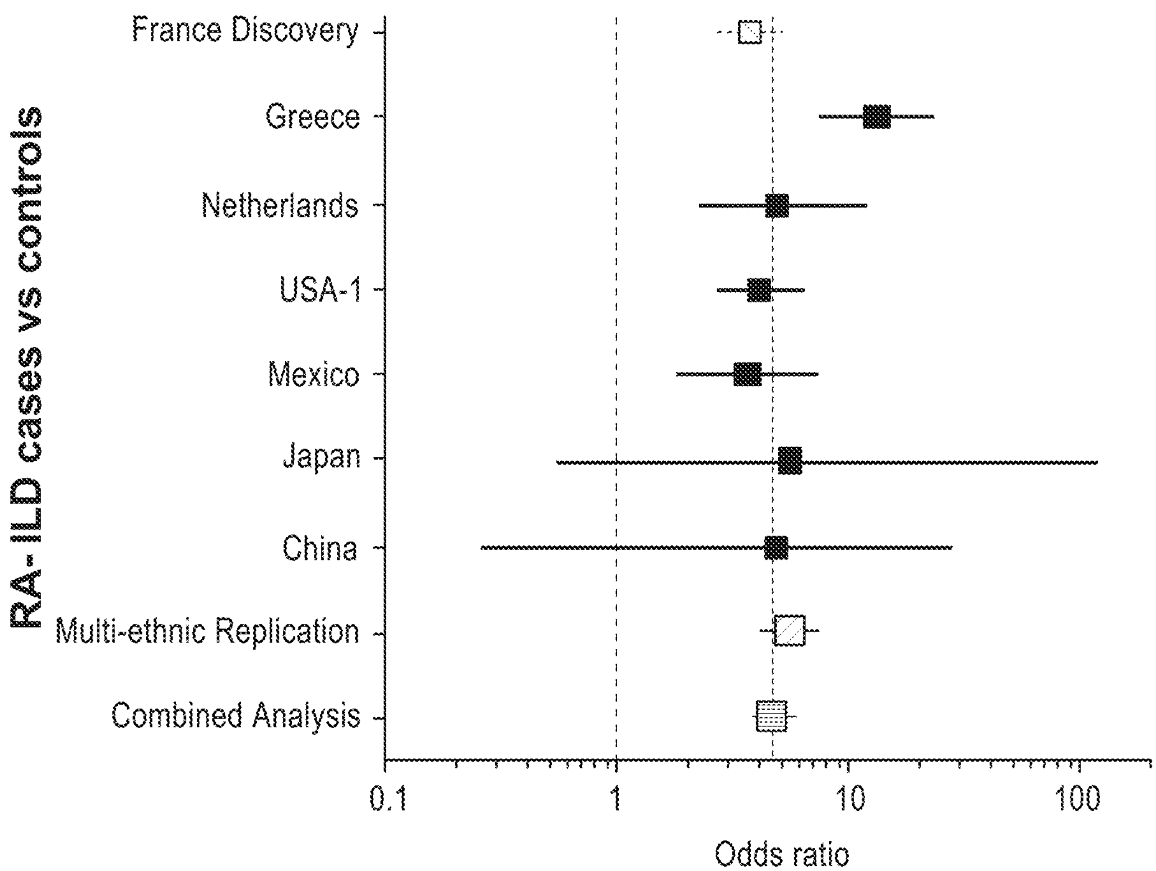

RA- ILD cases vs controls
France Discovery
Greece
Netherlands
USA-1
Mexico
Japan
China
Multi-ethnic Replication
Combined Analysis
0.1
1
10
100
Odds ratio

FIG. 16C

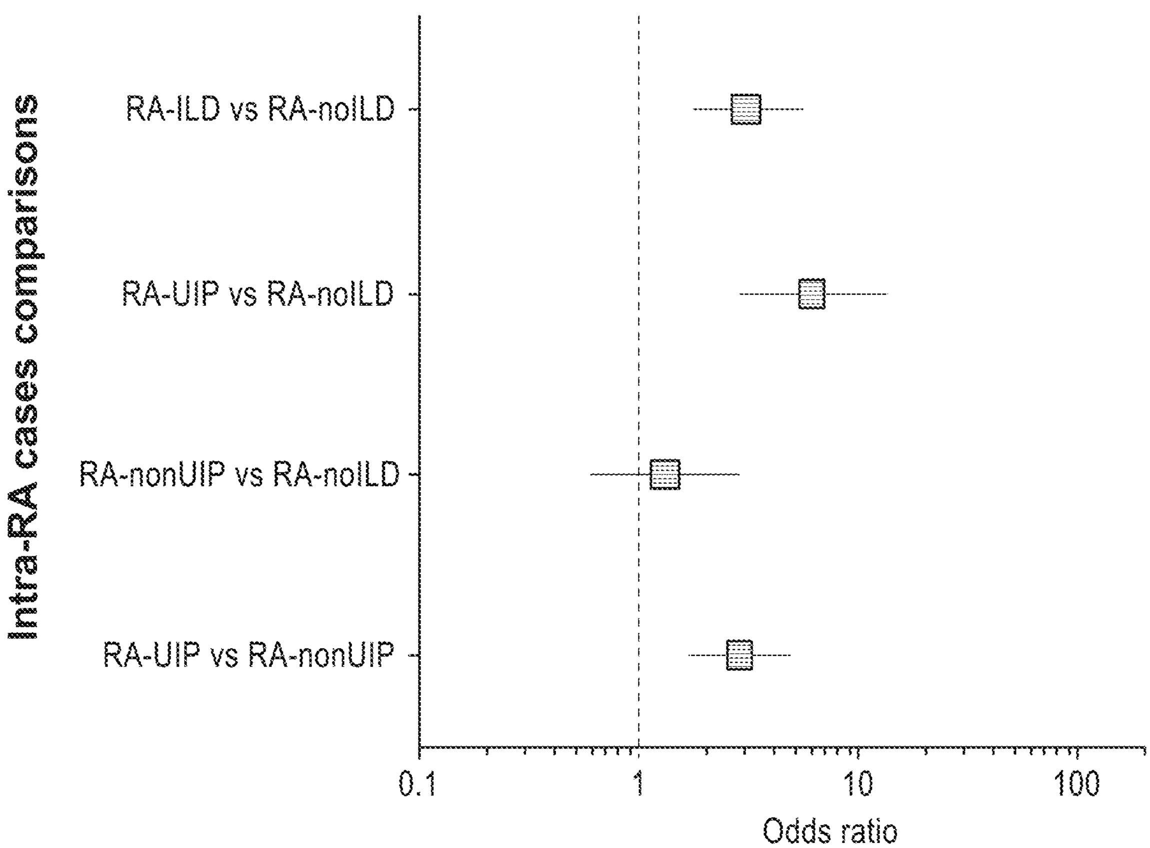

Intra-RA cases comparisons
RA-ILD vs RA-noILD
RA-UIP vs RA-noILD
RA-nonUIP vs RA-noILD
RA-UIP vs RA-nonUIP
0.1
1
10
100
Odds ratio

FIG. 17

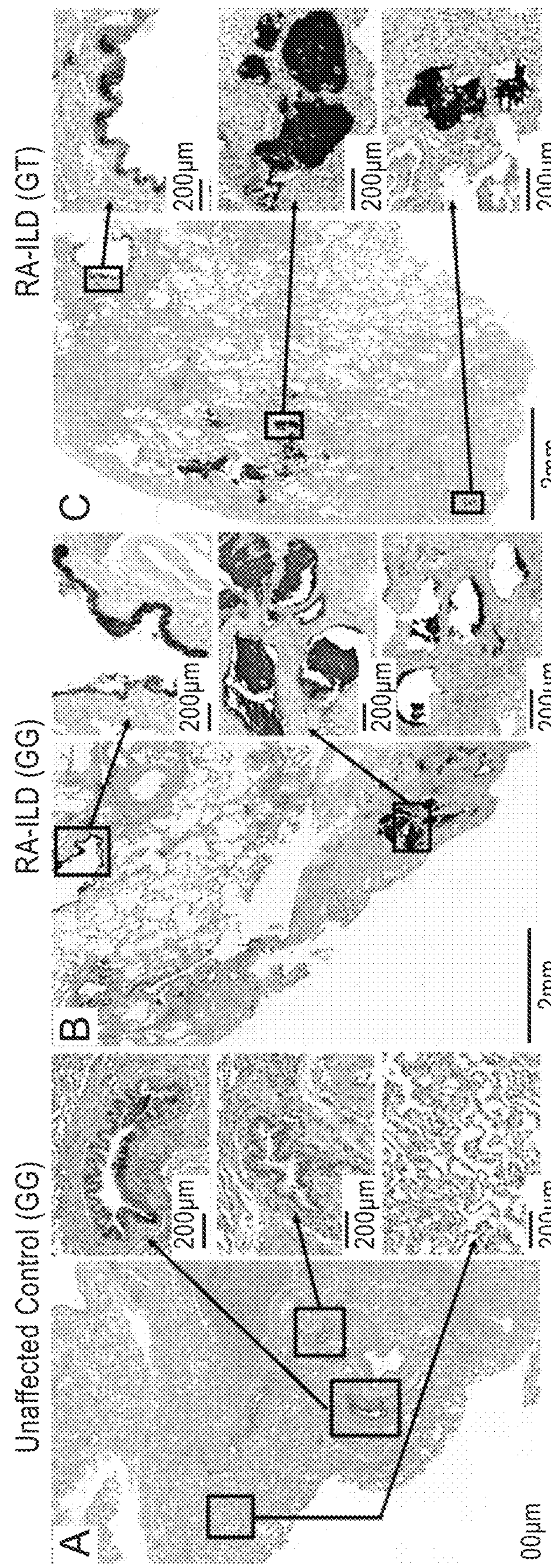
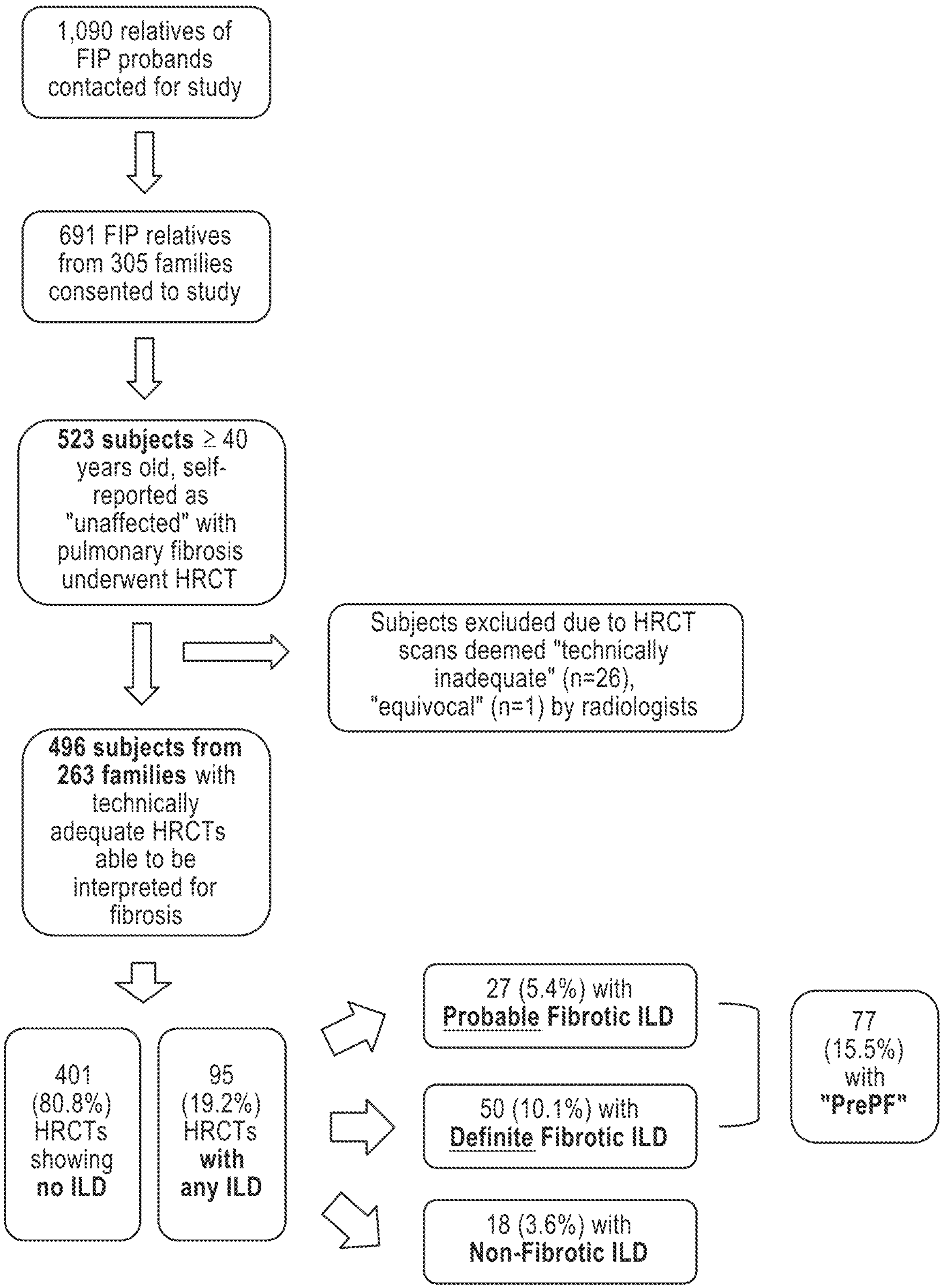

A
Unaffected Control (GG)
00µm
200µm
200µm
200µm
B
RA-ILD (GG)
2mm
200µm
200µm
200µm
C
RA-ILD (GT)
2mm
200µm
200µm
200µm

FIG. 18


1,090 relatives of FIP probands contacted for study
691 FIP relatives from 305 families consented to study
523 subjects ≥ 40 years old, self-reported as "unaffected" with pulmonary fibrosis underwent HRCT
Subjects excluded due to HRCT scans deemed "technically inadequate" (n=26), "equivocal" (n=1) by radiologists
496 subjects from 263 families with technically adequate HRCTs able to be interpreted for fibrosis
401 (80.8%) HRCTs showing no ILD
95 (19.2%) HRCTs with any ILD
27 (5.4%) with Probable Fibrotic ILD
50 (10.1%) with Definite Fibrotic ILD
18 (3.6%) with Non-Fibrotic ILD
77 (15.5%) with "PrePF"

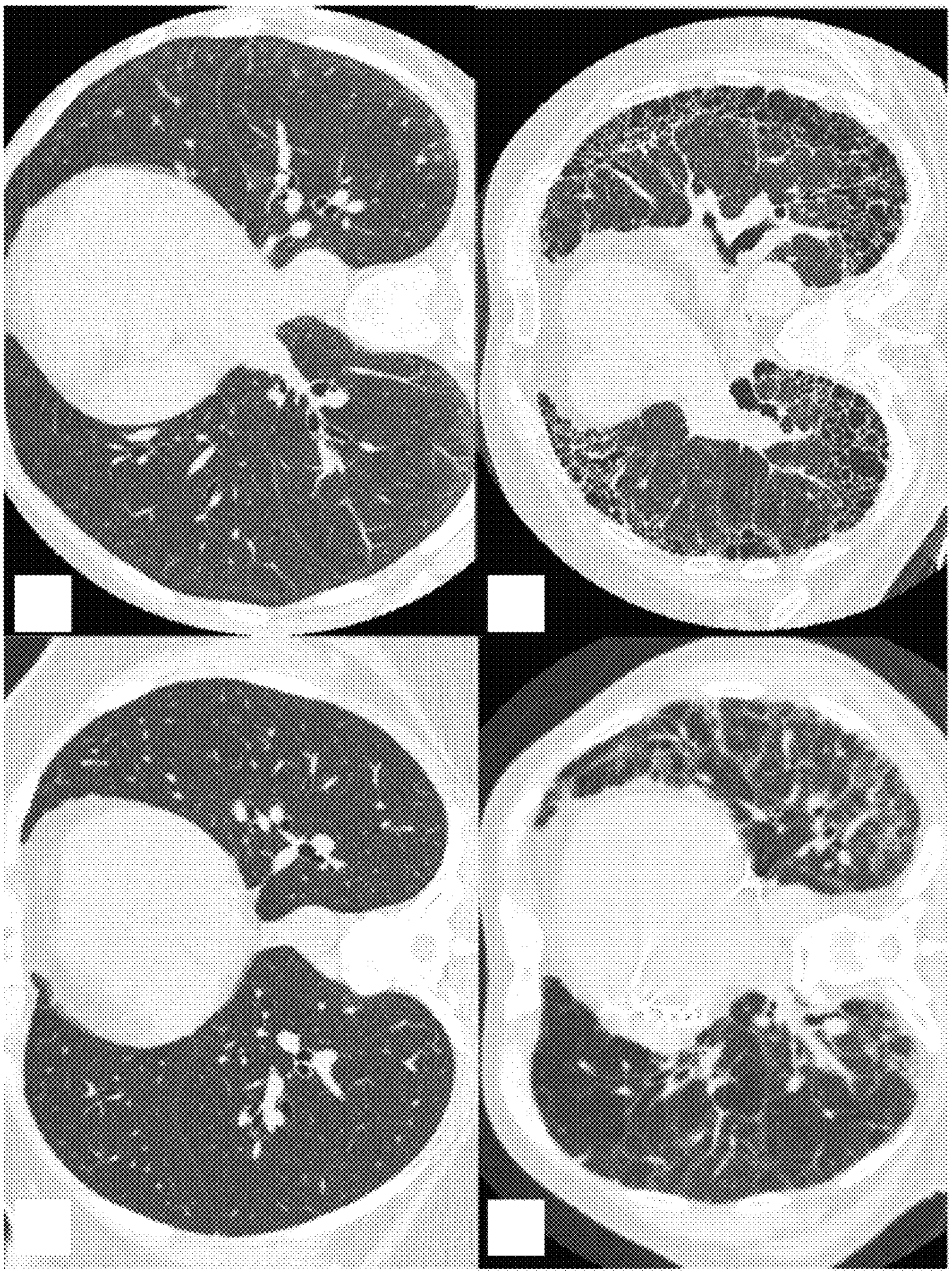


FIG. 19A

FIG. 19B

FIG. 19C

FIG. 19D

FIG. 20

	COPDGene Nonsmoking Controls (n=100)	FIP Relative in Dataset (n=402)
Age, mean (95%CI)	57.2 (55.3-59.0)	57.4 (56.5-58.3)
Male, n (%)	47 (47%)	153 (38%)
rs35705950 minor allele frequency*	NA	0-22

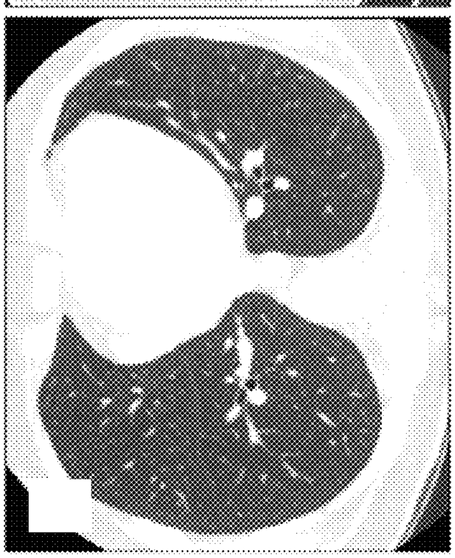


FIG. 21A

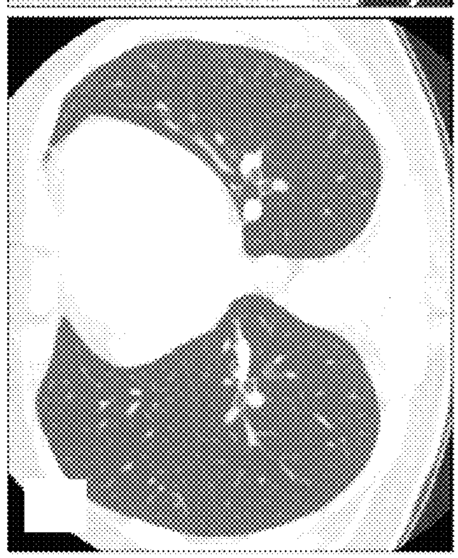


FIG. 21B

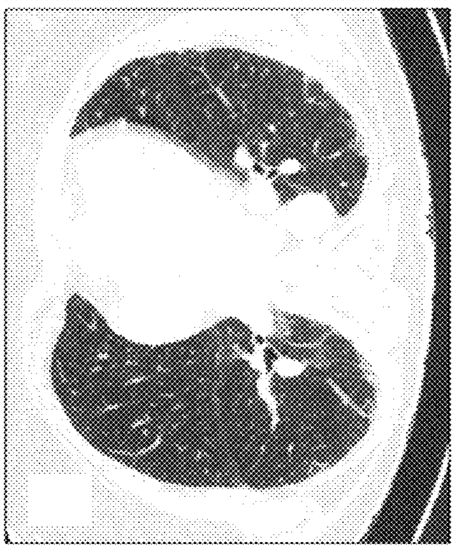


FIG. 21C

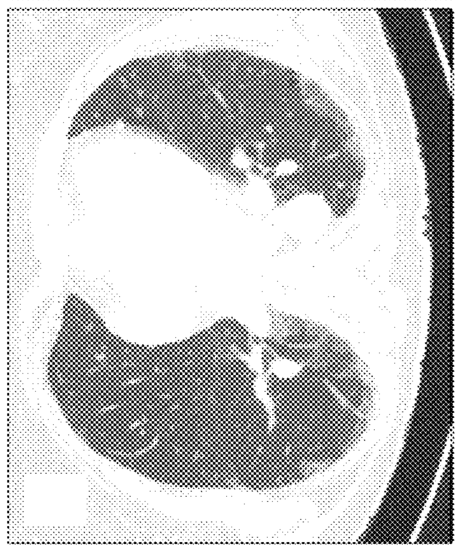


FIG. 21D

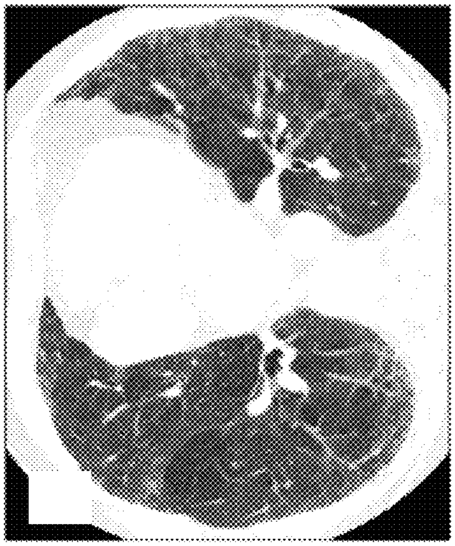


FIG. 21E

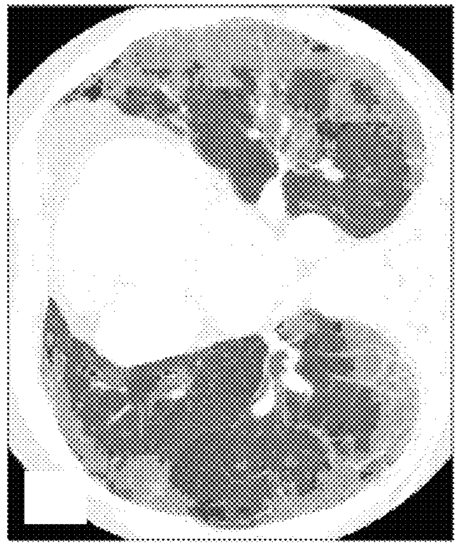


FIG. 21F

FIG. 22

	No Fibrosis (n=419)	PrePF (n=77)	p-value	OR (95% CI) controlling for family**	Mixed effects logistic regression p-value
Age, mean (95%CI)	55·8 (54·9-56·6)	65·8 (63.5-68.1)	$6·36 \times 10^{-13}$	*1·15 (1·09-1·22)*	*$7·34x10^{-7}$*
Male, %	36%	48%	0·05	*1·80 (0·88-3·68)*	*0·11*
Ever smoker, %	27%	48%	0·007	*1·46 (0·70-3·06)*	*0·31*
MUC5B Promoter Variant (rs35705950), MAF (% subjects with variant)*	0·21 (40%)	0·29 (53%)	0·03	*2·18 (1·00-4·73)*	*0·05****
TERT Common Variant (rs2736100), MAF (%subjects with variant)*	0·45 (68%)	0·45 (67%)	0·88	*NA*	*NA*

FIG. 23

Total with Fibrotic ILD	**77**
Cranio-caudal distribution	
Upper	11 (14%)
Middle	5 (7%)
Lower	54 (70%)
Diffuse	4 (5%)
Not noted	3 (4%)
Axial distribution	
Subpleural	67 (87%)
Diffuse	5 (6·5%)
Peribronchovasular	2 (2.5%)
Not noted	3 (4%)
Honeycombing?	**12 (15·6%)**
CT pattern*	
UIP	**59 (77·7%)**
Possible	41 (70%)
Probable	9 (15%)
Definite	9 (15%)
NSIP	**45 (58·4%)**
Possible	41 (91%)
Probable	2 (4%)
Definite	2 (4%)
Sarcoidosis	**3 (3·9%)**
Hypersensitivity Pneumonitis	**14 - Possible (18·2%)**

FIG. 24

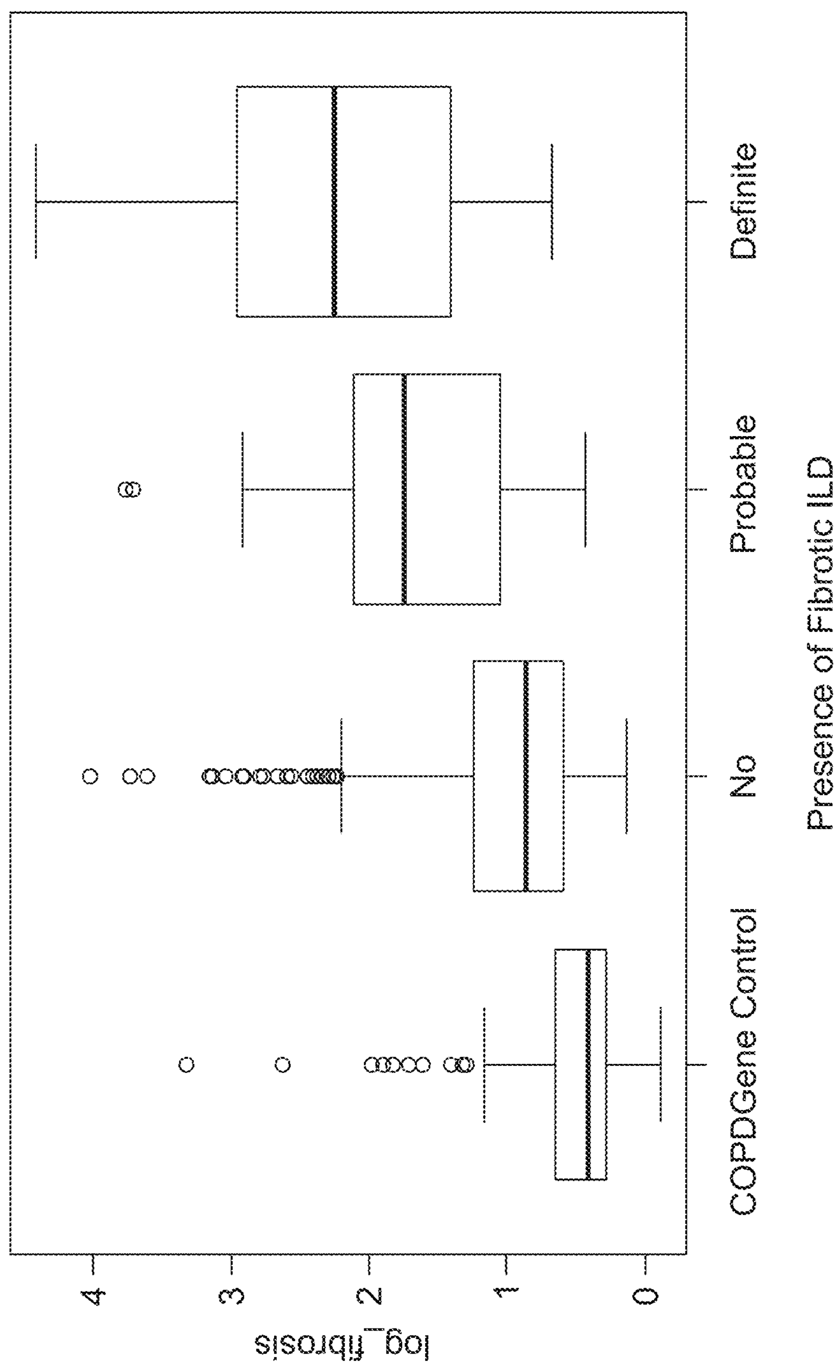

log_fibrosis
0
1
2
3
4
COPDGene Control
No
Probable
Definite
Presence of Fibrotic ILD

FIG. 25A

ROC for visual dx vs log HAA

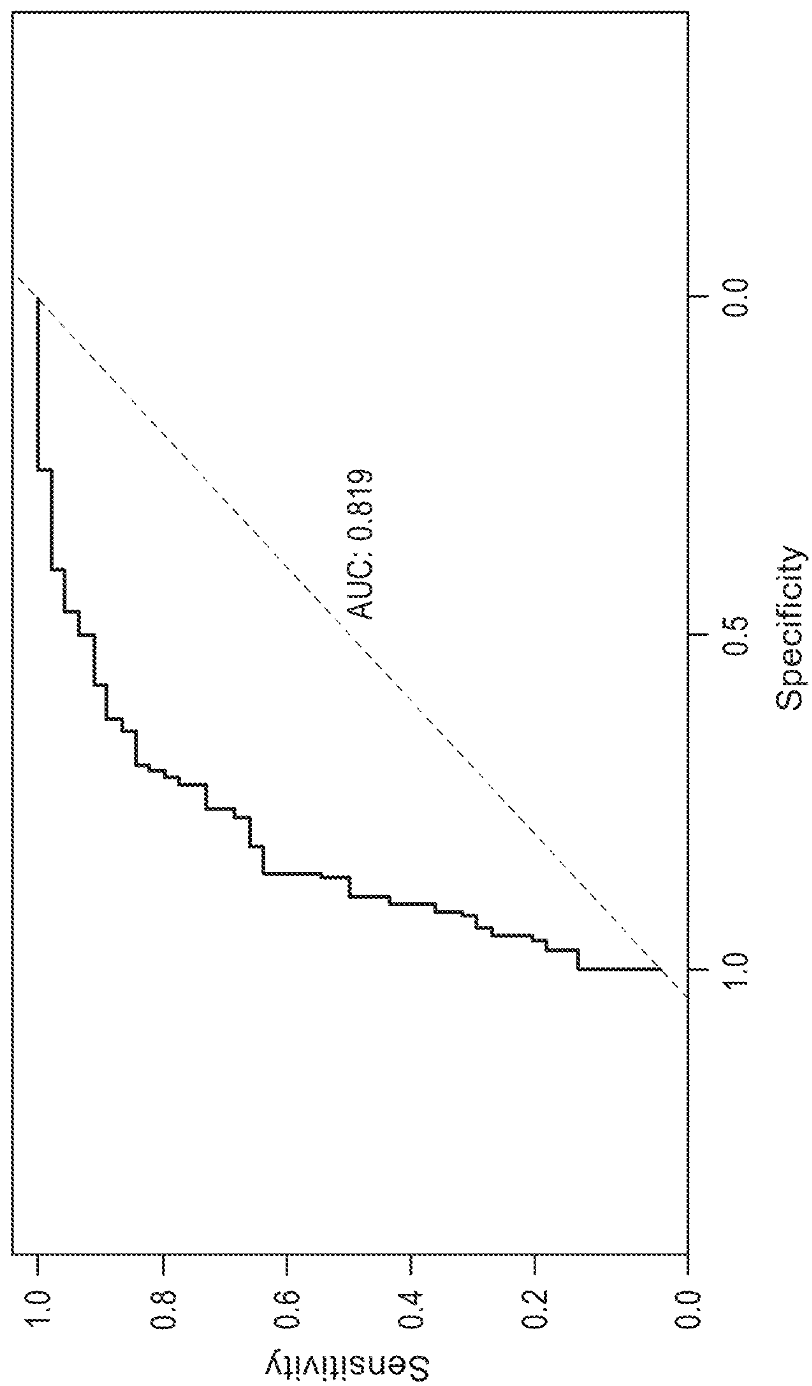


FIG. 25B

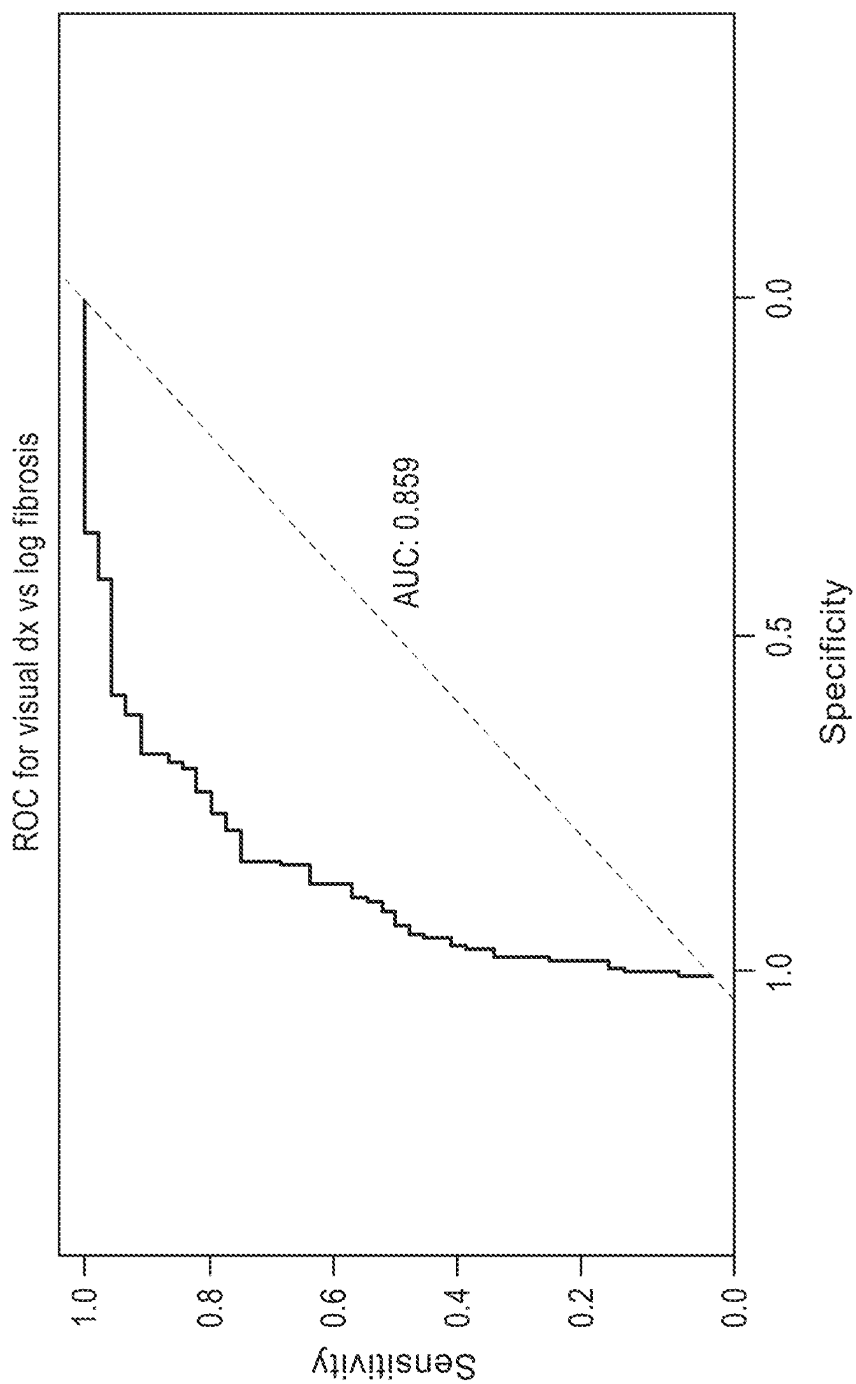

ROC for visual dx vs log fibrosis
AUC: 0.859
Sensitivity
1.0
0.8
0.6
0.4
0.2
0.0
Specificity
1.0
0.5
0.0

FIG. 25C

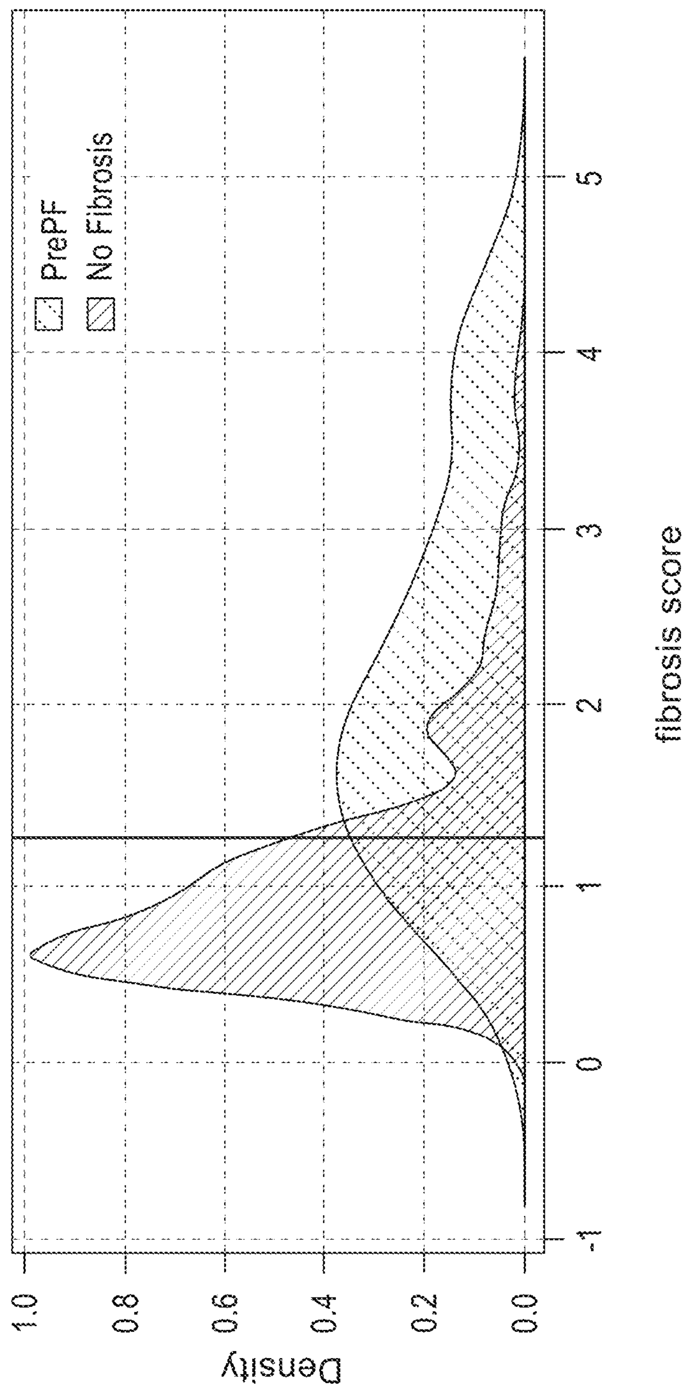

PrePF
No Fibrosis
Density
1.0
0.8
0.6
0.4
0.2
0.0
fibrosis score
-1
0
1
2
3
4
5

FIG. 26

A. Breathlessness Responses for Cohort:	Yes	No	No answer
Are you troubled by shortness of breath when hurrying on the level or walking up a slight hill?	121	344	31
Do you have to walk slower than people of your age on the level because of breathlessness?	46	422	28
Do you ever have to stop for breath when walking at your own pace on the level?	32	442	22
Do you ever have to stop for breath after walking about 100 yards (or after a few minutes)?	36	439	21
Are you too breathless to leave the house or breathless dressing or undressing?	7	470	19

B. Breathlessness Responses by Visual CT Diagnosis:	PrePF (n=77)	No Fibrosis (n=419)
Are you troubled by shortness of breath when hurrying on the level or walking up a slight hill?	43 no 26 yes (37%) 8 no answer	301 no 95 yes (24%) 23 no answer
Do you have to walk slower than people of your age on the level because of breathlessness?	60 no 9 yes (13%) 8 no answer	362 no 37 yes (9.3%) 20 no answer
Do you ever have to stop for breath when walking at your own pace on the level?	64 no 6 yes (8.6%) 7 no answer	378 no 26 yes (6.4%) 15 no answer
Do you ever have to stop for breath after walking about 100 yards (or after a few minutes)?	66 no 6 yes (8.3%) 5 no answer	373 no 30 yes (7.4%) 16 no answer
Are you to breathless to leave the house or breathless dressing or undressing?	71 no 1 yes (1.4%) 5 no answer	399 no 6 yes (1.4%) 14 no answer

FIG. 27

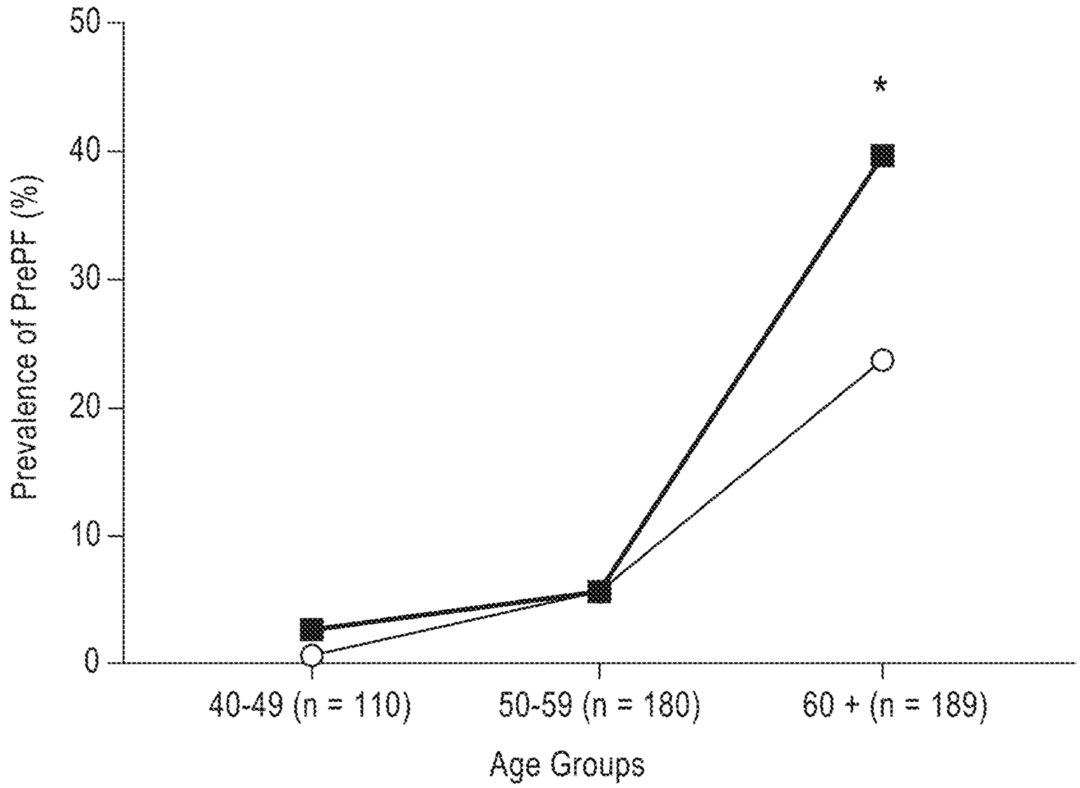

50
40
30
20
10
0
Prevalence of PrePF (%)
*
40-49 (n = 110)
50-59 (n = 180)
60 + (n = 189)
Age Groups

FIG. 28

	Fibrosis score negative (n=295)	Fibrosis score positive (n=107)	p-value*	Linear regression, p-value
Age, mean (95%CI)	55·4 (54·5-56·3)	62·9 (60·8-64·9)	$7·05 \times 10^{-10}$	$2·17 \times 10^{-9}$
Male, n (%)	112 (38%)	41 (38%)	0·95	0·63
Ever smoker, n (%)	87 (29.8%)	32 (29.9%)	0·98	0·94
MUC5B Promotor Variant (rs35705950), MAF (% subjects with variant)	0·21 (39·4%)	0·26 (48·6%)	0·12	0·03**

FIG. 29

SNP	Position*	Minor allele	Locus	Nearest gene	Minor allele frequency (%)						Genotypic association test			
					France		USA		Mexico		Crude		Adjusted**	
					RA-ILD n=118	RA-noILD n=105	RA-ILD n=99	RA-noILD n=68	RA-ILD n=55	RA-noILD n=69	Odds Ratio (95%CI)	P value	Odds Ratio (95%CI)	P value
rs6793295	169518455	C	3q26	*LRRC34*	21.9	21,2	29,7	32,1	60,8	60,9	0.89(0.68-1.15)	0.36	0.84(0.56-1.25)	0.39
rs2609255	89811195	G	4q22	*FAM/3A*	28.6	28.7	25.3	25.4	29.4	37.5	0.89(0.67-1.16)	0.38	0.89(0.62-1.29)	0.54
rs2736100	1286516	A	5p15	*TERT*	46.5	42.2	48.1	47.8	71.6	71.7	1.00(0.79-1.27)	0.99	1.18(0.84-1.67)	0.35
rs7887	31864547	T	6p21.3	*EHMT2*	42.3	38.6	36.7	42.5	58.8	72.5	0.82(0.64-1.04)	0.10	0.91(0.64-1.29)	0.59
rs2076295	7563232	G	6p24	*DSP*	50.0	50.0	44.3	46.3	31.0	34.0	1.01(0.80-1.28)	0.94	1.03(0.73-1.46)	0.85
rs4727443	99593346	A	7q22	*Intergenic*	35.2	46.0	41.8	35.8	50.0	53.7	0.83(0.65-1.07)	0.15	0.76(0.53-1.07)	0.11
rs11191865	105672842	G	10q24	*OBFC/*	49.5	46.8	38.0	55.4	37.3	43.4	0.81(0.62-1.05)	0.11	1.13(0.78-1.65)	0.52
rs35705950	**1241221**	**T**	**11p15.5**	***MUCSB***	**32.6**	**12.9**	**28.8**	**11.0**	**16.4**	**3.6**	**3.63(2.54-5.31)**	**7.31×10^{-12}**	**3.00(1.6-4.99)**	**$1.15E \times 10^{-05}$**
rs5743890	1325829	C	11p15.5	*TOLLIP*	21.5	14.2	16.7	15.1	10.0	4.5	1.61(1.06-2.48)	0.03	2.13(1.13-4.10)	0.02
rs111521887	1312706	G	11p15.5	*TOLLIP*	16.7	18.0	19.2	12.3	6.9	4.4	1.33(0.89-1.98)	0.16	0.96(0.55-1.69)	0.89
rs1278769	113536627	A	13q34	*ATP//A*	19.6	26.1	23.4	23.8	12.7	18.1	0.83(0.61-1.12)	0.22	0.64(0.41-0.99)	0.05
rs2034650	40717302	G	15q14-15	*IVD*	43.3	53.7	41.0	48.4	27.5	38.1	0.71(0.54-0.95)	0.02	0.59(0.38-0.89)	0.01
rs12610495	4717672	G	19p13	*DPP*	31.4	33.0	32.9	31.1	19.6	12.5	1.14(0.87-1.50)	0.33	1.19(0.82-1.75)	0.37

*Based on GRCh 37/hg19 database

***P* values and odds ratios in this category were adjusted for sex, age at inclusion, ever smoking status and case-series origin.

BIOMARKERS FOR THE DIAGNOSIS AND TREATMENT OF FIBROTIC LUNG DISEASE

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/624,500, filed Dec. 19, 2019, which application is a National Stage Application, filed under 35 U.S.C. § 371, of PCT/US2018/039573, filed Jun. 26, 2018, which claims the benefit of provisional application U.S. Ser. No. 62/525,087, filed Jun. 26, 2017 and U.S. Ser. No. 62/525,088, filed Jun. 26, 2017, the contents of each of which are herein incorporated by reference in their entirety.

GOVERNMENT SUPPORT

This invention was made with government support under grant number HL097163, HL123442, and HL138131 awarded by National Institutes of Health and grant number W81XWH-17-1-0597 awarded by Department of Defense. The government has certain rights in the invention.

INCORPORATION OF SEQUENCE LISTING

The Sequence Listing XML associated with this application is provided electronically in XML format and is hereby incorporated by reference into the specification. The name of the XML file containing the Sequence Listing XML is "UNCO-018_C01US_SeqList_ST26.xml". The XML file is 227,030 bytes, created on Sep. 7, 2022, and is being submitted electronically via USPTO Patent Center.

FIELD OF THE DISCLOSURE

The disclosure is directed to molecular biology, genetics, and therapeutics for fibrotic lung disease.

BACKGROUND

Fibrotic pulmonary diseases are progressive and irreversible. Standard therapies are mere palliative as they cannot address the underlying disease mechanism once the subject has progressed to a point at which symptoms are present. Thus, there is a long-felt but unmet need in the field for a method of treating asymptomatic subjects as well as those who are at risk of developing fibrotic pulmonary diseases to prevent onset of the disease, delay onset of the disease, or reduce the severity of disease symptoms. The methods of the disclosure provide a preventative or efficacious treatment, as opposed to a merely palliative treatment, for asymptomatic subjects as well as those subjects at risk of developing the disease.

SUMMARY

The disclosure provides a method of treating a fibrotic lung disease in a subject comprising administering to the subject an effective amount of a therapeutic agent, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease.

In some embodiments of the methods of the disclosure, the subject presents radiographic Usual Interstitial Pneumonia (UIP). In some embodiments, the subject has fibrotic interstitial lung disease (FILD). In some embodiments, the subject has a blood relative with familial interstitial pneumonia (FIP). In some embodiments, including those embodiments wherein the subject has a blood relative with familial interstitial pneumonia (FIP), the blood relative is a sibling. Alternatively, or in addition, in some embodiments, the subject has a mutation in a sequence encoding Mucin 5B (MUC5B), Telomerase RNA Component (TERC), Family with sequence similarity 13 member A (FAM13A), Telomerase Reverse Transcriptase (TERT), Desmoplakin (DSP), Zinc-alpha 2-Glycoprotein 1 (AZGP1), Oligonucleotide/oligosaccharide-binding Fold Containing 1 (OBFC1), ATPase Phospholipid Transporting 11A (ATP11A), Isovaleryl-CoA dehydrogenase (IVD)/Dispatched RND Transporter Family Member 2 (DISP2), Dipeptidyl Peptidase 9 (DPP9), Sialic Acid Binding Ig-Like Lectin 14 (SIGLEC14), Adrenomedullin 2 (ADM2), Tetraspanin 5 (TSPAN5), Calcium/Calmodulin-Dependent Protein Kinase 1 (CAMKK1), zinc finger with KRAB and SCAN domains 1 (ZKSCAN1), isovaleryl-CoA dehydrogenase (IVD), ATPase phospholipid transporting 11A (AK025511) or Matrix Metalloprotease-7 (MMP-7).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding a gene or gene product that is upregulated in a subject having a fibrotic pulmonary disease of the disclosure. In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), 5100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitrin (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metallopeptidase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNCSB), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metallopeptidase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF

Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding a gene or gene product that is down-regulated in a subject having a fibrotic pulmonary disease of the disclosure. In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding MUC5B. In some embodiments, the mutation is a polymorphism in a sequence encoding a MUC5B promoter. In some embodiments, the polymorphism is rs35705950 comprising (SEQ ID NO: 7).

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding TERC. In some embodiments, the mutation is a polymorphism in a sequence encoding TERC or a regulatory sequence thereof. In some embodiments the polymorphism is rs6793295 comprising (SEQ ID NO: 1).

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic FAM13A. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic FAM13A or a regulatory sequence thereof. In some embodiments, the polymorphism is rs2609260.

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic TERT. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic TERT or a regulatory sequence thereof. In some embodiments, the polymorphism is rs4449583.

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic DSP. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic DSP or a regulatory sequence thereof. In some embodiments, the polymorphism is rs2076295.

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic ZKSCAN1. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic ZKSCAN1 or a regulatory sequence thereof. In some embodiments, the polymorphism is rs6963345.

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic OBFC1. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic OBFC1 or a regulatory sequence thereof. In some embodiments, the polymorphism is rs2488000.

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding an AK025511 3' UTR. In some embodiments, the mutation is a polymorphism in a sequence encoding an AK025511 3' UTR or a regulatory sequence thereof. In some embodiments, the polymorphism is rs1278769.

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding IVD. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic IVD or a regulatory sequence thereof. In some embodiments, the polymorphism is rs35700143.

In some embodiments of the methods of the disclosure, the human subject has a mutation in a sequence encoding intronic DPP9. In some embodiments, the mutation is a polymorphism in a sequence encoding intronic DPP9 or a regulatory sequence thereof. In some embodiments, the polymorphism is rs12610495.

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding FAM13A. In some embodiments, the mutation is a polymorphism in a sequence encoding FAM13A or a regulatory sequence thereof. In some embodiments the polymorphism is rs2609255 comprising (SEQ ID NO: 2).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding TERT. In some embodiments, the mutation is a polymorphism in a sequence encoding TERT or a regulatory sequence thereof. In some embodiments the polymorphism is rs2736100 comprising (SEQ ID NO: 3).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding DSP. In some embodiments, the mutation is a polymorphism in a sequence encoding DSP or a regulatory sequence thereof. In some embodiments the polymorphism is rs2076295 comprising (SEQ ID NO: 4).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding AZGP1. In some embodiments, the mutation is a polymorphism in a sequence encoding AZGP1 or a regulatory sequence thereof. In some embodiments the polymorphism is rs4727443 comprising (SEQ ID NO: 5).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding OBFC1. In some embodiments, the mutation is a polymorphism in a sequence encoding OBFC1 or a regulatory sequence thereof. In some embodiments the polymorphism is rs11191865 comprising (SEQ ID NO: 6).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding ATP11A. In some embodiments, the mutation is a polymorphism in a sequence encoding ATP11A or a regulatory sequence thereof. In some embodiments the polymorphism is rs12787690 comprising (SEQ ID NO: 8).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding IVD/DISP2. In some embodiments, the mutation is a polymorphism in a sequence encoding IVD/DISP2 or a regulatory sequence thereof. In some embodiments the polymorphism is rs2034650 comprising (SEQ ID NO: 9).

In some embodiments of the methods of the disclosure, the subject has a mutation in a sequence encoding DPP9. In some embodiments, the mutation is a polymorphism in a sequence encoding DPP9 or a regulatory sequence thereof. In some embodiments the polymorphism is rs12610495 comprising (SEQ ID NO: 10).

In some embodiments of the methods of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA. In some embodiments, the fibrotic lung disease is pulmonary fibrosis or IPF. In some embodiments, the fibrotic lung disease is IPF.

In some embodiments of the methods of the disclosure, the therapeutic agent comprises a N-acetylcysteine, pirfenidone, and nintedanib.

In some embodiments of the methods of the disclosure, the therapeutic agent comprises pirfenidone. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the effective dosage is about 2400 mg/day. In some embodiments, the effective dosage is administered according to an escalating dosage regimen. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the escalating dosage regimen comprises (a) administering to the subject about 800 mg of pirfenidone per day for a first week; (b) administering to the subject about 1600 mg of pirfenidone per day for a second week; and (c) administering to the subject about 2400 mg of pirfenidone per day for the remainder of the treatment. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the escalating dosage regimen comprises (a) administering to the subject a capsule or tablet comprising about 250 mg of pirfenidone three times a day for a first week; (b) administering to the subject two capsules or tablets comprising about 250 mg of pirfenidone three times a day for a second week; and (c) administering to the subject three capsules or tablets comprising about 250 mg of pirfenidone three times a day for the remainder of the treatment. In some embodiments of the escalating dosage regimen, the capsule or tablet comprises 267 mg of pirfenidone.

In some embodiments of the methods of the disclosure, the therapeutic agent comprises nintedanib. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is about 300 mg/day. In some embodiments, the effective dosage is about 150 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is about 200 mg/day. In some embodiments, the effective dosage is about 100 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is administered according to a modified or interrupted dosage regimen. In some embodiments, the modified or interrupted dosage regimen comprises (a) administering to the subject about 300 mg of nintedanib per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; (b) administering to the subject about 200 mg of nintedanib per day until the subject presents the control level of liver enzymes; and (c) administering to the subject about 300 mg of nintedanib per day for the remainder of the treatment; wherein the control level of liver enzymes is a level detected in the subject prior to an initiation of the treatment. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the modified or interrupted regimen comprises (a) administering to the subject a capsule or tablet comprising about 150 mg of nintedanib twice per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; (b) administering to the subject two capsules or tablets comprising about 100 mg twice per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; and (c) administering to the subject a capsule or tablet comprising about 150 mg of nintedanib twice per day for the remainder of the treatment; wherein the control level of liver enzymes is a level detected in the subject prior to an initiation of the treatment.

In some embodiments of the methods of the disclosure, the therapeutic agent prevents the onset or development of a sign or symptom of the fibrotic lung disease.

In some embodiments of the methods of the disclosure, the therapeutic agent delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom in the absence of treatment with the therapeutic agent.

In some embodiments of the methods of the disclosure, the therapeutic agent reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the therapeutic agent.

In some embodiments of the methods of the disclosure, the therapeutic agent reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the therapeutic agent.

In some embodiments of the methods of the disclosure, the at least one sign of the fibrotic lung disease is detectable before the subject presents a symptom of the fibrotic lung disease. In some embodiments, the at least one sign comprises gradual or unintended weight loss, clubbing of the fingers or toes, rapid and shallow breathing, fibrotic lesions in one or both lungs detectable by radiography, or a cough. In some embodiments, the symptom comprises shortness of breath during exercise, shortness of breath at rest, a dry and hacking cough, repeated bouts of coughing, and uncontrollable bouts of coughing.

In some embodiments of the methods of the disclosure, the method prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease. In some embodiments, a secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

The disclosure provides a method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, comprising administering to a non-human subject a dose of a composition that modifies transcription or translation of a sequence encoding Mucin 5B (MUC5B), Telomerase RNA Component (TERC), Family with sequence similarity 13 member A (FAM13A), Telomerase Reverse Transcriptase (TERT), Desmoplakin (DSP), Zinc-alpha 2-Glycoprotein 1 (AZGP1), Oligonucleotide/oligosaccharide-binding Fold Containing 1 (OBFC1), ATPase Phospholipid Transporting 11A (ATP11A), Isovaleryl-CoA dehydrogenase (IVD)/Dispatched RND Transporter Family Member 2 (DISP2), Dipeptidyl Peptidase 9 (DPP9), Sialic Acid Binding Ig-Like Lectin 14 (SIGLEC14), Adrenomedullin 2 (ADM2), Tetraspanin 5 (TSPAN5), Calcium/Calmodulin-Dependent Protein Kinase Kinase 1 (CAMKK1) or Matrix Metalloprotease-7 (MMP-7), wherein the dose of the composition is tolerable to the non-human subject and wherein the dose of the composition is therapeutically effective.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, comprising administering to a non-human subject a composition that modifies an activity of a product of a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, wherein the dose of the composition is tolerable to the non-human subject and wherein the dose of the composition is therapeutically effective.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition that modifies transcription or translation decreases or inhibits transcription or translation.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition decreases or inhibits transcription or translation of a sequence encoding a gene selected from the group consisting of Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), S100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitrin (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metallopeptidase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNCSB), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metallopeptidase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition that modifies transcription or translation increases or activates transcription or translation.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition increases or activates transcription or translation of a sequence encoding a gene selected from the group consisting of Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition that modifies an activity decreases or inhibits the activity.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition decreases or inhibits the activity of a sequence encoding a gene selected from Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), 5100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitrin (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metallopeptidase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNCSB), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metallopeptidase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition that modifies an activity increases or activates the activity.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition increases or activates the activity of a sequence encoding Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the non-human subject is a mammal.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the mammal is genetically-modified.

In some embodiments of the methods of the disclosure, the genetically-modified mammal is a model organism for the fibrotic lung disease.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the fibrotic lung disease is pulmonary fibrosis or IPF.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the fibrotic lung disease is IPF.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the non-human subject carries a mutation in a sequence encoding MUC5B.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the mutation comprises a polymorphism in a sequence encoding a MUC5B promoter.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the polymorphism is rs35705950.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the non-human subject carries a mutation in a sequence encoding TERC, FAM13A, TERT, DSP, ZKSCAN1, AZGP1, OBFC1, MUC5B, AK025511, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition prevents the onset or development of a sign or symptom of the fibrotic lung disease.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom in the absence of treatment with the composition.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom when treated using a standard therapeutic intervention.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the composition.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the composition reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom when treated using a standard therapeutic intervention.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the standard therapeutic intervention comprises a N-acetylcysteine, pirfenidone, and nintedanib.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the standard therapeutic intervention comprises pirfenidone.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, an effective dosage of pirfenidone is about 2400 mg/day.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is administered orally as a capsule or a tablet.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is administered three times per day.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is administered according to an escalating dosage regimen.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the escalating dosage regimen comprises, administering to the non-human subject about 800 mg of pirfenidone per day for a first week; administering to the non-human subject about 1600 mg of pirfenidone per day for a second week; and administering to the non-human subject about 2400 mg of pirfenidone per day for the remainder of the treatment.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the escalating dosage regimen comprises, administering to the non-human subject a capsule or tablet comprising about 250 mg of pirfenidone three times a day for a first week; administering to the non-human subject two capsules or tablets comprising about 250 mg of pirfenidone three times a day for a second week; and administering to the non-human subject three capsules or tablets comprising about 250 mg of pirfenidone three times a day for the remainder of the treatment.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the capsule or tablet comprises 267 mg of pirfenidone.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the standard therapeutic intervention comprises nintedanib.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, an effective dosage of nintedanib is administered orally as a capsule or a tablet.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is about 300 mg/day.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is about 150 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is about 200 mg/day.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the effective dosage is about 100 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the non-human subject presents at least one sign of the fibrotic lung disease.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the at least one sign comprises gradual or unintended weight loss, clubbing of the fingers or toes, rapid and shallow breathing, fibrotic lesions in one or both lungs detectable by radiography, or a cough.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the compound prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, the compound prevents the onset for at 1 year, 2 years, 3 years, 4 years, 5 years or any whole or fractional number of years in between.

In some embodiments of the methods of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure, secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

The disclosure provides a composition for the treatment of a fibrotic lung disease identified by a method of the disclosure, including, a method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease of the disclosure.

The disclosure provides a method of treating fibrotic lung disease in a human subject of the disclosure comprising administering a therapeutically effective amount of a composition identified by a method of the disclosure, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease. In some embodiments, the subject is wild type (e.g. does not comprises a mutation or a sequence variation) with respect to a nucleic acid or amino acid sequence encoding one or more of TERC, FAM13A, TERT, DSP, ZKSCAN1, AZGP1, OBFC1, MUC5B, AK025511, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the human subject presents radiographic Usual Interstitial Pneumonia (UIP).

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, wherein the human subject has fibrotic interstitial lung disease (FILD).

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, wherein the human subject has a blood relative with familial interstitial pneumonia (FIP).

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, wherein the blood relative is a sibling.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, wherein the human subject has a mutation or a sequence variation in a nucleic acid or an amino acid sequence encoding TERC, FAM13A, TERT, DSP, ZKSCAN1, AZGP1, OBFC1, MUC5B, AK025511, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the mutation comprises a polymorphism in a sequence encoding a MUC5B promoter.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the polymorphism is rs35705950.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the fibrotic lung disease is pulmonary fibrosis or IPF.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, the fibrotic lung disease is IPF.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosuresure, the method prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease.

In some embodiments of the methods of treating fibrotic lung disease in a human subject of the disclosure by administering a composition identified by a method of the disclosure, a secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a map depicting an exemplary hierarchical clustering of differentially expressed genes for pre-pulmonary fibrosis subjects and normal subjects.

FIG. **2**A-B is a pair of volcano plots showing serum sample quality control using Principal component analysis (PCA). FIG. **2**A shows before outlier exclusion and FIG. **2**B shows after outlier exclusion.

FIG. **3** is a volcano plot of 3315 plasma proteins, comparing results from 70 patients with established IPF and 70 controls. Solid red symbols represent 57 proteins that were significantly up-regulated and solid blue symbols 12 proteins that were significantly downregulated in patients with IPF after controlling for multiple comparisons and age/gender/smoking.

FIG. **4** is a survival plot showing receiver operator curves of predictive model for PrePF in asymptomatic relatives from FIP families. Area Under Curve (AUC) values for each model are as follows: Gene Expression alone (red)=0.83, Clinical Predictors (blue)=0.87, Clinical Predictors+MUC5B genotype (green)=0.87, Clinical Predictors+Gene Expression Score (yellow)=0.95, Clinical Predictors+MUC5B genotype+Gene Expression Score (black)=0.95, indicating that a peripheral blood biomarker panel may improve the diagnostic power of a predictive model for PrePF in an at-risk population.

FIG. **5** is a graph showing MUC5B expression in IPF (N=203) and unaffected subjects (N=139) stratified by MUC5B promoter variant (r535705950) genotype.

FIG. **6**A is a microscopic image demonstrating that MUC5B is produced in bronchoalveolar epithelia of patients with IPF (brown staining in photomicrographs). Staining is increased in the airways of patients positive for rs35705950 (TT) compared to WT (GG).

FIG. **6**B is a graph showing the percentage of MUC5B positive area of bronchiolar epithelium. Unbiased stereological assessment of staining demonstrates that the volume fraction of stained airways (% positive area) is significantly greater in both the GT heterozygotes and the TT homozygotes.

FIG. **7**A-B is series of bar graphs showing that Scgb1a1- and SFPTC promoter show significant worsening of fibrosis (hydroxyproline) after bleomycin while Muc5b−/− mice are protected. FIG. **7**A is a series of graphs and FIG. **7**B is a series of confocal images showing that the concentration of Muc5b is directly related to the fibroproliferative response to bleomycin. Representative images from second harmonic generation (SHG) demonstrate increased lung collagen (red) in transgenic mice following bleomycin injury.

FIG. **8** is bar graph showing that the baseline expression of ER stress genes in lung tissue from WT and Scgb1a1 Muc5bTg mice. Muc5bTg mice have greater ER stress gene expression than their WT littermates (all genes in the ER stress pathway, with p<0.05). Bleomycin also induces ER stress (data not shown).

FIG. **9** is a pair of microscopic images showing enhanced CHOP (Ddit3) protein in wild type (WT, top photograph) and Scgb1a1-Muc5bTg mice (bottom photograph) after repeat bleomycin.

FIG. **10** is a pair of microscopic images and corresponding graphs showing the expanded mucus layer and decreased mucociliary transport in SFTPC-Muc5bTg mice compared to littermate wild-type mice. Statistical differences were assessed by Mann-Whitney U Test.

FIG. **11** is a series of schematic diagram showing that the MUC5B variant and other biomarkers can identify an at-risk population or those with PrePF, establishing the opportunity for primary and secondary prevention of IPF. The 'at-risk' population and the population with PrePF is large (19% with the MUC5B promoter variant and 1.8% of individuals ≥50 years of age respectively), IPF is diagnosed in a small population with established, end-stage disease and PrePF can be identified using the MUC5B variant rs35705950. Results indicate that PrePF (detected via chest CT scan) is associated with a poor prognosis suggesting that PrePF may be a harbinger of IPF.

FIG. **12** is a schematic diagram showing a method of screening at-risk populations (family members of patients with IPF) to identify individuals with PrePF. Focus is placed on identifying the genetic variants and biomarkers that increase the yield of PrePF on HRCT scan, in addition to gender, age, and physiology scores.

FIG. **13** is a table describing the baseline characteristics of patients with rheumatoid arthritis.

FIG. **14** is a table describing the genotypic association of MUC5B rs35705950 single nucleotide polymorphism in patients with RA, with and without interstitial lung disease

FIG. **15** is a table describing the dominant genotypic association of MUC5B rs35705950 single nucleotide polymorphism in patients with RA-ILD and a usual interstitial pneumonia or possible usual interstitial pneumonia pattern (RA-UIP) and in patients with RA-ILD and a pattern inconsistent with usual interstitial pneumonia (RA non-UIP).

FIG. **16**A is a forest plot of odds ratios {OR) and 95% confidence intervals {Cl) depicting the lack of association of the MUC5B rs35705950 promoter variant with RA without 1LD {RA-nolLD). The boxes indicate OR, and the horizontal lines indicate 95% Cl for the best-fitting genetic model for each association test. The black dotted line represents a mean OR value of 1. The red boxes and red lines indicate the overall OR and 95% Cl, respectively. For comparisons between RA cases and controls, the associations were adjusted for the country of origin and sex. For intra-RA cases comparisons, the associations were adjusted for the country of origin, sex, age at inclusion and smoking.

FIG. **16**B is a forest plot of odds ratios (OR) and 95% confidence intervals {Cl) depicting the additive genotypic association of the MUC5B rs 35705950 promoter variant with RA-ILD. The red dotted line represent the mean value of overall OR value. The boxes indicate OR, and the horizontal lines indicate 95% Cl for the best-fitting genetic model for each association test. The black dotted line represents a mean OR value of 1. The red boxes and red lines indicate the overall OR and 95% Cl, respectively. For comparisons between RA cases and controls, the associations were adjusted for the country of origin and sex. For intra-RA cases comparisons, the associations were adjusted for the country of origin, sex, age at inclusion and smoking.

FIG. **16**C is a forest plot of odds ratios {OR) and 95% confidence intervals {Cl) depicting dominant genotypic association of the MUC5B re35705950 promoter variant with ILD among patients with RA and those with the usual interstitial pneumonia or possible usual interstitial pneumonia (UIP) pattern. The boxes indicate OR, and the horizontal lines indicate 95% Cl for the best-fitting genetic model for each association test. The red dotted line represent the mean value of overall OR value. The black dotted line represents a mean OR value of 1. The red boxes and red lines indicate the overall OR and 95% Cl, respectively. For comparisons between RA cases and controls, the associations were adjusted for the country of origin and sex. For intra-RA cases comparisons, the associations were adjusted for the country of origin, sex, age at inclusion and smoking.

FIG. **17** is a series of photographs depicting MUC5B expression in explanted lung tissue from rheumatoid arthritis associates interstitial lung disease. Representative lung tissue images from unaffected control (GG genotype, Panel A), RA-ILD case #1 (GG genotype, Panel B), and RA-ILD case #2 (GT genotype, Panel C). Low power views with high power view insets identified. Panel A—low power view of normal lung; top and middle insets with high power view of bronchiole with MUC5B staining; bottom inset with high power view of alveolar epithelia. Panel B and C—low power view of the usual interstitial pneumonia pattern in explanted lung tissue of RA-ILD; top inset with high power view of bronchiole with MUC5B staining; middle and bottom insets with high power view of MUC5B staining in metaplastic epithelia lining honeycomb cysts and MUC5B staining of mucous in honeycomb cysts.

FIG. **18** is a flow chart depicting the screening and enrollment process for study subjects.

FIG. **19**A-D is a series of photographs depicting High-resolution CT (HRCT) images of: **19**A) chest from a study subject whose scan was read as normal, without signs of interstitial lung disease or fibrosis. **19**B) HRCT image from subject who was categorized as having "Probable Fibrotic ILD." **19**C) Representative HRCT image from subject who was characterized as having "Definite Fibrotic ILD." **19**D) HRCT image from a case of previously diagnosed, established Idiopathic Pulmonary Fibrosis (IPF) in one of the study families.

FIG. **20** is a table depicting a summary of characteristics of study subjects used in quantitative CT Analyses.

FIG. **21**A-F is a series of photographs depicting representative axial HRCT images visually assessed as "No Fibrosis" (**21**A), "Probable Fibrotic ILD" (**21**C) and "Definite Fibrotic ILD" (E). Below each is the corresponding quantitative HRCT results for the above scan: (**21**B) "No Fibrosis" fibrosis extent 1.7% (fibrosis score=0.55), (**21**D) "Probable Fibrotic ILD" fibrosis extent 18.5% (fibrosis score 2.92), (F) "Definite Fibrotic ILD" fibrosis extent 35.5% (fibrosis score 3.60), Classification results color coded as follows: green=normal lung, blue=airway, yellow=reticular abnormality, magenta=ground glass opacity, red=honeycombing.

FIG. **22** is a table depicting Screening Cohort Subject Characteristics. *DNA available on a total of 489 subjects (404 No Fibrosis and 75 PrePF subjects). **Odds ratios reported in this table were calculated from a mixed effects logistic regression model including age (as a continuous variable), male sex, ever smoker (yes/no), and MUC5B promoter variant (r535705950) genotype. ***In the reported model, rs35705950 coded as a dominant allele; in log-additive genetic model, p=0.05, as well.

FIG. **23** is a table depicting patterns of CT abnormalities in scans with probable or definite fibrotic ILD. *Because a confident single diagnosis was relatively uncommon, most cases included consideration of several patterns. For this reason, the percentages add up to more than 100%.

FIG. **24** is a box plot depicting fibrosis score by visual diagnosis. Boxplots of fibrosis scores based on quantitative HRCT assessment for each visual diagnosis category. Fibrosis score means were significantly different (ANOVA, p<0.0001) across groups defined by visual diagnosis. Comparison of fibrosis score between groups showed significant differences for all comparisons (p<0.01 for all).

FIG. **25**A-C is a series of graphs depicting Receiver Operating Characteristic (ROC) curves for quantitative imaging measures of Fibrosis and PrePF. FIG. **5**A depicts ROC curves for visual diagnosis compared to log HAA scores. FIG. **5**B depicts ROC Curves for visual diagnosis compared to fibrosis scores. ROC analysis showed that fibrosis score discriminates subjects with visual diagnosis of PrePF. Average area under the curve (AUC) in fivefold cross validation was 0.85 (range 0.83-0.87) and average accuracy, sensitivity, and specificity in the test partitions were 0.83 (range 0.74-0.86), 0.74 (range 0.56-0.92) and 0.84 (range 0.76-0.89) respectively. Optimal threshold for fibrosis score ranged from 1.40-1.42. FIG. **5**C depicts Density plots of fibrosis scores for visually diagnosed PrePF (pink) and No Fibrosis (blue) scans—the fibrosis score optimal threshold is indicated with the red line (1.40).

FIG. **26** is a series of tables depicting Dyspnea questionnaire data. FIG. **26**A depicts breathlessness responses for the cohort. FIG. **26**B depicts breathlessness responses by Visual CT diagnosis.

FIG. **27** is a graph that depicts the prevalence of PrePF in FIP Siblings Cohort by Age and MUC5B Genotype. PrePF prevalence in this FIP siblings cohort increases by age, as shown in this graph. By age >60 years, the prevalence of PrePF differed significantly based on MUC5B genotype (*p=0.02). Subjects with the variant are depicted by the red line, while those without it are depicted with the blue line.

FIG. **28** is a table depicting subject characteristics based on Quantitative Fibrosis Score. Clinical characteristics and genotype breakdown of subjects with quantitative HRCT analyses. The cutoff of 1.4 for the logarithm of fibrosis score is based on analyses presented in the text. *p-value compares characteristic between groups. Linear regression values regress fibrosis score on age, male sex, smoking history, and MUC5B promoter variant. **In the reported model, rs35705950 coded as a dominant allele given small number of TT subjects.

FIG. **29** is a table depicting an exploratory genetic association study of 13 pulmonary fibrosis susceptibility variants in RA-ILD.

DETAILED DESCRIPTION OF THE DISCLOSURE

The present disclosure provides a method of treating a fibrotic lung disease in a subject comprising administering to the subject an effective amount of a therapeutic agent, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease.

Methods of Identifying a Therapeutic Agent of the Disclosure or Target Thereof

The disclosure provides a method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, comprising administering to a non-human subject a dose of a composition that modifies transcription or translation of a sequence encoding Mucin 5B (MUC5B), Telomerase RNA Component (TERC), Family with sequence similarity 13 member A (FAM13A), Telomerase Reverse Transcriptase (TERT), Desmoplakin (DSP), Zinc-alpha 2-Glycoprotein 1 (AZGP1), Oligonucleotide/oligosaccharide-binding Fold Containing 1 (OBFC1), ATPase Phospholipid Transporting 11A (ATP11A), Isovaleryl-CoA dehydrogenase (IVD)/Dispatched RND Transporter Family Member 2 (DISP2), Dipeptidyl Peptidase 9 (DPP9), Sialic Acid Binding Ig-Like Lectin 14 (SIGLEC14), Adrenomedullin 2 (ADM2), Tetraspanin 5 (TSPAN5), Calcium/Calmodulin-Dependent Protein Kinase Kinase 1 (CAMKK1) or Matrix Metalloprotease-7 (MMP-7), wherein the dose of the composition is tolerable to the non-human subject and wherein the dose of the composition is therapeutically effective.

The disclosure provides method of identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, comprising administering to a non-human subject a composition that modifies an activity of a product of a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, wherein the dose of the composition is tolerable to the non-human subject and wherein the dose of the composition is therapeutically effective.

In some embodiments of the methods of the disclosure, the composition that modifies transcription or translation decreases or inhibits transcription or translation. In some embodiments, the composition decreases or inhibits transcription or translation of a sequence encoding a gene selected from the group consisting of Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), 5100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitrin (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metallopeptidase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNCSB), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metallopeptidase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

In some embodiments of the methods of the disclosure, the composition that modifies transcription or translation increases or activates transcription or translation. In some embodiments, the composition increases or activates transcription or translation of a sequence encoding a gene selected from the group consisting of Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

In some embodiments of the methods of the disclosure, the composition that modifies an activity decreases or inhibits the activity. In some embodiments, the composition decreases or inhibits the activity of a sequence encoding a gene selected from Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), S100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitrin (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metallopeptidase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNC5B), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase (APRT), Matrix Metallopeptidase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

In some embodiments of the methods of the disclosure, the composition that modifies an activity increases or activates the activity. In some embodiments, the composition increases or activates the activity of a sequence encoding

Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

In some embodiments of the methods of the disclosure, the non-human subject is a mammal. In some embodiments, mammal is genetically-modified. In some embodiments, the genetically-modified mammal is a model organism for the fibrotic lung disease.

In some embodiments of the methods of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA. In some embodiments, the fibrotic lung disease is pulmonary fibrosis or IPF. In some embodiments, the fibrotic lung disease is IPF.

In some embodiments of the methods of the disclosure, the non-human subject carries a mutation in a sequence encoding MUC5B. In some embodiments, the mutation comprises a polymorphism in a sequence encoding a MUC5B promoter. In some embodiments, the polymorphism is rs35705950. Alternatively, or in addition, in some embodiments, the non-human subject carries a mutation in a sequence encoding TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

In some embodiments of the methods of the disclosure, the composition prevents the onset or development of a sign or symptom of the fibrotic lung disease.

In some embodiments of the methods of the disclosure, the composition delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom in the absence of treatment with the composition. In some embodiments, the composition delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom when treated using a standard therapeutic intervention.

In some embodiments of the methods of the disclosure, the composition reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the composition. In some embodiments, the composition reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom when treated using a standard therapeutic intervention.

In some embodiments of the methods of the disclosure, the standard therapeutic intervention comprises a N-acetylcysteine, pirfenidone, and nintedanib.

In some embodiments of the methods of the disclosure, the standard therapeutic intervention comprises pirfenidone. In some embodiments, an effective dosage of pirfenidone is about 2400 mg/day. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, the effective dosage is administered three times per day. In some embodiments, the effective dosage is administered according to an escalating dosage regimen. In some embodiments, the escalating dosage regimen comprises (a) administering to the non-human subject about 800 mg of pirfenidone per day for a first week; (b) administering to the non-human subject about 1600 mg of pirfenidone per day for a second week; and (c) administering to the non-human subject about 2400 mg of pirfenidone per day for the remainder of the treatment. In some embodiments, the escalating dosage regimen comprises (a) administering to the non-human subject a capsule or tablet comprising about 250 mg of pirfenidone three times a day for a first week; (b) administering to the non-human subject two capsules or tablets comprising about 250 mg of pirfenidone three times a day for a second week; and (c) administering to the non-human subject three capsules or tablets comprising about 250 mg of pirfenidone three times a day for the remainder of the treatment. In some embodiments, the capsule or tablet comprises 267 mg of pirfenidone.

In some embodiments of the methods of the disclosure, the standard therapeutic intervention comprises nintedanib. In some embodiments, an effective dosage of nintedanib is administered orally as a capsule or a tablet. In some embodiments, the effective dosage is about 300 mg/day. In some embodiments, the effective dosage is about 150 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, the effective dosage is about 200 mg/day. In some embodiments, the effective dosage is about 100 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another.

In some embodiments of the methods of the disclosure, the non-human subject presents at least one sign of the fibrotic lung disease. In some embodiments, the at least one sign comprises gradual or unintended weight loss, clubbing of the fingers or toes, rapid and shallow breathing, fibrotic lesions in one or both lungs detectable by radiography, or a cough.

In some embodiments of the methods of the disclosure, the compound prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease. In some embodiments, the compound prevents the onset for at 1 year, 2 years, 3 years, 4 years, 5 years or any whole or fractional number of years in between. In some embodiments, the secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

The disclosure provides a composition for the treatment of a fibrotic lung disease identified by a method of the disclosure for identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease.

Subjects of the Disclosure

The disclosure provides a method of treating a fibrotic lung disease in a human subject comprising administering to the subject the composition for the treatment of a fibrotic lung disease identified by a method of the disclosure for identifying a therapeutic agent or target thereof for the treatment of a fibrotic lung disease, wherein the subject is asymptomatic and wherein the subject is at risk of developing the fibrotic lung disease.

In some embodiments of the methods of treating a fibrotic lung disease in a human subject of the disclosure, the human subject presents radiographic Usual Interstitial Pneumonia (UIP). In some embodiments, the human subject has fibrotic interstitial lung disease (FILD). In some embodiments, the human subject has a blood relative with familial interstitial pneumonia (FIP). In some embodiments, the blood relative is a sibling. Alternatively, or in addition, in some embodiments, the human subject has a mutation in a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7. In some embodiments, the mutation comprises a polymorphism in a sequence encoding a MUC5B promoter. In some embodiments, the polymorphism is rs35705950.

In some embodiments of the methods of treating a fibrotic lung disease in a human subject of the disclosure, the fibrotic lung disease is pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), or an asymptomatic ILA. In some embodiments, the fibrotic lung disease is pulmonary fibrosis or IPF. In some embodiments, the fibrotic lung disease is IPF.

In some embodiments of the methods of treating a fibrotic lung disease in a human subject of the disclosure, the method prevents the onset of a secondary condition associated with a severe form of the fibrotic lung disease. In some embodiments, the secondary condition comprises a collapsed lung, an infected lung, a blood clot in a lung, lung cancer, respiratory failure, pulmonary hypertension, heart failure or death.

Idiopathic Pulmonary Fibrosis (IPF)

IPF is localized to the lung and is characterized by a pattern of heterogeneous, subpleural patches of fibrotic, remodeled lung, and often results in death within 3-5 years of diagnosis. IPF affects 5 million people worldwide, disproportionately affects men, is associated with cigarette smoking, increases with age, is inexplicably increasing in prevalence, and is likely underdiagnosed. Most patients with IPF are discovered in the advanced stage when little can be done to influence survival. There is a critical unmet need in idiopathic pulmonary fibrosis (IPF) for an early detection and prevention of IPF. Earlier diagnosis of IPF detects subjects with a lower burden of fibrotic lung disease providing an opportunity for secondary prevention of this progressive disease and changes the clinical approach to patients with IPF from palliative to preventive.

Early detection and prevention of idiopathic pulmonary fibrosis (IPF) is critical. As demonstrated herein, treatment of subjects at risk for developing PrePF is based on two central concepts of first, understanding that PrePF is essential for primary and secondary prevention of IPF and second, that similar to asymptomatic family members of familial IPF (FIP; ≥2 family members with IPF), asymptomatic family members of sporadic IPF represent an at-risk population for PrePF. These central concepts are supported by the observation that 1) IPF has a pre-symptomatic phase and PrePF appears to be a harbinger of IPF, 2) familial and sporadic IPF are similar etiologically, 3) MUC5B promoter variant is critical to early disease recognition and 4) identification of PrePF represents an opportunity to prevent extensive lung fibrosis. As shown herein, a common gain-of-function MUC5B promoter variant rs35705950 is a strong risk factor (genetic and otherwise), accounting for at least 30% of the total risk of developing IPF. The MUC5B promoter variant rs35705950 may be used to identify individuals with PrePF. MUC5B promoter variant rs35705950 is also predictive of radiographic progression of PrePF and is present in over 50% of non-Hispanic white patients with IPF and is also associated with unique clinical and biological IPF phenotypes. PrePF can be predicted using a combination of clinical risk factors, the MUC5B promoter variant rs35705950, and a panel of biomarkers. This disclosure provides methods of treating subjects with Preclinical Pulmonary Fibrosis (PrePF) and who may also be at risk for developing IPF. The methods of the disclosure fundamentally change the clinical approach to treating subjects with IPF, shifting the focus from a merely palliative to a proactive and preventive therapy.

Rheumatoid Arthritis-Associated Interstitial Lung Disease (RA-ILD)

Rheumatoid arthritis (RA) is a common inflammatory and autoimmune disease that is associated with progressive impairment, systemic complications and increased mortality. Interstitial lung disease (RA-ILD) is detected in up to 60% of patients with RA on high-resolution computed-tomography (HRCT), is clinically significant in 10%, and is a leading cause of morbidity and mortality in patients with RA.

RA-ILD shares several characteristics with idiopathic pulmonary fibrosis (IPF), including common environmental risk factors, the high prevalence of the usual interstitial pneumonia (UIP) pattern, the progressive nature of the disease, and poor survival. The hypothesis of a shared genetic background between IPF and RA-ILD was recently suggested by a whole-exome sequencing (WES) genetic association study in patients with RA-ILD, revealing an excess of mutations in genes in RA-ILD previously associated with familial interstitial pneumonia (FIP) including TERT, RTEL1, PARN and SFTPC.

The common gain-of-function promoter variant rs3570595013 of the gene encoding mucin5B (MUC5B) is the strongest genetic risk factor for IPF, observed in at least 50% of the cases of IPF and accounting for 30% of the risk of developing this disease. The MUC5B promoter variant is associated with increased expression of MUC5B in lung parenchyma of unaffected controls and cases of IPF. Consequently, it is hypothesized that the MUC5B promoter variant rs35705950 would also contribute to the occurrence of RA-ILD. To test this hypothesis, a multi-ethnic association study of the MUC5B promoter variant and RA-ILD in seven distinct case series was performed.

The MUC5B promoter variant rs35705950, the strongest genetic risk factor for IPF, is also a strong risk factor for RA-ILD, especially among those with radiographic evidence of UIP. Of note, the effect of the MUC5B promoter variant on the development of ILD associated with RA was similar in magnitude and direction to that observed in IPF.

The relationship between the MUC5B promoter variant and RA-ILD may be specific to UIP and may not be generalizable to other autoimmune conditions of the lung. The MUC5B promoter variant has not been found to be associated with risk of ILDs linked to systemic sclerosis or autoimmune myositis. Unlike these other types of ILD, RA-ILD shares more characteristics with IPF, notably the increased frequency of the UIP pattern (both radiologic and histologic), an increased prevalence of male sex and older age, and genetic susceptibility as assessed by an excess of mutations in genes linked to FIP in a cohort of RA-ILD, and now the MUC5B promoter variant rs35705950.

The disclosure demonstrates that the MUC5B promoter variant is a risk factor for UIP, and not simply limited to IPF and RA-ILD. In fact, emerging studies have identified the MUC5B promoter variant as a risk factor for chronic hypersensitivity pneumonitis, another condition known to have a sub-phenotype of UIP. Further, since HRCT underestimates the presence of ILD and the UIP pattern of fibrosis, our point estimates for association with the MUC5B variant are likely conservative. Similar to IPF, early forms of RA-ILD can be identified using the MUC5B promoter variant as biomarker.

The disclosure demonstrates that Muc5b is overexpressed by the bronchoalveolar epithelia and MUC5B mRNA is co-expressed by cells expressing surfactant protein C, as has been shown in IPF. These findings suggest either type 2 alveolar epithelial cells can express MUC5B or that in patients with RA-ILD, the cells in the distal airspace dedifferentiate. Importantly, the disclosure demonstrates for the first time that cells that overexpress MUC5B are undergoing ER stress, a recognized mechanism of cell injury and repair. In aggregate, these findings indicate that the gain-of-function MUC5B promoter variant rs35705950 injures alveolar epithelia by inducing ER stress.

RA-ILD is a complex genetic phenotype with the minor allele of the MUC5B promoter variant rs35705950 identified as a risk factor for the disease. The odds ratios for the association of MUC5B promoter variant with RA-ILD is equivalent to that observed with IPF and substantively higher than those for the most other common risk variants for RA-ILD, including cigarette smoking and the human leukocyte antigen locus for RA.

The MUC5B promoter variant is a risk factor for UIP in general and may prove relevant beyond RA-ILD and IPF.

Expression of MUC5B in the bronchoalveolar epithelia co-incident with markers of ER stress suggest that the MUC5B promoter variant may be causing pulmonary fibrosis by initiating microscopic foci of injury and repair.

The MUC5B promoter variant appears to predict ILD in the RA population, identifying potential opportunities for early ILD detection in patients with RA.

Preclinical Idiopathic Pulmonary Fibrosis

Better understanding and recognition of early pulmonary fibrosis is critical because medical therapies have been shown to slow progression, not to reverse or even stabilize established fibrosis—therefore, intervention before irreversible fibrosis has become extensive has the potential to improve quality of life and decrease morbidity. While IPF affects approximately 5 million people worldwide, between 1.8 and 14% of the general population ≥50 years of age have radiologic findings of undiagnosed pulmonary fibrosis. Large cohort studies indicate that interstitial lung abnormalities, postulated to represent early pulmonary fibrosis, are associated with increased mortality, and that most of these abnormalities progress over time. Members of families with 2 or more cases of pulmonary fibrosis (FIP, Familial Interstitial Pneumonia) have been identified as an "at-risk" population. In a previous study of FIP relatives, 14% had interstitial lung abnormalities on high resolution computed tomography (HRCT), and 35% had an abnormal transbronchial biopsy indicating interstitial lung disease.

HRCT provides visualization of the lung parenchyma and plays a key role in the diagnosis of the Idiopathic Interstitial Pneumonias (IIPs), including IPF. Currently, visual diagnosis by thoracic radiologists, in conjunction with multidisciplinary clinical conference, is the gold standard for diagnosing TIPS. However, visual assessment is imprecise and hampered by inter-observer variation. Quantitative HRCT (qHRCT) evaluation provides measures of fibrosis extent that, in subjects diagnosed with IPF, correlate with degree of physiologic impairment at baseline, and may be more sensitive to subtle changes in disease status than routinely used physiological metrics. The design and utility of quantitative methods in the context of early forms of fibrotic ILD requires further study. Deep learning methods have been increasingly used in imaging to identify and classify CT patterns, and may be particularly valuable in detection of early lung fibrosis.

PrePF is prevalent among FIP relatives, and a texture-based quantitative method of HRCT analyses is useful in identifying these abnormalities in this population, and key risk factors, including the MUC5B promoter variant, predict those at risk of this disease. PrePF subjects are older, more likely to be male, and more likely to have smoked than the unaffected subjects; additionally, the gain-of-function MUC5B promoter variant rs35705950, which has been shown in prior studies to be associated with pulmonary fibrosis, is more common in PrePF subjects when compared to their unaffected family members. Given the subtlety of the fibrotic change in many of these cases of PrePF, the high prevalence of potential UIP pattern on HRCT scan suggests that PrePF subjects may progress to IPF over time.

Methods for Detecting a Genetic Variant

The present disclosure also provides methods of detecting the biomarkers of the present disclosure. Methods of detecting a genetic variant are further described in US Application US 2016-0060701A1 (the contents of which are incorporated herein by reference in their entirety). The practice of the present disclosure employs, unless otherwise indicated, conventional methods of analytical biochemistry, microbiology, molecular biology and recombinant DNA techniques within the skill of the art. Such techniques are explained fully in the literature. (See, e.g., Sambrook, J. et al. Molecular Cloning: A Laboratory Manual. 3rd, ed., Cold Spring Harbor Laboratory, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N. Y., 2000; DNA Cloning: A Practical Approach, Vol. I & II (D. Glover, ed.); Oligonucleotide Synthesis (N. Gait, ed., Current Edition); Nucleic Acid Hybridization (B. Hames & S. Higgins, eds., Current Edition); Transcription and Translation (B. Hames & S. Higgins, eds., Current Edition); CRC Handbook of Parvoviruses, Vol. I & II (P. Tijessen, ed.); Fundamental Virology, 2nd Edition, Vol. I & II (B. N. Fields and D. M. Knipe, eds.)).

The methods of the invention are not limited to any particular way of detecting the presence or absence of a genetic variant (e.g. SNP) and can employ any suitable method to detect the presence or absence of a variant(s), of which numerous detection methods are known in the art. Dynamic allele-specific hybridization (DASH) can be used to detect a genetic variant. DASH genotyping takes advantage of the differences in the melting temperature in DNA that results from the instability of mismatched base pairs. The process can be vastly automated and encompasses a few simple principles. Thus, the aspects and embodiments described herein provide methods for assessing the presence or absence of SNPs in a sample (e.g. biological sample) from a subject suspected of having or developing an interstitial lung disease (e.g., because of family history). In certain embodiments, one or more SNPs are screened in one or more samples from a subject. The SNPs can be associated with one or more genes, e.g., one or more genes or other genes associated with mucous secretions as disclosed herein.

Typically, the target genomic segment is amplified and separated from non-target sequence, e.g., through use of a biotinylated primer and chromatography. A probe that is specific for the particular allele is added to the amplification product. The probe can be designed to hybridize specifically to a variant sequence or to the dominant allelic sequence. The probe can be either labeled with or added in the presence of a molecule that fluoresces when bound to double-stranded DNA. The signal intensity is then measured as temperature is increased until the Tm can be determined. A non-matching sequence (either genetic variant or dominant allelic sequence, depending on probe design), will result in a lower than expected Tm.

DASH genotyping relies on a quantifiable change in Tm, and is thus capable of measuring many types of mutations, not just SNPs. Other benefits of DASH include its ability to work with label free probes and its simple design and performance conditions.

Molecular beacons can also be used to detect a genetic variant. This method makes use of a specifically engineered single-stranded oligonucleotide probe. The oligonucleotide is designed such that there are complementary regions at each end and a probe sequence located in between. This design allows the probe to take on a hairpin, or stem-loop, structure in its natural, isolated state. Attached to one end of the probe is a fluorophore and to the other end a fluorescence quencher. Because of the stem-loop structure of the probe, the fluorophore is in close proximity to the quencher, thus preventing the molecule from emitting any fluorescence. The molecule is also engineered such that only the probe sequence is complementary to the targeted genomic DNA sequence.

If the probe sequence of the molecular beacon encounters its target genomic DNA sequence during the assay, it will anneal and hybridize. Because of the length of the probe sequence, the hairpin segment of the probe will be denatured in favor of forming a longer, more stable probe-target hybrid. This conformational change permits the fluorophore and quencher to be free of their tight proximity due to the hairpin association, allowing the molecule to fluoresce.

If on the other hand, the probe sequence encounters a target sequence with as little as one non-complementary nucleotide, the molecular beacon will preferentially stay in its natural hairpin state and no fluorescence will be observed, as the fluorophore remains quenched. The unique design of these molecular beacons allows for a simple diagnostic assay to identify SNPs at a given location. If a molecular beacon is designed to match a wild-type allele and another to match a mutant of the allele, the two can be used to identify the genotype of an individual. If only the first probe's fluorophore wavelength is detected during the assay then the individual is homozygous to the wild type. If only the second probe's wavelength is detected then the individual is homozygous to the mutant allele. Finally, if both wavelengths are detected, then both molecular beacons must be hybridizing to their complements and thus the individual must contain both alleles and be heterozygous.

A microarray can also be used to detect genetic variants. Hundreds of thousands of probes can be arrayed on a small chip, allowing for many genetic variants or SNPs to be interrogated simultaneously. Because SNP alleles only differ in one nucleotide and because it is difficult to achieve optimal hybridization conditions for all probes on the array, the target DNA has the potential to hybridize to mismatched probes. This can be addressed by using several redundant probes to interrogate each SNP. Probes can be designed to have the SNP site in several different locations as well as containing mismatches to the SNP allele. By comparing the differential amount of hybridization of the target DNA to each of these redundant probes, it is possible to determine specific homozygous and heterozygous alleles.

Restriction fragment length polymorphism (RFLP) can be used to detect genetic variants and SNPs. RFLP makes use of the many different restriction endonucleases and their high affinity to unique and specific restriction sites. By performing a digestion on a genomic sample and determining fragment lengths through a gel assay it is possible to ascertain whether or not the enzymes cut the expected restriction sites. A failure to cut the genomic sample results in an identifiably larger than expected fragment implying that there is a mutation at the point of the restriction site which is rendering it protected from nuclease activity.

PCR- and amplification-based methods can be used to detect genetic variants. For example, tetra-primer PCR employs two pairs of primers to amplify two alleles in one PCR reaction. The primers are designed such that the two primer pairs overlap at a SNP location but each matches perfectly to only one of the possible alleles. As a result, if a given allele is present in the PCR reaction, the primer pair specific to that allele will produce product but not the alternative allele with a different allelic sequence. The two primer pairs can be designed such that their PCR products are of a significantly different length allowing for easily distinguishable bands by gel electrophoresis, or such that they are differently labeled.

Primer extension can also be used to detect genetic variants. Primer extension first involves the hybridization of a probe to the bases immediately upstream of the SNP nucleotide followed by a 'mini-sequencing' reaction, in which DNA polymerase extends the hybridized primer by adding a base that is complementary to the SNP nucleotide. The incorporated base that is detected determines the presence or absence of the SNP allele. Because primer extension is based on the highly accurate DNA polymerase enzyme, the method is generally very reliable. Primer extension is able to genotype most SNPs under very similar reaction conditions making it also highly flexible. The primer extension method is used in a number of assay formats, and can be detected using e.g., fluorescent labels or mass spectrometry.

Primer extension can involve incorporation of either fluorescently labeled ddNTP or fluorescently labeled deoxynucleotides (dNTP). With ddNTPs, probes hybridize to the target DNA immediately upstream of SNP nucleotide, and a single, ddNTP complementary to the SNP allele is added to the 3' end of the probe (the missing 3'-hydroxyl in didioxynucleotide prevents further nucleotides from being added). Each ddNTP is labeled with a different fluorescent signal allowing for the detection of all four alleles in the same reaction. With dNTPs, allele-specific probes have 3' bases which are complementary to each of the SNP alleles being interrogated. If the target DNA contains an allele complementary to the 3' base of the probe, the target DNA will completely hybridize to the probe, allowing DNA polymerase to extend from the 3' end of the probe. This is detected by the incorporation of the fluorescently labeled dNTPs onto the end of the probe. If the target DNA does not contain an allele complementary to the probe's 3' base, the target DNA will produce a mismatch at the 3' end of the probe and DNA polymerase will not be able to extend from the 3' end of the probe.

The iPLEX® SNP genotyping method takes a slightly different approach, and relies on detection by mass spectrometer. Extension probes are designed in such a way that many different SNP assays can be amplified and analyzed in a PCR cocktail. The extension reaction uses ddNTPs as above, but the detection of the SNP allele is dependent on the actual mass of the extension product and not on a fluorescent molecule. This method is for low to medium high throughput, and is not intended for whole genome scanning.

Primer extension methods are, however, amenable to high throughput analysis. Primer extension probes can be arrayed on slides allowing for many SNPs to be genotyped at once. Broadly referred to as arrayed primer extension (APEX), this technology has several benefits over methods based on differential hybridization of probes. Comparatively, APEX methods have greater discriminating power than methods using differential hybridization, as it is often impossible to obtain the optimal hybridization conditions for the thousands of probes on DNA microarrays (usually this is addressed by having highly redundant probes).

Oligonucleotide ligation assays can also be used to detect genetic variants. DNA ligase catalyzes the ligation of the 3' end of a DNA fragment to the 5' end of a directly adjacent DNA fragment. This mechanism can be used to interrogate a SNP by hybridizing two probes directly over the SNP polymorphic site, whereby ligation can occur if the probes are identical to the target DNA. For example, two probes can be designed; an allele-specific probe which hybridizes to the target DNA so that its 3' base is situated directly over the SNP nucleotide and a second probe that hybridizes the template upstream (downstream in the complementary strand) of the SNP polymorphic site providing a 5' end for the ligation reaction. If the allele-specific probe matches the target DNA, it will fully hybridize to the target DNA and ligation can occur. Ligation does not generally occur in the presence of a mismatched 3' base. Ligated or unligated products can be detected by gel electrophoresis, MALDI-TOF mass spectrometry or by capillary electrophoresis.

The 5'-nuclease activity of Taq DNA polymerase can be used for detecting genetic variants. The assay is performed concurrently with a PCR reaction and the results can be read in real-time. The assay requires forward and reverse PCR primers that will amplify a region that includes the SNP polymorphic site. Allele discrimination is achieved using FRET, and one or two allele-specific probes that hybridize to the SNP polymorphic site. The probes have a fluorophore linked to their 5' end and a quencher molecule linked to their 3' end. While the probe is intact, the quencher will remain in close proximity to the fluorophore, eliminating the fluorophore's signal. During the PCR amplification step, if the allele-specific probe is perfectly complementary to the SNP allele, it will bind to the target DNA strand and then get degraded by 5'-nuclease activity of the Taq polymerase as it extends the DNA from the PCR primers. The degradation of the probe results in the separation of the fluorophore from the quencher molecule, generating a detectable signal. If the allele-specific probe is not perfectly complementary, it will have lower melting temperature and not bind as efficiently. This prevents the nuclease from acting on the probe.

Förster resonance energy transfer (FRET) detection can be used for detection in primer extension and ligation reactions where the two labels are brought into close proximity to each other. It can also be used in the 5'-nuclease reaction, the molecular beacon reaction, and the invasive cleavage reactions where the neighboring donor/acceptor pair is separated by cleavage or disruption of the stem-loop structure that holds them together. FRET occurs when two conditions are met. First, the emission spectrum of the fluorescent donor dye must overlap with the excitation wavelength of the acceptor dye. Second, the two dyes must be in close proximity to each other because energy transfer drops off quickly with distance. The proximity requirement is what makes FRET a good detection method for a number of allelic discrimination mechanisms.

A variety of dyes can be used for FRET, and are known in the art. The most common ones are fluorescein, cyanine dyes (Cy3 to Cy7), rhodamine dyes (e.g. rhodamine 6G), the Alexa series of dyes (Alexa 405 to Alexa 730). Some of these dyes have been used in FRET networks (with multiple donors and acceptors). Optics for imaging all of these require detection from UV to near IR (e.g. Alex 405 to Cy7), and the Atto series of dyes (Atto-Tec GmbH). The Alexa series of dyes from Invitrogen cover the whole spectral range. They are very bright and photostable.

Example dye pairs for FRET labeling include Alexa-405/Alex-488, Alexa-488/Alexa-546, Alexa-532/Alexa-594, Alexa-594/Alexa-680, Alexa-594/Alexa-700, Alexa-700/Alexa-790, Cy3/Cy5, Cy3.5/Cy5.5, and Rhodamine-Green/Rhodamine-Red, etc. Fluorescent metal nanoparticles such as silver and gold nanoclusters can also be used (Richards et al. (2008) J Am Chem Soc 130:5038-39; Vosch et al. (2007) Proc Natl Acad Sci USA 104:12616-21; Petty and Dickson (2003) J Am Chem Soc 125:7780-81 Available filters, dichroics, multichroic mirrors and lasers can affect the choice of dye.

In Vitro Complexes

Provided herein are nucleic acid complexes, e.g., formed in in vitro assays to indicate the presence of a genetic variant sequence. One of skill will understand that a nucleic acid complex can also be formed to detect the presence of a dominant allelic sequence, depending on the design of the probe or primer, e.g., in assays to distinguish homozygous and heterozygous subjects.

In some embodiments, the complex comprises a first nucleic acid hybridized to a genetic variant nucleic acid, wherein the genetic variant nucleic acid is a genetic variant in a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7. In some embodiments, the genetic variant nucleic acid is an amplification product. In some embodiments, the genetic variant nucleic acid is on genomic DNA, e.g., from a subject that has or is suspected of having an interstitial lung disease. In some embodiments, the first nucleic acid is an amplification product or a primer extension product. In some embodiments, the first nucleic acid is labeled. In some embodiments, the nucleic acid complex further comprises a second nucleic acid hybridized to the genetic variant nucleic acid. In some embodiments, the second nucleic acid is labeled e.g., with a FRET or other fluorescent label. In some embodiments, the first and second nucleic acids form a FRET pair when hybridized to a genetic variant sequence.

In some embodiments, the nucleic acid complex further comprises an enzyme, such as a DNA polymerase (e.g., standard DNA polymerase or thermostable polymerase such as Taq) or ligase.

The present disclosure includes but is not limited to the following embodiments:

A method for determining if an individual is predicted to develop and/or progress rapidly with an interstitial pneumonia comprising: detecting in a biological sample from the individual, at least one of: a) the presence of a marker polymorphism selected from the group consisting of: rs35705950; and/or, b) a level of gene expression of a marker gene or plurality of marker genes selected from the group consisting of: a marker gene having at least 95% sequence identity with at least one sequence selected from the group consisting of MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof; c) polypeptides encoded by the marker genes of b) d) fragments of polypeptides of c); and e) a polynucleotide which is fully complementary to at least a portion of a marker gene of b); wherein the presence of the plurality of markers is indicative of whether an individual will develop a disease. In some embodiments, the genes detected share 100% sequence identity with the corresponding marker gene in b). In some embodiments, the presence or level of at least one of the plurality of markers is determined and compared to a standard level or reference set. In some embodiments, the standard level or reference set is determined according to a statistical procedure for risk prediction. In some embodiments, the statistical procedure for risk prediction comprises using the sum of the gene expression of the marker or markers or the presence or absence of a set of markers, weighted by a Proportional Hazards coefficient. In some embodiments, the presence of the at least one marker is determined by detecting the presence or absence or expression level of a polypeptide. In some embodiments, the method further comprises detecting the presence of the polypeptide using a reagent that specifically binds to the polypeptide or a fragment thereof. In some embodiments, the reagent is selected from the group consisting of an antibody, an antibody derivative, and an antibody fragment. In some embodiments, the presence of the marker is determined by obtaining the sequence of genomic DNA at the locus of the polymorphism. In some embodiments, the presence of the marker is determined by obtaining RNA from the biological sample; generating cDNA from the RNA; amplifying the cDNA with probes or primers for marker genes; obtaining from the amplified cDNA the expression levels of the genes or gene expression products in the sample. In some embodiments, the individual is a human.

In some embodiments, the method further comprises: a) comparing the expression level of the marker gene or plurality of marker genes in the biological sample to a control level of the marker gene(s) selected from the group consisting of: a control level of the marker gene that has been correlated with interstitial lung disease, the risk of developing interstitial lung disease, or having a interstitial lung disease; and a control level of the marker that has been correlated with slow or no progression of interstitial lung disease, or low risk of developing an interstitial lung disease; and b) selecting the individual as being predicted to progress rapidly in the development of interstitial pneumonia, if the expression level of the marker gene in the individual's biological sample is statistically similar to, or greater than, the control level of expression of the marker gene that has been correlated with interstitial lung disease, or c) selecting the individual as being predicted to not develop interstitial lung disease, or to progress slowly, if the level of the marker gene in the individual's biological sample is statistically less than the control level of the marker gene that has been correlated with interstitial lung disease.

In some embodiments, the method further comparing the presence of a polymorphism, in the biological sample to a set of genetic variants or polymorphic markers from an individual or control group having developed interstitial lung disease, and, selecting the individual as being predicted to develop or to progress with interstitial pneumonia if the polymorphic markers present in the biological sample are identical to or statistically similar to a set of polymorphic markers from the individual or control group or, selecting the individual as being predicted to develop or rapidly progress with interstitial pneumonia, if the polymorphic markers present in the biological sample are not identical to or statistically similar to the set of genetic variants or polymorphic markers from the individual or control group.

A method for monitoring the progression of interstitial lung disease in a subject, comprising: i) measuring expression levels of a plurality of gene markers in a first biological sample obtained from the subject, wherein the plurality of markers comprise a plurality of markers selected from the group consisting of: a marker gene having at least 95% sequence identity with a sequence selected from the group consisting of a) MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof; b) polypeptides encoded by the marker genes of a), c) fragments of polypeptides of d); and e) a polynucleotide which is fully complementary to at least a portion of a marker gene of b); ii) measuring expression levels of the plurality of markers in a second biological sample obtained from the subject; and iii) comparing the expression level of the marker measured in the first sample with the level of the marker measured in the second sample. In some embodiments, the marker genes detected share 100% sequence identity with the corresponding marker gene in a). In some embodiments, the method further comprises performing a follow-up step selected from the group consisting of CT scan of the chest and pathological examination of lung tissues from the subject. In some embodiments, the first biological sample from the subject is obtained at a time t_0, and the second biological sample from the subject is obtained at a later time t_1. In some embodiments, the first biological sample and the second biological sample are obtained from the subject are obtained more than once over a range of times.

A method of assessing the efficacy of a treatment for interstitial lung disease or interstitial pneumonia in a subject, the method comprising comparing: i) the expression level of a marker measured in a first sample obtained from the subject at a time to, wherein the marker is selected from the group consisting of a) a marker gene having at least 95% sequence identity with a sequence selected from the group consisting of MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof; b) polypeptides encoded by the marker genes of a) c) fragments of polypeptides of b); and d) a polynucleotide which is fully complementary to at least a portion of a marker gene of a); ii) the level of the marker in a second sample obtained from the subject at time t_1; and, iii) performing a follow-up step selected from CT scan of the chest and pathological examination of lung tissues from the subject; wherein a decrease in the level of the marker in the second sample relative to the first sample is an indication that the treatment is efficacious for treating interstitial pneumonia in the subject. In some embodiments, the genes detected share 100% sequence identity with the corresponding marker gene in a). In some embodiments, the time t0 is before the treatment has been administered to the subject, and the time t1 is after the treatment has been administered to the subject. In some embodiments, the comparing is repeated over a range of times.

An assay system for predicting individual prognosis therapy for interstitial pneumonia comprising a means to detect at least one of: a) the presence of a marker polymorphism selected from the group consisting of: rs35705950; and/or, b) a level of gene expression of a marker gene or plurality of marker genes selected from the group consisting of: a marker gene having at least 95% sequence identity with a sequence selected from the group consisting of MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof; c) polypeptides encoded by the marker genes of b) d) fragments of polypeptides of c); and e) a polynucleotide which is fully complementary to at least a portion of a marker gene of b). In some embodiments, the means to detect comprises nucleic acid probes comprising at least 10 to 50 contiguous nucleic acids of the marker polymorphisms or gene(s), or complementary nucleic acid sequences thereof. In some embodiments, the means to detect comprises binding ligands that specifically detect polypeptides encoded by the marker genes. In some embodiments, the genes detected share 100% sequence identity with the corresponding marker gene in b).

In some embodiments, the means to detect comprises at least one of nucleic acid probe and binding ligands disposed on an assay surface. In some embodiments, the assay surface comprises a chip, array, or fluidity card. In some embodiments, the probes comprise complementary nucleic acid sequences to at least 10 to 50 nucleic acid sequences of the marker genes. In some embodiments, the binding ligands comprise antibodies or binding fragments thereof. In some embodiments, the assay system further comprises: a control selected from information containing a predetermined control level or set of genetic variants or polymorphic markers that has been correlated with diagnosis, development, progression, or life expectancy in interstitial lung disease patients.

A method of detecting a level of gene expression of one or more marker genes in a human subject with interstitial pneumonia, comprising, optionally, obtaining a biological sample from a human individual with interstitial pneumonia; detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof, in one or more cells from the biological sample from the individual. In some embodiments, the method further comprises detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof, in one or more cells from the biological sample from the individual. In some embodiments, the method further comprises detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof in one or more cells from the biological sample from the individual.

A method of treating an interstitial lung disease in a subject in need of such treatment, comprising: detecting a level of one or more marker genes selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof in a biological sample obtained from the human subject; and, administering an effective amount of an effective treatment. In some embodiments, the method further comprises detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof, in one or more cells from the biological sample from the individual. In some embodiments, the method further comprises detecting the level of expression of a gene selected from MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7, or homologs or variants thereof, in one or more cells from the biological sample from the individual.

Detection of Genetic Variants

Methods of detecting a genetic variant are further described, for example, in U.S. Pat. No. 8,673,565 (the contents of which are herein incorporated by reference in their entirety). Genetic variations in the mucin genes are associated with pulmonary diseases. These genetic variations can be found in any part of the gene, e.g., in the regulatory regions, introns, or exons. Relevant genetic variations may also be found the intergene regions, e.g., in sequences between mucin genes. Insertions, substitutions, and deletions are included in genetic variants. Single nucleotide polymorphisms (SNPs) are exemplary genetic variants.

In particular, 14 independent SNPs are associated with pulmonary disorders (e.g. FIP or IPF). The studies disclosed herein demonstrate that presence of one or more of these SNPs associated with MUC5B can lead to predisposition to a pulmonary disorder. In addition, in some embodiments, if present, some of these SNPs are related to a transcription factor binding site. The transcription factor binding site can effect modulation of MUC5B expression, for example E2F3 loss, and HOXA9 and PAX-2 generation.

The disclosure thus provides methods for assessing the presence or absence of SNPs in a sample from a subject suspected of having or developing a pulmonary disorder (e.g., because of family history). In certain embodiments, one or more SNPs are screened in one or more samples from a subject. The SNPs can be associated with one or more genes, e.g., one or more MUC genes or other genes associated with mucous secretion. In some embodiments, a MUC gene associated SNP is associated with MUC5B and/or another MUC gene, such as MUC5AC or MUC1. SNPs contemplated for diagnostic, treatment, or prognosis can include SNPs found within a MUC gene and/or within a regulatory or promoter region associated with a MUC gene. For example, one or more SNPs can include, but are not limited to, detection of the SNPs of MUC5B alone or in combination with other genetic variations or SNPs and/or other diagnostic or prognostic methods.

Methods for detecting genetic variants such as a SNP are known in the art, e.g., Southern or Northern blot, nucleotide array, amplification methods, etc. Primers or probes are designed to hybridize to a target sequence. For example, genomic DNA can be screened for the presence of an identified genetic element of using a probe based upon one or more sequences, e.g., using a probe with substantial identity to a subsequence of the MUC5B gene. Expressed RNA can also be screened, but may not include all relevant genetic variations. Various degrees of stringency of hybridization may be employed in the assay. As the conditions for hybridization become more stringent, there must be a greater degree of complementarity between the probe and the target for duplex formation to occur. Thus, high stringency conditions are typically used for detecting a SNP.

Thus, in some embodiments, a genetic variant MUC5B gene in a subject is detected by contacting a nucleic acid in a sample from the subject with a probe having substantial identity to a subsequence of the MUC5B gene, and determining whether the nucleic acid indicates that the subject has a genetic variant MUC5B gene. In some cases, the sample can be processed prior to amplification, e.g., to separate genomic DNA from other sample components. In some cases, the probe has at least 90, 92, 94, 95, 96, 98, 99, or 100% identity to the MUC5B gene subsequence. Typically, the probe is between 10-500 nucleotides in length, e.g., 10-100, 10-40, 10-20, 20-100, 100-400, etc. In the case of detecting a SNP, the probe can be even shorter, e.g., 8-20 nucleotides in length. In some cases, the MUC5B gene sequence to be detected includes at least 8 contiguous nucleotides, e.g., at least 10, 15, 20, 25, 30, 35 or more contiguous nucleotides. In some embodiments, the sequence to be detected includes 8 contiguous nucleotides, e.g., at least 10, 15, 20, 25, 30, 35 or more contiguous nucleotides.

The degree of stringency can be controlled by temperature, ionic strength, pH and/or the presence of a partially denaturing solvent such as formamide. For example, the stringency of hybridization is conveniently varied by changing the concentration of formamide within the range up to and about 50%. The degree of complementarity (sequence identity) required for detectable binding will vary in accordance with the stringency of the hybridization medium and/or wash medium. In certain embodiments, in particular for detection of a particular SNP, the degree of complementarity is about 100 percent. In other embodiments, sequence variations can result in <100% complementarity, <90% complementarity probes, <80% complementarity probes, etc., in particular, in a sequence that does not involve a SNP. In some examples, e.g., detection of species homologs, primers may be compensated for by reducing the stringency of the hybridization and/or wash medium.

High stringency conditions for nucleic acid hybridization are well known in the art. For example, conditions may comprise low salt and/or high temperature conditions, such as provided by about 0.02 M to about 0.15 M NaCl at temperatures of about 50° C. to about 70° C. Other exemplary conditions are disclosed in the following Examples. It is understood that the temperature and ionic strength of a desired stringency are determined in part by the length of the particular nucleic acid(s), the length and nucleotide content of the target sequence(s), the charge composition of the nucleic acid(s), and by the presence or concentration of formamide, tetramethylammonium chloride or other solvent(s) in a hybridization mixture. Nucleic acids can be completely complementary to a target sequence or exhibit one or more mismatches.

Nucleic acids of interest can also be amplified using a variety of known amplification techniques. For instance, polymerase chain reaction (PCR) technology may be used to amplify target sequences (e.g., genetic variants) directly from DNA, RNA, or cDNA. In some embodiments, a stretch of nucleic acids is amplified using primers on either side of a targeted genetic variation, and the amplification product is then sequenced to detect the targeted genetic variation (using, e.g., Sanger sequencing, Pyrosequencing, Nextgen® sequencing technologies). For example, the primers can be designed to hybridize to either side of the upstream regulatory region of the MUC5B gene, and the intervening sequence determined to detect a SNP in the promoter region. In some embodiments, one of the primers can be designed to hybridize to the targeted genetic variant. In some cases, a genetic variant nucleotide can be identified using RT-PCR, e.g., using labeled nucleotide monomers. In this way, the identity of the nucleotide at a given position can be detected as it is added to the polymerizing nucleic acid. The Scorpion™ system is a commercially available example of this technology.

Thus, in some embodiments, a genetic variant MUC5B gene in a subject is detected by amplifying a nucleic acid in a sample from the subject to form an amplification product, and determining whether the amplification product indicates a genetic variant MUC5B gene. In some cases, the sample can be processed prior to amplification, e.g., to separate genomic DNA from other sample components. In some cases, amplifying comprises contacting the sample with amplification primers having substantial identity to MUC5B genomic subsequences, e.g., at least 90, 92, 94, 95, 96, 98, 99, or 100% identity. Typically, the sequence to be amplified is between 30-1000 nucleotides in length, e.g., 50-500, 50-400, 100-400, 50-200, 100-300, etc. In some cases, the sequence to be amplified or detected includes at least 8 contiguous nucleotides, e.g., at least 10, 15, 20, 25, 30, 35 or more contiguous nucleotides. In some embodiments, the sequence to be amplified or detected includes 8 contiguous nucleotides, e.g., at least 10, 15, 20, 25, 30, 35 or more contiguous nucleotides. In some aspects, the contiguous nucleotides include nucleotide 28.

Amplification techniques can also be useful for cloning nucleic acid sequences, to make nucleic acids to use as probes for detecting the presence of a target nucleic acid in samples, for nucleic acid sequencing, for control samples, or for other purposes. Probes and primers are also readily available from commercial sources, e.g., from Invitrogen, Clonetech, etc.

Detection of Expression Levels

Expression of a given gene, e.g., MUC5B or another mucin, pulmonary disease marker, or standard (control), is typically detected by detecting the amount of RNA (e.g., mRNA) or protein. Sample levels can be compared to a control level.

Methods for detecting RNA are largely cumulative with the nucleic acid detection assays described above. RNA to be detected can include mRNA. In some embodiments, a reverse transcriptase reaction is carried out and the targeted sequence is then amplified using standard PCR. Quantitative PCR (qPCR) or real time PCR (RT-PCR) is useful for determining relative expression levels, when compared to a control. Quantitative PCR techniques and platforms are known in the art, and commercially available (see, e.g., the qPCR Symposium website, available at qpersymposium.com). Nucleic acid arrays are also useful for detecting nucleic acid expression. Customizable arrays are available from, e.g., Affimatrix. An exemplary human MUC5B mRNA sequence, e.g., for probe and primer design, can be found at GenBank Accession No. AF086604.1.

Protein levels can be detected using antibodies or antibody fragments specific for that protein, natural ligands, small molecules, aptamers, etc. An exemplary human MUC5B sequence, e.g., for screening a targeting agent, can be found at UniProt Accession No. 000446.

Antibody based techniques are known in the art, and described, e.g., in Harlow & Lane (1988) Antibodies: A Laboratory Manual and Harlow (1998) Using Antibodies: A Laboratory Manual; Wild, The Immunoassay Handbook, 3d edition (2005) and Law, Immunoassay: A Practical Guide (1996). The assay can be directed to detection of a molecular target (e.g., protein or antigen), or a cell, tissue, biological sample, liquid sample or surface suspected of carrying an antibody or antibody target.

A non-exhaustive list of immunoassays includes: competitive and non-competitive formats, enzyme linked immunosorption assays (ELISA), microspot assays, Western blots, gel filtration and chromatography, immunochromatography, immunohistochemistry, flow cytometry or fluorescence activated cell sorting (FACS), microarrays, and more. Such techniques can also be used in situ, ex vivo, or in vivo, e.g., for diagnostic imaging.

Aptamers are nucleic acids that are designed to bind to a wide variety of targets in a non-Watson Crick manner. An aptamer can thus be used to detect or otherwise target nearly any molecule of interest, including a pulmonary disease associated protein. Methods of constructing and determining the binding characteristics of aptamers are well known in the art. For example, such techniques are described in U.S. Pat. Nos. 5,582,981, 5,595,877 and 5,637,459. Aptamers are typically at least 5 nucleotides, 10, 20, 30 or 40 nucleotides in length, and can be composed of modified nucleic acids to improve stability. Flanking sequences can be added for structural stability, e.g., to form 3-dimensional structures in the aptamer.

Protein detection agents described herein can also be used as a treatment and/or diagnosis of pulmonary disease or predictor of disease progression, e.g., propensity for survival, in a subject having or suspected of developing a pulmonary disorder. In certain embodiments, MUC5B antibodies can be used to assess MUC5B protein levels in a subject having or suspected of developing a pulmonary disorder. It is contemplated herein that antibodies or antibody fragments may be used to modulate MUC5B production in a subject having or suspected of developing a pulmonary disease. In certain embodiments, one or more agents capable of modulating MUC5B may be used to treat a subject having or suspected of developing a pulmonary disorder. One or more antibodies or antibody fragments may be generated to detect one or more of the SNPs disclosed herein by any method known in the art.

In certain embodiments, MUC5B diagnostic tests may include, but are not limited to, alone or in combination, analysis of rs35705950 SNP in MUC5B gene, MUC5B mRNA levels, and/or MUC5B protein levels.

Additional Pulmonary Disease Markers

The above methods of detection can be applied to additional pulmonary disease markers. That is, the expression level or presence of genetic variants of at least one additional pulmonary disease marker gene can be determined, or the activity of the marker protein can be determined, and compared to a standard control for the pulmonary disease marker. The examination of additional pulmonary disease markers can be used to confirm a diagnosis of pulmonary disease, monitor disease progression, or determine the efficacy of a course of treatment in a subject.

In some cases, pulmonary disease is indicated by an increased number of lymphocytes, e.g., CD4+CD28− cells.

Genetic variations in the following genes are associated with pulmonary disease: Surfactant Protein A2, Surfactant Protein B, Surfactant Protein C, TERC, TERT, IL-1RN, IL-1α, IL-1β, TNF, Lymphotoxin a, TNF-RII, IL-10, IL-6, IL-12, IFNγ, TGFβ, CR1, ACE, IL-8, CXCR1, CXCR2, MUC1 (KL6), or MUC5AC. Thus, the invention further includes methods of determining whether the genome of a subject comprises a genetic variant of at least one gene selected from these genes. The presence of a genetic variant indicates that the subject has or is at risk of developing pulmonary disease. Said determining can optionally be combined with determining whether the genome of the subject comprises a genetic variant MUC5B gene, or determining whether the subject has an elevated level of MUC5B RNA or protein to confirm or strengthen the diagnosis or prognosis.

Abnormal expression in the following genes can also be indicative of pulmonary disease: Surfactant Protein A, Surfactant Protein D, KL-6/MUC1, CC16, CK-19, Ca 19-9, SLX, MCP-1, MIP-1a, ITAC, glutathione, type III procollagen peptide, sIL-2R, ACE, neopterin, beta-glucuronidase, LDH, CCL-18, CCL-2, CXCL12, MMP7, and osteopontin. Thus, the expression of one of these genes can be detected and compared to a control, wherein an abnormal expression level indicates that the subject has or is at risk of developing pulmonary disease. Said determining can optionally be combined with determining whether the genome of the subject comprises a genetic variant MUC5B gene, or determining whether the subject has an elevated level of MUC5B RNA or protein to confirm or strengthen the diagnosis or prognosis.

Biomarkers

The present disclosure provides a peripheral blood biomarker profile for IPF to demonstrate the use of a predictive biomarker profile in cases of preclinical pulmonary fibrosis (PrePF) derived from families with familial IPF. The present disclosure also provides biomarker identification for association between each genetic, epigenetic or protein (gene product) biomarker with PrePF and the predictive value of the combination of biomarkers associated with PrePF.

A large cohort of families with familial IPF for genetic research was established, including 937 families with ≥2 cases of IPF, and 2375 family members that have been previously phenotyped as unaffected. This study focuses on subjects with PrePF to elucidate the processes active in early disease pathogenesis and to predict or prevent the irreversible fibroproliferative process. Genetic risk factors, especially the MUC5B promoter variant, identifies individuals with preclinical interstitial changes on chest CT scan that progress and are associated with reduced survival. Biomarkers may be used to identify those subjects with PrePF among those at-risk for IPF. Given the irreversible nature of IPF, even approved treatments (pirfenidone and nintedanib) only modestly slow progression and have not been shown to alter the 3-5 year survival. Pirfenidone and nintedanib are effective in patients with mild disease, suggesting that patients with PrePF may be targeted for early intervention, before most of the lung has been irreversibly remodeled.

Table 1 below shows additional gene expression changes present in subjects with IPF compared to controls. Specifically, the expression of the genes listed in Table 1 are upregulated in IPF compared to the expression of these same genes in control subjects. Accordingly, the discovery of elevated expression levels of one or more genes listed in Table 1 compared to a control in an asymptomatic subject may indicate that the subject has PrePF and/or that the subject is at risk for developing IPF.

In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding a gene or gene product that is upregulated in a subject having a fibrotic pulmonary disease of the disclosure. In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding Leukotriene A4 Hydrolase (LTA4H), Surfactant Protein B (SFTPB), Breast Cancer Anti-Estrogen Resistance 3 (BCAR3), C-X-C motif Chemokine Ligand 13 (CXCL13), EPH Receptor A2 (EPHA2), Serum Amyloid A1 (SAA1), Phospholipase A2 Group IIA (PLA2G2A), Insulin-Like Growth Factor Binding Protein 3 (IGFBP3), C-C Motif Chemokine Ligand 28 (CCL28), 5100 Calcium Binding Protein A12 (S100A12), Thromboxane A Synthase 1 (TBXAS1), Leukocyte Cell Derived Chemotaxin 1 (LECT1), Complement C3 (C3), Gastrin Releasing Peptide (GRP), C-Reactive Protein (CRP), Vitrin (VIT), Insulin-Like Growth Factor Binding Protein 1 (IGFBP1), Family with Sequence Similarity 173 Member A (FAM173A), Natriuretic Peptide A (NPPA), Secreted Frizzled Related Protein 1 (SFRP1), Ezrin (EZR), Inter-Alpha-Trypsin Inhibitor Heavy Chain Family Member 5 (ITIH5), Pleckstrin and Sec7 Domain Containing 2 (PSD2), Galectin 3 Binding Protein (LGALS3BP), Catenin Beta 1 (CTNNB1), Chromodomain Y Like 2 (CDYL2), Matrix Metallopeptidase 7 (MMP7), Apolipoprotein B (APOB), Proline and Arginine Rich End Leucine Rich Repeat Protein (PRELP), Eukaryotic Translation Initiation Factor 1A, X-linked (EIF1AX), Mesencephalic Astrocyte Derived Neurotrophic Factor (MANF), TNF Receptor Superfamily Member 13C (TNFRSF13C), Deformed Epidermal Autoregulatory Factor 1 transcription factor (DEAF1), Tumor Protein Translationally-Controlled 1 (TPT1), Unc-5 Netrin Receptor B (UNCSB), Phosphatidylethanolamine Binding Protein 1 (PEBP1), Syntaxin 8 (STX8), Polymeric Immunoglobulin Receptor (PIGR), Adenine Phosphoribosyltransferase

(APRT), Matrix Metallopeptidase 3 (MMP3), Galectin 7 (LGALS7), Bruton Tyrosine Kinase (BTK), NSFL1 Cofactor (NSFL1C), FER Tyrosine Kinase (FER), Regenerating Family Member 1 Beta (REG1B), SMAD Family Member 2 (SMAD2), Interleukin 1 Receptor Like 1 (IL1RL1), C-C Motif Chemokine Ligand 18 (CCL18), Acid Phosphatase 2 Lysosomal (ACP2), Eukaryotic Translation Initiation Factor 4E Family Member 2 (EIF4E2), Neurexin 3 (NRXN3), IGF Like Family Member 1 (IGFL1), NME/NM23 Nucleoside Diphosphate Kinase 1 (NME1), Potassium Voltage-Gated Channel Isk-Related Family Member 1-Like (KCNE1L) or Neurexophilin 2 (NXPH2).

TABLE 1

TARGET_GENE_SYMBOL	ORGANISM	p-value	B-H q-value	Fold Change
LTA4H	Human	8.70E−43	3.13E−39	3.912
SFTP8	Human	1.17E−37	2.10E−34	3.399
BCAR3	Human	4.28E−25	3.85E−22	2.906
CXCL13	Human	1.30E−29	1.56E−26	2.904
EPHA2	Human	9.62E−23	6.93E−20	2.651
SAA1	Human	6.01E−07	7.84E−06	2.631
PLA2G2A	Human	8.19E−21	2.95E−18	2.171
Igfbp3	Mouse	1.18E−18	2.66E−16	2.149
CCL28	Human	1.22E−22	7.30E−20	2.135
S100A12	Human	1.06E−20	3.45E−18	2.125
TBXAS1	Human	1.60E−21	7.20E−19	2.11
LECT1	Human	4.17E−19	1.00E−16	2.082
C3	Human	7.08E−07	8.95E−06	2.062
GRP	Human	8.35E−09	1.66E−07	1.988
CSP	Human	1.36E−08	2.61E−07	1.957
VIT	Human	2.47E−17	4.45E−15	1.929
IGFBP1	Human	4.32E−11	1.56E−09	1.914
FAM173A	Human	2.19E−13	1.84E−11	1.904
NPPA	Human	5.02E−12	2.58E−10	1.877
SFRP1	Human	1.74E−20	5.23E−18	1.866
EZR	Human	6.41E−10	1.72E−08	1.809
ITIH5	Human	5.11E−21	2.04E−18	1.705
PSD2	Human	5.38E−18	1.08E−15	1.689
LGALS38P	Human	8.06E−22	4.15E−19	1.678
		1.18E−05	0.000102	1.668
CTNNB1	Human	5.66E−12	2.87E−10	1.625
CDYL2	Human	4.11E−07	5.59E−06	1.622
MMP7	Human	1.56E−19	4.02E−17	1.621
APOB	Human	8.73E−13	6.42E−11	1.597
PRELP	Human	1.13E−10	3.53E−09	1.595
EIF1AX	Human	2.13E−06	2.31E−05	1.59
MANF	Human	0.00458	0.015006	1.585
TNFRSF13C	Human	1.77E−11	7.31E−10	1.573
C3	Human	2.40E−16	3.93E−14	1.566
DEAF1	Human	0.000221	0.001192	1.565
TPT1	Human	1.22E−12	7.82E−11	1.548
UNC5B	Human	2.06E−34	2.18E−12	1.547
PEBP1	Human	4.92E−11	1.72E−09	1.544
STX8	Human	8.82E−12	4.13E−10	1.537
PIGR	Human	1.29E−09	3.19E−08	1.532
APRT	Human	1.51E−07	2.26E−06	1.525
MMP3	Human	9.50E−07	1.15E−05	1.524
LGALS7	Human	7.51E−05	0.000474	1.514
BTK	Human	1.47E−09	3.52E−08	1.511
NSFL1C	Human	7.33E−11	2.40E−09	1.506
FER	Human	2.24E−07	3.24E−06	1.503
REG1B	Human	6.68E−11	2.25E−09	1.502
SMAD2	Human	4.39E−10	1.25E−08	1.493
IL1RL1	Human	9.55E−07	1.15E−05	1.492
CCL18	Human	1.25E−13	1.07E−11	1.491
ACP2	Human	3.73E−08	6.33E−07	1.488
EIF4E2	Human	1.67E−12	1.02E−10	1.483
NRXN3	Human	2.33E−17	4.42E−15	1.48
IGFL1	Human	5.07E−10	1.40E−08	1.474
NME1	Human	1.43E−10	4.39E−09	1.463
KCNE1L	Human	3.93E−20	1.09E−17	1.462
NXPH2	Human	9.66E−30	2.47E−08	1.451

Table 2 below shows additional gene expression changes present in subjects with IPF compared to controls. Specifically, the expression of the genes listed in Table 2 are downregulated in IPF compared to the expression of these same genes in control subjects. Accordingly, the discovery of decreased expression levels of one or more genes listed in Table 2 compared to a control in an asymptomatic subject may indicate that the subject has PrePF and/or that the subject is at risk for developing IPF.

In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding a gene or gene product that is downregulated in a subject having a fibrotic pulmonary disease of the disclosure. In some embodiments of the methods of the disclosure, the subject has a mutation in a nucleic acid or amino acid sequence encoding Surfactant Protein D (SFTPD), Glyceraldehyde-3-Phosphate Dehydrogenase (GAPDH), Histone Cluster 1 H1 Family Member C (HIST1H1C), YTH Domain Containing 1 (YTHDC1), Plexin A1 (PLXNA1), Serine Peptidase Inhibitor Kazal Type 6 (SPINK6), LDL Receptor Related Protein Associated Protein 1 (LRPAP1), Secretoglobin Family 3A Member 1 (SCGB3A1), H2A Histone Family Member Z (H2AFZ) or Chromosome 1 Open Reading Frame 162 (C1orf162).

TABLE 2

TARGET_GENE_SYMBOL	ORGANISM	p-value	B-H q-value	Fold Change
SFTPD	Human	8.19E−15	9.83E−13	−2.262
GAPDH	Human	1.46E−09	3.52E−08	−2.096
HIST1H1C	Human	3.68E−18	7.80E−16	−2.011
		3.63E−16	5.69E−14	−1.964
YTHDC1	Human	1.19E−11	5.38E−10	−1.699
PLXNA1	Human	1.64E−12	1.02E−10	−1.64
SPINK6	Human	3.68E−07	5.04E−06	−1.635
LRPAP1	Human	2.65E−15	3.53E−13	−1.521
SCGB3A1	Human	3.35E−07	4.61E−06	−1.518
H2AFZ	Human	3.91E−14	3.91E−12	−1.501
		2.95E−11	1.16E−09	−1.493
C1orf162	Human	1.29E−84	7.52E−04	−1.458

In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding MUC5B, TERC, FAM13A, TERT, DSP, AZGP1, OBFC1, ATP11A, IVD/DISP2, DPP9, SIGLEC14, ADM2, TSPAN5, CAMKK1 or MMP-7.

In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Telomerase RNA Component (TERC). In some embodiments the polymorphism is rs6793295 comprising (SEQ ID NO: 1).

```
                                    (SEQ ID NO: 1)
AGAAAGAAGT CATGAAAGTA GGAACCACAT

TTTTACTCAT CTTTCTGTCT CCAGCAAGCA

GCTTACTGCT TTTCATACAC ATTTTGCTTT

TATTACTCAT GATTTCAAAG GTGTAATGGT

TCAGCCACAT CAATGTAACA AACAGTTCAC

ACTGGGCTCT TATAGTCTGG CCTTTAAAAC

CTTCACTATT TATGCTTTCA TCTTAACTAC

TTTGACCCTC ACAGGTTTAC TCACTAAGAA

CTTGAGTTTC AAGAGAAAAG ATGACATGTT
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TGCTGCTTAA ACAAGCAATA TCTAAAAGCA

TATTTAGTTA TAAACGTCTT ACCAAGAATT

GATATAATTT TCATTTAAAC ATTTTTATAA

ATAGTAGTTT ACAAGATATA GTAAGTACAT

CTCTAAAAAT ACAGTGTATT CATGTACCTT

GACATAAACT TGTAGTAGTA CCTTAGTTTT

ATTCATGTTG TTATATTAAC TACCATCACT

TTGAATACAT ACCTGTTCAC

B

GTACAGTATA GGTCGGTTTA GGTTTATTGC

CTTAATTGCT TGGTTTTGAG TTAGTACTGT

AGCAAATGCT ATCACACTTT GCATTCCCTA

AAAACAGGTA AATTCATTAA GGAAACAGAC

AAAGTATATA ATAATCTCGC TACATAAATA

TTTCAAGATC AGCTATCTGC ATTCTGATAA

AATTGTTTTT AAAATTTAAG CATTCCTTGG

ACTTTGAATT GTAAGTTGAT CAAATTCAAA

AATGAATTGT TACTGTATTC TTCTCTCCTG

GCCCTAAAAT CTATCTAAAA CATGGCATGG

GGAGTTTCTT AATGTTTCAG TGTCCATTTC

CTGGGTGTTT CCCTCTAGGT TTTTTTTCCT

CACCCCTCAA GCTTCTATGT GGATCCCAGC

TAGAGCTCAT ACTACTTATC CAACACACAT

CATTGTGCAA GCACTCTTTT ATATTCATAC

TAGTACTTTT AAGTGTGTGT GCGGTGGGAA

AAGGTTACCA ATCACATTTT

In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Family with sequence similarity 13 member A (FAM13A). In some embodiments the polymorphism is rs2609255 comprising (SEQ ID NO: 2).

(SEQ ID NO: 2)

GTATTCATCA ACTCCTATTT CATTCCCTCT TCCTGTGCTC

ACTGGAAGAT GACATTTCCC AGACTTCCAA GAATGTTACT

GAGTTCTGGA ATGTAAGTAG AAGGGATAAG TATCACTTCT

GTGCTGTGGC GGTTATGGAC CTGTGAACTT TGCACACGCC

TTCTATCTTC TTTTTCAGTG TCCATTTCAG AGGGCATGTT

TTCAGATGAA ACCAGTAGAA GATGGAAGCA GCCTGTGACT

AGAATCACTG CTTAGGGTCT TGCTGCCTAG GAATCCCACT

CTACCTGCAA CAGACTGTGA AAGAACCGAG AAATACACTG

ATTTTGAACA TAGCCCATAC TATAATGGGG ATGTTTGTTA

-continued

CAGCAGTTAG CATTAAAAAC CTTGGCTAGG CATTGGTCAT

AATTGTAGAA CACAGCAAAT GAAGGGAAAC TGGAACATAG

AGGCCAGTGA GAACTTTAGG GTTAATGAAA AATGAGGGCA

ACCAGGATAA TTTGGTTCTT

K

GCCAAATAGG AAGGTGAAAC CAAAGGTAGA CTGGAGGTCA

GAAAATCAGT CCAGCACATG TGATGTTTTC ATTTAGTTGC

CTGTATGTCT GTCTGGTCTC CAGCTCAGCC TGGCTCCTTG

AGGTAAGAGG CAGTGGCTGT TCACCTTTGC ATCCCAGCAC

CTGGCATACA ATAGATGGGA TGAAATGTTC AAACTGAGCC

TAAGCTTCAG GGTGCTTATC AAAGCAGGGA AGATACACAA

GAGGAGATGA TTCAGGTCCA GGGCAGGTCA GGTATCTAAA

CCCAGTCTCT TAGGAAGCTG GATCCTCCGA ACCAGGGAGA

ACAAGCTGGA TATGCACTGG ATTTCCCAGC AGTACTGATC

TAGAGACTCT CATAGAGTCC CTTTTATTCC TTGGCCTAGG

GTTACAACTG CTTATAGCAT CTGGAAAGAC TCAACACCTC

AAAAGAGACT TTCAGTAGAT ACAGCAAATA CACTCATGGA

ATTGATAATT AAGCTTCAAT

In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Telomerase Reverse Transcriptase (TERT). In some embodiments the polymorphism is rs2736100 comprising (SEQ ID NO: 3).

(SEQ ID NO: 3)

ATTGTCGTTG TTTGCTTTTG TTTATTGAGA CAGTCTCACT

CTGTCACCCA GGCTGGAGTG TAATGGCACA ATCTCGGCTC

ACTGCAACCT CTGCCTCCTC GGTTCAAGCA GTTCTCATTC

CTCAACCTCA TGAGTAGCTG GGATTACAGG CGCCCACCAC

CACGCCTGGC TAATTTTTGT ATTTTTAGTA GAGATAGGCT

TTCACCATGT TGGCCAGGCT GGTCTCAAAC TCCTGACCTC

AAGTGATCTG CCCGCCTTGG CCTCCCACAG TGCTGGGATT

ACAGGTGCAA GCCACCGTGC CCGGCATACC TTGATCTTTT

AAAATGAAGT CTGAAACATT GCTACCCTTG TCCTGAGCAA

TAAGACCCTT AGTGTATTTT AGCTCTGGCC ACCCCCCAGC

CTGTGTGCTG TTTTCCCTGC TGACTTAGTT CTATCTCAGG

CATCTTGACA CCCCCACAAG CTAAGCATTA TTAATATTGT

TTTCCGTGTT GAGTGTTTCT

K

TAGCTTTGCC CCCGCCCTGC TTTTCCTCCT TTGTTCCCCG

TCTGTCTTCT GTCTCAGGCC CGCCGTCTGG GGTCCCCTTC

CTTGTCCTTT GCGTGGTTCT TCTGTCTTGT TATTGCTGGT

AAACCCCAGC TTTACCTGTG CTGGCCTCCA TGGCATCTAG

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CGACGTCCGG GGACCTCTGC TTATGATGCA CAGATGAAGA
TGTGGAGACT CACGAGGAGG GCGGTCATCT TGGCCCGTGA
GTGTCTGGAG CACCACGTGG CCAGCGTTCC TTAGCCAGTG
AGTGACAGCA ACGTCCGCTC GGCCTGGGTT CAGCCTGGAA
AACCCCAGGC ATGTCGGGGT CTGGTGGCTC CGCGGTGTCG
AGTTTGAAAT CGCGCAAACC TGCGGTGTGG CGCCAGCTCT
GACGGTGCTG CCTGGCGGGG GAGTGTCTGC TTCCTCCCTT
CTGCTTGGGA ACCAGGACAA AGGATGAGGC TCCGAGCCGT
TGTCGCCCAA CAGGAGCATG
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In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Desmoplakin (DSP). In some embodiments the polymorphism is rs2076295 comprising (SEQ ID NO: 4).

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                                         (SEQ ID NO: 4)
ATTTGGGAAC CTTTAAAAAA TATTCTGGCT TCAAAAATAC
TCCATATTTA CATCTTTGGT TCTATCTGAA GTAAAGCCGT
GATGGTGTGC GTAAGTGAAA CAGGTGCAAA GGGGCAACAA
CAAAGGGCGC CTCTCTTTGT CTTTGTGTCG CAGGCGGAGA
TGGACATGGT GGCCTGGGGT GTGGACCTGG CCTCAGTGGA
GCAGCACATT AACAGCCACC GGGGCATCCA CAACTCCATC
GGCGACTATC GCTGGCAGCT GGACAAAATC AAAGCCGACC
TGGTACTTGT CTGTGTTTCA TTTTAGAGTC TTCAAAATAT
CTACCGAAGG ATCGTGTAAT TACTCAATCC CAGGGAGTTT
CTTCTGAAAC ATTGCTATTA TTTCTTTCCC AGAAGACTGG
AAATGTTTAG AAATCCCACT TCTTAAATGG GGAAGTGGAA
TCAGTAGCCC TATTAGAGAT TATGTTAACA CTTGAAGAGG
AGTTAAACCA GAGGCTGAGG
K
TGTGCAAACA CTCATTTGCA GTTTGTGAAT AAGTCTCTTT
AGGGGTGGCA GTTTGTTTCT GCGGTAAGCA GAACATCTTT
TTGAATAGGG GAAATGCAAC AGTCTTATAC AGTAGTTTGT
GTCATTGGTG AATCCTTTCC TAGGTGGTAA TTAAAACATT
ATTTCTACTG AGCAAAGCCA TATGTCATCC CGACACCCGC
TCCCATGCTG AAAAAAGTCA GACTTGAAAC TGGGTTGAGA
ATTACAGCAT AAAATCATAA CTGATCTTAA GTGCTTAGTT
TCCCGCAGGT CTCTACACTT GTAAATCACT AAACTTTTTT
TTTTTTTTTT TACCTGAGAC CATAGCTTCT CATCCTCATT
TCTTCTTCTG GCTTTTTGGG GCTTACTTTT GTCCACCTGA
GCCCCTGACC AACTTTCTCC TTCATTTCTC TAAGACCTAG
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GGAATCCTAA ATGATGTCTT TAAACTTTAA GACAATTTTC
TAACACGTGA GTCTTTAAGT
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In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Zinc-alpha 2-Glycoprotein 1 (AZGP1). In some embodiments the polymorphism is rs4727443 comprising (SEQ ID NO: 5).

```
                                         (SEQ ID NO: 5)
CCCAACCCAA ATAAGCACTA TAACCTCTTG TTATTCACTT
CTCATGCAAC CAGTCTTCTG TTCTCTGTGA GTCTTTAGGA
AATGAGGAGC ATGATCTTCT AGCAGTAAAA CACCTGTAGA
GAATTGCCTT ATGTTTTTTG TTTGTTTATT TGTTTGTGTG
CTTTGGTTTG GTTTGCTTTT TTTTTTTTTT TTTTTTTTTT
TTTGAGATGG AGTCTCGCCC TGTTGCCCAG GCTGGAGTGT
AGTGGCGAAA TCTCGGCTCA CTGCAACCTC CACCTCCCTG
GTTCAAGCAA TTCCCCTGTC TCAGCCTCCC GAGTAGCTGA
GATTACAGGT GCACACCACC ACGCCCGGCT AATTTTTTTG
TATTTTTAGT AGAGATGGGG TTTCACCATG TTGGCCAGAC
TGGTCTCGAA CTTCTGACCT CAGGCAATCC GCCTGCCTCA
GCCTCCCAAA GCGCTGGGAT TACAGGCATG AGCCACTGCG
CCCCGCCTCC ATGTTAATCA
M
TCTTTCTGAT TTCAAATAAC TCATTATCCC CATGACCTTA
TGGATTTGTT TTTCCTCTTC ATCCACAAAA TTCTCCAGAG
AAGTCTCCCT TGTTATCTCT TGGCTGTGCT TTCTATCTCA
CCAGTTATCT TTCTCCAAAG AGCTTCCTCT GCAAAGAAGC
TTTGTATATG AAGACCATGT GGGGGCTGAA TCAAGACCAA
GTTTCACAAC CTAAAAGTAG TTCACAAAGC TTCCTTGCCT
CTATTCTCTG CAAATCTGTA AACTCTTCAG CTGACCCAAT
TTCTCTCTTT AGCCTTCAGA GATTATTTTA TTTTATTTTA
TTTCATTTCA TTTCATTTCA TTTTGACAGA ATCTAGCTCT
GTCGCCCAGG CTGGAGTGCA GTGGCACCAT CTTTGCTCAC
TGCAACCTCC CCCTCACAGG TTCAAGCAAC TGTCCTGCCT
CAGCCTCCCG AGTAGCTGGG ATTACAGGCG TGAGCCACCA
CGCCCAGCTG ATTTTTTTTT
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In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Oligonucleotide/oligosaccharide-binding Fold Containing 1 (OBFC1). In some embodiments the polymorphism is rs11191865 comprising (SEQ ID NO: 6).

(SEQ ID NO: 6)
CCTCTACTGC CGTACACCCC ACCACTCAGC CTTGGAGTGC
CTGTGTGCAG AGCAGGGCTG AGGCATGGTG CTGCTTTGGT
GGTCTAGGTT TGCTGCAGGG CCAGGTGGCC TGAGCTCCAG
GCAGGATCTC TGGCTGCACT CAGCCCTTTC TGCCTCCCCA
AATGCTCTAT ATCACTATTT GTACACTGAG CAGAGTAAAG
TTAGAGAGAA CTGTTTTATA GAATAGGGCT GGCCCCCGCT
CCCCTGGCCT ACGTGATGGT CCTTCCTGGC TGCCAGGTAC
TTGTTTGTAT TAGAGACAGA CACTCCACAG GGTCTGTTGT
GGCCCACAGC ACATAGGCAA TCAGAGGCAG AAAGCAGAGC
TGTTTGGACC CACAGAGGGC CGGCTGTCTG CCACTGAAAT
GTCTTTCCAG TTGGTTGAGA AGCAGCAGGA TGCTCTGCTG
GTGATGTCTG AAAGTCCCAG GATTCTTTGG GTCTCCAAGG
AGATCCTAGC ATATACCACT
R
TCGTGGTTTT AATAAAGAGC AAAAACACTT TCAGATGGGG
AGAAGAGTGG AACAAAAGGT ATTCTTCCTG GGTTGAAGTC
TGGGGGAAAG GCATTGAGAA GACTGGGCTA ATGGCACAAA
CCAATGAAGT ACTCAAGTCA CCTGTGATGG AGGCCAGTCA
TCCAATGGTA TCAACTTTGT ATGTGGCAAC ACTTAATAAA
AATCTGAACA GGTCTTCACT TGTGGACACA GTAGACTTTC
TTGAAAAAGG ACAGAAAAGT GAGCCCTGTG AATTTTCATC
TCACGGACTG ACAACAATGA CTTGCCTTTA AGGACAGTCA
CTCAAGATGA AGATGCAACA AAACCCTTCC AGTTCCAAGT
GGCTGATGAA AAAAAAAAAA TCTTAAAAGC ATCACAGAAC
AACGGAGAAA GAGATCAGAA GACTATAACA GATAGTTTGA
ATTTTAAAAC TCAGAGAAAA GCAACTGAGG AGGAAATACA
CTGCTTAGAA AGAAGAAACT

In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Mucin 5B (MUC5B). In some embodiments the polymorphism is rs35705950 comprising (SEQ ID NO: 7).

SEQ ID NO: 7)
TGGACGGCCT CTGAAGGGGT CTGTGGGGTC CTGGACGGGT
CCCCATTCAT GGCAGGATTA ACCCCCCTCG GGTTCTGTGT
GGTCCAGGCC GCCCCTTTGT CTCCACTGCC CCCTGGCCAG
AATGAGGGAC AGTGACCCAC CCAGGGCTGG GCCTGGCTCA
GACTCCGTCA GAGCCGCAGG GCAAGTTCCT GGCACGTCCG
AGGTGGGAGG CTCCTCTGCG CTCCAGGAGG CTGTGCCTGG

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CCCCCCTTCC CGGCAGGAAC CGGCTGTGTC CCTTTCCTTC
CTTTATCTTC TGTTTTCAGC
D
CCTTCAACTG TGAAGAGGTG AACTCTTCAA ACACGCTGAG
CAAACAGGCC CGACTCCCAG GGCCGCATCC GGGATGTCTC
AATAGCTGTG GCCTTGACGT CCACCTCGGA CCCCTGCCCC
GGACCCAGCC CAGTTCCCAA TGGGCCCTCT GCCCGGGGAG
GTGCCTAGTG GGAGGGACGA GGGCAAAGTC GGGGCCCCCA
CTTGTTTGGT GTCACTGTGT GCCAGCGGCC ACTGGCGGGC
GAGGCTGTTC CAGGGTGGAG GCGGGGAGGG TTGGACCACA
GGCACTGAGC GGGGACAGAG

In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding ATPase Phospholipid Transporting 11A (ATP11A). In some embodiments the polymorphism is rs12787690 comprising (SEQ ID NO: 8).

(SEQ ID NO: 8)
GTCATTGGTC AAATGTGGCC TGTATCTAAA TTCCAACTGT
TAGAATCATA GACATCTAGA GCTTACGTCA GTTTTAGATA
TTTCTTATGA ATTCTCAGAA TTCATAGATT CTCATTTTTA
TTCTTAGACT TCTCAGATAT TCCGTTTTTG ATAGTATACC
CTTCTGAGTC TAATATGTCC TAAAGTGCGA ACTTGTACAA
TTTttttttt tttttttttt tttttttttt t
K
tgataaggag ttttactctg tcacccaggc tggagtgcag
tgacccgatc tcggctcact gcaacctctg cctcccgggt
tcaagtgatt gtgatgtctc agtctcccaa gtagctggga
ttacaggctc ctgccaccac atgcctagct aattgttata
ctttagtaga aatggggctt cgccgtgtta gtcaggctgg
tcttgtactc ctgacctcag ttgatctgcc taccttggcc
cccaaggtgc tgggattaca ggcatgagcc accgcgcctg
accCAGCTTC TTAAATTATT CTGGGCCACC AGTAATGTGA
ATCATGtaaa ttaaaatata taattaaaCA AAATCATATA
GCGATTAGAG ATAATAGTTG TGAAATGCTT GAAAAATCAT
AGGCATTTAA TAAATAGAAG CCATTCCAAT TAGGATTCTT
CTTGATTTTT TTTCAAGACC AAAAAAATAC TCttttaaat
atttattata ataCTCCATG

In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Isovaleryl-CoA dehydrogenase (IVD)/Dispatched RND Transporter Family Member 2 (DISP2). In some embodiments the polymorphism is rs2034650 comprising (SEQ ID NO: 9).

(SEQ ID NO: 9)

aggctgcagt tagtcatgac tgcgcgctgc actccagcct

gggtgacaaa gtgaggccct gtctcaaaaa caataaaaaa

TTTAAAAGAG CTGAGCATGG AGGCcacttt gggaggctga

ggcaggcaga tctcttaagc ccaggagtct gagaccagcc

tgggcgacat gatgaagccc catctctaca aaaaatacaa

aaaaattagc tgagctttat ggcaaatccc tgtaatccca

gttacctagg aggcccaggc aggaagatgg cttgagccca

aaaggttgag gctgtagtga gctgtgatca tgaacagagt

gagaccctgt ttcaaaacaa aatgaaaaac aaacaaacaa

aaaaaCCAAG AAAACAAGAA AACAAAAACT ATACAATGAT

GAGCCAAAAA GCAAGATATG GAAGAatata tatatatata

tatatatata tataGTATGA GTCCAGCTAT AGAAAGTTTG

AAATCAGGCA ACCTAAACAA TATTGTTCAG GGATCTATAC

AGAGGCAGGA AGCCATTGAG AAAGGTAAGG GGAGGATTAT

CACCAAATTC AGGATGGTGG CTCCCCTGGG GAGAATATGT

CAAGGAGGGG CACATGGGCT TGGAATACTG TCTTCATTGA

CCTGCGTGTT GGGTACACAG GAGTTTGTTA TTTTTCACAC

TGCATATGTG CATGTATATA CTCTCCCATA TATACCATGC

ATTTCACACA AGAACACAAA GGCTGTGTGG CTCTGCTCTG

CCCCTTTCCC CTTCCAGCTC CCATTCTCGT C

Y

TCAGCTAGCA GAGGAGGGTC AGGGTCTTTT AGCACAGCTT

CCTTCTGTCT CTGAGTGGGT CAGAGGAGTA CGGGGATGAG

GGCCTCCCTT CTGCGGCTGG GCTCTGGCCA CTCCAGGGTG

GGAAGGCCTG GAGAAAACAG GGCCAGGCAA AGCCGGCTGG

CCCTGCTGTT TCTGCCAATG CTGGGATTAG GCCAGGGCTC

TGGCCCACCT GTCATTTCAC TCATTCAGCA TGAACATAGC

CACTGAGCAC TTACTGTGAG CCCCGGGTGC TATTGGGAGA

GTTCAGATAA GTGAGAGAGG GTCTTTGACC TCAAAGATCT

TACAGAGAGG ACCGTATACA CAAATAACAG TATACCAGCA

AAATGTGAGC TAAGTGTCAT GTGACTACTC atctactctt

tcaataaata tttgttgtgc acctattaca tgccaggaac

tgtgctggat ggtgatcatg taaagacagt caaatcacag

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tcctagctct cagattcaca gcctgcctaa tgctggggaa

acTGGAAT

In some embodiments of the methods of the disclosure, the subject having PrePF or at risk of developing IPF has a mutation in a sequence encoding Dipeptidyl Peptidase 9 (DPP9). In some embodiments the polymorphism is rs12610495 comprising (SEQ ID NO: 10).

(SEQ ID NO: 10)

CCAGCCAGAA GGGGCGCAGT TTGTTAGTTC AGCTCCTCCT

GAGACAGAAA TAAAGACACG AACCAAAGGA CATCAGCACT

TACAGGGCTC TCAGGTCACA CACAGGATGT CCGCGCCCAC

TGCAGAGCTG CAGGTCCCCT CCAGGGCAGT GGGGAGCCAC

AAGCAGCGTT AGGCAGCGGC TGGGACCAGG ACCGCCTGAG

CACTCAAGAA CCCCCACTGC CCCAAGCACT GCTGGCAGCA

AGCCCAGAAA ACTGAGCCCG GGGAGCTCCT CTGAGCGGCC

TAAGCACCCC TCTAAGCTGT GCTGCCCCAA TTCAAGCCTG

GCTCACGGCA GCAAAGAAAA AATGTGACCT TCGGAGCTCC

CAAAGGGGCC ACCCATAAGC TGAGAGCCTG CCCGGAAGCA

CTTATAGACC CGCGTGGCTT GTTTTCATTG CAAAGAACAA

TAAAAATTAT CTTGCCTCTG ATCACCACTG ATAGCCCAAG

AAGCAAAAAT TCGATCCCGG

D

GATGAGAAAT GAAATGAAAC ATCGCGAGAA ACTTCCAGGA

ATCTTCTGGA TGTGGCTAGA CTCTTTAGCT TGAGCTTCCA

GACAGGCCGA GGCTTGGTGC TGGAGCCTGG CCCTCCGCTG

ACCTCTCTTC TACCCGGGGG CACAGCCCGG ATTGCAGAGA

GGCTGGCGCA AGAGTGAGGG AGCGAGGGCT AGCCTGTGAT

GGGCTTTCTC CACCTAGCAC CACCCTATGC TGTGGCTCAG

GGGAGTCAAG AGTTTACACA GCTGCAGAGA TGGATTCCAG

GCCACTTACT CAAGTCTACC TACTCCTTCC TTCGGCCAAT

CAGCTGGGTG CCTCTGCGGC CTGTGACACC ACCAGCAAAC

AGCTCCAGAC CTCCTAGCAT GGTCTCTGTC AAGGCTGGGT

GGCAGATCTG TGATCTCCTT TTTAAATTTT TCATTTTTTT

TAAGAGATGG GGTCTTGCTA TATTGCCCAG GCTGGTCTCA

AACTCCTGGG CTCCAGCGAT

In some embodiments of the methods of the disclosure, the wild type human MUC5B gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_002458.2):

(SEQ ID NO: 11)

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   1 cacccggccc ggctccctcc ctgcccgtcc ccgtcccccc acccgtgcca gcccccagga
  61 tgggtgcccc gagcgcgtgc cggacgctgg tgttggctct ggcggccatg ctcgtggtgc
 121 cgcaggcaga gacccagggc cctgtggagc cgagctggga gaatgcaggg cacaccatgg
 181 atggcggtgc cccgacgtcc tcgcccaccc ggcgcgtgag ctttgttcca cccgtcactg
 241 tcttccccag cctgagcccc ctgaacccgg cgcacaatgg gcgggtgtgc agcacctggg
 301 gtgacttcca ctacaagacc ttcgacggcg acgtcttccg cttccctggc ctttgcaact
 361 acgtgttctc tgagcactgc cgcgccgcct acgaggactt caacgtccag ctacgccgag
 421 gcctagtggg ctccaggcct gtggtcaccc gtgttgtcat caaggcccag ggctggtgc
 481 tggaggcgtc caacggctcc gtcctcatca atgggcagcg ggaggagctg ccttacagcc
 541 gcactggcct cctggtggag cagagcgggg actacatcaa ggtcagcatc cggctggtgc
 601 tgacattcct gtggaacgga gaggacagtg ccctgctgga gctggatccc aaatacgcca
 661 accagacctg tggcctgtgt ggggacttca acggcctccc ggccttcaac gagttctatg
 721 cccacaacgc caggctgacc ccgctccagt ttgggaacct gcagaagttg gatgggccca
 781 cggagcagtg cccggacccg ctgcccttgc cggccggcaa ctgcacggac gaggagggca
 841 tctgccaccg caccctgctg gggccggcct ttgcggagtg ccacgcactg gtggacagca
 901 ctgcgtacct ggccgcctgc gcccaggacc tgtgccgctg ccccacctgc ccgtgtgcca
 961 cctttgtgga atactcacgc cagtgcgccc acgcgggggg ccagccgcgg aactggaggt
1021 gccctgagct ctgcccccgg acctgccccc tcaacatgca gcaccaggag tgtggctcac
1081 cctgcacgga cacctgctcc aacccccagc gcgcgcagct ctgcgaggac cactgtgtgg
1141 acggctgctt ctgcccccca ggcacggtgc tggatgacat cacgcactct ggctgcctgc
1201 ccctcgggca gtgcccctgc acccacggcg gccgcaccta cagcccgggc acctccttca
1261 acaccacctg cagctcctgc acctgctccg gggggctatg gcagtgccag gacctgccgt
1321 gccctggcac ctgctctgtg cagggcgggg cccacatctc cacctatgat gagaaactct
1381 acgacctgca tggtgactgc agctacgttc tgtccaagaa atgtgccgac agcagcttca
1441 ccgtgctggc tgagctgcgg aagtgcggcc tgacggacaa cgagaactgc ctgaaagcgg
1501 tgacgctcag cctggacggc ggggacacgg ccatccgggt ccaagcggac ggcggcgtgt
1561 tcctcaactc catctacacg cagctgcccc tgtcggcagc caacatcacc ctgttcacac
1621 cctcgagctt cttcatcgtg gtgcagacag gcctggggct gcagctgctg gtgcagctgg
1681 tgccactcat gcaggtgttt gtcaggctgg accccgccca ccagggccag atgtgcggcc
1741 tgtgtgggaa cttcaaccag aaccaggctg acgacttcac ggccctcagc ggggtggtgg
1801 aggccacggg cgcagccttc gccaacacct ggaaggccca ggctgcctgt gccaatgcca
1861 ggaacagctt tgaggacccc tgctccctca gtgtggagaa tgagaactac gcccggcact
1921 ggtgctcgcg cctgaccgat cccaacagtg ccttctcgcg ctgccactcc atcatcaacc
1981 ccaagccctt ccactcgaac tgcatgtttg acacctgcaa ctgtgagcgg agcgaggact
2041 gcctgtgcgc cgcgctgtcc tcctatgtgc acgcctgtgc cgccaagggc gtacagctca
2101 gcgactggag ggacggcgtc tgcaccaagt acatgcagaa ctgccccaag tcccagcgct
2161 acgcctacgt ggtggatgcc tgccagccca cttgccgcgg cctgagtgag gccgacgtca
2221 cctgcagcgt ttccttcgtg cctgtggacg gctgcacctg ccccgcgggc accttcctca
2281 atgacgcggg cgcctgtgtg cccgcccagg agtgcccctg ctacgctcac ggcaccgtgc
2341 tggctcctgg agaggtggtg cacgacgagg gcgccgtgtg ttcatgtacg ggtgggaagc
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2401 taagctgcct gggagcctct ctgcagaaaa gcacagggtg tgcagccccc atggtgtacc

2461 tggactgcag caacagctcg gcgggcaccc ctggggccga gtgcctccgg agctgccaca

2521 cgctggacgt gggctgtttc agcacacact gcgtgtccgg ctgtgtctgt cccccggggc

2581 tggtgtcgga tgggagtggg ggctgcattg ccgaggagga ctgcccctgt gtgcacaacg

2641 aggccaccta caagcctgga gagaccatca gggtcgactg caacacctgc acctgcagga

2701 accggaggtg ggagtgcagc caccggctct gcctgggcac ctgcgtggcc tacggggatg

2761 gccacttcat cacctttgat ggcgatcgct acagctttga aggcagctgc gagtacatct

2821 tggcccagga ctactgtggg gacaacacca cccacgggac cttccgcatc gtcaccgaga

2881 acatcccctg tgggaccacc ggcaccacct gctccaaggc catcaagctc ttcgtggaga

2941 gctacgagct gatcctccaa gaggggacct ttaaggcggt ggcgagaggg ccgggtgggg

3001 acccacccta caagatacgc tacatgggga tcttcctggt catcgagacc cacgggatgg

3061 ccgtgtcctg ggaccggaag accagcgtgt tcatccgact gcaccaggac tacaagggca

3121 gggtctgcgg cctgtgcggg aacttcgacg acaatgccat caatgacttt gccacgcgta

3181 gccggtccgt ggtgggggac gcactggagt ttgggaacag ctggaagctc tccccctcct

3241 gcccggacgc cctggcaccc aaggacccct gcacggccaa ccccttccgc aagtcctggg

3301 cccagaagca gtgcagcatc ctccacggcc ccaccttcgc cgcctgccgc tcccaggttg

3361 actccaccaa gtactacgag gcctgcgtga acgacgcgtg tgcctgcgac tcgggtggcg

3421 actgcgagtg tttctgcacg gctgtggctg cctacgccca ggcctgccac gacgcgggcc

3481 tgtgtgtgtc ctggcggact ccggacacct gccccttgtt ctgtgacttc tacaacccac

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3601 accccagtgg gcactgcctg gtggacctgc ctggcctgga aggctgctac ccgaagtgcc

3661 cacccagcca gcccttcttc aatgaggacc agatgaagtg cgtggcccag tgtggctgct

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4381 gccctcagcc ctccctcagt gccagcacgg agcctgctgt gcctacccca acccagacca

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4501 cctcgcagac tgggtccagc tcaggccccg tgacggtcac cccctcggcc ccaggtacca

4561 ccacctgcca gccccggtgt cagtggacag agtggtttga tgaggactac cccaagtctg

4621 aacaacttgg aggggacgtt gagtcctacg ataagatcag ggccgctgga gggcacttat

4681 gccagcagcc taaggacata gagtgccagg ccgagagctt ccccaactgg accctggcac

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4861 accactgcag gggacgtgcc acaaccccgc caccgaccac agagctggag acggccacca
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7141 gtgcccaggc ccagcctggt gtcccccctgc gggagttggg ccaggtcgtg gaatgcagcc
7201 tggactttgg cctggtctgc aggaaccgtg agcaggtggg gaagttcaag atgtgcttca

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7261 actatgaaat ccgtgtgttc tgctgcaact acggccactg ccccagcacc ccggccacca

7321 gctctacggc catgccctcc tccactccgg ggacgacctg gatcctcaca gagctgacca

7381 caacagccac tacgactgag tccactggat ccacggccac cccgtcctcc accccaggga

7441 ccacctggat cctcacagag ccgagcacta cagccaccgt gacggtgccc accggatcca

7501 cggccaccgc ctcctccacc caggcaactg ctggcacccc acatgtgagc accacggcca

7561 cgacacccac agtcaccagc tccaaagcca ctcccttctc cagtccaggg actgcaaccg

7621 cccttccagc actgagaagc acagccacca cacccacagc taccagcttt acagccatcc

7681 cctcctcctc cctgggcacc acctggaccc gcctatcaca gaccaccaca cccacggcca

7741 ccatgtccac agccacaccc tcctccactc cagagactgt ccacacctcc acagtgctta

7801 ccaccacggc caccacaacc ggggccaccg gctctgtggc cacccccctcc tccaccccag

7861 gaacagctca cactaccaaa gtgctgacta ccacaaccac gggcttcaca gccacccccct

7921 cctccagccc agggacggca cgcacgcttc cagtgtggat cagcacaacc accacaccca

7981 caaccagagg ttccacggtg acccccctcct ccatcccggg gaccacccac acccccacag

8041 tgctgaccac caccaccaca actgtggcca ctggttctat ggcaacaccc tcctctagca

8101 cacagaccag tggtactccc ccatcactga ccaccacggc cactacgatc acggccaccg

8161 gctccaccac caacccctcc tcaactccag ggacaacacc tatcccccca gtgctgacca

8221 ccaccgccac cacacctgca gccaccagca gcacagtgac tccctcctct gccctaggga

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8341 ccagcagtcc ccacacggtg cgcacagcct ggacttcggc cacctcaggc accttgggca

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8461 ccacccagca ctcgactcca gccctgtcca gccctcaccc tagcagcagg accaccgagt

8521 caccccccttc tccagggacg accaccccgg gccacaccag ggccacctcc aggaccacgg

8581 ccacggccac acccagcaag acccgcacct cgaccctgct gcccagcagc cccacatcgg

8641 ccccaataac cacggtggtg accatgggct gtgagcccca gtgtgcctgg tcagagtggc

8701 tggactacag ctaccccatg ccggggccct ctggcgggga ctttgacacc tactccaaca

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8821 cccagcctgg tgtcccccctg cgggagttgg gccaggtcgt ggaatgcagc ctggactttg

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In some embodiments of the methods of the disclosure, the wild type human MUC5B gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_002449.2):

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                                                 (SEQ ID NO: 12)
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 121 glvgsrpvvt rvvikaqglv leasngsvli ngqreelpys rtgllveqsg dyikvsirlv
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 361 dgcfcppgtv lddithsgcl plgqcpcthg grtyspgtsf nttcssctcs gglwqcqdlp
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 541 vplmqvfvrl dpahqgqmcg lcgnfnqnqa ddftalsgvv eatgaafant wkaqaacana
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4201 crnreqvgkf kmcfnyeirv fccnyghcps tpatsstamp sstpgttwil telttttattt
4261 astgstatps stpgtapppk vltspattpt atsskatsss sprtattlpv ltstatksta
4321 tsvtpipsst lgttgtlpeq tttpvatmst ihpsstpett htstvlttka tttratssts
4381 tpsstpgttw iltelttaat ttaatgptat psstpgttwi ltelttttatt tastgstatp
4441 sstpgttwil tepsttatvt vptgstatas stqatagtph vsttattptv tsskatpsss
4501 pgtatalpal rstattptat sftaipsssl gttwtrlsqt ttptatmsta tpsstpetvh
4561 tstvltatat ttgatgsvat psstpgtaht tkvptttttg ftatpssspg taltppvwis
4621 ttttpttttp ttsgstvtps sipgtthtar vlttttttva tgsmatpsss tqtsgtppsl
4681 tttattitat gsttnpsstp gttpitpvlt stattpaats skatsssspr tattlpvlts
4741 tatkstatsf tpipsstlwt twtvpaqttt pmstmstiht sstpetthts tvltttatmt
4801 ratnstatps stlgttrilt eltttattta atgstatlss tpgttwilte pstiatvmvp
4861 tgstatasst lgtahtpkvv ttmatmptat astvpssstv gttrtpavlp sslptfsvst
4921 vsssvlttlr ptgfpsshfs tpcfcrafgq ffspgeviyn ktdragchfy avcnqhcdid
4981 rfqgacptsp ppvssaplss pspapgcdna iplrqvnetw tlenctvarc vgdnrvvlld
5041 pkpvanvtcv nkhlpikvsd psqpcdfhye cecicsmwgg shvstfdgts ytfrgnctyv
5101 lmreiharfg nlslyldnhy ctasataaaa rcpralsihy ksmdivltvt mvhgkeegli
5161 lfdqipvssg fskngvlvsv lgtttmrvdi palgvsvtfn gqvfqarlpy slfhnntegq
5221 cgtctnnqrd dclqrdgtta asckdmaktw lvpdsrkdgc waptgtppta spaapvsstp
5281 tptpcppqpl cdlmlsqvfa echnlvppgp ffnacisdhc rgrlevpcqs leayaelcra
5341 rgvcsdwrga tgglcdltcp ptkvykpcgp iqpatcnsrn qspqlegmae gcfcpedqil
5401 fnahmgicvq acpcvgpdgf pkfpgerwvs ncqscvcdeg svsvqckplp cdaqgqpppc
5461 nrpgfvtvtr praenpccpe tvcvcntttc pqslpvcppg qesictqeeg dccptfrcrp
5521 qlcsyngtfy gvgatfpgal pchmctclsg dtqdptvqcq edacnnttcp qgfeykrvag
5581 qccgecvqta cltpdgqpvq lnetwvnshv dnctvylcea eggvhlltpq pascpdvssc
5641 rgslrktgcc ysceedscqv rinttilwhq gcetevnitf cegscpgask ysaeaqamqh
5701 qctccqerry heetvplhcp ngsailhtyt hvdecgctpf cvpapmapph trgfpaqeat
5761 av
```

In some embodiments of the methods of the disclosure, the wild type human TERT gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_198253.2, transcript variant 1):

(SEQ ID NO: 13)

```
   1 caggcagcgc tgcgtcctgc tgcgcacgtg ggaagccctg gccccggcca cccccgcgat
  61 gccgcgcgct ccccgctgcc gagccgtgcg ctccctgctg cgcagccact accgcgaggt
 121 gctgccgctg gccacgttcg tgcggcgcct ggggccccag ggctggcggc tggtgcagcg
 181 cggggacccg gcggctttcc gcgcgctggt ggcccagtgc ctggtgtgcg tgccctggga
 241 cgcacggccg ccccccgccg ccccctcctt ccgccaggtg tcctgcctga aggagctggt
 301 ggcccgagtg ctgcagaggc tgtgcgagcg cggcgcgaag aacgtgctgg ccttcggctt
 361 cgcgctgctg gacggggccc gcgggggccc ccccgaggcc ttcaccacca gcgtgcgcag
 421 ctacctgccc aacacggtga ccgacgcact gcgggggagc ggggcgtggg ggctgctgct
 481 gcgccgcgtg ggcgacgacg tgctggttca cctgctggca cgctgcgcgc tctttgtgct
 541 ggtggctccc agctgcgcct accaggtgtg cgggccgccg ctgtaccagc tcggcgctgc
 601 cactcaggcc cggcccccgc cacacgctag tggaccccga aggcgtctgg gatgcgaacg
 661 ggcctggaac catagcgtca gggaggccgg ggtccccctg ggcctgccag ccccgggtgc
 721 gaggaggcgc gggggcagtg ccagccgaag tctgccgttg cccaagaggc ccaggcgtgg
 781 cgctgcccct gagccggagc ggacgcccgt tgggcagggg tcctgggccc acccgggcag
 841 gacgcgtgga ccgagtgacc gtggtttctg tgtggtgtca cctgccagac ccgccgaaga
 901 agccacctct ttggagggtg cgctctctgg cacgcgccac tcccacccat ccgtgggccg
 961 ccagcaccac gcgggccccc catccacatc gcggccacca cgtccctggg acacgccttg
1021 tcccccggtg tacgccgaga ccaagcactt cctctactcc tcaggcgaca aggagcagct
1081 gcggccctcc ttcctactca gctctctgag gcccagcctg actggcgctc ggaggctcgt
1141 ggagaccatc tttctgggtt ccaggccctg gatgccaggg actccccgca ggttgccccg
1201 cctgccccag cgctactggc aaatgcggcc cctgtttctg gagctgcttg ggaaccacgc
1261 gcagtgcccc tacggggtgc tcctcaagac gcactgcccg ctgcgagctg cggtcacccc
1321 agcagccggt gtctgtgccc gggagaagcc ccagggctct gtggcggccc ccgaggagga
1381 ggacacagac ccccgtcgcc tggtgcagct gctccgccag cacagcagcc cctggcaggt
1441 gtacggcttc gtgcgggcct gcctgcgccg gctggtgccc ccaggcctct ggggctccag
1501 gcacaacgaa cgccgcttcc tcaggaacac caagaagttc atctccctgg ggaagcatgc
1561 caagctctcg ctgcaggagc tgacgtggaa gatgagcgtg cgggactgcg cttggctgcg
1621 caggagccca ggggttggct gtgttccggc cgcagagcac cgtctgcgtg aggagatcct
1681 ggccaagttc ctgcactggc tgatgagtgt gtacgtcgtc gagctgctca ggtctttctt
1741 ttatgtcacg gagaccacgt ttcaaaagaa caggctcttt ttctaccgga agagtgtctg
1801 gagcaagttg caaagcattg gaatcagaca gcacttgaag agggtgcagc tgcgggagct
1861 gtcggaagca gaggtcaggc agcatcggga agccaggccc gccctgctga cgtccagact
1921 ccgcttcatc cccaagcctg acgggctgcg gccgattgtg aacatggact acgtcgtggg
1981 agccagaacg ttccgcagag aaaagagggc cgagcgtctc acctcgaggg tgaaggcact
2041 gttcagcgtg ctcaactacg agcgggcgcg gcgccccggc ctcctgggcg cctctgtgct
2101 gggcctggac gatatccaca gggcctggcg caccttcgtg ctgcgtgtgc gggcccagga
2161 cccgccgcct gagctgtact ttgtcaaggt ggatgtgacg ggcgcgtacg acaccatccc
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2221 ccaggacagg ctcacggagg tcatcgccag catcatcaaa ccccagaaca cgtactgcgt

2281 gcgtcggtat gccgtggtcc agaaggccgc ccatgggcac gtccgcaagg ccttcaagag

2341 ccacgtctct accttgacag acctccagcc gtacatgcga cagttcgtgg ctcacctgca

2401 ggagaccagc ccgctgaggg atgccgtcgt catcgagcag agctcctccc tgaatgaggc

2461 cagcagtggc ctcttcgacg tcttcctacg cttcatgtgc caccacgccg tgcgcatcag

2521 gggcaagtcc tacgtccagt gccaggggat cccgcagggc tccatcctct ccacgctgct

2581 ctgcagcctg tgctacggcg acatggagaa caagctgttt gcggggattc ggcgggacgg

2641 gctgctcctg cgtttggtgg atgatttctt gttggtgaca cctcacctca cccacgcgaa

2701 aaccttcctc aggaccctgg tccgaggtgt ccctgagtat ggctgcgtgg tgaacttgcg

2761 gaagacagtg gtgaacttcc ctgtagaaga cgaggccctg ggtggcacgg cttttgttca

2821 gatgccggcc cacggcctat tcccctggtg cggcctgctg ctggataccc ggaccctgga

2881 ggtgcagagc gactactcca gctatgcccg gacctccatc agagccagtc tcaccttcaa

2941 ccgcggcttc aaggctggga ggaacatgcg tcgcaaactc tttggggtct tgcggctgaa

3001 gtgtcacagc ctgtttctgg atttgcaggt gaacagcctc cagacggtgt gcaccaacat

3061 ctacaagatc ctcctgctgc aggcgtacag gtttcacgca tgtgtgctgc agctcccatt

3121 tcatcagcaa gtttggaaga accccacatt tttcctgcgc gtcatctctg acacggcctc

3181 cctctgctac tccatcctga aagccaagaa cgcagggatg tcgctggggg ccaagggcgc

3241 cgccggccct ctgccctccg aggccgtgca gtggctgtgc caccaagcat tcctgctcaa

3301 gctgactcga caccgtgtca cctacgtgcc actcctgggg tcactcagga cagcccagac

3361 gcagctgagt cggaagctcc cggggacgac gctgactgcc ctggaggccg cagccaaccc

3421 ggcactgccc tcagacttca agaccatcct ggactgatgg ccacccgccc acagccaggc

3481 cgagagcaga caccagcagc cctgtcacgc cgggctctac gtcccaggga gggaggggcg

3541 gcccacaccc aggcccgcac cgctgggagt ctgaggcctg agtgagtgtt tggccgaggc

3601 ctgcatgtcc ggctgaaggc tgagtgtccg gctgaggcct gagcgagtgt ccagccaagg

3661 gctgagtgtc cagcacacct gccgtcttca cttccccaca ggctggcgct cggctccacc

3721 ccagggccag cttttcctca ccaggagccc ggcttccact ccccacatag gaatagtcca

3781 tccccagatt cgccattgtt cacccctcgc cctgccctcc tttgccttcc acccccacca

3841 tccaggtgga gaccctgaga aggaccctgg gagctctggg aatttggagt gaccaaaggt

3901 gtgccctgta cacaggcgag gaccctgcac ctggatgggg gtccctgtgg gtcaaattgg

3961 ggggaggtgc tgtgggagta aaatactgaa tatatgagtt tttcagtttt gaaaaaaa
```

In some embodiments of the methods of the disclosure, the wild type human TERT gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_937983.2, transcript variant 1):

```
                                                    (SEQ ID NO: 14)
  1 mpraprcrav rsllrshyre vlplatfvrr lgpqgwrlvq rgdpaafral vaqclvcvpw

 61 darpppaaps frqvsclkel varvlqrlce rgaknvlafg falldgargg ppeafttsvr

121 sylpntvtda lrgsgawgll lrrvgddvlv hllarcalfv lvapscayqv cgpplyqlga

181 atqarpppha sgprrrlgce rawnhsvrea gvplglpapg arrrggsasr slplpkrprr

241 gaapepertp vgqgswahpg rtrgpsdrgf cvvsparpae eatslegals gtrhshpsvg

301 rqhhagppst srpprpwdtp cppvyaetkh flyssgdkeq lrpsfllssl rpsltgarrl
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 361 vetiflgsrp wmpgtprrlp rlpqrywqmr plflellgnh aqcpygvllk thcplraavt

 421 paagvcarek pqgsvaapee edtdprrlvq llrqhsspwq vygfvraclr rlvppglwgs

 481 rhnerrflrn tkkfislgkh aklslqeltw kmsvrdcawl rrspgvgcvp aaehrlreei

 541 lakflhwlms vyvvellrsf fyvtettfqk nrlffyrksv wsklqsigir qhlkrvqlre

 601 lseaevrqhr earpalltsr lrfipkpdgl rpivnmdyvv gartfrrekr aerltsrvka

 661 lfsvinyera rrpgllgasv lglddihraw rtfvlrvraq dpppelyfvk vdvtgaydti

 721 pqdrltevia siikpqntyc vrryavvqka ahghvrkafk shvstltdlq pymrqfvahl

 781 qetsplrdav vieqssslne assglfdvfl rfmchhavri rgksyvqcqg ipqgsilstl

 841 lcslcygdme nklfagirrd glllrlvddf llvtphltha ktflrtivrg vpeygcvvnl

 901 rktvvnfpve dealggtafv qmpahglfpw cglildtrtl evqsdyssya rtsirasltf

 961 nrgfkagrnm rrklfgvlrl kchslfldlq vnslqtvctn iykillllqay rfhacvlqlp

1021 fhqqvwknpt fflrvisdta slcysilkak nagmslgakg aagplpseav qwlchqafll

1081 kltrhrvtyv pllgslrtaq tqlsrklpgt tltaleaaan palpsdfkti ld
```

In some embodiments of the methods of the disclosure, the wild type human TERT gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001193376.1, transcript variant 2):

```
                                                       (SEQ ID NO: 15)
   1 caggcagcgc tgcgtcctgc tgcgcacgtg ggaagccctg gccccggcca cccccgcgat

  61 gccgcgcgct ccccgctgcc gagccgtgcg ctccctgctg cgcagccact accgcgaggt

 121 gctgccgctg gccacgttcg tgcggcgcct ggggccccag ggctggcggc tggtgcagcg

 181 cggggacccg gcggctttcc gcgcgctggt ggcccagtgc ctggtgtgcg tgccctggga

 241 cgcacggccg ccccccgccg ccccctcctt ccgccaggtg tcctgcctga aggagctggt

 301 ggcccgagtg ctgcagaggc tgtgcgagcg cggcgcgaag aacgtgctgg ccttcggctt

 361 cgcgctgctg gacggggccc gcgggggccc ccccgaggcc ttcaccacca gcgtgcgcag

 421 ctacctgccc aacacggtga ccgacgcact gcgggggagc ggggcgtggg ggctgctgct

 481 gcgccgcgtg ggcgacgacg tgctggttca cctgctggca cgctgcgcgc tctttgtgct

 541 ggtggctccc agctgcgcct accaggtgtg cgggccgccg ctgtaccagc tcggcgctgc

 601 cactcaggcc cggcccccgc cacacgctag tggaccccga aggcgtctgg gatgcgaacg

 661 ggcctggaac catagcgtca gggaggccgg ggtccccctg ggcctgccag ccccgggtgc

 721 gaggaggcgc gggggcagtg ccagccgaag tctgccgttg cccaagaggc ccaggcgtgg

 781 cgctgcccct gagccggagc ggacgcccgt tgggcagggg tcctgggccc acccgggcag

 841 gacgcgtgga ccgagtgacc gtggtttctg tgtggtgtca cctgccagac ccgccgaaga

 901 agccacctct ttggagggtg cgctctctgg cacgcgccac tcccacccat ccgtgggccg

 961 ccagcaccac gcgggccccc catccacatc gcggccacca cgtccctggg acacgccttg

1021 tcccccggtg tacgccgaga ccaagcactt cctctactcc tcaggcgaca aggagcagct

1081 gcggccctcc ttcctactca gctctctgag gcccagcctg actggcgctc ggaggctcgt

1141 ggagaccatc tttctgggtt ccaggccctg gatgccaggg actccccgca ggttgccccg

1201 cctgccccag cgctactggc aaatgcggcc cctgtttctg gagctgcttg ggaaccacgc

1261 gcagtgcccc tacggggtgc tcctcaagac gcactgcccg ctgcgagctg cggtcacccc

1321 agcagccggt gtctgtgccc gggagaagcc ccagggctct gtggcggccc ccgaggagga
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1381 ggacacagac ccccgtcgcc tggtgcagct gctccgccag cacagcagcc cctggcaggt
1441 gtacggcttc gtgcgggcct gcctgcgccg gctggtgccc ccaggcctct ggggctccag
1501 gcacaacgaa cgccgcttcc tcaggaacac caagaagttc atctccctgg ggaagcatgc
1561 caagctctcg ctgcaggagc tgacgtggaa gatgagcgtg cgggactgcg cttggctgcg
1621 caggagccca ggggttggct gtgttccggc cgcagagcac cgtctgcgtg aggagatcct
1681 ggccaagttc ctgcactggc tgatgagtgt gtacgtcgtc gagctgctca ggtctttctt
1741 ttatgtcacg gagaccacgt ttcaaaagaa caggctcttt ttctaccgga agagtgtctg
1801 gagcaagttg caaagcattg gaatcagaca gcacttgaag agggtgcagc tgcgggagct
1861 gtcggaagca gaggtcaggc agcatcggga agccaggccc gccctgctga cgtccagact
1921 ccgcttcatc cccaagcctg acgggctgcg gccgattgtg aacatggact acgtcgtggg
1981 agccagaacg ttccgcagag aaaagagggc cgagcgtctc acctcgaggg tgaaggcact
2041 gttcagcgtg ctcaactacg agcgggcgcg gcgccccggc ctcctgggcg cctctgtgct
2101 gggcctggac gatatccaca gggcctggcg caccttcgtg ctgcgtgtgc gggcccagga
2161 cccgccgcct gagctgtact ttgtcaaggt ggatgtgacg ggcgcgtacg acaccatccc
2221 ccaggacagg ctcacggagg tcatcgccag catcatcaaa ccccagaaca cgtactgcgt
2281 gcgtcggtat gccgtggtcc agaaggccgc ccatgggcac gtccgcaagg ccttcaagag
2341 ccacgtctct accttgacag acctccagcc gtacatgcga cagttcgtgg ctcacctgca
2401 ggagaccagc ccgctgaggg atgccgtcgt catcgagcag agctcctccc tgaatgaggc
2461 cagcagtggc ctcttcgacg tcttcctacg cttcatgtgc caccacgccg tgcgcatcag
2521 gggcaagtcc tacgtccagt gccaggggat cccgcagggc tccatcctct ccacgctgct
2581 ctgcagcctg tgctacggcg acatggagaa caagctgttt gcggggattc ggcgggacgg
2641 gctgctcctg cgtttggtgg atgatttctt gttggtgaca cctcacctca cccacgcgaa
2701 aaccttcctc agctatgccc ggacctccat cagagccagt ctcaccttca accgcggctt
2761 caaggctggg aggaacatgc gtcgcaaact ctttggggtc ttgcggctga agtgtcacag
2821 cctgtttctg gatttgcagg tgaacagcct ccagacggtg tgcaccaaca tctacaagat
2881 cctcctgctg caggcgtaca ggtttcacgc atgtgtgctg cagctcccat ttcatcagca
2941 agtttggaag aaccccacat ttttcctgcg cgtcatctct gacacggcct ccctctgcta
3001 ctccatcctg aaagccaaga acgcagggat gtcgctgggg gccaagggcg ccgccggccc
3061 tctgccctcc gaggccgtgc agtggctgtg ccaccaagca ttcctgctca agctgactcg
3121 acaccgtgtc acctacgtgc cactcctggg gtcactcagg acagcccaga cgcagctgag
3181 tcggaagctc ccggggacga cgctgactgc cctggaggcc gcagccaacc cggcactgcc
3241 ctcagacttc aagaccatcc tggactgatg gccacccgcc cacagccagg ccgagagcag
3301 acaccagcag ccctgtcacg ccgggctcta cgtcccaggg agggaggggc ggcccacacc
3361 caggcccgca ccgctgggag tctgaggcct gagtgagtgt ttggccgagg cctgcatgtc
3421 cggctgaagg ctgagtgtcc ggctgaggcc tgagcgagtg tccagccaag ggctgagtgt
3481 ccagcacacc tgccgtcttc acttccccac aggctggcgc tcggctccac cccagggcca
3541 gcttttcctc accaggagcc cggcttccac tccccacata ggaatagtcc atccccagat
3601 tcgccattgt tcacccctcg ccctgccctc ctttgccttc cacccccacc atccaggtgg
3661 agaccctgag aaggaccctg ggagctctgg gaatttggag tgaccaaagg tgtgccctgt
3721 acacaggcga ggaccctgca cctggatggg ggtccctgtg ggtcaaattg gggggaggtg
3781 ctgtgggagt aaaatactga atatatgagt ttttcagttt tgaaaaaaa
```

In some embodiments of the methods of the disclosure, the wild type human TERT gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001180305.1, transcript variant 2):

(SEQ ID NO: 16)

```
   1 mpraprcrav rsllrshyre vlplatfvrr lgpqgwrlvq rgdpaafral vaqclvcvpw
  61 darpppaaps frqvsclkel varvlqrlce rgaknvlafg falldgargg ppeafttsvr
 121 sylpntvtda lrgsgawgll lrrvgddvlv hllarcalfv lvapscayqv cgpplyqlga
 181 atqarpppha sgprrrlgce rawnhsvrea gvplglpapg arrrggsasr slplpkrprr
 241 gaapepertp vgqgswahpg rtrgpsdrgf cvvsparpae eatslegals gtrhshpsvg
 301 rqhhagppst srpprpwdtp cppvyaetkh flyssgdkeq lrpsfllssl rpsltgarrl
 361 vetiflgsrp wmpgtprrlp rlpqrywqmr plflellgnh aqcpygvllk thcplraavt
 421 paagvcarek pqgsvaapee edtdprrlvq llrqhsspwq vygfvraclr rlvppglwgs
 481 rhnerrflrn tkkfislgkh aklslqeltw kmsvrdcawl rrspgvgcvp aaehrlreei
 541 lakflhwlms vyvvellrsf fyvtettfqk nrlffyrksv wsklqsigir qhlkrvqlre
 601 lseaevrqhr earpalltsr lrfipkpdgl rpivnmdyvv gartfrrekr aerltsrvka
 661 lfsvlnyera rrpgllgasv lglddihraw rtfvlrvraq dpppelyfvk vdvtgaydti
 721 pqdrltevia siikpqntyc vrryavvqka ahghvrkafk shvstltdlq pymrqfvahl
 781 qetsplrdav vieqssslne assglfdvfl rfmchhavri rgksyvqcqg ipqgsilstl
 841 lcslcygdme nklfagirrd glllrlvddf llvtphltha ktflsyarts irasltfnrg
 901 fkagrnmrrk lfgvlrlkch slfldlqvns lqtvctniyk illlqayrfh acvlqlpfhq
 961 qvwknptffl rvisdtaslc ysilkaknag mslgakgaag plpseavqwl chqafllklt
1021 rhrvtyvpll gslrtaqtql srklpgttlt aleaaanpal psdfktild
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_014883.3, transcript variant 1):

(SEQ ID NO: 36)

```
   1 atcaaatttc aactccaggc agtccttcca gccatgtggg ttcagcggaa agagaagcaa
  61 aaccactctt cctaaaatgt tagaagctgc tcttcgctta ccttggggcc tttgcattgg
 121 gagctgtttt tcacatcaaa gaatatgtgc tgaatggaat tttagtattt tgctgtcgtt
 181 ttaatatttt cgtctggtct tcctcagttc ttccagacgc tttctgagag aatgggggca
 241 ggagctctag ccatctgtca aagtaaagca gcggttcggc tgaaagaaga catgaaaaag
 301 atagtggcag tgccattaaa tgaacagaag gattttacct atcagaagtt atttggagtc
 361 agtctccaag aacttgaacg gcaggggctc accgagaatg gcattccagc agtagtgtgg
 421 aatatagtgg aatatttgac gcagcatgga cttacccaag aaggtctttt tagggtgaat
 481 ggtaacgtga aggtggtgga acaacttcga ctgaagttcg agagtggagt gcccgtggag
 541 ctcgggaagg acggtgatgt ctgctcagca gccagtctgt tgaagctgtt tctgagggag
 601 ctgcctgaca gtctgatcac ctcagcgttg cagcctcgat tcattcaact ctttcaggat
 661 ggcagaaatg atgttcagga gagtagctta agagacttaa taaaagagct gccagacacc
 721 cactactgcc tcctcaagta cctttgccag ttcttgacaa aagtagccaa gcatcatgtg
 781 cagaatcgca tgaatgttca caatctcgcc actgtatttg ggccaaattg ctttcatgtg
 841 ccacctgggc ttgaaggcat gaaggaacag gacctgtgca acaagataat ggctaaaatt
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```
 901 ctagaaaatt acaataccct gtttgaagta gagtatacag aaaatgatca tctgagatgt
 961 gaaaacctgg ctaggcttat catagtaaaa gaggtctatt ataagaactc cctgcccatc
1021 cttttaacaa gaggcttaga aagagacatg ccaaaaccac ctccaaaaac caagatccca
1081 aaatccagga gtgagggatc tattcaggcc cacagagtac tgcaaccaga gctatctgat
1141 ggcattcctc agctcagctt gcggctaagt tatagaaaag cctgcttgga agacatgaat
1201 tcagcagagg gtgctattag tgccaagttg gtacccagtt cacaggaaga tgaaagacct
1261 ctgtcacctt tctatttgag tgctcatgta ccccaagtca gcaatgtgtc tgcaaccgga
1321 gaactcttag aaagaaccat ccgatcagct gtagaacaac atcttttga tgttaataac
1381 tctggaggtc aaagttcaga ggactcagaa tctggaacac tatcagcatc ttctgccaca
1441 tctgccagac agcgccgccg ccagtccaag gagcaggatg aagttcgaca tgggagagac
1501 aagggactta tcaacaaaga aaatactcct tctgggttca accaccttga tgattgtatt
1561 ttgaatactc aggaagtcga aaaggtacac aaaaatactt ttggttgtgc tggagaaagg
1621 agcaagccta aacgtcagaa atccagtact aaactttctg agcttcatga caatcaggac
1681 ggtcttgtga atatggaaag tctcaattcc acacgatctc atgagagaac tggacctgat
1741 gattttgaat ggatgtctga tgaaaggaaa ggaaatgaaa aagatggtgg acacactcag
1801 cattttgaga gccccacaat gaagatccag gagcatccca gcctatctga caccaaacag
1861 cagagaaatc aagatgccgg tgaccaggag gagagctttg tctccgaagt gccccagtcg
1921 gacctgactg cattgtgtga tgaaaagaac tgggaagagc ctatccctgc tttctcctcc
1981 tggcagcggg agaacagtga ctctgatgaa gcccacctct cgccgcaggc tgggcgcctg
2041 atccgtcagc tgctggacga agacagcgac cccatgctct ctcctcggtt ctacgcttat
2101 gggcagagca ggcaatacct ggatgacaca gaagtgcctc cttccccacc aaactcccat
2161 tctttcatga ggcggcgaag ctcctctctg gggtcctatg atgatgagca agaggacctg
2221 acacctgccc agctcacacg aaggattcag agccttaaaa agaagatccg gaagtttgaa
2281 gatagattcg aagaagagaa gaagtacaga ccttcccaca gtgacaaagc agccaatccg
2341 gaggttctga aatggacaaa tgaccttgcc aaattccgga gacaacttaa agaatcaaaa
2401 ctaaagatat ctgaagagga cctaactccc aggatgcggc agcgaagcaa cacactcccc
2461 aagagttttg gttcccaact tgagaaagaa gatgagaaga agcaagagct ggtggataaa
2521 gcaataaagc ccagtgttga agccacattg gaatctattc agaggaagct ccaggagaag
2581 cgagcggaaa gcagccgccc tgaggacatt aaggatatga ccaaagacca gattgctaat
2641 gagaaagtgg ctctgcagaa agctctgtta tattatgaaa gcattcatgg acggccggta
2701 acaaagaacg aacggcaggt gatgaagcca ctatacgaca ggtaccggct ggtcaaacag
2761 atcctctccc gagctaacac catacccatc attggttccc cctccagcaa gcggagaagc
2821 cctttgctgc agccaattat cgagggcgaa actgcttcct tcttcaagga gataaaggaa
2881 gaagaggagg ggtcagaaga cgatagcaat gtgaagccag acttcatggt cactctgaaa
2941 accgatttca gtgcacgatg ctttctggac caattcgaag atgacgctga tggatttatt
3001 tccccaatgg atgataaaat accatcaaaa tgcagccagg acacagggct ttcaaatctc
3061 catgctgcct caatacctga actcctggaa cacctccagg aaatgagaga agaaaagaaa
3121 aggattcgaa agaaacttcg ggattttgaa gacaactttt tcagacagaa tggaagaaat
3181 gtccagaagg aagaccgcac tcctatggct gaagaataca gtgaatataa gcacataaag
3241 gcgaaactga ggctcctgga ggtgctcatc agcaagagag acactgattc caagtccatg
3301 tgaggggcat ggccaagcac agggggctgg cagctgcggt gagagtttac tgtccccaga
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3361 gaaagtgcag ctctggaagg cagccttggg gctggccctg caaagcatgc agcccttctg
3421 cctctagacc atttggcatc ggctcctgtt tccattgcct gccttagaaa ctggctggaa
3481 gaagacaatg tgacctgact taggcatttt gtaattggaa agtcaagact gcagtatgtg
3541 cacatgcgca cgcgcatgca cgcacacaca cacacagtag tggagctttc ctaacactag
3601 cagagattaa tcactacatt agacaacact catctacaga gaatatacac tgttcttccc
3661 tggataactg agaaacaaga gaccattctc tgtctaactg tgataaaaac aagctcagga
3721 ctttattcta tagagcaaac ttgctgtgga gggccatgct ctccttggac ccagttaact
3781 gcaaacgtgc attggagccc tatttgctgc cgctgccatt ctagtgacct ttccacagag
3841 ctgcgccttc ctcacgtgtg tgaaaggttt tccccttcag ccctcaggta gatggaagct
3901 gcatctgccc acgatggcag tgcagtcatc atcttcagga tgtttcttca ggacttcctc
3961 agctgacaag gaattttggt ccctgcctag gaccgggtca tctgcagagg acagagagat
4021 ggtaagcagc tgtatgaatg ctgattttaa aaccaggtca tgggagaaga gcctggagat
4081 tctttcctga acactgactg cacttaccag tctgatttta tcgtcaaaca ccaagccagg
4141 ctagcatgct catggcaatc tgtttggggc tgttttgttg tggcactagc caaacataaa
4201 ggggcttaag tcagcctgca tacagaggat cggggagaga aggggcctgt gttctcagcc
4261 tcctgagtac ttaccagagt ttaatttttt taaaaaaaat ctgcactaaa atccccaaac
4321 tgacaggtaa atgtagccct cagagctcag cccaaggcag aatctaaatc acactatttt
4381 cgagatcatg tataaaaaga aaaaaaagaa gtcatgctgt gtggccaatt ataatttttt
4441 tcaaagactt tgtcacaaaa ctgtctatat tagacatttt ggagggacca ggaaatgtaa
4501 gacaccaaat cctccatctc ttcagtgtgc ctgatgtcac ctcatgattt gctgttactt
4561 ttttaactcc tgcgccaagg acagtgggtt ctgtgtccac ctttgtgctt tgcgaggccg
4621 agcccaggca tctgctcgcc tgccacggct gaccagagaa ggtgcttcag gagctctgcc
4681 ttagacgacg tgttacagta tgaacacaca gcagaggcac cctcgtatgt tttgaaagtt
4741 gccttctgaa agggcacagt tttaaggaaa agaaaaagaa tgtaaaacta tactgacccg
4801 ttttcagttt taaagggtcg tgagaaactg gctggtccaa tgggatttac agcaacattt
4861 tccattgctg aagtgaggta gcagctctct tctgtcagct gaatgttaag gatggggaaa
4921 aagaatgcct ttaagtttgc tcttaatcgt atggaagctt gagctatgtg ttggaagtgc
4981 cctggtttta atccatacac aaagacggta cataatccta caggtttaaa tgtacataaa
5041 aatatagttt ggaattcttt gctctactgt ttacattgca gattgctata atttcaagga
5101 gtgagattat aaataaaatg atgcacttta ggatgtttcc tatttttgaa atctgaacat
5161 gaatcattca catgaccaaa aattgtgttt ttttaaaaat acatgtctag tctgtccttt
5221 aatagctctc ttaaataagc tatgatatta atcagatcat taccagttag cttttaaagc
5281 acatttgttt aagactatgt ttttggaaaa atacgctaca gaattttttt ttaagctaca
5341 aataaatgag atgctactaa ttgttttgga atctgttgtt tctgccaaag gtaaattaac
5401 taaagattta ttcaggaatc cccatttgaa tttgtatgat tcaataaaag aaaacaccaa
5461 gtaagttata taaaataaat tgtgtatgag atgttgtgtt ttcctttgta atttccacta
5521 actaactaac taacttatat tcttcatgga atggagccca gaagaaatga gaggaagccc
5581 ttttcacact agatcttatt tgaagaaatg tttgttagtc agtcagtcag tggtttctgg
5641 ctctgccgag ggagatgtgt tccccagcaa ccatttctgc agcccagaat ctcaaggcac
5701 tagaggcggt gtcttaatta attggcttca caaagacaaa atgctctgga ctgggatttt

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5761 tcctttgctg tgttgggaat atgtgtttat taattagcac atgccaacaa aataaatgtc
5821 aagagttatt tcataagtgt aagtaaactt aagaattaaa gagtgcagac ttataatttt
5881 ca
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_055698.2, transcript variant 1):

(SEQ ID NO: 37)

```
   1 mgagalaicq skaavrlked mkkivavpln eqkdftyqkl fgvslqeler qgltengipa
  61 vvwniveylt qhgltqeglf rvngnvkvve qlrlkfesgv pvelgkdgdv csaasllklf
 121 lrelpdslit salqprfiql fqdgrndvqe sslrdlikel pdthycllky lcqfltkvak
 181 hhvqnrmnvh nlatvfgpnc fhvppglegm keqdlcnkim akilenyntl feveytendh
 241 lrcenlarli ivkevyykns lpilltrgle rdmpkpppkt kipksrsegs iqahrvlqpe
 301 lsdgipqlsl rlsyrkacle dmnsaegais aklvpssqed erplspfyls ahvpqvsnvs
 361 atgellerti rsaveqhlfd vnnsggqsse dsesgtlsas satsarqrrr qskeqdevrh
 421 grdkglinke ntpsgfnhld dcilntqeve kvhkntfgca gerskpkrqk sstklselhd
 481 nqdglvnmes lnstrshert gpddfewmsd erkgnekdgg htqhfesptm kigehpslsd
 541 tkqqrnqdag dqeesfvsev pqsdltalcd eknweepipa fsswqrensd sdeahlspqa
 601 grlirqllde dsdpmlsprf yaygqsrqyl ddtevppspp nshsfmrrrs sslgsyddeq
 661 edltpaqltr riqslkkkir kfedrfeeek kyrpshsdka anpevlkwtn dlakfrrqlk
 721 esklkiseed ltprmrqrsn tlpksfgsql ekedekkqel vdkaikpsve atlesiqrkl
 781 qekraessrp edikdmtkdq ianekvalqk allyyesihg rpvtknerqv mkplydryrl
 841 vkqilsrant ipiigspssk rrspllqpii egetasffke ikeeeegsed dsnvkpdfmv
 901 tlktdfsarc fldqfeddad gfispmddki pskcsqdtgl snlhaasipe llehlqemre
 961 ekkrirkklr dfednffrqn grnvqkedrt pmaeeyseyk hikaklrlle vliskrdtds
1021 ksm
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001015045.2, transcript variant 2):

(SEQ ID NO: 17)

```
   1 attgaggagc agaaggagta gggtgcgggg gaggaggagg agcgccttta gtgctgcagc
  61 agctgctgct ctgattggcc cggtggttca gctgcttccc tggaacaaaa ggtcaaagtg
 121 gactgcagtg taaatgtaga gaagcagccg ataaaatagc attgcctgaa gaagtttgga
 181 ggctgagagc agcagtagac tggccaactg cagagcaagt tgtttctcca gccgtgcggt
 241 gcagcctcat gcccccaacc cagcttagcc actgtaagaa gacgttcact gtacagacga
 301 ccaaacttgc cgtggaagag acagttgtga gattcccttg caaatttaca tacgagaatg
 361 gcttgtgaaa tcatgcctct gcaaagttca caggaagatg aaagacctct gtcacctttc
 421 tatttgagtg ctcatgtacc ccaagtcagc aatgtgtctg caaccggaga actcttagaa
 481 agaaccatcc gatcagctgt agaacaacat ctttttgatg ttaataactc tggaggtcaa
 541 agttcagagg actcagaatc tggaacacta tcagcatctt ctgccacatc tgccagacag
 601 cgccgccgcc agtccaagga gcaggatgaa gttcgacatg ggagagacaa gggacttatc
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661 aacaaagaaa atactccttc tgggttcaac caccttgatg attgtatttt gaatactcag
721 gaagtcgaaa aggtacacaa aaatactttt ggttgtgctg gagaaaggag caagcctaaa
781 cgtcagaaat ccagtactaa actttctgag cttcatgaca atcaggacgg tcttgtgaat
841 atggaaagtc tcaattccac acgatctcat gagagaactg gacctgatga ttttgaatgg
901 atgtctgatg aaaggaaagg aaatgaaaaa gatggtggac acactcagca ttttgagagc
961 cccacaatga agatccagga gcatcccagc ctatctgaca ccaaacagca gagaaatcaa
1021 gatgccggtg accaggagga gagctttgtc tccgaagtgc cccagtcgga cctgactgca
1081 ttgtgtgatg aaaagaactg ggaagagcct atccctgctt tctcctcctg gcagcgggag
1141 aacagtgact ctgatgaagc ccacctctcg ccgcaggctg ggcgcctgat ccgtcagctg
1201 ctggacgaag acagcgaccc catgctctct cctcggttct acgcttatgg gcagagcagg
1261 caatacctgg atgacacaga agtgcctcct tccccaccaa actcccattc tttcatgagg
1321 cggcgaagct cctctctggg gtcctatgat gatgagcaag aggacctgac acctgcccag
1381 ctcacacgaa ggattcagag ccttaaaaag aagatccgga agtttgaaga tagattcgaa
1441 gaagagaaga agtacagacc ttcccacagt gacaaagcag ccaatccgga ggttctgaaa
1501 tggacaaatg accttgccaa attccggaga caacttaaag aatcaaaact aaagatatct
1561 gaagaggacc taactcccag gatgcggcag cgaagcaaca cactccccaa gagttttggt
1621 tcccaacttg agaaagaaga tgagaagaag caagagctgg tggataaagc aataaagccc
1681 agtgttgaag ccacattgga atctattcag aggaagctcc aggagaagcg agcggaaagc
1741 agccgccctg aggacattaa ggatatgacc aaagaccaga ttgctaatga gaaagtggct
1801 ctgcagaaag ctctgttata ttatgaaagc attcatggac ggccggtaac aaagaacgaa
1861 cggcaggtga tgaagccact atacgacagg taccggctgg tcaaacagat cctctcccga
1921 gctaacacca tacccatcat tggttccccc tccagcaagc ggagaagccc tttgctgcag
1981 ccaattatcg agggcgaaac tgcttccttc ttcaaggaga taaaggaaga agaggagggg
2041 tcagaagacg atagcaatgt gaagccagac ttcatggtca ctctgaaaac cgatttcagt
2101 gcacgatgct ttctggacca attcgaagat gacgctgatg gatttatttc cccaatggat
2161 gataaaatac catcaaaatg cagccaggac acagggcttt caaatctcca tgctgcctca
2221 atacctgaac tcctggaaca cctccaggaa atgagagaag aaaagaaaag gattcgaaag
2281 aaacttcggg attttgaaga caactttttc agacagaatg gaagaaatgt ccagaaggaa
2341 gaccgcactc ctatggctga agaatacagt gaatataagc acataaaggc gaaactgagg
2401 ctcctggagg tgctcatcag caagagagac actgattcca agtccatgtg aggggcatgg
2461 ccaagcacag gggggctggca gctgcggtga gagtttactg tccccagaga aagtgcagct
2521 ctggaaggca gccttggggc tggccctgca aagcatgcag cccttctgcc tctagaccat
2581 ttggcatcgg ctcctgtttc cattgcctgc cttagaaact ggctggaaga agacaatgtg
2641 acctgactta ggcattttgt aattggaaag tcaagactgc agtatgtgca catgcgcacg
2701 cgcatgcacg cacacacaca cacagtagtg gagctttcct aacactagca gagattaatc
2761 actacattag acaacactca tctacagaga atatacactg ttcttccctg gataactgag
2821 aaacaagaga ccattctctg tctaactgtg ataaaaacaa gctcaggact ttattctata
2881 gagcaaactt gctgtggagg gccatgctct ccttggaccc agttaactgc aaacgtgcat
2941 tggagcccta tttgctgccg ctgccattct agtgaccttt ccacagagct gcgccttcct
3001 cacgtgtgtg aaaggttttc cccttcagcc ctcaggtaga tggaagctgc atctgcccac
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3061 gatggcagtg cagtcatcat cttcaggatg tttcttcagg acttcctcag ctgacaagga
3121 attttggtcc ctgcctagga ccgggtcatc tgcagaggac agagagatgg taagcagctg
3181 tatgaatgct gattttaaaa ccaggtcatg ggagaagagc ctggagattc tttcctgaac
3241 actgactgca cttaccagtc tgattttatc gtcaaacacc aagccaggct agcatgctca
3301 tggcaatctg tttggggctg ttttgttgtg gcactagcca aacataaagg ggcttaagtc
3361 agcctgcata cagaggatcg gggagagaag gggcctgtgt tctcagcctc ctgagtactt
3421 accagagttt aattttttta aaaaaaatct gcactaaaat ccccaaactg acaggtaaat
3481 gtagccctca gagctcagcc caaggcagaa tctaaatcac actattttcg agatcatgta
3541 taaaaagaaa aaaaagaagt catgctgtgt ggccaattat aatttttttc aaagactttg
3601 tcacaaaact gtctatatta gacattttgg agggaccagg aaatgtaaga caccaaatcc
3661 tccatctctt cagtgtgcct gatgtcacct catgatttgc tgttactttt ttaactcctg
3721 cgccaaggac agtgggttct gtgtccacct ttgtgctttg cgaggccgag cccaggcatc
3781 tgctcgcctg ccacggctga ccagagaagg tgcttcagga gctctgcctt agacgacgtg
3841 ttacagtatg aacacacagc agaggcaccc tcgtatgttt tgaaagttgc cttctgaaag
3901 ggcacagttt taaggaaaag aaaaagaatg taaaactata ctgacccgtt ttcagtttta
3961 aagggtcgtg agaaactggc tggtccaatg ggatttacag caacattttc cattgctgaa
4021 gtgaggtagc agctctcttc tgtcagctga atgttaagga tggggaaaaa gaatgccttt
4081 aagtttgctc ttaatcgtat ggaagcttga gctatgtgtt ggaagtgccc tggttttaat
4141 ccatacacaa agacggtaca taatcctaca ggtttaaatg tacataaaaa tatagtttgg
4201 aattctttgc tctactgttt acattgcaga ttgctataat ttcaaggagt gagattataa
4261 ataaaatgat gcactttagg atgtttccta tttttgaaat ctgaacatga atcattcaca
4321 tgaccaaaaa ttgtgttttt ttaaaaatac atgtctagtc tgtcctttaa tagctctctt
4381 aaataagcta tgatattaat cagatcatta ccagttagct tttaaagcac atttgtttaa
4441 gactatgttt ttggaaaaat acgctacaga attttttttt aagctacaaa taaatgagat
4501 gctactaatt gttttggaat ctgttgtttc tgccaaaggt aaattaacta aagatttatt
4561 caggaatccc catttgaatt tgtatgattc aataaaagaa aacaccaagt aagttatata
4621 aaataaattg tgtatgagat gttgtgtttt cctttgtaat ttccactaac taactaacta
4681 acttatattc ttcatggaat ggagcccaga agaaatgaga ggaagccctt ttcacactag
4741 atcttatttg aagaaatgtt tgttagtcag tcagtcagtg gtttctggct ctgccgaggg
4801 agatgtgttc cccagcaacc atttctgcag cccagaatct caaggcacta gaggcggtgt
4861 cttaattaat tggcttcaca aagacaaaat gctctggact gggatttttc ctttgctgtg
4921 ttgggaatat gtgtttatta attagcacat gccaacaaaa taaatgtcaa gagttatttc
4981 ataagtgtaa gtaaacttaa gaattaaaga gtgcagactt ataattttca
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001015045.1, transcript variant 2):

(SEQ ID NO: 18)

```
  1 maceimplqs sqederplsp fylsahvpqv snvsatgell ertirsaveq hlfdvnnsgg
 61 qssedsesgt lsassatsar qrrrqskeqd evrhgrdkgl inkentpsgf nhlddcilnt
121 qevekvhknt fgcagerskp krqksstkls elhdnqdglv nmeslnstrs hertgpddfe
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181 wmsderkgne kdgghtqhfe sptmkigehp slsdtkqqrn qdagdqeesf vsevpqsdlt

241 alcdeknwee pipafsswqr ensdsdeahl spqagrlirq lldedsdpml sprfyaygqs

301 rqylddtevp psppnshsfm rrrssslgsy ddeqedltpa qltrriqslk kkirkfedrf

361 eeekkyrpsh sdkaanpevl kwtndlakfr rqlkesklki seedltprmr qrsntlpksf

421 gsqlekedek kqelvdkaik psveatlesi qrklqekrae ssrpedikdm tkdqianekv

481 alqkallyye sihgrpvtkn erqvmkplyd ryrlvkqils rantipiigs psskrrspll

541 qpiiegetas ffkeikeeee gseddsnvkp dfmvtlktdf sarcfldqfe ddadgfispm

601 ddkipskcsq dtglsnlhaa sipellehlq emreekkrir kklrdfednf frqngrnvqk

661 edrtpmaeey seykhikakl rllevliskr dtdsksm
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001265578.1, transcript variant 3):

```
                                                          (SEQ ID NO: 38)
   1 attgaggagc agaaggagta gggtgcgggg gaggaggagg agcgccttta gtgctgcagc

  61 agctgctgct ctgattggcc cggtggttca gctgcttccc tggaacaaaa ggtcaaagtg

 121 gactgcagtg taaatgtaga gaagcagccg ataaaatagc attgcctgaa gaagtttgga

 181 ggctgagagc agcagtagac tggccaactg cagagcaagt tgtttctcca gccgtgcggt

 241 gcagcctcat gcccccaacc cagcttagcc actgtaagaa gacgttcact gtacagacga

 301 ccaaacttgc cgtggaagag acagttgtga gattcccttg caaatttaca tacgagaatg

 361 gcttgtgaaa tcatgcctct gcaaagtgct catgtacccc aagtcagcaa tgtgtctgca

 421 accggagaac tcttagaaag aaccatccga tcagctgtag aacaacatct tttgatgtt

 481 aataactctg gaggtcaaag ttcagaggac tcagaatctg gaacactatc agcatcttct

 541 gccacatctg ccagacagcg ccgccgccag tccaaggagc aggatgaagt tcgacatggg

 601 agagacaagg gacttatcaa caaagaaaat actccttctg ggttcaacca ccttgatgat

 661 tgtattttga atactcagga agtcgaaaag gtacacaaaa atacttttgg ttgtgctgga

 721 gaaaggagca agcctaaacg tcagaaatcc agtactaaac tttctgagct tcatgacaat

 781 caggacggtc ttgtgaatat ggaaagtctc aattccacac gatctcatga gagaactgga

 841 cctgatgatt ttgaatggat gtctgatgaa aggaaaggaa atgaaaaaga tggtggacac

 901 actcagcatt ttgagagccc cacaatgaag atccaggagc atcccagcct atctgacacc

 961 aaacagcaga gaaatcaaga tgccggtgac caggaggaga gctttgtctc cgaagtgccc

1021 cagtcggacc tgactgcatt gtgtgatgaa aagaactggg aagagcctat ccctgctttc

1081 tcctcctggc agcgggagaa cagtgactct gatgaagccc acctctcgcc gcaggctggg

1141 cgcctgatcc gtcagctgct ggacgaagac agcgacccca tgctctctcc tcggttctac

1201 gcttatgggc agagcaggca atacctggat gacacagaag tgcctccttc cccaccaaac

1261 tcccattctt tcatgaggcg gcgaagctcc tctctggggt cctatgatga tgagcaagag

1321 gacctgacac ctgcccagct cacacgaagg attcagagcc ttaaaaagaa gatccggaag

1381 tttgaagata gattcgaaga agagaagaag tacagacctt cccacagtga caaagcagcc

1441 aatccggagg ttctgaaatg gacaaatgac cttgccaaat tccggagaca acttaaagaa

1501 tcaaaactaa agatatctga agaggaccta actcccagga tgcggcagcg aagcaacaca

1561 ctccccaaga gttttggttc ccaacttgag aaagaagatg agaagaagca agagctggtg
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1621 gataaagcaa taaagcccag tgttgaagcc acattggaat ctattcagag gaagctccag
1681 gagaagcgag cggaaagcag ccgccctgag gacattaagg atatgaccaa agaccagatt
1741 gctaatgaga aagtggctct gcagaaagct ctgttatatt atgaaagcat tcatggacgg
1801 ccggtaacaa agaacgaacg gcaggtgatg aagccactat acgacaggta ccggctggtc
1861 aaacagatcc tctcccgagc taacaccata cccatcattg gttccccctc cagcaagcgg
1921 agaagccctt tgctgcagcc aattatcgag ggcgaaactg cttccttctt caaggagata
1981 aaggaagaag aggaggggtc agaagacgat agcaatgtga agccagactt catggtcact
2041 ctgaaaaccg atttcagtgc acgatgcttt ctggaccaat tcgaagatga cgctgatgga
2101 tttatttccc caatggatga taaaatacca tcaaaatgca gccaggacac agggctttca
2161 aatctccatg ctgcctcaat acctgaactc ctggaacacc tccaggaaat gagagaagaa
2221 aagaaaagga ttcgaaagaa acttcgggat tttgaagaca actttttcag acagaatgga
2281 agaaatgtcc agaaggaaga ccgcactcct atggctgaag aatacagtga atataagcac
2341 ataaaggcga aactgaggct cctggaggtg ctcatcagca agagagacac tgattccaag
2401 tccatgtgag ggcatggcc aagcacaggg ggctggcagc tgcggtgaga gtttactgtc
2461 cccagagaaa gtgcagctct ggaaggcagc cttggggctg gccctgcaaa gcatgcagcc
2521 cttctgcctc tagaccattt ggcatcggct cctgtttcca ttgcctgcct tagaaactgg
2581 ctggaagaag acaatgtgac ctgacttagg cattttgtaa ttggaaagtc aagactgcag
2641 tatgtgcaca tgcgcacgcg catgcacgca cacacacaca cagtagtgga gctttcctaa
2701 cactagcaga gattaatcac tacattagac aacactcatc tacagagaat atacactgtt
2761 cttccctgga taactgagaa acaagagacc attctctgtc taactgtgat aaaaacaagc
2821 tcaggacttt attctataga gcaaacttgc tgtggagggc catgctctcc ttggacccag
2881 ttaactgcaa acgtgcattg gagccctatt tgctgccgct gccattctag tgacctttcc
2941 acagagctgc gccttcctca cgtgtgtgaa aggttttccc cttcagccct caggtagatg
3001 gaagctgcat ctgcccacga tggcagtgca gtcatcatct tcaggatgtt tcttcaggac
3061 ttcctcagct gacaaggaat tttggtccct gcctaggacc gggtcatctg cagaggacag
3121 agagatggta agcagctgta tgaatgctga ttttaaaacc aggtcatggg agaagagcct
3181 ggagattctt tcctgaacac tgactgcact taccagtctg attttatcgt caaacaccaa
3241 gccaggctag catgctcatg gcaatctgtt tggggctgtt ttgttgtggc actagccaaa
3301 cataaagggg cttaagtcag cctgcataca gaggatcggg gagagaaggg gcctgtgttc
3361 tcagcctcct gagtacttac cagagtttaa ttttttttaaa aaaaatctgc actaaaatcc
3421 ccaaactgac aggtaaatgt agccctcaga gctcagccca aggcagaatc taaatcacac
3481 tattttcgag atcatgtata aaaagaaaaa aaagaagtca tgctgtgtgg ccaattataa
3541 ttttttcaa agactttgtc acaaaactgt ctatattaga cattttggag ggaccaggaa
3601 atgtaagaca ccaaatcctc catctcttca gtgtgcctga tgtcacctca tgatttgctg
3661 ttactttttt aactcctgcg ccaaggacag tgggttctgt gtccaccttt gtgctttgcg
3721 aggccgagcc caggcatctg ctcgcctgcc acggctgacc agagaaggtg cttcaggagc
3781 tctgccttag acgacgtgtt acagtatgaa cacacagcag aggcaccctc gtatgttttg
3841 aaagttgcct tctgaaaggg cacagtttta aggaaaagaa aaagaatgta aaactatact
3901 gacccgtttt cagttttaaa gggtcgtgag aaactggctg gtccaatggg atttacagca
3961 acattttcca ttgctgaagt gaggtagcag ctctcttctg tcagctgaat gttaaggatg
4021 gggaaaaaga atgcctttaa gtttgctctt aatcgtatgg aagcttgagc tatgtgttgg
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4081 aagtgccctg gttttaatcc atacacaaag acggtacata atcctacagg tttaaatgta
4141 cataaaaata tagtttggaa ttctttgctc tactgtttac attgcagatt gctataattt
4201 caaggagtga gattataaat aaaatgatgc actttaggat gtttcctatt tttgaaatct
4261 gaacatgaat cattcacatg accaaaaatt gtgttttttt aaaaatacat gtctagtctg
4321 tcctttaata gctctcttaa ataagctatg atattaatca gatcattacc agttagcttt
4381 taaagcacat ttgtttaaga ctatgttttt ggaaaaatac gctacagaat tttttttaa
4441 gctacaaata aatgagatgc tactaattgt tttggaatct gttgtttctg ccaaaggtaa
4501 attaactaaa gatttattca ggaatcccca tttgaatttg tatgattcaa taaaagaaaa
4561 caccaagtaa gttatataaa ataaattgtg tatgagatgt tgtgttttcc tttgtaattt
4621 ccactaacta actaactaac ttatattctt catggaatgg agcccagaag aaatgagagg
4681 aagccctttt cacactagat cttatttgaa gaaatgtttg ttagtcagtc agtcagtggt
4741 ttctggctct gccgagggag atgtgttccc cagcaaccat ttctgcagcc cagaatctca
4801 aggcactaga ggcggtgtct taattaattg gcttcacaaa gacaaaatgc tctggactgg
4861 gatttttcct ttgctgtgtt gggaatatgt gtttattaat tagcacatgc caacaaaata
4921 aatgtcaaga gttatttcat aagtgtaagt aaacttaaga attaaagagt gcagacttat
4981 aattttca
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001252507.1, transcript variant 3):

```
                                                    (SEQ ID NO: 39)
  1 maceimplqs ahvpqvsnvs atgellerti rsaveqhlfd vnnsggqsse dsesgtlsas
 61 satsarqrrr qskeqdevrh grdkglinke ntpsgfnhld dcilntqeve kvhkntfgca
121 gerskpkrqk sstklselhd nqdglvnmes lnstrshert gpddfewmsd erkgnekdgg
181 htqhfesptm kigehpslsd tkqqrnqdag dqeesfvsev pqsdltalcd eknweepipa
241 fsswqrensd sdeahlspqa grlirqllde dsdpmlsprf yayggsrqyl ddtevppspp
301 nshsfmrrrs sslgsyddeq edltpaqltr riqslkkkir kfedrfeeek kyrpshsdka
361 anpevlkwtn dlakfrrqlk esklkiseed ltprmrqrsn tlpksfgsql ekedekkqel
421 vdkaikpsve atlesiqrkl qekraessrp edikdmtkdq ianekvalqk allyyesihg
481 rpvtknerqv mkplydryrl vkqilsrant ipiigspssk rrspllqpii egetasffke
541 ikeeeegsed dsnvkpdfmv tlktdfsarc fldqfeddad gfispmddki pskcsqdtgl
601 snlhaasipe llehlqemre ekkrirkklr dfednffrqn grnvqkedrt pmaeeyseyk
661 hikaklrlle vliskrdtds ksm
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001265579.1, transcript variant 4):

```
                                                         (SEQ ID NO: 40)
  1 attgaggagc agaaggagta gggtgcgggg gaggaggagg agcgccttta gtgctgcagc
 61 agctgctgct ctgattggcc cggtggttca gctgcttccc tggaacaaaa ggtcaaagtg
121 gactgcagtg taaatgtaga gaagcagccg ataaaatagc attgcctgaa gaagtttgga
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181 ggctgagagc agcagtagac tggccaactg cagagcaagt tgtttctcca gccgtgcggt
241 gcagcctcat gcccccaacc cagcttagcc actgtaagaa gacgttcact gtacagacga
301 ccaaacttgc cgtggaagag acagttgtga gattcccttg caaatttaca tacgagaatg
361 gcttgtgaaa tcatgcctct gcaaagttca caggaagatg aaagacctct gtcacctttc
421 tatttgagtg ctcatgtacc ccaagtcagc aatgtgtctg caaccggaga actcttagaa
481 agaaccatcc gatcagctgt agaacaacat ctttttgatg ttaataactc tggaggtcaa
541 agttcagagg actcagaatc tggaacacta tcagcatctt ctgccacatc tgccagacag
601 cgccgccgcc agtccaagga gcaggatgaa gttcgacatg ggagagacaa gggacttatc
661 aacaaagaaa atactccttc tgggttcaac caccttgatg attgtatttt gaatactcag
721 gaagtcgaaa aggtacacaa aaatactttt ggttgtgctg gagaaaggag caagcctaaa
781 cgtcagaaat ccagtactaa actttctgag cttcatgaca atcaggacgg tcttgtgaat
841 atggaaagtc tcaattccac acgatctcat gagagaactg gacctgatga ttttgaatgg
901 atgtctgatg aaaggaaagg aaatgaaaaa gatggtggac acactcagca ttttgagagc
961 cccacaatga agatccagga gcatcccagc ctatctgaca ccaaacagca gagaaatcaa
1021 gatgccggtg accaggagga gagctttgtc tccgaagtgc cccagtcgga cctgactgca
1081 ttgtgtgatg aaaagaactg ggaagagcct atccctgctt tctcctcctg gcagcgggag
1141 aacagtgact ctgatgaagc ccacctctcg ccgcaggctg ggcgcctgat ccgtcagctg
1201 ctggacgaag acagcgaccc catgctctct cctcggttct acgcttatgg gcagagcagg
1261 caatacctgg atgacacaga agtgcctcct tccccaccaa actcccattc tttcatgagg
1321 cggcgaagct cctctctggg gtcctatgat gatgagcaag aggacctgac acctgcccag
1381 ctcacacgaa ggattcagag ccttaaaaag aagatccgga agtttgaaga tagattcgaa
1441 gaagagaaga agtacagacc ttcccacagt gacaaagcag ccaatccgga ggttctgaaa
1501 tggacaaatg accttgccaa attccggaga caacttaaag aatcaaaact aaagatatct
1561 gaagaggacc taactcccag gatgcggcag cgaagcaaca cactccccaa gagttttggt
1621 tcccaacttg agaaagaaga tgagaagaag caagagctgg tggataaagc aataaagccc
1681 agtgttgaag ccacattgga atctattcag aggaagctcc aggagaagcg agcggaaagc
1741 agccgccctg aggacattaa ggatatgacc aaagaccaga ttgctaatga gaaagtggct
1801 ctgcagaaag ctctgttata ttatgaaagc attcatggac ggccggtaac aaagaacgaa
1861 cggcaggtga tgaagccact atacgacagg taccggctgg tcaaacagat cctctcccga
1921 gctaacacca tacccatcat tgaagaagag gaggggtcag aagacgatag caatgtgaag
1981 ccagacttca tggtcactct gaaaaccgat ttcagtgcac gatgctttct ggaccaattc
2041 gaagatgacg ctgatggatt tatttcccca atggatgata aaataccatc aaaatgcagc
2101 caggacacag ggctttcaaa tctccatgct gcctcaatac ctgaactcct ggaacacctc
2161 caggaaatga gagaagaaaa gaaaaggatt cgaaagaaac ttcgggattt tgaagacaac
2221 tttttcagac agaatggaag aaatgtccag aaggaagacc gcactcctat ggctgaagaa
2281 tacagtgaat ataagcacat aaaggcgaaa ctgaggctcc tggaggtgct catcagcaag
2341 agagacactg attccaagtc catgtgaggg gcatggccaa gcacaggggg ctggcagctg
2401 cggtgagagt ttactgtccc cagagaaagt gcagctctgg aaggcagcct tggggctggc
2461 cctgcaaagc atgcagccct tctgcctcta gaccatttgg catcggctcc tgtttccatt
2521 gcctgcctta gaaactggct ggaagaagac aatgtgacct gacttaggca ttttgtaatt
2581 ggaaagtcaa gactgcagta tgtgcacatg cgcacgcgca tgcacgcaca cacacacaca
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2641 gtagtggagc tttcctaaca ctagcagaga ttaatcacta cattagacaa cactcatcta
2701 cagagaatat acactgttct tccctggata actgagaaac aagagaccat tctctgtcta
2761 actgtgataa aaacaagctc aggactttat tctatagagc aaacttgctg tggagggcca
2821 tgctctcctt ggacccagtt aactgcaaac gtgcattgga gccctatttg ctgccgctgc
2881 cattctagtg acctttccac agagctgcgc cttcctcacg tgtgtgaaag gttttcccct
2941 tcagccctca ggtagatgga agctgcatct gcccacgatg gcagtgcagt catcatcttc
3001 aggatgtttc ttcaggactt cctcagctga caaggaattt tggtccctgc ctaggaccgg
3061 gtcatctgca gaggacagag agatggtaag cagctgtatg aatgctgatt ttaaaaccag
3121 gtcatgggag aagagcctgg agattctttc ctgaacactg actgcactta ccagtctgat
3181 tttatcgtca aacaccaagc caggctagca tgctcatggc aatctgtttg gggctgtttt
3241 gttgtggcac tagccaaaca taaaggggct taagtcagcc tgcatacaga ggatcgggga
3301 gagaaggggc ctgtgttctc agcctcctga gtacttacca gagtttaatt tttttaaaaa
3361 aaatctgcac taaaatcccc aaactgacag gtaaatgtag ccctcagagc tcagcccaag
3421 gcagaatcta aatcacacta ttttcgagat catgtataaa aagaaaaaaa agaagtcatg
3481 ctgtgtggcc aattataatt tttttcaaag actttgtcac aaaactgtct atattagaca
3541 ttttggaggg accaggaaat gtaagacacc aaatcctcca tctcttcagt gtgcctgatg
3601 tcacctcatg atttgctgtt actttttttaa ctcctgcgcc aaggacagtg ggttctgtgt
3661 ccacctttgt gctttgcgag gccgagccca ggcatctgct cgcctgccac ggctgaccag
3721 agaaggtgct tcaggagctc tgccttagac gacgtgttac agtatgaaca cacagcagag
3781 gcaccctcgt atgttttgaa agttgccttc tgaaagggca cagttttaag gaaaagaaaa
3841 agaatgtaaa actatactga cccgttttca gttttaaagg gtcgtgagaa actggctggt
3901 ccaatgggat ttacagcaac attttccatt gctgaagtga ggtagcagct ctcttctgtc
3961 agctgaatgt taaggatggg gaaaaagaat gcctttaagt ttgctcttaa tcgtatggaa
4021 gcttgagcta tgtgttggaa gtgccctggt tttaatccat acacaaagac ggtacataat
4081 cctacaggtt taaatgtaca taaaaatata gtttggaatt ctttgctcta ctgtttacat
4141 tgcagattgc tataatttca aggagtgaga ttataaataa aatgatgcac tttaggatgt
4201 ttcctatttt tgaaatctga acatgaatca ttcacatgac caaaaattgt gtttttttaa
4261 aaatacatgt ctagtctgtc ctttaatagc tctcttaaat aagctatgat attaatcaga
4321 tcattaccag ttagctttta aagcacattt gtttaagact atgtttttgg aaaaatacgc
4381 tacagaattt ttttttaagc tacaaataaa tgagatgcta ctaattgttt tggaatctgt
4441 tgtttctgcc aaaggtaaat taactaaaga tttattcagg aatccccatt tgaatttgta
4501 tgattcaata aaagaaaaca ccaagtaagt tatataaaat aaattgtgta tgagatgttg
4561 tgttttcctt tgtaatttcc actaactaac taactaactt atattcttca tggaatggag
4621 cccagaagaa atgagaggaa gcccttttca cactagatct tatttgaaga aatgtttgtt
4681 agtcagtcag tcagtggttt ctggctctgc cgagggagat gtgttcccca gcaaccattt
4741 ctgcagccca gaatctcaag gcactagagg cggtgtctta attaattggc ttcacaaaga
4801 caaaatgctc tggactggga tttttccttt gctgtgttgg gaatatgtgt ttattaatta
4861 gcacatgcca acaaaataaa tgtcaagagt tatttcataa gtgtaagtaa acttaagaat
4921 taaagagtgc agacttataa ttttca
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001252508.1, transcript variant 4):

```
                                                   (SEQ ID NO: 41)
  1 maceimplqs sqederplsp fylsahvpqv snvsatgell ertirsaveq hlfdvnnsgg
 61 qssedsesgt lsassatsar qrrrqskeqd evrhgrdkgl inkentpsgf nhlddcilnt
121 qevekvhknt fgcagerskp krqksstkls elhdnqdglv nmeslnstrs hertgpddfe
181 wmsderkgne kdgghtqhfe sptmkigehp slsdtkqqrn qdagdqeesf vsevpqsdlt
241 alcdeknwee pipafsswqr ensdsdeahl spqagrlirq lldedsdpml sprfyaygqs
301 rqylddtevp psppnshsfm rrrssslgsy ddeqedltpa qltrriqslk kkirkfedrf
361 eeekkyrpsh sdkaanpevl kwtndlakfr rqlkesklki seedltprmr qrsntlpksf
421 gsqlekedek kqelvdkaik psveatlesi qrklqekrae ssrpedikdm tkdqianekv
481 alqkallyye sihgrpvtkn erqvmkplyd ryrlvkqils rantipiiee eegseddsnv
541 kpdfmvtlkt dfsarcfldq feddadgfis pmddkipskc sqdtglsnlh aasipelleh
601 lqemreekkr irkklrdfed nffrqngrnv qkedrtpmae eyseykhika klrllevlis
661 krdtdsksm
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001265580.1, transcript variant 5):

```
                                                      (SEQ ID NO: 42)
   1 attgaggagc agaaggagta gggtgcgggg gaggaggagg agcgccttta gtgctgcagc
  61 agctgctgct ctgattggcc cggtggttca gctgcttccc tggaacaaaa ggtcaaagtg
 121 gactgcagtg taaatgtaga gaagcagccg ataaaatagc attgcctgaa gaagtttgga
 181 ggctgagagc agcagtagac tggccaactg cagagcaagt tgtttctcca gccgtgcggt
 241 gcagcctcat gcccccaacc cagcttagcc actgtaagaa gacgttcact gtacagacga
 301 ccaaacttgc cgtggaagag acagttgtga gattcccttg caaatttaca tacgagaatg
 361 gcttgtgaaa tcatgcctct gcaaagactc ttagaaagaa ccatccgatc agctgtagaa
 421 caacatcttt ttgatgttaa taactctgga ggtcaaagtt cagaggactc agaatctgga
 481 acactatcag catcttctgc cacatctgcc agacagcgcc gccgccagtc caaggagcag
 541 gatgaagttc gacatgggag agacaaggga cttatcaaca aagaaaatac tccttctggg
 601 ttcaaccacc ttgatgattg tattttgaat actcaggaag tcgaaaaggt acacaaaaat
 661 acttttggtt gtgctggaga aaggagcaag cctaaacgtc agaaatccag tactaaactt
 721 tctgagcttc atgacaatca ggacggtctt gtgaatatgg aaagtctcaa ttccacacga
 781 tctcatgaga gaactggacc tgatgatttt gaatggatgt ctgatgaaag gaaaggaaat
 841 gaaaaagatg gtggacacac tcagcatttt gagagcccca caatgaagat ccaggagcat
 901 cccagcctat ctgacaccaa acagcagaga aatcaagatg ccggtgacca ggaggagagc
 961 tttgtctccg aagtgcccca gtcggacctg actgcattgt gtgatgaaaa gaactgggaa
1021 gagcctatcc ctgctttctc ctcctggcag cgggagaaca gtgactctga tgaagcccac
1081 ctctcgccgc aggctgggcg cctgatccgt cagctgctgg acgaagacag cgaccccatg
1141 ctctctcctc ggttctacgc ttatgggcag agcaggcaat acctggatga cacagaagtg
1201 cctccttccc caccaaactc ccattctttc atgaggcggc gaagctcctc tctggggtcc
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1261 tatgatgatg agcaagagga cctgacacct gcccagctca cacgaaggat tcagagcctt
1321 aaaaagaaga tccggaagtt tgaagataga ttcgaagaag agaagaagta cagaccttcc
1381 cacagtgaca aagcagccaa tccggaggtt ctgaaatgga caaatgacct tgccaaattc
1441 cggagacaac ttaaagaatc aaaactaaag atatctgaag aggacctaac tcccaggatg
1501 cggcagcgaa gcaacacact ccccaagagt tttggttccc aacttgagaa agaagatgag
1561 aagaagcaag agctggtgga taaagcaata aagcccagtg ttgaagccac attggaatct
1621 attcagagga agctccagga gaagcgagcg gaaagcagcc gccctgagga cattaaggat
1681 atgaccaaag accagattgc taatgagaaa gtggctctgc agaaagctct gttatattat
1741 gaaagcattc atggacggcc ggtaacaaag aacgaacggc aggtgatgaa gccactatac
1801 gacaggtacc ggctggtcaa acagatcctc tcccgagcta acaccatacc catcattggt
1861 tccccctcca gcaagcggag aagccctttg ctgcagccaa ttatcgaggg cgaaactgct
1921 tccttcttca aggagataaa ggaagaagag gaggggtcag aagacgatag caatgtgaag
1981 ccagacttca tggtcactct gaaaaccgat ttcagtgcac gatgctttct ggaccaattc
2041 gaagatgacg ctgatggatt tatttcccca atggatgata aaataccatc aaaatgcagc
2101 caggacacag ggctttcaaa tctccatgct gcctcaatac ctgaactcct ggaacacctc
2161 caggaaatga gagaagaaaa gaaaaggatt cgaaagaaac ttcgggattt tgaagacaac
2221 tttttcagac agaatggaag aaatgtccag aaggaagacc gcactcctat ggctgaagaa
2281 tacagtgaat ataagcacat aaaggcgaaa ctgaggctcc tggaggtgct catcagcaag
2341 agagacactg attccaagtc catgtgaggg gcatggccaa gcacaggggg ctggcagctg
2401 cggtgagagt ttactgtccc cagagaaagt gcagctctgg aaggcagcct tggggctggc
2461 cctgcaaagc atgcagccct tctgcctcta gaccatttgg catcggctcc tgtttccatt
2521 gcctgcctta gaaactggct ggaagaagac aatgtgacct gacttaggca ttttgtaatt
2581 ggaaagtcaa gactgcagta tgtgcacatg cgcacgcgca tgcacgcaca cacacacaca
2641 gtagtggagc tttcctaaca ctagcagaga ttaatcacta cattagacaa cactcatcta
2701 cagagaatat acactgttct tccctggata actgagaaac aagagaccat tctctgtcta
2761 actgtgataa aaacaagctc aggactttat tctatagagc aaacttgctg tggagggcca
2821 tgctctcctt ggacccagtt aactgcaaac gtgcattgga gccctatttg ctgccgctgc
2881 cattctagtg acctttccac agagctgcgc cttcctcacg tgtgtgaaag gttttcccct
2941 tcagccctca ggtagatgga agctgcatct gcccacgatg gcagtgcagt catcatcttc
3001 aggatgtttc ttcaggactt cctcagctga caaggaattt tggtccctgc ctaggaccgg
3061 gtcatctgca gaggacagag agatggtaag cagctgtatg aatgctgatt ttaaaaccag
3121 gtcatgggag aagagcctgg agattctttc ctgaacactg actgcactta ccagtctgat
3181 tttatcgtca aacaccaagc caggctagca tgctcatggc aatctgtttg gggctgtttt
3241 gttgtggcac tagccaaaca taaaggggct taagtcagcc tgcatacaga ggatcgggga
3301 gagaaggggc ctgtgttctc agcctcctga gtacttacca gagtttaatt tttttaaaaa
3361 aaatctgcac taaaatcccc aaactgacag gtaaatgtag ccctcagagc tcagcccaag
3421 gcagaatcta aatcacacta ttttcgagat catgtataaa aagaaaaaaa agaagtcatg
3481 ctgtgtggcc aattataatt tttttcaaag actttgtcac aaaactgtct atattagaca
3541 ttttggaggg accaggaaat gtaagacacc aaatcctcca tctcttcagt gtgcctgatg
3601 tcacctcatg atttgctgtt acttttttaa ctcctgcgcc aaggacagtg ggttctgtgt
3661 ccacctttgt gctttgcgag gccgagccca ggcatctgct cgcctgccac ggctgaccag
```

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```
3721 agaaggtgct tcaggagctc tgccttagac gacgtgttac agtatgaaca cacagcagag
3781 gcaccctcgt atgttttgaa agttgccttc tgaaagggca cagttttaag gaaaagaaaa
3841 agaatgtaaa actatactga cccgttttca gttttaaagg gtcgtgagaa actggctggt
3901 ccaatgggat ttacagcaac attttccatt gctgaagtga ggtagcagct ctcttctgtc
3961 agctgaatgt taaggatggg gaaaaagaat gcctttaagt ttgctcttaa tcgtatggaa
4021 gcttgagcta tgtgttggaa gtgccctggt tttaatccat acacaaagac ggtacataat
4081 cctacaggtt taaatgtaca taaaaatata gtttggaatt ctttgctcta ctgtttacat
4141 tgcagattgc tataatttca aggagtgaga ttataaataa aatgatgcac tttaggatgt
4201 ttcctatttt tgaaatctga acatgaatca ttcacatgac caaaaattgt gttttttaa
4261 aaatacatgt ctagtctgtc ctttaatagc tctcttaaat aagctatgat attaatcaga
4321 tcattaccag ttagctttta aagcacattt gtttaagact atgtttttgg aaaaatacgc
4381 tacagaattt ttttttaagc tacaaataaa tgagatgcta ctaattgttt tggaatctgt
4441 tgtttctgcc aaaggtaaat taactaaaga tttattcagg aatccccatt tgaatttgta
4501 tgattcaata aaagaaaaca ccaagtaagt tatataaaat aaattgtgta tgagatgttg
4561 tgttttcctt tgtaatttcc actaactaac taactaactt atattcttca tggaatggag
4621 cccagaagaa atgagaggaa gccctttca cactagatct tatttgaaga aatgtttgtt
4681 agtcagtcag tcagtggttt ctggctctgc cgagggagat gtgttcccca gcaaccattt
4741 ctgcagccca gaatctcaag gcactagagg cggtgtctta attaattggc ttcacaaaga
4801 caaaatgctc tggactggga tttttccttt gctgtgttgg gaatatgtgt ttattaatta
4861 gcacatgcca acaaaataaa tgtcaagagt tatttcataa gtgtaagtaa acttaagaat
4921 taaagagtgc agacttataa ttttca
```

In some embodiments of the methods of the disclosure, the wild type human FAM13A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001252509.1, transcript variant 5):

```
                                                        (SEQ ID NO: 43)
  1 maceimplqr llertirsav eqhlfdvnns ggqssedses gtlsassats arqrrrqske
 61 qdevrhgrdk glinkentps gfnhlddcil ntqevekvhk ntfgcagers kpkrqksstk
121 lselhdnqdg lvnmeslnst rshertgpdd fewmsderkg nekdgghtqh fesptmkiqe
181 hpslsdtkqq rnqdagdqee sfvsevpqsd ltalcdeknw eepipafssw qrensdsdea
241 hlspqagrli rqlldedsdp mlsprfyayg qsrqylddte vppsppnshs fmrrrssslg
301 syddeqedlt paqltrriqs lkkkirkfed rfeeekkyrp shsdkaanpe vlkwtndlak
361 frrqlkeskl kiseedltpr mrqrsntlpk sfgsqleked ekkqelvdka ikpsveatle
421 siqrklqekr aessrpedik dmtkdqiane kvalqkally yesihgrpvt knerqvmkpl
481 ydryrlvkqi lsrantipii gspsskrrsp llqpiieget asffkeikee eegseddsnv
541 kpdfmvtlkt dfsarcfldq feddadgfis pmddkipskc sqdtglsnlh aasipelleh
601 lqemreekkr irkklrdfed nffrqngrnv qkedrtpmae eyseykhika klrllevlis
661 krdtdsksm
```

In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_004415.3, transcript variant 1):

(SEQ ID NO: 44)

```
   1 aagaaaccgg ccaggtgtgg cctaggcgcc cagtgccagc ggggaggaga ctcgctccgc
  61 cgccgaccaa caccaacacc cagctccgac gcagctcctc tgcgcccttg ccgccctccg
 121 agccacagct ttcctcccgc tcctgccccc ggcccgtcgc cgtctccgcg ctcgcagcgg
 181 cctcgggagg gcccaggtag cgagcagcga cctcgcgagc cttccgcact cccgcccggt
 241 tccccggccg tccgcctatc cttggccccc tccgctttct ccgcgccggc ccgcctcgct
 301 tatgcctcgg cgctgagccg ctctcccgat tgcccgccga catgagctgc aacggaggct
 361 cccacccgcg gatcaacact ctgggccgca tgatccgcgc cgagtctggc ccggacctgc
 421 gctacgaggt gaccagcggc ggcgggggca ccagcaggat gtactattct cggcgcggcg
 481 tgatcaccga ccagaactcg gacggctact gtcaaaccgg cacgatgtcc aggcaccaga
 541 accagaacac catccaggag ctgctgcaga actgctccga ctgcttgatg cgagcagagc
 601 tcatcgtgca gcctgaattg aagtatggag atggaataca actgactcgg agtcgagaat
 661 tggatgagtg ttttgcccag gccaatgacc aaatggaaat cctcgacagc ttgatcagag
 721 agatgcggca gatgggccag ccctgtgatg cttaccagaa aaggcttctt cagctccaag
 781 agcaaatgcg agccctttat aaagccatca gtgtccctcg agtccgcagg gccagctcca
 841 agggtggtgg aggctacact tgtcagagtg gctctggctg ggatgagttc accaaacatg
 901 tcaccagtga atgtttgggg tggatgaggc agcaaagggc ggagatggac atggtggcct
 961 ggggtgtgga cctggcctca gtggagcagc acattaacag ccaccggggc atccacaact
1021 ccatcggcga ctatcgctgg cagctggaca aaatcaaagc cgacctgcgc gagaaatctg
1081 cgatctacca gttggaggag gagtatgaaa acctgctgaa agcgtccttt gagaggatgg
1141 atcacctgcg acagctgcag aacatcattc aggccacgtc cagggagatc atgtggatca
1201 atgactgcga ggaggaggag ctgctgtacg actggagcga caagaacacc aacatcgctc
1261 agaaacagga ggccttctcc atacgcatga gtcaactgga agttaaagaa aaagagctca
1321 ataagctgaa acaagaaagt gaccaacttg tcctcaatca gcatccagct tcagacaaaa
1381 ttgaggccta tatggacact ctgcagacgc agtggagttg gattcttcag atcaccaagt
1441 gcattgatgt tcatctgaaa gaaaatgctg cctactttca gtttttgaa gaggcgcagt
1501 ctactgaagc atacctgaag ggctccagg actccatcag gaagaagtac ccctgcgaca
1561 agaacatgcc cctgcagcac ctgctggaac agatcaagga gctggagaaa gaacgagaga
1621 aaatccttga atacaagcgt caggtgcaga acttggtaaa caagtctaag aagattgtac
1681 agctgaagcc tcgtaaccca gactacagaa gcaataaacc cattattctc agagctctct
1741 gtgactacaa acaagatcag aaaatcgtgc ataaggggga tgagtgtatc ctgaaggaca
1801 acaacgagcg cagcaagtgg tacgtgacgg gcccgggagg cgttgacatg cttgttccct
1861 ctgtggggct gatcatccct cctccgaacc cactggccgt ggacctctct tgcaagattg
1921 agcagtacta cgaagccatc ttggctctgt ggaaccagct ctacatcaac atgaagagcc
1981 tggtgtcctg gcactactgc atgattgaca tagagaagat cagggccatg acaatcgcca
2041 agctgaaaac aatgcggcag gaagattaca tgaagacgat agccgacctt gagttacatt
2101 accaagagtt catcagaaat agccaaggct cagagatgtt tggagatgat gacaagcgga
2161 aaatacagtc tcagttcacc gatgcccaga agcattacca gaccctggtc attcagctcc
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2221 ctggctatcc ccagcaccag acagtgacca caactgaaat cactcatcat ggaacctgcc
2281 aagatgtcaa ccataataaa gtaattgaaa ccaacagaga aaatgacaag caagaaacat
2341 ggatgctgat ggagctgcag aagattcgca ggcagataga gcactgcgag ggcaggatga
2401 ctctcaaaaa cctccctcta gcagaccagg gatcttctca ccacatcaca gtgaaaatta
2461 acgagcttaa gagtgtgcag aatgattcac aagcaattgc tgaggttctc aaccagctta
2521 aagatatgct tgccaacttc agaggttctg aaaagtactg ctatttacag aatgaagtat
2581 ttggactatt tcagaaactg gaaaatatca atggtgttac agatggctac ttaaatagct
2641 tatgcacagt aagggcactg ctccaggcta ttctccaaac agaagacatg ttaaaggttt
2701 atgaagccag gctcactgag gaggaaactg tctgcctgga cctggataaa gtggaagctt
2761 accgctgtgg actgaagaaa ataaaaaatg acttgaactt gaagaagtcg ttgttggcca
2821 ctatgaagac agaactacag aaagcccagc agatccactc tcagacttca cagcagtatc
2881 cactttatga tctggacttg ggcaagttcg gtgaaaaagt cacacagctg acagaccgct
2941 ggcaaaggat agataaacag atcgacttta ggttatggga cctggagaaa caaatcaagc
3001 aattgaggaa ttatcgtgat aactatcagg ctttctgcaa gtggctctat gatgctaaac
3061 gccgccagga ttccttagaa tccatgaaat ttggagattc caacacagtc atgcggtttt
3121 tgaatgagca gaagaacttg cacagtgaaa tatctggcaa acgagacaaa tcagaggaag
3181 tacaaaaaat tgctgaactt tgcgccaatt caattaagga ttatgagctc cagctggcct
3241 catacacctc aggactggaa actctgctga acatacctat caagaggacc atgattcagt
3301 ccccttctgg ggtgattctg caagaggctg cagatgttca tgctcggtac attgaactac
3361 ttacaagatc tggagactat tacaggttct taagtgagat gctgaagagt ttggaagatc
3421 tgaagctgaa aaataccaag atcgaagttt tggaagagga gctcagactg gcccgagatg
3481 ccaactcgga aaactgtaat aagaacaaat tcctggatca gaacctgcag aaataccagg
3541 cagagtgttc ccagttcaaa gcgaagcttg cgagcctgga ggagctgaag agacaggctg
3601 agctggatgg gaagtcggct aagcaaaatc tagacaagtg ctacggccaa ataaaagaac
3661 tcaatgagaa gatcacccga ctgacttatg agattgaaga tgaaaagaga agaagaaaat
3721 ctgtggaaga cagatttgac caacagaaga atgactatga ccaactgcag aaagcaaggc
3781 aatgtgaaaa ggagaacctt ggttggcaga aattagagtc tgagaaagcc atcaaggaga
3841 aggagtacga gattgaaagg ttgagggttc tactgcagga agaaggcacc cggaagagag
3901 aatatgaaaa tgagctggca aaggtaagaa accactataa tgaggagatg agtaatttaa
3961 ggaacaagta tgaaacagag attaacatta cgaagaccac catcaaggag atatccatgc
4021 aaaaagagga tgattccaaa aatcttagaa accagcttga tagactttca agggaaaatc
4081 gagatctgaa ggatgaaatt gtcaggctca atgacagcat cttgcaggcc actgagcagc
4141 gaaggcgagc tgaagaaaac gcccttcagc aaaaggcctg tggctctgag ataatgcaga
4201 agaagcagca tctggagata gaactgaagc aggtcatgca gcagcgctct gaggacaatg
4261 cccggcacaa gcagtccctg gaggaggctg ccaagaccat tcaggacaaa aataaggaga
4321 tcgagagact caaagctgag tttcaggagg aggccaagcg ccgctgggaa tatgaaaatg
4381 aactgagtaa ggtaagaaac aattatgatg aggagatcat tagcttaaaa aatcagtttg
4441 agaccgagat caacatcacc aagaccacca tccaccagct caccatgcag aaggaagagg
4501 ataccagtgg ctaccgggct cagatagaca atctcacccg agaaaacagg agcttatctg
4561 aagaaataaa gaggctgaag aacactctaa cccagaccac agagaatctc aggagggtgg
4621 aagaagacat ccaacagcaa aaggccactg gctctgaggt gtctcagagg aaacagcagc

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4681 tggaggttga gctgagacaa gtcactcaga tgcgaacaga ggagagcgta agatataagc

4741 aatctcttga tgatgctgcc aaaaccatcc aggataaaaa caaggagata gaaaggttaa

4801 aacaactgat cgacaaagaa acaaatgacc ggaaatgcct ggaagatgaa aacgcgagat

4861 tacaaagggt ccagtatgac ctgcagaaag caaacagtag tgcgacggag acaataaaca

4921 aactgaaggt tcaggagcaa gaactgacac gcctgaggat cgactatgaa agggtttccc

4981 aggagaggac tgtgaaggac caggatatca cgcggttcca gaactctctg aaagagctgc

5041 agctgcagaa gcagaaggtg gaagaggagc tgaatcggct gaagaggacc gcgtcagaag

5101 actcctgcaa gaggaagaag ctggaggaag agctggaagg catgaggagg tcgctgaagg

5161 agcaagccat caaaatcacc aacctgaccc agcagctgga gcaggcatcc attgttaaga

5221 agaggagtga ggatgacctc cggcagcaga gggacgtgct ggatggccac ctgagggaaa

5281 agcagaggac ccaggaagag ctgaggaggc tctcttctga ggtcgaggcc ctgaggcggc

5341 agttactcca ggaacaggaa agtgtcaaac aagctcactt gaggaatgag catttccaga

5401 aggcgataga agataaaagc agaagcttaa atgaaagcaa aatagaaatt gagaggctgc

5461 agtctctcac agagaacctg accaaggagc acttgatgtt agaagaagaa ctgcggaacc

5521 tgaggctgga gtacgatgac ctgaggagag gacgaagcga agcggacagt gataaaaatg

5581 caaccatctt ggaactaagg agccagctgc agatcagcaa caaccggacc ctggaactgc

5641 aggggctgat taatgattta cagagagaga gggaaaattt gagacaggaa attgagaaat

5701 tccaaaagca ggctttagag gcatctaata ggattcagga atcaaagaat cagtgtactc

5761 aggtggtaca ggaaagagag agccttctgg tgaaaatcaa agtcctggag caagacaagg

5821 caaggctgca gaggctggag gatgagctga atcgtgcaaa atcaactcta gaggcagaaa

5881 ccagggtgaa acagcgcctg gagtgtgaga aacagcaaat tcagaatgac ctgaatcagt

5941 ggaagactca atattcccgc aaggaggagg ctattaggaa gatagaatcg gaaagagaaa

6001 agagtgagag agagaagaac agtcttagga gtgagatcga aagactccaa gcagagatca

6061 agagaattga agagaggtgc aggcgtaagc tggaggattc taccagggag acacagtcac

6121 agttagaaac agaacgctcc cgatatcaga gggagattga taaactcaga cagcgcccat

6181 atgggtccca tcgagagacc cagactgagt gtgagtggac cgttgacacc tccaagctgg

6241 tgtttgatgg gctgaggaag aaggtgacag caatgcagct ctatgagtgt cagctgatcg

6301 acaaaacaac cttggacaaa ctattgaagg ggaagaagtc agtggaagaa gttgcttctg

6361 aaatccagcc attccttcgg ggtgcaggat ctatcgctgg agcatctgct tctcctaagg

6421 aaaaatactc tttggtagag gccaagagaa agaaattaat cagcccagaa tccacagtca

6481 tgcttctgga ggcccaggca gctacaggtg gtataattga tccccatcgg aatgagaagc

6541 tgactgtcga cagtgccata gctcgggacc tcattgactt cgatgaccgt cagcagatat

6601 atgcagcaga aaaagctatc actggttttg atgatccatt ttcaggcaag acagtatctg

6661 tttcagaagc catcaagaaa aatttgattg atagagaaac cggaatgcgc ctgctggaag

6721 cccagattgc ttcagggggt gtagtagacc ctgtgaacag tgtctttttg ccaaaagatg

6781 tcgccttggc ccgggggctg attgatagag atttgtatcg atccctgaat gatccccgag

6841 atagtcagaa aaactttgtg gatccagtca ccaaaaagaa ggtcagttac gtgcagctga

6901 aggaacggtg cagaatcgaa ccacatactg gtctgctctt gctttcagta cagaagagaa

6961 gcatgtcctt ccaaggaatc agacaacctg tgaccgtcac tgagctagta gattctggta

7021 tattgagacc gtccactgtc aatgaactgg aatctggtca gatttcttat gacgaggttg

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7081 gtgagagaat taaggacttc ctccagggtt caagctgcat agcaggcata tacaatgaga
7141 ccacaaaaca gaagcttggc atttatgagg ccatgaaaat tggcttagtc cgacctggta
7201 ctgctctgga gttgctggaa gcccaagcag ctactggctt tatagtggat cctgttagca
7261 acttgaggtt accagtggag gaagcctaca agagaggtct ggtgggcatt gagttcaaag
7321 agaagctcct gtctgcagaa cgagctgtca ctgggtataa tgatcctgaa acaggaaaca
7381 tcatctcttt gttccaagcc atgaataagg aactcatcga aaagggccac ggtattcgct
7441 tattagaagc acagatcgca accgggggga tcattgaccc aaaggagagc catcgtttac
7501 cagttgacat agcatataag aggggctatt tcaatgagga actcagtgag attctctcag
7561 atccaagtga tgataccaaa ggatttttg accccaacac tgaagaaaat cttacctatc
7621 tgcaactaaa agaaagatgc attaaggatg aggaaacagg gctctgtctt ctgcctctga
7681 aagaaaagaa gaaacaggtg cagacatcac aaaagaatac cctcaggaag cgtagagtgg
7741 tcatagttga cccagaaacc aataaagaaa tgtctgttca ggaggcctac aagaagggcc
7801 taattgatta tgaaaccttc aaagaactgt gtgagcagga atgtgaatgg gaagaaataa
7861 ccatcacggg atcagatggc tccaccaggg tggtcctggt agatagaaag acaggcagtc
7921 agtatgatat tcaagatgct attgacaagg gccttgttga caggaagttc tttgatcagt
7981 accgatccgg cagcctcagc ctcactcaat ttgctgacat gatctccttg aaaaatggtg
8041 tcggcaccag cagcagcatg ggcagtggtg tcagcgatga tgtttttagc agctcccgac
8101 atgaatcagt aagtaagatt tccaccatat ccagcgtcag gaatttaacc ataaggagca
8161 gctctttttc agacaccctg gaagaatcga gccccattgc agccatcttt gacacagaaa
8221 acctggagaa aatctccatt acagaaggta tagagcgggg catcgttgac agcatcacgg
8281 gtcagaggct tctggaggct caggcctgca caggtggcat catccaccca accacgggcc
8341 agaagctgtc acttcaggac gcagtctccc agggtgtgat tgaccaagac atggccacca
8401 ggctgaagcc tgctcagaaa gccttcatag gcttcgaggg tgtgaaggga aagaagaaga
8461 tgtcagcagc agaggcagtg aaagaaaaat ggctcccgta tgaggctggc cagcgcttcc
8521 tggagttcca gtacctcacg ggaggtcttg ttgacccgga agtgcatggg aggataagca
8581 ccgaagaagc catccggaag gggttcatag atggccgcgc cgcacagagg ctgcaagaca
8641 ccagcagcta tgccaaaatc ctgacctgcc ccaaaaccaa attaaaaata tcctataagg
8701 atgccataaa tcgctccatg gtagaagata tcactgggct gcgccttctg gaagccgcct
8761 ccgtgtcgtc caagggctta cccagccctt acaacatgtc ttcggctccg ggtcccgct
8821 ccggctcccg ctcgggatct cgctccggat ctcgctccgg gtcccgcagt gggtcccgga
8881 gaggaagctt tgacgccaca gggaattctt cctactctta ttcctactca tttagcagta
8941 gttctattgg gcactagtag tcagttggga gtggttgcta taccttgact tcatttatat
9001 gaatttccac tttattaaat aatagaaaag aaaatcccgg tgcttgcagt agagtgatag
9061 gacattctat gcttacagaa aatatagcca tgattgaaat caaatagtaa aggctgttct
9121 ggctttttat cttcttagct catcttaaat aagcagtaca cttggatgca gtgcgtctga
9181 agtgctaatc agttgtaaca atagcacaaa tcgaacttag gatttgtttc ttctcttctg
9241 tgtttcgatt tttgatcaat tctttaattt tggaagccta taatacagtt ttctattctt
9301 ggagataaaa attaaatgga tcactgatat tttagtcatt ctgcttctca tctaaatatt
9361 tccatattct gtattaggag aaaattaccc tcccagcacc agcccccctc tcaaaccccc
9421 aacccaaaac caagcatttt ggaatgagtc tcctttagtt tcagagtgtg gattgtataa
9481 cccatatact cttcgatgta cttgtttggt ttggtattaa tttgactgtg catgacagcg
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9541 gcaatctttt ctttggtcaa agttttctgt ttattttgct tgtcatattc gatgtacttt
9601 aaggtgtctt tatgaagttt gctattctgg caataaactt ttagactttt gaagtgtttg
9661 tgttttaatt taatatgttt ataagcatgt ataaacattt agcatatttt tatcataggt
9721 ctaaaaatat ttgtttacta aatacctgtg aagaaatacc attaaaaaac tatttggttc
9781 tgaattctta ctagaaaaaa aa
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In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_004406.2, transcript variant 1):

(SEQ ID NO: 45)

```
   1 mscnggshpr intlgrmira esgpdlryev tsggggtsrm yysrrgvitd qnsdgycqtg
  61 tmsrhqnqnt iqellqncsd clmraelivq pelkygdgiq ltrsreldec faqandqmei
 121 ldsliremrq mgqpcdayqk rllqlqeqmr alykaisvpr vrrasskggg gytcqsgsgw
 181 deftkhvtse clgwmrqqra emdmvawgvd lasveqhins hrgihnsigd yrwqldkika
 241 dlreksaiyq leeeyenllk asfermdhlr qlqniiqats reimwindce eeellydwsd
 301 kntniaqkqe afsirmsqle vkekelnklk qesdqlvinq hpasdkieay mdtlqtqwsw
 361 ilqitkcidv hlkenaayfq ffeeaqstea ylkglqdsir kkypcdknmp lqhlleqike
 421 lekerekile ykrqvqnlvn kskkivqlkp rnpdyrsnkp iilralcdyk qdqkivhkgd
 481 ecilkdnner skwyvtgpgg vdmlvpsvgl iipppnplav dlsckieqyy eailalwnql
 541 yinmkslvsw hycmidieki ramtiaklkt mrqedymkti adlelhyqef irnsqgsemf
 601 gdddkrkiqs qftdaqkhyq tlviqlpgyp qhqtvtttei thhgtcqdvn hnkvietnre
 661 ndkqetwmlm elqkirrqie hcegrmtlkn lpladqgssh hitvkinelk svqndsgqia
 721 evlnqlkdml anfrgsekyc ylqnevfglf qkleningvt dgylnslctv rallqailqt
 781 edmlkvyear lteeetvcld ldkveayrcg lkkikndlnl kksllatmkt elqkaqqihs
 841 qtsqqyplyd ldlgkfgekv tqltdrwqri dkqidfrlwd lekqikqlrn yrdnyqafck
 901 wlydakrrqd slesmkfgds ntvmrflneq knlhseisgk rdkseevqki aelcansikd
 961 yelqlasyts gletllnipi krtmiqspsg vilqeaadvh aryielltrs gdyyrflsem
1021 lksledlklk ntkievleee lrlardanse ncnknkfldq nlqkyqaecs qfkaklasle
1081 elkrqaeldg ksakqnldkc ygqikelnek itrltyeied ekrrrksved rfdqqkndyd
1141 qlqkarqcek enlgwqkles ekaikekeye ierlrvllqe egtrkreyen elakvrnhyn
1201 eemsnlrnky eteinitktt ikeismqked dsknlrnqld rlsrenrdlk deivrlndsi
1261 lqateqrrra eenalqqkac gseimqkkqh leielkqvmq qrsednarhk qsleeaakti
1321 qdknkeierl kaefqeeakr rweyenelsk vrnnydeeii slknqfetei nitkttihql
1381 tmqkeedtsg yraqidnltr enrslseeik rlkntltqtt enlrrveedi qqqkatgsev
1441 sqrkqqleve lrqvtqmrte esvrykqsld daaktiqdkn keierlkqli dketndrkcl
1501 edenarlqry qydlqkanss atetinklkv qeqeltrlri dyervsgert vkdqditrfq
1561 nslkelqlqk qkveeelnrl krtasedsck rkkleeeleg mrrslkeqai kitnitqqle
1621 qasivkkrse ddlrqqrdvl dghlrekqrt qeelrrlsse vealrrqllq eqesvkqahl
1681 rnehfqkaie dksrslnesk ieierlqslt enitkehlml eeelrnlrle yddlrrgrse
1741 adsdknatil elrsqlqisn nrtlelqgli ndlqrerenl rqeiekfqkq aleasnriqe
1801 sknqctqvvq eresllvkik vleqdkarlq rledelnrak stleaetrvk qrlecekqqi
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1861 qndlnqwktq ysrkeeairk ieserekser eknslrseie rlqaeikrie ercrrkleds
1921 tretqsqlet ersrygreid klrqrpygsh retqtecewt vdtsklvfdg lrkkvtamql
1981 yecqlidktt ldkllkgkks veevaseiqp flrgagsiag asaspkekys lveakrkkli
2041 spestvmlle aqaatggiid phrnekltvd saiardlidf ddrqqiyaae kaitgfddpf
2101 sgktvsvsea ikknlidret gmrlleaqia sggvvdpvns vflpkdvala rglidrdlyr
2161 slndprdsqk nfvdpvtkkk vsyvqlkerc riephtglll lsvqkrsmsf qgirqpvtvt
2221 elvdsgilrp stvnelesgq isydevgeri kdflqgssci agiynettkq klgiyeamki
2281 glvrpgtale lleaqaatgf ivdpvsnlrl pveeaykrgl vgiefkekll saeravtgyn
2341 dpetgniisl fqamnkelie kghgirllea qiatggiidp keshrlpvdi aykrgyfnee
2401 lseilsdpsd dtkgffdpnt eenltylqlk ercikdeetg lcllplkekk kqvqtsqknt
2461 lrkrrvvivd petnkemsvq eaykkglidy etfkelceqe ceweeititg sdgstrvvlv
2521 drktgsqydi qdaidkglvd rkffdqyrsg slsltqfadm islkngvgts ssmgsgvsdd
2581 vfsssrhesv skistissvr nltirsssfs dtleesspia aifdtenlek isitegierg
2641 ivdsitgqrl leaqactggi ihpttgqkls lqdaysqgvi dqdmatrlkp aqkafigfeg
2701 vkgkkkmsaa eavkekwlpy eagqrflefq yltgglvdpe vhgristeea irkgfidgra
2761 aqrlqdtssy akiltcpktk lkisykdain rsmveditgl rlleaasvss kglpspynms
2821 sapgsrsgsr sgsrsgsrsg srsgsrrgsf datgnssysy sysfssssig h
```

In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001008844.2, transcript variant 2):

```
                                                        (SEQ ID NO: 19)
   1 aagaaaccgg ccaggtgtgg cctaggcgcc cagtgccagc ggggaggaga ctcgctccgc
  61 cgccgaccaa caccaacacc cagctccgac gcagctcctc tgcgcccttg ccgccctccg
 121 agccacagct ttcctcccgc tcctgccccc ggcccgtcgc cgtctccgcg ctcgcagcgg
 181 cctcgggagg gcccaggtag cgagcagcga cctcgcgagc cttccgcact cccgcccggt
 241 tccccggccg tccgcctatc cttggccccc tccgctttct ccgcgccggc ccgcctcgct
 301 tatgcctcgg cgctgagccg ctctcccgat tgcccgccga catgagctgc aacggaggct
 361 cccacccgcg gatcaacact ctgggccgca tgatccgcgc cgagtctggc ccggacctgc
 421 gctacgaggt gaccagcggc ggcgggggca ccagcaggat gtactattct cggcgcggcg
 481 tgatcaccga ccagaactcg gacggctact gtcaaaccgg cacgatgtcc aggcaccaga
 541 accagaacac catccaggag ctgctgcaga actgctccga ctgcttgatg cgagcagagc
 601 tcatcgtgca gcctgaattg aagtatggag atggaataca actgactcgg agtcgagaat
 661 tggatgagtg ttttgcccag gccaatgacc aaatggaaat cctcgacagc ttgatcagag
 721 agatgcggca gatgggccag ccctgtgatg cttaccagaa aaggcttctt cagctccaag
 781 agcaaatgcg agccctttat aaagccatca gtgtccctcg agtccgcagg gccagctcca
 841 agggtggtgg aggctacact tgtcagagtg gctctggctg ggatgagttc accaaacatg
 901 tcaccagtga atgtttgggg tggatgaggc agcaaagggc ggagatggac atggtggcct
 961 ggggtgtgga cctggcctca gtggagcagc acattaacag ccaccggggc atccacaact
1021 ccatcggcga ctatcgctgg cagctggaca aaatcaaagc cgacctgcgc gagaaatctg
1081 cgatctacca gttggaggag gagtatgaaa acctgctgaa agcgtccttt gagaggatgg
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1141 atcacctgcg acagctgcag aacatcattc aggccacgtc cagggagatc atgtggatca

1201 atgactgcga ggaggaggag ctgctgtacg actggagcga caagaacacc aacatcgctc

1261 agaaacagga ggccttctcc atacgcatga gtcaactgga agttaaagaa aaagagctca

1321 ataagctgaa acaagaaagt gaccaacttg tcctcaatca gcatccagct tcagacaaaa

1381 ttgaggccta tatggacact ctgcagacgc agtggagttg gattcttcag atcaccaagt

1441 gcattgatgt tcatctgaaa gaaaatgctg cctactttca gtttttgaa gaggcgcagt

1501 ctactgaagc atacctgaag gggctccagg actccatcag gaagaagtac ccctgcgaca

1561 agaacatgcc cctgcagcac ctgctggaac agatcaagga gctggagaaa gaacgagaga

1621 aaatccttga atacaagcgt caggtgcaga acttggtaaa caagtctaag aagattgtac

1681 agctgaagcc tcgtaaccca gactacagaa gcaataaacc cattattctc agagctctct

1741 gtgactacaa acaagatcag aaaatcgtgc ataaggggga tgagtgtatc ctgaaggaca

1801 acaacgagcg cagcaagtgg tacgtgacgg gcccgggagg cgttgacatg cttgttccct

1861 ctgtggggct gatcatccct cctccgaacc cactggccgt ggacctctct tgcaagattg

1921 agcagtacta cgaagccatc ttggctctgt ggaaccagct ctacatcaac atgaagagcc

1981 tggtgtcctg gcactactgc atgattgaca tagagaagat cagggccatg acaatcgcca

2041 agctgaaaac aatgcggcag gaagattaca tgaagacgat agccgacctt gagttacatt

2101 accaagagtt catcagaaat agccaaggct cagagatgtt tggagatgat gacaagcgga

2161 aaatacagtc tcagttcacc gatgcccaga agcattacca gaccctggtc attcagctcc

2221 ctggctatcc ccagcaccag acagtgacca caactgaaat cactcatcat ggaacctgcc

2281 aagatgtcaa ccataataaa gtaattgaaa ccaacagaga aaatgacaag caagaaacat

2341 ggatgctgat ggagctgcag aagattcgca ggcagataga gcactgcgag ggcaggatga

2401 ctctcaaaaa cctccctcta gcagaccagg gatcttctca ccacatcaca gtgaaaatta

2461 acgagcttaa gagtgtgcag aatgattcac aagcaattgc tgaggttctc aaccagctta

2521 aagatatgct tgccaacttc agaggttctg aaaagtactg ctatttacag aatgaagtat

2581 ttggactatt tcagaaactg gaaaatatca atggtgttac agatggctac ttaaatagct

2641 tatgcacagt aagggcactg ctccaggcta ttctccaaac agaagacatg ttaaaggttt

2701 atgaagccag gctcactgag gaggaaactg tctgcctgga cctggataaa gtggaagctt

2761 accgctgtgg actgaagaaa ataaaaaatg acttgaactt gaagaagtcg ttgttggcca

2821 ctatgaagac agaactacag aaagcccagc agatccactc tcagacttca cagcagtatc

2881 cactttatga tctggacttg ggcaagttcg gtgaaaaagt cacacagctg acagaccgct

2941 ggcaaaggat agataaacag atcgacttta ggttatggga cctggagaaa caaatcaagc

3001 aattgaggaa ttatcgtgat aactatcagg ctttctgcaa gtggctctat gatgctaaac

3061 gccgccagga ttccttagaa tccatgaaat ttggagattc caacacagtc atgcggtttt

3121 tgaatgagca gaagaacttg cacagtgaaa tatctggcaa acgagacaaa tcagaggaag

3181 tacaaaaaat tgctgaactt tgcgccaatt caattaagga ttatgagctc cagctggcct

3241 catacacctc aggactggaa actctgctga acatacctat caagaggacc atgattcagt

3301 ccccttctgg ggtgattctg caagaggctg cagatgttca tgctcggtac attgaactac

3361 ttacaagatc tggagactat tacaggttct taagtgagat gctgaagagt ttggaagatc

3421 tgaagctgaa aaataccaag atcgaagttt tggaagagga gctcagactg gcccgagatg

3481 ccaactcgga aaactgtaat aagaacaaat tcctggatca gaacctgcag aaataccagg

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3541 cagagtgttc ccagttcaaa gcgaagcttg cgagcctgga ggagctgaag agacaggctg

3601 agctggatgg gaagtcggct aagcaaaatc tagacaagtg ctacggccaa ataaaagaac

3661 tcaatgagaa gatcacccga ctgacttatg agattgaaga tgaaaagaga agaagaaaat

3721 ctgtggaaga cagatttgac caacagaaga atgactatga ccaactgcag aaagcaaggc

3781 aatgtgaaaa ggagaacctt ggttggcaga aattagagtc tgagaaagcc atcaaggaga

3841 aggagtacga gattgaaagg ttgagggttc tactgcagga agaaggcacc cggaagagag

3901 aatatgaaaa tgagctggca aaggcatcta ataggattca ggaatcaaag aatcagtgta

3961 ctcaggtggt acaggaaaga gagagccttc tggtgaaaat caaagtcctg gagcaagaca

4021 aggcaaggct gcagaggctg gaggatgagc tgaatcgtgc aaaatcaact ctagaggcag

4081 aaaccagggt gaaacagcgc ctggagtgtg agaaacagca aattcagaat gacctgaatc

4141 agtggaagac tcaatattcc cgcaaggagg aggctattag gaagatagaa tcggaaagag

4201 aaaagagtga gagagagaag aacagtctta ggagtgagat cgaaagactc caagcagaga

4261 tcaagagaat tgaagagagg tgcaggcgta agctggagga ttctaccagg gagacacagt

4321 cacagttaga aacagaacgc tcccgatatc agagggagat tgataaactc agacagcgcc

4381 catatgggtc ccatcgagag acccagactg agtgtgagtg gaccgttgac acctccaagc

4441 tggtgtttga tgggctgagg aagaaggtga cagcaatgca gctctatgag tgtcagctga

4501 tcgacaaaac aaccttggac aaactattga aggggaagaa gtcagtggaa gaagttgctt

4561 ctgaaatcca gccattcctt cggggtgcag gatctatcgc tggagcatct gcttctccta

4621 aggaaaaata ctctttggta gaggccaaga gaaagaaatt aatcagccca gaatccacag

4681 tcatgcttct ggaggcccag gcagctacag gtggtataat tgatccccat cggaatgaga

4741 agctgactgt cgacagtgcc atagctcggg acctcattga cttcgatgac cgtcagcaga

4801 tatatgcagc agaaaaagct atcactggtt ttgatgatcc attttcaggc aagacagtat

4861 ctgtttcaga agccatcaag aaaaatttga ttgatagaga aaccggaatg cgcctgctgg

4921 aagcccagat tgcttcaggg ggtgtagtag accctgtgaa cagtgtcttt ttgccaaaag

4981 atgtcgcctt ggcccggggg ctgattgata gagatttgta tcgatccctg aatgatcccc

5041 gagatagtca gaaaaacttt gtggatccag tcaccaaaaa gaaggtcagt tacgtgcagc

5101 tgaaggaacg gtgcagaatc gaaccacata ctggtctgct cttgctttca gtacagaaga

5161 gaagcatgtc cttccaagga atcagacaac ctgtgaccgt cactgagcta gtagattctg

5221 gtatattgag accgtccact gtcaatgaac tggaatctgg tcagatttct tatgacgagg

5281 ttggtgagag aattaaggac ttcctccagg gttcaagctg catagcaggc atatacaatg

5341 agaccacaaa acagaagctt ggcatttatg aggccatgaa aattggctta gtccgacctg

5401 gtactgctct ggagttgctg gaagcccaag cagctactgg ctttatagtg gatcctgtta

5461 gcaacttgag gttaccagtg gaggaagcct acaagagagg tctggtgggc attgagttca

5521 aagagaagct cctgtctgca gaacgagctg tcactgggta taatgatcct gaaacaggaa

5581 acatcatctc tttgttccaa gccatgaata aggaactcat cgaaaagggc cacggtattc

5641 gcttattaga agcacagatc gcaaccgggg ggatcattga cccaaaggag agccatcgtt

5701 taccagttga catagcatat aagaggggct atttcaatga ggaactcagt gagattctct

5761 cagatccaag tgatgatacc aaaggatttt ttgaccccaa cactgaagaa aatcttacct

5821 atctgcaact aaaagaaaga tgcattaagg atgaggaaac agggctctgt cttctgcctc

5881 tgaaagaaaa gaagaaacag gtgcagacat cacaaaagaa taccctcagg aagcgtagag

5941 tggtcatagt tgacccagaa accaataaag aaatgtctgt tcaggaggcc tacaagaagg

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6001 gcctaattga ttatgaaacc ttcaaagaac tgtgtgagca ggaatgtgaa tgggaagaaa

6061 taaccatcac gggatcagat ggctccacca gggtggtcct ggtagataga aagacaggca

6121 gtcagtatga tattcaagat gctattgaca agggccttgt tgacaggaag ttctttgatc

6181 agtaccgatc cggcagcctc agcctcactc aatttgctga catgatctcc ttgaaaaatg

6241 gtgtcggcac cagcagcagc atgggcagtg gtgtcagcga tgatgttttt agcagctccc

6301 gacatgaatc agtaagtaag atttccacca tatccagcgt caggaattta accataagga

6361 gcagctcttt ttcagacacc ctggaagaat cgagccccat tgcagccatc tttgacacag

6421 aaaacctgga gaaaatctcc attacagaag gtatagagcg gggcatcgtt gacagcatca

6481 cgggtcagag gcttctggag gctcaggcct gcacaggtgg catcatccac ccaaccacgg

6541 gccagaagct gtcacttcag gacgcagtct cccagggtgt gattgaccaa gacatggcca

6601 ccaggctgaa gcctgctcag aaagccttca taggcttcga gggtgtgaag ggaaagaaga

6661 agatgtcagc agcagaggca gtgaaagaaa aatggctccc gtatgaggct ggccagcgct

6721 tcctggagtt ccagtacctc acgggaggtc ttgttgaccc ggaagtgcat gggaggataa

6781 gcaccgaaga agccatccgg aaggggttca tagatggccg cgccgcacag aggctgcaag

6841 acaccagcag ctatgccaaa atcctgacct gccccaaaac caaattaaaa atatcctata

6901 aggatgccat aaatcgctcc atggtagaag atatcactgg gctgcgcctt ctggaagccg

6961 cctccgtgtc gtccaagggc ttacccagcc cttacaacat gtcttcggct ccggggtccc

7021 gctccggctc ccgctcggga tctcgctccg gatctcgctc cgggtcccgc agtgggtccc

7081 ggagaggaag ctttgacgcc acagggaatt cttcctactc ttattcctac tcatttagca

7141 gtagttctat tgggcactag tagtcagttg ggagtggttg ctataccttg acttcattta

7201 tatgaatttc cactttatta aataatagaa aagaaaatcc cggtgcttgc agtagagtga

7261 taggacattc tatgcttaca gaaaatatag ccatgattga aatcaaatag taaaggctgt

7321 tctggctttt tatcttctta gctcatctta aataagcagt acacttggat gcagtgcgtc

7381 tgaagtgcta atcagttgta acaatagcac aaatcgaact taggatttgt ttcttctctt

7441 ctgtgtttcg atttttgatc aattctttaa ttttggaagc ctataataca gttttctatt

7501 cttggagata aaaattaaat ggatcactga tattttagtc attctgcttc tcatctaaat

7561 atttccatat tctgtattag gagaaaatta ccctcccagc accagccccc ctctcaaacc

7621 cccaacccaa aaccaagcat tttggaatga gtctccttta gtttcagagt gtggattgta

7681 taacccatat actcttcgat gtacttgttt ggtttggtat taatttgact gtgcatgaca

7741 gcggcaatct tttctttggt caaagttttc tgtttatttt gcttgtcata ttcgatgtac

7801 tttaaggtgt ctttatgaag tttgctattc tggcaataaa cttttagact tttgaagtgt

7861 ttgtgtttta atttaatatg tttataagca tgtataaaca tttagcatat ttttatcata

7921 ggtctaaaaa tatttgttta ctaaatacct gtgaagaaat accattaaaa aactatttgg

7981 ttctgaattc ttactagaaa aaaaa

In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001008844.1, transcript variant 2):

(SEQ ID NO: 20)

```
   1 mscnggshpr intlgrmira esgpdlryev tsggggtsrm yysrrgvitd qnsdgycqtg
  61 tmsrhqnqnt iqellqncsd clmraelivq pelkygdgiq ltrsreldec faqandqmei
 121 ldslirемrq mgqpcdayqk rllqlqeqmr alykaisvpr vrrasskggg gytcqsgsgw
 181 deftkhvtse clgwmrqqra emdmvawgvd lasveqhins hrgihnsigd yrwqldkika
 241 dlreksaiyq leeeyenllk asfermdhlr qlqniiqats reimwindce eeellydwsd
 301 kntniaqkqe afsirmsqle vkekelnklk qesdqlvinq hpasdkieay mdtlqtqwsw
 361 ilqitkcidv hlkenaayfq ffeeaqstea ylkglqdsir kkypcdknmp lqhlleqike
 421 lekerekile ykrqvqnlvn kskkivqlkp rnpdyrsnkp iilralcdyk qdqkivhkgd
 481 ecilkdnner skwyvtgpgg vdmlvpsvgl iipppnplav dlsckieqyy eailalwnql
 541 yinmkslvsw hycmidieki ramtiaklkt mrqedymkti adlelhyqef irnsqgsemf
 601 gdddkrkiqs qftdaqkhyq tiviqlpgyp qhqtvtttei thhgtcqdvn hnkvietnre
 661 ndkqetwmlm elqkirrqie hcegrmtlkn lpladqgssh hitvkinelk svqndsgaia
 721 evinqlkdml anfrgsekyc ylqnevfglf qkleningvt dgylnslctv rallqailqt
 781 edmlkvyear lteeetvcld ldkveayrcg lkkikndlnl kksllatmkt elqkaqqihs
 841 qtsqqyplyd ldlgkfgekv tqltdrwqri dkqidfrlwd lekqikqlrn yrdnyqafck
 901 wlydakrrqd slesmkfgds ntvmrflneq knlhseisgk rdkseevqki aelcansikd
 961 yelqlasyts gletllnipi krtmiqspsg vilqeaadvh aryielltrs gdyyrflsem
1021 lksledlklk ntkievleee lrlardanse ncnknkfldq nlqkyqaecs qfkaklasle
1081 elkrqaeldg ksakqnldkc ygqikelnek itrltyeied ekrrrksved rfdqqkndyd
1141 qlqkarqcek enlgwqkles ekaikekeye ierlrvllqe egtrkreyen elakasnriq
1201 esknqctqvv qeresllvki kvleqdkarl qrledelnra kstleaetrv kqrlecekqq
1261 iqndlnqwkt qysrkeeair kieserekse reknslrsei erlqaeikri eercrrkled
1321 stretqsqle tersrygrei dklrqrpygs hretqtecew tvdtsklvfd glrkkvtamq
1381 lyecqlidkt tldkllkgkk sveevaseiq pflrgagsia gasaspkeky slveakrkkl
1441 ispestvmll eaqaatggii dphrnekltv dsaiardlid fddrqqiyaa ekaitgfddp
1501 fsgktvsvse aikknlidre tgmrlleaqi asggvvdpvn svflpkdval arglidrdly
1561 rslndprdsq knfvdpvtkk kvsyvqlker criephtgll llsvqkrsms fqgirqpvtv
1621 telvdsgilr pstvnelesg qisydevger ikdflqgssc iagiynettk qklgiyeamk
1681 iglvrpgtal elleaqaatg fivdpvsnlr lpveeaykrg lvgiefkekl lsaeravtgy
1741 ndpetgniis lfqamnkeli ekghgirlle aqiatggiid pkeshrlpvd iaykrgyfne
1801 elseilsdps ddtkgffdpn teenitylql kercikdeet glcllplkek kkqvqtsqkn
1861 tlrkrrvviv dpetnkemsv qeaykkglid yetfkelceq eceweeitit gsdgstrvvl
1921 vdrktgsqyd iqdaidkglv drkffdqyrs gslsltqfad mislkngvgt sssmgsgvsd
1981 dvfsssrhes vskistissv rnltirsssf sdtleesspi aaifdtenle kisitegier
2041 givdsitgqr lleaqactgg iihpttgqkl slqdavsqgv idqdmatrlk paqkafigfe
2101 gvkgkkkmsa aeavkekwlp yeagqrflef qyltgglvdp evhgristee airkgfidgr
2161 aaqrlqdtss yakiltcpkt klkisykdai nrsmveditg lrlleaasvs skglpspynm
2221 ssapgsrsgs rsgsrsgsrs gsrsgsrrgs fdatgnssys ysysfsssi gh
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In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001319034.1, transcript variant 3):

(SEQ ID NO: 46)

```
   1 aagaaaccgg ccaggtgtgg cctaggcgcc cagtgccagc ggggaggaga ctcgctccgc
  61 cgccgaccaa caccaacacc cagctccgac gcagctcctc tgcgcccttg ccgccctccg
 121 agccacagct ttcctcccgc tcctgccccc ggcccgtcgc cgtctccgcg ctcgcagcgg
 181 cctcgggagg gcccaggtag cgagcagcga cctcgcgagc cttccgcact cccgcccggt
 241 tccccggccg tccgcctatc cttggccccc tccgctttct ccgcgccggc ccgcctcgct
 301 tatgcctcgg cgctgagccg ctctcccgat tgcccgccga catgagctgc aacggaggct
 361 cccacccgcg gatcaacact ctgggccgca tgatccgcgc cgagtctggc ccggacctgc
 421 gctacgaggt gaccagcggc ggcgggggca ccagcaggat gtactattct cggcgcggcg
 481 tgatcaccga ccagaactcg gacggctact gtcaaaccgg cacgatgtcc aggcaccaga
 541 accagaacac catccaggag ctgctgcaga actgctccga ctgcttgatg cgagcagagc
 601 tcatcgtgca gcctgaattg aagtatggag atggaataca actgactcgg agtcgagaat
 661 tggatgagtg ttttgcccag gccaatgacc aaatggaaat cctcgacagc ttgatcagag
 721 agatgcggca gatgggccag ccctgtgatg cttaccagaa aaggcttctt cagctccaag
 781 agcaaatgcg agccctttat aaagccatca gtgtccctcg agtccgcagg gccagctcca
 841 agggtggtgg aggctacact tgtcagagtg gctctggctg ggatgagttc accaaacatg
 901 tcaccagtga atgtttgggg tggatgaggc agcaaagggc ggagatggac atggtggcct
 961 ggggtgtgga cctggcctca gtggagcagc acattaacag ccaccggggc atccacaact
1021 ccatcggcga ctatcgctgg cagctggaca aaatcaaagc cgacctgcgc gagaaatctg
1081 cgatctacca gttggaggag gagtatgaaa acctgctgaa agcgtccttt gagaggatgg
1141 atcacctgcg acagctgcag aacatcattc aggccacgtc cagggagatc atgtggatca
1201 atgactgcga ggaggaggag ctgctgtacg actggagcga caagaacacc aacatcgctc
1261 agaaacagga ggccttctcc atacgcatga gtcaactgga agttaaagaa aaagagctca
1321 ataagctgaa acaagaaagt gaccaacttg tcctcaatca gcatccagct tcagacaaaa
1381 ttgaggccta tatggacact ctgcagacgc agtggagttg gattcttcag atcaccaagt
1441 gcattgatgt tcatctgaaa gaaaatgctg cctactttca gtttttgaa gaggcgcagt
1501 ctactgaagc atacctgaag ggctccagg actccatcag gaagaagtac ccctgcgaca
1561 agaacatgcc cctgcagcac ctgctggaac agatcaagga gctggagaaa gaacgagaga
1621 aaatccttga atacaagcgt caggtgcaga acttggtaaa caagtctaag aagattgtac
1681 agctgaagcc tcgtaaccca gactacagaa gcaataaacc cattattctc agagctctct
1741 gtgactacaa acaagatcag aaaatcgtgc ataaggggga tgagtgtatc ctgaaggaca
1801 acaacgagcg cagcaagtgg tacgtgacgg gcccgggagg cgttgacatg cttgttccct
1861 ctgtggggct gatcatccct cctccgaacc cactggccgt ggacctctct tgcaagattg
1921 agcagtacta cgaagccatc ttggctctgt ggaaccagct ctacatcaac atgaagagcc
1981 tggtgtcctg gcactactgc atgattgaca tagagaagat cagggccatg acaatcgcca
2041 agctgaaaac aatgcggcag gaagattaca tgaagacgat agccgacctt gagttacatt
2101 accaagagtt catcagaaat agccaaggct cagagatgtt tggagatgat gacaagcgga
2161 aaatacagtc tcagttcacc gatgcccaga agcattacca gaccctggtc attcagctcc
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2221 ctggctatcc ccagcaccag acagtgacca caactgaaat cactcatcat ggaacctgcc

2281 aagatgtcaa ccataataaa gtaattgaaa ccaacagaga aaatgacaag caagaaacat

2341 ggatgctgat ggagctgcag aagattcgca ggcagataga gcactgcgag ggcaggatga

2401 ctctcaaaaa cctccctcta gcagaccagg gatcttctca ccacatcaca gtgaaaatta

2461 acgagcttaa gagtgtgcag aatgattcac aagcaattgc tgaggttctc aaccagctta

2521 aagatatgct tgccaacttc agaggttctg aaaagtactg ctatttacag aatgaagtat

2581 ttggactatt tcagaaactg gaaaatatca atggtgttac agatggctac ttaaatagct

2641 tatgcacagt aagggcactg ctccaggcta ttctccaaac agaagacatg ttaaaggttt

2701 atgaagccag gctcactgag gaggaaactg tctgcctgga cctggataaa gtggaagctt

2761 accgctgtgg actgaagaaa ataaaaaatg acttgaactt gaagaagtcg ttgttggcca

2821 ctatgaagac agaactacag aaagcccagc agatccactc tcagacttca cagcagtatc

2881 cactttatga tctggacttg ggcaagttcg gtgaaaaagt cacacagctg acagaccgct

2941 ggcaaaggat agataaacag atcgacttta ggttatggga cctggagaaa caaatcaagc

3001 aattgaggaa ttatcgtgat aactatcagg ctttctgcaa gtggctctat gatgctaaac

3061 gccgccagga ttccttagaa tccatgaaat ttggagattc caacacagtc atgcggtttt

3121 tgaatgagca gaagaacttg cacagtgaaa tatctggcaa acgagacaaa tcagaggaag

3181 tacaaaaaat tgctgaactt tgcgccaatt caattaagga ttatgagctc cagctggcct

3241 catacacctc aggactggaa actctgctga acatacctat caagaggacc atgattcagt

3301 ccccttctgg ggtgattctg caagaggctg cagatgttca tgctcggtac attgaactac

3361 ttacaagatc tggagactat tacaggttct taagtgagat gctgaagagt ttggaagatc

3421 tgaagctgaa aaataccaag atcgaagttt tggaagagga gctcagactg gcccgagatg

3481 ccaactcgga aaactgtaat aagaacaaat tcctggatca gaacctgcag aaataccagg

3541 cagagtgttc ccagttcaaa gcgaagcttg cgagcctgga ggagctgaag agacaggctg

3601 agctggatgg gaagtcggct aagcaaaatc tagacaagtg ctacggccaa ataaaagaac

3661 tcaatgagaa gatcacccga ctgacttatg agattgaaga tgaaaagaga agaagaaaat

3721 ctgtggaaga cagatttgac caacagaaga atgactatga ccaactgcag aaagcaaggc

3781 aatgtgaaaa ggagaacctt ggttggcaga aattagagtc tgagaaagcc atcaaggaga

3841 aggagtacga gattgaaagg ttgagggttc tactgcagga agaaggcacc cggaagagag

3901 aatatgaaaa tgagctggca aaggtaagaa accactataa tgaggagatg agtaatttaa

3961 ggaacaagta tgaaacagag attaacatta cgaagaccac catcaaggag atatccatgc

4021 aaaaagagga tgattccaaa aatcttagaa accagcttga tagactttca agggaaaatc

4081 gagatctgaa ggatgaaatt gtcaggctca atgacagcat cttgcaggcc actgagcagc

4141 gaaggcgagc tgaagaaaac gcccttcagc aaaaggcctg tggctctgag ataatgcaga

4201 agaagcagca tctggagata gaactgaagc aggtcatgca gcagcgctct gaggacaatg

4261 cccggcacaa gcagtccctg gaggaggctg ccaagaccat tcaggacaaa aataaggaga

4321 tcgagagact caaagctgag tttcaggagg aggccaagcg ccgctgggaa tatgaaaatg

4381 aactgagtaa ggcatctaat aggattcagg aatcaaagaa tcagtgtact caggtggtac

4441 aggaaagaga gagccttctg gtgaaaatca aagtcctgga gcaagacaag gcaaggctgc

4501 agaggctgga ggatgagctg aatcgtgcaa aatcaactct agaggcagaa accagggtga

4561 aacagcgcct ggagtgtgag aaacagcaaa ttcagaatga cctgaatcag tggaagactc

4621 aatattcccg caaggaggag gctattagga agatagaatc ggaaagagaa aagagtgaga

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4681 gagagaagaa cagtcttagg agtgagatcg aaagactcca agcagagatc aagagaattg
4741 aagagaggtg caggcgtaag ctggaggatt ctaccaggga gacacagtca cagttagaaa
4801 cagaacgctc ccgatatcag agggagattg ataaactcag acagcgccca tatgggtccc
4861 atcgagagac ccagactgag tgtgagtgga ccgttgacac ctccaagctg gtgtttgatg
4921 ggctgaggaa gaaggtgaca gcaatgcagc tctatgagtg tcagctgatc gacaaaacaa
4981 ccttggacaa actattgaag gggaagaagt cagtggaaga agttgcttct gaaatccagc
5041 cattccttcg gggtgcagga tctatcgctg gagcatctgc ttctcctaag gaaaaatact
5101 ctttggtaga ggccaagaga aagaaattaa tcagcccaga atccacagtc atgcttctgg
5161 aggcccaggc agctacaggt ggtataattg atccccatcg gaatgagaag ctgactgtcg
5221 acagtgccat agctcgggac ctcattgact tcgatgaccg tcagcagata tatgcagcag
5281 aaaaagctat cactggtttt gatgatccat tttcaggcaa gacagtatct gtttcagaag
5341 ccatcaagaa aaatttgatt gatagagaaa ccggaatgcg cctgctggaa gcccagattg
5401 cttcaggggg tgtagtagac cctgtgaaca gtgtcttttt gccaaaagat gtcgccttgg
5461 cccgggggct gattgataga gatttgtatc gatccctgaa tgatccccga gatagtcaga
5521 aaaactttgt ggatccagtc accaaaaaga aggtcagtta cgtgcagctg aaggaacggt
5581 gcagaatcga accacatact ggtctgctct tgctttcagt acagaagaga agcatgtcct
5641 tccaaggaat cagacaacct gtgaccgtca ctgagctagt agattctggt atattgagac
5701 cgtccactgt caatgaactg gaatctggtc agatttctta tgacgaggtt ggtgagagaa
5761 ttaaggactt cctccagggt tcaagctgca tagcaggcat atacaatgag accacaaaac
5821 agaagcttgg catttatgag gccatgaaaa ttggcttagt ccgacctggt actgctctgg
5881 agttgctgga agcccaagca gctactggct ttatagtgga tcctgttagc aacttgaggt
5941 taccagtgga ggaagcctac aagagaggtc tggtgggcat tgagttcaaa gagaagctcc
6001 tgtctgcaga acgagctgtc actgggtata atgatcctga aacaggaaac atcatctctt
6061 tgttccaagc catgaataag gaactcatcg aaaagggcca cggtattcgc ttattagaag
6121 cacagatcgc aaccgggggg atcattgacc caaaggagag ccatcgttta ccagttgaca
6181 tagcatataa gaggggctat ttcaatgagg aactcagtga gattctctca gatccaagtg
6241 atgataccaa aggatttttt gaccccaaca ctgaagaaaa tcttacctat ctgcaactaa
6301 aagaaagatg cattaaggat gaggaaacag ggctctgtct tctgcctctg aaagaaaaga
6361 agaaacaggt gcagacatca caaaagaata ccctcaggaa gcgtagagtg gtcatagttg
6421 acccagaaac caataaagaa atgtctgttc aggaggccta caagaagggc ctaattgatt
6481 atgaaacctt caaagaactg tgtgagcagg aatgtgaatg ggaagaaata accatcacgg
6541 gatcagatgg ctccaccagg gtggtcctgg tagatagaaa gacaggcagt cagtatgata
6601 ttcaagatgc tattgacaag ggccttgttg acaggaagtt ctttgatcag taccgatccg
6661 gcagcctcag cctcactcaa tttgctgaca tgatctcctt gaaaaatggt gtcggcacca
6721 gcagcagcat gggcagtggt gtcagcgatg atgtttttag cagctcccga catgaatcag
6781 taagtaagat ttccaccata tccagcgtca ggaatttaac cataaggagc agctcttttt
6841 cagacaccct ggaagaatcg agccccattg cagccatctt tgacacagaa aacctggaga
6901 aaatctccat tacagaaggt atagagcggg gcatcgttga cagcatcacg ggtcagaggc
6961 ttctggaggc tcaggcctgc acaggtggca tcatccaccc aaccacgggc cagaagctgt
7021 cacttcagga cgcagtctcc cagggtgtga ttgaccaaga catggccacc aggctgaagc
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7081 ctgctcagaa agccttcata ggcttcgagg gtgtgaaggg aaagaagaag atgtcagcag
7141 cagaggcagt gaaagaaaaa tggctcccgt atgaggctgg ccagcgcttc ctggagttcc
7201 agtacctcac gggaggtctt gttgacccgg aagtgcatgg gaggataagc accgaagaag
7261 ccatccggaa ggggttcata gatggccgcg ccgcacagag gctgcaagac accagcagct
7321 atgccaaaat cctgacctgc cccaaaacca aattaaaaat atcctataag gatgccataa
7381 atcgctccat ggtagaagat atcactgggc tgcgccttct ggaagccgcc tccgtgtcgt
7441 ccaagggctt acccagccct tacaacatgt cttcggctcc ggggtcccgc tccggctccc
7501 gctcgggatc tcgctccgga tctcgctccg ggtcccgcag tgggtcccgg agaggaagct
7561 ttgacgccac agggaattct tcctactctt attcctactc atttagcagt agttctattg
7621 ggcactagta gtcagttggg agtggttgct ataccttgac ttcatttata tgaatttcca
7681 ctttattaaa taatagaaaa gaaaatcccg gtgcttgcag tagagtgata ggacattcta
7741 tgcttacaga aaatatagcc atgattgaaa tcaaatagta aaggctgttc tggcttttta
7801 tcttcttagc tcatcttaaa taagcagtac acttggatgc agtgcgtctg aagtgctaat
7861 cagttgtaac aatagcacaa atcgaactta ggatttgttt cttctcttct gtgtttcgat
7921 ttttgatcaa ttctttaatt ttggaagcct ataatacagt tttctattct tggagataaa
7981 aattaaatgg atcactgata ttttagtcat tctgcttctc atctaaatat ttccatattc
8041 tgtattagga gaaaattacc ctcccagcac cagcccccct ctcaaacccc caacccaaaa
8101 ccaagcattt tggaatgagt ctcctttagt ttcagagtgt ggattgtata acccatatac
8161 tcttcgatgt acttgtttgg tttggtatta atttgactgt gcatgacagc ggcaatcttt
8221 tctttggtca aagtttctg tttattttgc ttgtcatatt cgatgtactt taaggtgtct
8281 ttatgaagtt tgctattctg gcaataaact tttagacttt tgaagtgttt gtgttttaat
8341 ttaatatgtt tataagcatg tataaacatt tagcatattt ttatcatagg tctaaaaata
8401 tttgtttact aaatacctgt gaagaaatac cattaaaaaa ctatttggtt ctgaattctt
8461 actagaaaaa aaa
```

In some embodiments of the methods of the disclosure, the wild type human DSP gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001305963.1, transcript variant 3):

```
                                                        (SEQ ID NO: 47)
  1 mscnggshpr intlgrmira esgpdlryev tsggggtsrm yysrrgvitd qnsdgycqtg
 61 tmsrhqnqnt iqellqncsd clmraelivq pelkygdgiq ltrsreldec faqandqmei
121 ldslirемrq mgqpcdayqk rllqlqeqmr alykaisvpr vrrasskggg gytcqsgsgw
181 deftkhvtse clgwmrqqra emdmvawgvd lasveqhins hrgihnsigd yrwqldkika
241 dlreksaiyq leeeyenllk asfermdhlr qlqniiqats reimwindce eeellydwsd
301 kntniaqkqe afsirmsqle vkekelnklk qesdqlvlnq hpasdkieay mdtlqtqwsw
361 ilqitkcidv hlkenaayfq ffeeaqstea ylkglqdsir kkypcdknmp lqhlleqike
421 lekerekile ykrqvqnlvn kskkivqlkp rnpdyrsnkp iilralcdyk qdqkivhkgd
481 ecilkdnner skwyvtgpgg vdmlvpsvgl iipppnplav dlsckieqyy eailalwnql
541 yinmkslvsw hycmidieki ramtiaklkt mrqedymkti adlelhyqef irnsqgsemf
601 gdddkrkiqs qftdaqkhyq tlviqlpgyp qhqtvtttei thhgtcqdvn hnkvietnre
661 ndkqetwmlm elqkirrqie hcegrmtlkn lpladqgssh hitvkinelk svqndsqaia
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 721 evlnqlkdml anfrgsekyc ylqnevfglf qkleningvt dgylnslctv rallqailqt

 781 edmlkvyear lteeetvcld ldkveayrcg lkkikndlnl kksllatmkt elqkaqqihs

 841 qtsqqyplyd ldlgkfgekv tqltdrwqri dkqidfrlwd lekqikqlrn yrdnyqafck

 901 wlydakrrqd slesmkfgds ntvmrflneq knlhseisgk rdkseevqki aelcansikd

 961 yelqlasyts gletllnipi krtmiqspsg vilqeaadvh aryielltrs gdyyrflsem

1021 lksledlklk ntkievleee lrlardanse ncnknkfldq nlqkyqaecs qfkaklasle

1081 elkrqaeldg ksakqnldkc ygqikelnek itrltyeied ekrrrksved rfdqqkndyd

1141 qlqkarqcek enlgwqkles ekaikekeye ierlrvllqe egtrkreyen elakvrnhyn

1201 eemsnlrnky eteinitktt ikeismqked dsknlrnqld rlsrenrdlk deivrlndsi

1261 lqateqrrra eenalqqkac gseimqkkqh leielkqvmq qrsednarhk qsleeaakti

1321 qdknkeierl kaefqeeakr rweyenelsk asnriqeskn qctqvvqere sllvkikvle

1381 qdkarlqrle delnrakstl eaetrvkqrl ecekqqiqnd lnqwktqysr keeairkies

1441 erekserekn slrseierlq aeikrieerc rrkledstre tqsqleters ryqreidklr

1501 qrpygshret qtecewtvdt sklvfdglrk kvtamqlyec qlidkttldk llkgkksvee

1561 vaseiqpflr gagsiagasa spkekyslve akrkklispe stvmlleaqa atggiidphr

1621 nekltvdsai ardlidfddr qqiyaaekai tgfddpfsgk tvsyseaikk nlidretgmr

1681 lleaqiasgg vvdpvnsvfl pkdvalargl idrdlyrsln dprdsqknfv dpvtkkkvsy

1741 vqlkercrie phtglllllsv qkrsmsfqgi rqpvtvtelv dsgilrpstv nelesgqisy

1801 devgerikdf lqgssciagi ynettkqklg iyeamkiglv rpgtalelle aqaatgfivd

1861 pvsnlrlpve eaykrglvgi efkekllsae ravtgyndpe tgniislfqa mnkeliekgh

1921 girlleaqia tggiidpkes hrlpvdiayk rgyfneelse ilsdpsddtk gffdpnteen

1981 ltylqlkerc ikdeetglcl lplkekkkqv qtsqkntlrk rrvvivdpet nkemsvqeay

2041 kkglidyetf kelceqecew eeititgsdg strvvlvdrk tgsqydiqda idkglvdrkf

2101 fdqyrsgsls ltqfadmisl kngvgtsssm gsgvsddvfs ssrhesvski stissvrnlt

2161 irsssfsdtl eesspiaaif dtenlekisi tegiergivd sitgqrllea qactggiihp

2221 ttgqklslqd aysqgvidqd matrlkpaqk afigfegvkg kkkmsaaeav kekwlpyeag

2281 qrflefqylt gglvdpevhg risteeairk gfidgraaqr lqdtssyaki ltcpktklki

2341 sykdainrsm veditglrll eaasysskgl pspynmssap gsrsgsrsgs rsgsrsgsrs

2401 gsrrgsfdat gnssysysys fssssigh
```

In some embodiments of the methods of the disclosure, the wild type human AZGP1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001185.3):

```
                                                  (SEQ ID NO: 21)
   1 ccattggcct gtagattcac ctcccctggg cagggcccca ggacccagga taatatctgt

  61 gcctcctgcc cagaaccctc caagcagaca caatggtaag aatggtgcct gtcctgctgt

 121 ctctgctgct gcttctgggt cctgctgtcc cccaggagaa ccaagatggt cgttactctc

 181 tgacctatat ctacactggg ctgtccaagc atgttgaaga cgtccccgcg tttcaggccc

 241 ttggctcact caatgacctc cagttcttta gatacaacag taaagacagg aagtctcagc

 301 ccatgggact ctggagacag gtggaaggaa tggaggattg gaagcaggac agccaacttc

 361 agaaggccag ggaggacatc tttatggaga ccctgaaaga catcgtggag tattacaacg

 421 acagtaacgg gtctcacgta ttgcagggaa ggtttggttg tgagatcgag aataacagaa

 481 gcagcggagc attctggaaa tattactatg atggaaagga ctacattgaa ttcaacaaag

 541 aaatcccagc ctgggtcccc ttcgacccag cagcccagat aaccaagcag aagtgggagg

 601 cagaaccagt ctacgtgcag cgggccaagg cttacctgga ggaggagtgc cctgcgactc

 661 tgcggaaata cctgaaatac agcaaaaata tcctggaccg gcaagatcct ccctctgtgg

 721 tggtcaccag ccaccaggcc ccaggagaaa agaagaaact gaagtgcctg gcctacgact

 781 tctacccagg gaaaattgat gtgcactgga ctcgggccgg cgaggtgcag gagcctgagt

 841 tacggggaga tgttcttcac aatggaaatg gcacttacca gtcctgggtg gtggtggcag

 901 tgcccccgca ggacacagcc ccctactcct gccacgtgca gcacagcagc ctggcccagc

 961 ccctcgtggt gccctgggag gccagctagg aagcaagggt tggaggcaat gtgggatctc

1021 agacccagta gctgcccttc ctgcctgatg tgggagctga accacagaaa tcacagtcaa

1081 tggatccaca aggcctgagg agcagtgtgg ggggacagac aggaggtgga tttggagacc

1141 gaagactggg atgcctgtct tgagtagact tggacccaaa aaatcatctc accttgagcc

1201 cacccccacc ccattgtcta atctgtagaa gctaataaat aatcatccct ccttgcctag

1261 cataaaaaaa aaaaaaaa
```

In some embodiments of the methods of the disclosure, the wild type human AZGP1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NP_001176.1):

```
                                                  (SEQ ID NO: 22)
  1  mvrmvpvlls lllllgpavp qenqdgrysl tyiytglskh vedvpafqal gslndlqffr

 61  ynskdrksqp mglwrqvegm edwkqdsqlq karedifmet lkdiveyynd sngshvlqgr

121  fgceiennrs sgafwkyyyd gkdyiefnke ipawvpfdpa aqitkqkwea epvyvqraka

181  yleeecpatl rkylkyskni ldrqdppsvv vtshqapgek kklkclaydf ypgkidvhwt

241  ragevqepel rgdvlhngng tyqswvvvav ppqdtapysc hvqhsslaqp lvvpweas
```

In some embodiments of the methods of the disclosure, the wild type human OBFC1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_024928):

(SEQ ID NO: 23)

```
   1 aaatgcgctg gcggggagac cggggttggt ccctggcggg gcagggggcg ggctcaggcc
  61 ggaactccag agacgacctc agccaactgc tcctgcgccg ggcggggtcg tcgccgccag
 121 cggctccgag cgccggaagg gccaggtctc agggctcctg gagctgcagg cggcgggagg
 181 ggctacaaat gcttgactca gtgatgcaga acctttcaga gttagctgga agccacagcc
 241 ctgcctcttg atgcagcctg gatccagccg gtgtgaagag gagacccctt ccctcttgtg
 301 gggtttggat cctgtgtttc tagcctttgc aaaactctac atcagggata tcctggacat
 361 gaaggagtcc cgccaggtgc caggtgtatt tttgtacaat ggacatccaa taaaacaggt
 421 agatgtcttg ggaactgtca ttggagtgag agaaagagat gctttctaca gttatggagt
 481 ggatgacagc actggagtta taaactgcat ctgctggaaa aagttgaata ctgagtctgt
 541 atcagctgct ccaagtgcag caagagagct cagcttaacc tcacaactta agaagctaca
 601 agagaccatt gagcagaaaa caaagataga gatcggggac acgatccgag tcagaggcag
 661 tatccgcaca tacagagaag agcgagagat tcatgccacc acttactata aagtggacga
 721 cccagtgtgg aacattcaaa ttgcaaggat gcttgagctg cccactatct acaggaaagt
 781 ttatgaccag ccttttcaca gctcagccct agagaaagaa gaggcactaa gcaatccagg
 841 cgccctggac ctccccagtc tcacgagttt gctgagtgaa aaagccaaag aattcctcat
 901 ggagaacaga gtgcagagct tttaccagca ggagctggaa atggtggagt ctttgctgtc
 961 ccttgccaat cagcctgtga ttcacagtgc ctcctccgac caagtgaatt ttaagaagga
1021 caccacttcc aaggcaattc atagtatatt taagaatgct atacaactgc tgcaggaaaa
1081 aggacttgtt ttccagaaag atgatggttt tgataaccta tactatgtaa ccagagaaga
1141 caaagacctg cacagaaaga tccaccggat cattcagcag gactgccaga aaccaaatca
1201 catggagaag ggctgtcact tcctgcacat cttggcctgt gctcgcctga gcatccgccc
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2101 ccaattgcct ggcaggtggt agatgttatt ttcagggttg ccgctgagtg tgcaggatgt
2161 gctgacacca tccagacaaa gactcggtat gtgcccagac aggtgatgga gtcatgcttt
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2221 tgctcagaat gacaaggtaa aggaaaaaca tctgaggtat gttgtaggcc tgttctgaca

2281 gcaaaatgac aaatccagcc agcaaaaata aagtgtggag aaagatttgg agttaattac

2341 agtcatttca cagaaggcac tgccttcgtc tgctgcattt gctcttgatg tgataagctc

2401 ttcgtggctc agctggagat cctttaggcc tggagagttg ctcctctctc cgtggaaaca

2461 ggacagtctt tatacgcaga agtccgctgc agctcgatac gtcaggctga gagctagaac

2521 cagtagattg cctcctgtca tagacttttg taatgatgca aacctttgct gatttctaac

2581 agtgattatg tagtggctgc cctgcatctt ctctgtgtac agaagggtcc ctagcataga

2641 gtctgcctgg aatgatgtcc tgggcagttc ttccttgagg tcagcagctg ttccacgttg

2701 aatgcatctg attagtgggg ctgcccagga aggagttcag aatcagaagg taaaaagggc

2761 ataccсttgc ctatagcaac tctgctctta ggggtttatc tcaaggagat ggctacacaa

2821 gtgtgaaagg atggttgcac aaggtgttca ttgctgtata atctagaatt ctatattggg

2881 gaaaatacct atagggaaaa agttaattac ggttcttggg cacaatgaaa tactatgcag

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3121 tataaatgta gcacatgtac atatatatat atatacacac acacagtcga ccactcccтt

3181 ctcctggaag tactttccgc gtttggcttt caggacacca agctctctgg ttgctccttc

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3541 caaatgccta ggcagagagg ggagggtccc aaatgaaacc ccacgttcaa gcaaagatca

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3781 taaaaagacc cagactcccc catattaagg ggactttcct gcctttgggt agggggacca

3841 cccccacgtc tcctctctgt tgaaaactgt ttcatcactc aataaaactc ccagctttgc

3901 tcactcttcc actgtcagca cattctcatt cttctttggt gctgggcaag aactcaacca

3961 gtgtggaagc catacttggc ccaggcgggt gaagtgggcg ggccgtctcc tgcagcaggt

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6361 gttttcattg taatccagaa tgtttttaaa tcaccctact tctattgaac ttgcactatc
6421 atctgttaac ctctactgta tttattaaat aaacctgaat aggtaaatca cagtacagca
6481 aaa
```

In some embodiments of the methods of the disclosure, the wild type human OBFC1 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_079204.2):

(SEQ ID NO: 24)

```
  1 mqpgssrcee etpsllwgld pvflafakly irdildmkes rqvpgvflyn ghpikqvdvl
 61 gtvigvrerd afysygvdds tgvincicwk klntesvsaa psaarelslt sqlkklqeti
121 eqktkieigd tirvrgsirt yreereihat tyykvddpvw niqiarmlel ptiyrkvydq
181 pfhssaleke ealsnpgald lpsltsllse kakeflmenr vqsfyqqele mvesllslan
241 qpvihsassd qvnfkkdtts kaihsifkna iqllqekglv fqkddgfdnl yyvtredkdl
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301  hrkihriiqq dcqkpnhmek gchflhilac arlsirpgls eavlqqvlel ledqsdivst

361  mehyytaf
```

In some embodiments of the methods of the disclosure, the wild type human ATP11A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_015205.2, transcript variant 1):

```
                                                          (SEQ ID NO: 25)
   1 gcggccgcac tagtaccccg gagcccatgg gcgcgccgag ccgggcgcgg gggcgctgaa

  61 cggcggagcg ggagcggccg gaggagccat ggactgcagc ctcgtgcgga cgctcgtgca

 121 cagatactgt gcaggagaag agaattgggt ggacagcagg accatctacg tgggacacag

 181 ggagccacct ccgggcgcag aggcctacat cccacagaga tacccagaca acaggatcgt

 241 ctcgtccaag tacacatttt ggaactttat acccaagaat ttatttgaac aattcagaag

 301 agtagccaac ttttatttcc ttatcatatt tctggtgcag ttgattattg atacacccac

 361 aagtccagtg acaagcggac ttccactctt ctttgtcatt actgtgacgg ctatcaaaca

 421 gggttatgaa gactggcttc gacataaagc agacaatgcc atgaaccagt gtcctgttca

 481 tttcattcag cacggcaagc tcgttcggaa acaaagtcga aagctgcgag ttggggacat

 541 tgtcatggtt aaggaggacg agacctttcc ctgcgacttg atcttccttt ccagcaaccg

 601 gggagatggg acgtgccacg tcaccaccgc cagcttggat ggagaatcca gccataaaac

 661 gcattacgcg gtccaggaca ccaaaggctt ccacacagag gaggatatcg gcggacttca

 721 cgccaccatc gagtgtgagc agccccagcc cgacctctac aagttcgtgg gtcgcatcaa

 781 cgtttacagt gacctgaatg accccgtggt gaggccctta ggatcggaaa acctgctgct

 841 tagaggagct acactgaaga acactgagaa aatctttggt gtggctattt acacgggaat

 901 ggaaaccaag atggcattaa attatcaatc aaaatctcag aagcgatctg ccgtggaaaa

 961 atcgatgaat gcgttcctca ttgtgtatct ctgcattctg atcagcaaag ccctgataaa

1021 cactgtgctg aaatacatgt ggcagagtga gccctttcgg gatgagccgt ggtataatca

1081 gaaaacggag tcggaaaggc agaggaatct gttcctcaag gcattcacgg acttcctggc

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1201 gaagttcctc ggctcttact tcatcacctg ggacgaagac atgtttgacg aggagactgg

1261 cgaggggcct ctggtgaaca cgtcggacct caatgaagag ctgggacagg tggagtacat

1321 cttcacagac aagaccggca ccctcacgga aaacaacatg gagttcaagg agtgctgcat

1381 cgaaggccat gtctacgtgc cccacgtcat ctgcaacggg caggtcctcc cagagtcgtc

1441 aggaatcgac atgattgact cgtcccccag cgtcaacggg agggagcgcg aggagctgtt

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1561 ccccaggaaa tcgccggacg gggggaaatc ctgtgtgtac atctcatcct cgcccgacga

1621 ggtggcgctg gtcgaaggtg tccagagact tggctttacc tacctaaggc tgaaggacaa

1681 ttacatggag atattaaaca gggagaacca catcgaaagg tttgaattgc tggaaatttt

1741 gagttttgac tcagtcagaa ggagaatgag tgtaattgta aaatctgcta caggagaaat

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1861 tgaccagatc cgagccagag tggagcgtaa cgcagtggag gggctccgaa ctttgtgtgt

1921 tgcttataaa aggctgatcc aagaagaata tgaaggcatt tgtaagctgc tgcaggctgc

1981 caaagtggcc cttcaagatc gagagaaaaa gttagcagaa gcctatgagc aaatagagaa
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7441 gcctccctgg gatgaagagt ccccctcctg gcagaatgtc tgggctttgc agagcaggcc
7501 ccgggggtga agtcgcagct tcacttacac cagctgctct gtgagcaagg cttggtgccc
7561 tggacaaggc ccttcccctt tagggaggtc cagcctcgca agctgaaacc tcccctcggc
7621 tcagccctat accaggcggc cacagcagga ctggccacac ccacgccgca cctcatccgt
7681 gcacgcgtcg gagcacggcc agccttccgc cacgagccag ctgggaaggg ccgcggccgc
7741 ctaaagcccc agtcaaccca gcctgtgtct gagcagacag ggcgaacaag caggccacac
7801 cgtctcgagg gaggaggcca gatgcggcca gcgtctccaa cagggtgacc atccgctcgg
7861 cttgctgagc gtttaaacaa atgtttagac aggctgtggg gactcccctg agttgagcct
7921 tggccagggg tccggtgctg tcgcgggaaa cctccagcct tgttcttcaa accactcagc
7981 tcatgtgttt tgcactgact agtactgaat aatacaacca ctcttattta atgttagtat
8041 tatttatttg acaactcagt gtctaacagc ttgatatgca ggtccttgca tcctacattt
8101 ctttaggaag ttacccattt gtaactttaa aaacaggaaa aatatcagtt ggcaaatgca
8161 atcttttttt tttttaagct aaaggtgggt gaactggaat gaaaatcttt ctgatgttgt
8221 gtctataagc agccttgatg ggatatgtta gaagtgtcat gaaagtgtga ttctactttt
8281 gcagaaaaat ctaaagatca atttatatag ctttattttt tactttatca aagtatacag
8341 aattttaata tgcatatatt gtgtctgact taaaattata atgtctgcgt caccatttaa
8401 aatgtctgtt cattatgtaa tgtaataaaa gaaggtcttc aaaaatgtat ttaacatgaa
8461 tggtatccat agttgtcatc atcataaata ctggagttta tttttaaatt attaaacata
8521 gtaggtgcat taacataaat cagtctccac acagtaacat ttaactgata attcattaat
8581 cagctttgaa aaattaaatt gttaattaaa ccaatctaac atttcagtaa agtttatttt
8641 gtatgcttct gtttttaact tttatttctg tagataaact gactggataa tattatattg
8701 gacttttctc tagattatct aagcaggaga cctgaatctg cttgcaataa agaataaaag
8761 tctgcttcag tttctttata aagaaactca cacaa
```

In some embodiments of the methods of the disclosure, the wild type human ATP11A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NP_056020.2, transcript variant 1):

(SEQ ID NO: 26)

```
  1 mdcslvrtlv hrycageenw vdsrtiyvgh repppgaeay ipqrypdnri vsskytfwnf
 61 ipknlfeqfr rvanfyflii flvqliidtp tspvtsglpl ffvitvtaik qgyedwlrhk
121 adnamnqcpv hfiqhgklvr kqsrklrvgd ivmvkedetf pcdliflssn rgdgtchvtt
181 asldgesshk thyavqdtkg fhteediggl hatieceqpq pdlykfvgri nvysdlndpv
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 241 vrplgsenll lrgatlknte kifgvaiytg metkmalnyq sksqkrsave ksmnaflivy
 301 lciliskali ntvlkymwqs epfrdepwyn qkteserqrn lflkaftdfl afmvlfnyii
 361 pvsmyvtvem qkflgsyfit wdedmfdeet gegplvntsd lneelgqvey iftdktgtlt
 421 ennmefkecc ieghvyvphv icngqvlpes sgidmidssp svngrereel ffralclcht
 481 vqvkdddsvd gprkspdggk scvyissspd evalvegvqr lgftylrlkd nymeilnren
 541 hierfellei lsfdsvrrrm svivksatge iylfckgads sifprviegk vdqirarver
 601 naveglrtlc vaykrliqee yegickllqa akvalqdrek klaeayeqie kdltllgata
 661 vedrlqekaa dtiealqkag ikvwvltgdk metaaatcya cklfrrntql lelttkriee
 721 qslhdvlfel sktvlrhsgs ltrdnlsgls admqdyglii dgaalslimk predgsssgny
 781 relfleicrs csavlccrma plqkaqivkl ikfskehpit laigdgandv smileahvgi
 841 gvigkegrqa arnsdyaipk fkhlkkmllv hghfyyiris elvqyffykn vcfifpqfly
 901 qffcgfsqqt lydtayltly nisftslpil lyslmeqhvg idvlkrdptl yrdvaknall
 961 rwrvfiywtl lglfdalvff fgayfvfent tvtsngqifg nwtfgtlvft vmvftvtlkl
1021 aldthywtwi nhfviwgsll fyvvfsllwg gviwpflnyq rmyyvfiqml ssgpawlaiv
1081 llvtisllpd vlkkvlcrql wptatervqt ksqclsveqs tifmlsqtss slsf
```

In some embodiments of the methods of the disclosure, the wild type human ATP11A gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_032189.3, transcript variant 2):

(SEQ ID NO: 48)

```
   1 gcggccgcac tagtaccccg gagcccatgg gcgcgccgag ccgggcgcgg gggcgctgaa
  61 cggcggagcg ggagcggccg gaggagccat ggactgcagc ctcgtgcgga cgctcgtgca
 121 cagatactgt gcaggagaag agaattgggt ggacagcagg accatctacg tgggacacag
 181 ggagccacct ccgggcgcag aggcctacat cccacagaga tacccagaca acaggatcgt
 241 ctcgtccaag tacacatttt ggaactttat acccaagaat ttatttgaac aattcagaag
 301 agtagccaac ttttatttcc ttatcatatt tctggtgcag ttgattattg atacacccac
 361 aagtccagtg acaagcggac ttccactctt ctttgtcatt actgtgacgg ctatcaaaca
 421 gggttatgaa gactggcttc gacataaagc agacaatgcc atgaaccagt gtcctgttca
 481 tttcattcag cacggcaagc tcgttcggaa acaaagtcga aagctgcgag ttggggacat
 541 tgtcatggtt aaggaggacg agacctttcc ctgcgacttg atcttccttt ccagcaaccg
 601 gggagatggg acgtgccacg tcaccaccgc cagcttggat ggagaatcca gccataaaac
 661 gcattacgcg gtccaggaca ccaaaggctt ccacacagag gaggatatcg gcggacttca
 721 cgccaccatc gagtgtgagc agccccagcc cgacctctac aagttcgtgg gtcgcatcaa
 781 cgtttacagt gacctgaatg accccgtggt gaggccctta ggatcggaaa acctgctgct
 841 tagaggagct acactgaaga acactgagaa aatctttggt gtggctattt acacgggaat
 901 ggaaaccaag atggcattaa attatcaatc aaaatctcag aagcgatctg ccgtggaaaa
 961 atcgatgaat gcgttcctca ttgtgtatct ctgcattctg atcagcaaag ccctgataaa
1021 cactgtgctg aaatacatgt ggcagagtga gccctttcgg gatgagccgt ggtataatca
1081 gaaaacggag tcggaaaggc agaggaatct gttcctcaag gcattcacgg acttcctggc
1141 cttcatggtc ctctttaact acatcatccc tgtgtccatg tacgtcacgg tcgagatgca
1201 gaagttcctc ggctcttact tcatcacctg ggacgaagac atgtttgacg aggagactgg
```

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1261 cgaggggcct ctggtgaaca cgtcggacct caatgaagag ctgggacagg tggagtacat
1321 cttcacagac aagaccggca ccctcacgga aaacaacatg gagttcaagg agtgctgcat
1381 cgaaggccat gtctacgtgc cccacgtcat ctgcaacggg caggtcctcc cagagtcgtc
1441 aggaatcgac atgattgact cgtcccccag cgtcaacggg agggagcgcg aggagctgtt
1501 tttccgggcc ctctgtctct gccacaccgt ccaggtgaaa gacgatgaca gcgtagacgg
1561 ccccaggaaa tcgccggacg gggggaaatc ctgtgtgtac atctcatcct cgcccgacga
1621 ggtggcgctg gtcgaaggtg tccagagact tggctttacc tacctaaggc tgaaggacaa
1681 ttacatggag atattaaaca gggagaacca catcgaaagg tttgaattgc tggaaatttt
1741 gagttttgac tcagtcagaa ggagaatgag tgtaattgta aaatctgcta caggagaaat
1801 ttatctgttt tgcaaaggag cagattcttc gatattcccc cgagtgatag aaggcaaagt
1861 tgaccagatc cgagccagag tggagcgtaa cgcagtggag gggctccgaa ctttgtgtgt
1921 tgcttataaa aggctgatcc aagaagaata tgaaggcatt tgtaagctgc tgcaggctgc
1981 caaagtggcc cttcaagatc gagagaaaaa gttagcagaa gcctatgagc aaatagagaa
2041 agatcttact ctgcttggtg ctacagctgt tgaggaccgg ctgcaggaga aagctgcaga
2101 caccatcgag gccctgcaga aggccgggat caaagtctgg gttctcacgg gagacaagat
2161 ggagacggcc gcggccacgt gctacgcctg caagctcttc cgcaggaaca cgcagctgct
2221 ggagctgacc accaagagga tcgaggagca gagcctgcac gacgtcctgt tcgagctgag
2281 caagacggtc ctgcgccaca gcgggagcct gaccagagac aacctgtccg gactttcagc
2341 agatatgcag gactacggtt taattatcga cggagctgca ctgtctctga taatgaagcc
2401 tcgagaagac gggagttccg gcaactacag ggagctcttc ctggaaatct gccggagctg
2461 cagcgcggtg ctctgctgcc gcatggcgcc cttgcagaag gctcagattg ttaaattaat
2521 caaattttca aaagagcacc caatcacgtt agcaattggc gatggtgcaa atgatgtcag
2581 catgattctg gaagcgcacg tgggcatagg tgtcatcggc aaggaaggcc gccaggctgc
2641 caggaacagc gactatgcaa tcccaaagtt taagcatttg aagaagatgc tgcttgttca
2701 cgggcatttt tattacatta ggatctctga gctcgtgcag tacttcttct ataagaacgt
2761 ctgcttcatc ttccctcagt ttttatacca gttcttctgt gggttttcac aacagacttt
2821 gtacgacacc gcgtatctga ccctctacaa catcagcttc acctccctcc ccatcctcct
2881 gtacagcctc atggagcagc atgttggcat tgacgtgctc aagagagacc cgaccctgta
2941 cagggacgtc gccaagaatg ccctgctgcg ctggcgcgtg ttcatctact ggacgctcct
3001 gggactgttt gacgcactgg tgttcttctt tggtgcttat ttcgtgtttg aaaatacaac
3061 tgtgacaagc aacgggcaga tatttggaaa ctggacgttt ggaacgctgg tattcaccgt
3121 gatggtgttc acagttacac taaagcttgc attggacaca cactactgga cttggatcaa
3181 ccattttgtc atctgggggt cgctgctgtt ctacgttgtc ttttcgcttc tctggggagg
3241 agtgatctgg ccgttcctca actaccagag gatgtactac gtgttcatcc agatgctgtc
3301 cagcgggccc gcctggctgg ccatcgtgct gctggtgacc atcagcctcc ttcccgacgt
3361 cctcaagaaa gtcctgtgcc ggcagctgtg gccaacagca acagagagag tccagaatgg
3421 gtgcgcacag cctcgggacc gcgactcaga attcacccct cttgcctctc tgcagagccc
3481 aggctaccag agcacctgtc cctcggccgc ctggtacagc tcccactctc agcaggtgac
3541 actcgcggcc tggaaggaga aggtgtccac ggagccccca cccatcctcg gcggttccca
3601 tcaccactgc agttccatcc caagtcacag ctgccctagg tcccgtgtgg gaatgctcgt
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3661 gtgatggatg gtcctaagcc tgtggagact gtgcacgtgc ctcttcctgg cccccagcag
3721 gcaaggaggg gggtcacagg ccttgccctc gagcatggca ccctggccgc ctggacccag
3781 cactgtggtt gttgagccac accagtggcc tctgggcatt cggctcaacg caggagggac
3841 attctgctgg cccaccctgc gcgctgtcat gcagaggcca ttcccccagg cctgtgtctt
3901 cacccacctg ccatcattgg cctttgctgt cactgggaga gaagagccgt ccagggaccc
3961 atggtggccc acatgtggat gccacatgct gctgtttcct gcttgcccgg ccaccaccca
4021 tgccctccat agggtgaggt ggagccatgg tggtgcgtcc tttactcaac aaccctccaa
4081 tccggatgct gtgggaaggg ccgggtcact cggataccat catccctgcg gatgcaccgc
4141 cgtaccctgc tcatctggga gtggtttccc tgcggttacg tccaagcccg cctgccctgt
4201 gtgttggggc tggctgagtt tcggtctccc catcaccggc cgcctcgtgg agaaggcagt
4261 gccacgtggg aggacaaggc cacgccggca gcttccagcc ctgccgcaga agtgccagga
4321 tgtccatcag ccactcgcca gggcacggag ccgtcagtcc actgttacgg gagaatgttg
4381 atttcgcggg tgcgagggcc gggagacaga tacttggctg tgatgagcag acatcctctg
4441 tccccgtgga ggggtcaaca ccaaggtggt gttcgtgcac cagaacctgt ctcgggctga
4501 cgggggtggc acacaggaca cgggtggatc ccaacaggca gcaccgcacc tctgcccgcc
4561 tcccgcactg cagctccgcc cgccgggctc tgcgtcccca cgtcccctcg tcccatcccc
4621 acgtcccctc atcccgtcac ctcgtcccca catccccttg ccccgtcacc tcgtcctcat
4681 gtccccttgt cctgtcacct cgtccccacg tcccctcgtc tcctcatccc cacgtcctct
4741 cgtccccttg tcccgtcccc acataccctc gtccccatgt ccccacgcag ggctctcctt
4801 cgtcttagga tctgtccagc gctgctctgg gtgggttagc aaccccaggg ctgctgtgat
4861 aggaagtccc tgttgttctc cgtactggca tttctatttc tagaaataat atttgacata
4921 gccttaatgg tccttaaaga agacatttca gtgtgagatt cagacttcag acgctgaaac
4981 tgctgccttt caggaaagca ccaccaacgc tggaggagga gccggccctc acgcccgccc
5041 cgcgccacgc tgtggaacgg ggctccggca agtgaaaccc agagggtgtt tccgaggtgc
5101 tcgacagtag gtatttttgg aagctcagat ttcaccattt gattgtataa tcttttacct
5161 ataaaatatt tatttgaagt agagggtaaa tcagcggtaa gaacagtgaa cacagtggtt
5221 gggataaaat aaggtgacaa acatcacacc aaagatgagg gtagcgagca actggcttga
5281 gcagacagaa cggggaagac tccactctgt cccgaggggc cagccgcagg cgtccccagg
5341 gccaccctgc cctgaggtcc ttgtgtggcc gccctggctt ggcagccctg cccacgctgc
5401 ccccgcaaac aatggtgtgt gcgtttttac agcccttttt aggaacccaa tatgggcata
5461 aatgtaacac ctgtagcggg ggcagattct ctgtatgttc agttaacaaa ttatttgtaa
5521 tgtatttttt tagaaatctt aaaattgcct ttgcactgaa gtattttcat agctgtttat
5581 atctctttta ttcatttatt taacatactg tctaatttta aaaataggtt tttaaagctt
5641 tcatttttaa gtttatgaaa ttttggccac tttacattta gattctggtg agagttttga
5701 ctgaatgttc caatctctga tgaatgcgaa ttttcagatt tgattttatt ctctacacac
5761 acctcttctt ttcttggtat ttctggtggc agtgattagt tgaacagcac atttaaggca
5821 cgataatttg ctacactttt tctttacaat ttgttgcaat ttcatctgct ttctatgttt
5881 cattgttaat tgccatcctt cagccttaaa aatagaagat tctcacgtga aggtttagta
5941 agttgggtcc cagctctgcc tgtgtggaga tagtcaccat gtacctctga caacaagttt
6001 tagtgtgaaa gtcactaaac ttttacacac tcccaaacgt ctttttaaaa attgcttggg
6061 aaattattaa atgaatgtgc ctgatgattt gaaatagaca aggggcacga gataaaaaag
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6121 aaaaggatga gaagatcctc agtgaatgac gttgcagggt cttcatgcaa ttttccacct
6181 cgcagtagtt agtatttact tgccttaaac taactttgaa gcaagtaatg tcaactttga
6241 gcactttgtt gagttttgaa aaatcttatt tgttgctgca caggttaata aattatcaat
6301 ttgtaattca gcatgttggt cagagacacg gtcactgatt cacacccagt ccctgccaca
6361 gaccgtctca gacacgcaca gtgggcctgc tgcatgattc acacccagtc cctgccacag
6421 accgtctcag acacgcacag tgggcctgct gcatgattca cacccagtcc ctgccacaga
6481 ccgtctcaga cacgcacagt gggcctgctg catgcgtgtt acctggcttt tggctccacg
6541 ctcactcata gccatgtcca catgggggct tgcacacagg atcactcaca tatgtacatg
6601 tacccaccac aaacgtgcaa gctcctgcac acatgcatgc acacaaacgt gtacacaagt
6661 gtgagctcct acacgcatac acacacacac gtgtacatgc accaaagcat gtgtgaccta
6721 cagacatgca gaacatgcac gtgtacacat accacagaca cgcgtgtgca tgctcctaca
6781 caatacatat gcacatatca tgaacagcgt aagttcctac acacggacgt gtgatacaca
6841 catgcatgta caggtaagca cacatgtaca agctcctaca ggcttgctct cacacacgtg
6901 tatgcacagc agagagacgt atgagcttct actgcacaca tgcacacaca cacgcacacg
6961 tacattcact acaaacgtgc agcctcctgc acacgtgcac attcatgtgt acaccacaaa
7021 tgagttccca gacgtgtaaa cacacgtgca cacatcgtac acatgtgagc tcccacacgt
7081 acacacagat gcacatggac acaccccaaa cacgcacagg ctcctacaca catgcacaca
7141 cgtgtacacc acaaacgagc tcccagacat gtaaacacac gtctcccaca cgtgagctcc
7201 cacacgtaca catgcacatg tacgcaccac aaacacatgc gcaggctcct gcaggcgtga
7261 atacacacat gcacacacat atacacacat gtgccacaaa caagtgcaca ctgtcctggt
7321 gtcctgcact gcatcctgcc tccttgctga ggggcccctg tgagaggcct ctggatgggc
7381 atgggaagat gggctccctg gcccccagcc catgcctccc tgggatgaag agtccccctc
7441 ctggcagaat gtctgggctt tgcagagcag gccccggggg tgaagtcgca gcttcactta
7501 caccagctgc tctgtgagca aggcttggtg ccctggacaa ggcccttccc ctttagggag
7561 gtccagcctc gcaagctgaa acctcccctc ggctcagccc tataccaggc ggccacagca
7621 ggactggcca cacccacgcc gcacctcatc cgtgcacgcg tcggagcacg gccagccttc
7681 cgccacgagc cagctgggaa gggccgcggc cgcctaaagc cccagtcaac ccagcctgtg
7741 tctgagcaga cagggcgaac aagcaggcca caccgtctcg agggaggagg ccagatgcgg
7801 ccagcgtctc caacagggtg accatccgct cggcttgctg agcgtttaaa caaatgttta
7861 gacaggctgt ggggactccc ctgagttgag ccttggccag gggtccggtg ctgtcgcggg
7921 aaacctccag ccttgttctt caaaccactc agctcatgtg ttttgcactg actagtactg
7981 aataatacaa ccactcttat ttaatgttag tattatttat ttgacaactc agtgtctaac
8041 agcttgatat gcaggtcctt gcatcctaca tttctttagg aagttaccca tttgtaactt
8101 taaaaacagg aaaaatatca gttggcaaat gcaatctttt ttttttttaa gctaaaggtg
8161 ggtgaactgg aatgaaaatc tttctgatgt tgtgtctata agcagccttg atgggatatg
8221 ttagaagtgt catgaaagtg tgattctact tttgcagaaa aatctaaaga tcaatttata
8281 tagctttatt ttttacttta tcaaagtata cagaatttta atatgcatat attgtgtctg
8341 acttaaaatt ataatgtctg cgtcaccatt taaaatgtct gttcattatg taatgtaata
8401 aaagaaggtc ttcaaaaatg tatttaacat gaatggtatc catagttgtc atcatcataa
8461 atactggagt ttatttttaa attattaaac atagtaggtg cattaacata aatcagtctc
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8521 cacacagtaa catttaactg ataattcatt aatcagcttt gaaaaattaa attgttaatt

8581 aaaccaatct aacatttcag taaagtttat tttgtatgct tctgtttttta acttttattt

8641 ctgtagataa actgactgga taatattata ttggactttt ctctagatta tctaagcagg

8701 agacctgaat ctgcttgcaa taaagaataa aagtctgctt cagtttcttt ataaagaaac

8761 tcacacaa
```

In some embodiments of the methods of the disclosure, the wild type human ATP11A gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_115565.3, transcript variant 2):

```
                                                        (SEQ ID NO: 49)
   1 mdcslvrtlv hrycageenw vdsrtiyvgh repppgaeay ipqrypdnri vsskytfwnf

  61 ipknlfeqfr rvanfyflii flvqliidtp tspvtsglpl ffvitvtaik qgyedwlrhk

 121 adnamnqcpv hfiqhgklvr kqsrklrvgd ivmvkedetf pcdliflssn rgdgtchvtt

 181 asldgesshk thyavqdtkg fhteediggl hatieceqpq pdlykfvgri nvysdlndpv

 241 vrplgsenll lrgatlknte kifgvaiytg metkmalnyq sksqkrsave ksmnaflivy

 301 lciliskali ntvlkymwqs epfrdepwyn qkteserqrn lflkaftdfl afmvlfnyii

 361 pvsmyvtvem qkflgsyfit wdedmfdeet gegplvntsd lneelgqvey iftdktgtlt

 421 ennmefkecc ieghvyvphv icngqvlpes sgidmidssp svngrereel ffralclcht

 481 vqvkdddsvd gprkspdggk scvyissspd evalvegvqr lgftylrlkd nymeilnren

 541 hierfellei lsfdsvrrrm svivksatge iylfckgads sifprviegk vdqirarver

 601 naveglrtlc vaykrliqee yegickllqa akvalqdrek klaeayeqie kdltllgata

 661 vedrlqekaa dtiealqkag ikvwvltgdk metaaatcya cklfrrntql lelttkriee

 721 qslhdvlfel sktvlrhsgs ltrdnlsgls admqdyglii dgaalslimk predgssgny

 781 relfleicrs csavlccrma plqkaqivkl ikfskehpit laigdgandv smileahvgi

 841 gvigkegrqa arnsdyaipk fkhlkkmllv hghfyyiris elvqyffykn vcfifpqfly

 901 qffcgfsqqt lydtayltly nisftslpil lyslmeqhvg idvlkrdptl yrdvaknall

 961 rwrvfiywtl lglfdalvff fgayfvfent tvtsngqifg nwtfgtlvft vmvftvtlkl

1021 aldthywtwi nhfviwgsll fyvvfsllwg gviwpflnyq rmyyvfiqml ssgpawlaiv

1081 llvtisllpd vlkkvlcrql wptatervqn gcaqprdrds eftplaslqs pgyqstcpsa

1141 awysshsqqv tlaawkekvs tepppilggs hhhcssipsh scprsrvgml v
```

In some embodiments of the methods of the disclosure, the wild type human IVD/DISP2 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_002225.3, transcript variant 1):

```
                                                        (SEQ ID NO: 50)
   1 tttccgcagt taggggctgc tatttcaacg cagggagata aaaagaaaaa aacacttgct

  61 cttctacccc gctaaaaaca ctcatcctag ggagcacgcc agcatttgca gcgttcgggg

 121 cagggccact cggcctgcgg ccgttgcact ggctggaagc tggcaggcga tcacggttga

 181 ttggctcggg tgcggtccaa gggcagcaac gccttcggcg ggccgcctag ggtgattggc

 241 tgctgcagcc caccccctag ccggtttggt gggcggcgaa gcctggattg gtggagctaa

 301 gagctggctc agtttcagcg ctggctcttc gtgcatggca gagatggcga ctgcgactcg

 361 gctgctgggg tggcgtgtgg cgagctggag gctgcggccg ccgcttgccg gcttcgtttc
```

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421 ccagcgggcc cactcgcttt tgcccgtgga cgatgcaatc aatgggctaa gcgaggagca

481 gaggcagctt cgtcagacca tggctaagtt ccttcaggag cacctggccc ccaaggccca

541 ggagatcgat cgcagcaatg agttcaagaa cctgcgagaa ttttggaagc agctggggaa

601 cctgggcgta ttgggcatca cagcccctgt tcagtatggc ggctccggcc tgggctacct

661 ggagcatgtg ctggtgatgg aggagatatc ccgagcttcc ggagcagtgg ggctcagtta

721 cggtgcccac tccaacctct gcatcaacca gcttgtacgc aatgggaatg aggcccagaa

781 agagaagtat ctcccgaagc tgatcagtgg tgagtacatc ggagccctgg ccatgagtga

841 gcccaatgca ggctctgatg ttgtctctat gaagctcaaa gcggaaaaga aaggaaatca

901 ctacatcctg aatggcaaca agttctggat cactaatggc cctgatgctg acgtcctgat

961 tgtctatgcc aagacagatc tggctgctgt gccagcttct cggggcatca cagccttcat

1021 tgtggagaag ggtatgcctg gctttagcac ctctaagaag ctggacaagc tggggatgag

1081 gggctctaac acctgtgagc taatctttga agactgcaag attcctgctg ccaacatcct

1141 gggccatgag aataagggtg tctacgtgct gatgagtggg ctggacctgg agcggctggt

1201 gctggccggg gggcctcttg ggctcatgca agcggtcctg gaccacacca ttccctacct

1261 gcacgtgagg gaagcctttg gccagaagat cggccacttc cagttgatgc aggggaagat

1321 ggctgacatg tacacccgcc tcatggcgtg tcggcagtat gtctacaatg tcgccaaggc

1381 ctgcgatgag ggccattgca ctgctaagga ctgtgcaggt gtgattcttt actcagctga

1441 gtgtgccaca caggtagccc tggacggcat tcagtgtttt ggtggcaatg gctacatcaa

1501 tgactttccc atgggccgct ttcttcgaga tgccaagctg tatgagatag ggctgggacc

1561 cagcgaggtg aggcggctgg tcatcggcag agccttcaat gcagactttc actagtcctg

1621 agacccttcg cccccttttc ctgcacctag tggcctttct tgggaagtag agatgtggcg

1681 gctttcccac cctgcccaca gcaggccctc ctgcccagct gctcttgtca gccctctggc

1741 ctctggatga ggttgagttc tccacaacag ctcccaagca tcatgggcct cgcagccggg

1801 cctgtgccac ggctagtgtt gtgtgattta aaatggactc agcaggaagc atattgtctg

1861 gggattgttg ggacaggttt tggtgactct gtgcccttgc tctctaactt ctgagcccac

1921 ctcccagggt aggcacctgg gggcatgcag gtgcccacct cccagggtag gcacctgggg

1981 gcatgcaggt acccacctct ttctcttggg tgaggctctg gcaaggagat ctctctgctc

2041 aagcacagca gaatcatggc ccctctccat gaattggaac ttggtacagg ttaagtatcc

2101 ctaatcctga aatctgaaac acttgtggtt ccaagcattt tggataaggc aaattcaact

2161 ttcagtctct tttctggggg aaaaaaataa taaacctagc ctagccaggc gtggtggctc

2221 atgcttgtaa tcccagcact tcaggaggct gagatgggtg gatcacctga ggtcaggagt

2281 tcaagaccag cctggccaac atgtggaaac ctcgcctcaa ctaaaaatag aaaaaaatta

2341 gttgggcatg gtggtgggca cctgtaatcc cagctacttc aggaggctga ggcaggagaa

2401 ttacttgaac ccaggaggcg gacgttgcag tgagccgagc ttgtgccatt gcactccagc

2461 ctgggcgaca agagcaaaac tcttcaaaaa acaaaacaaa acaaaaaaac cctggccctt

2521 gtttcttcca gtttctagag gtatcagctc ctagcagctt atgaacacat atgcttgctt

2581 ggccaggcaa ggtggtgtgt gcctgtaatc ccagcacttt gggaggccaa ggcaggtgga

2641 tcacttgcag tcaggagttc aagaccagcc tgtccaacgt ggtgaaaccc catctctact

2701 aaaaatacaa aaattagcca ggggtggtgg tgcacgtctg taatcccagc tactcaggag

2761 gctgaggcag gagaatcact tgaacccggg aggtggaggt tgcaatgagc caatatgaca

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2821 ccgctgcagt ccagcctggg ccatagagtg agactctgtc tcaaaaaagg aaagaaaaat
2881 aggctgggca cagtgactca tgcctgtaat cccaacactt tgggaggccg aggcaggtgg
2941 atcacgaggt caggagttca agaccagcct ggccaagatg gtaaaacctc gtctctacta
3001 aaaatacaaa aattagccag gtgtggtggc aggctcctgt aatcccagct actcaggagg
3061 ctgaggcaga gaattgcttg aacccgggag gcagagtttg cagtgagcca agatcacacc
3121 actgcactcc agcttggacg acagagcgag actctgtctc aaaaaataat aggccaggca
3181 tggtggctca acgtctgtaa tcccagcact ttgggaggcc gaggcgggca gatcacaagg
3241 tcaggagttc gagaccagcc tgacgaccaa catggtgaaa cctcgtctct actaaaaata
3301 caaaaattag ccaggcctgg tggcacgcgc ctgtaatccc agttacacag aagactgagg
3361 caggagaatc gcttgaacgc aggaggcaga ggttgcagga gctgagatcg cgccattgca
3421 ctccagcctg ggcaacagag tgagactctg tctcaaaaaa taataataaa ataaatgaac
3481 acacatgctg ctgagtccgc aggggggca gagcagagga cagcgtgctt ttgtgtactg
3541 ttggaagact ggctcctcct gtacagcacc tctgagccct tgtgcaccgc cctgccacgg
3601 gcaccatcca gtcctggccg tgtgaccacc cacagctgac tgggcagcag gcacaggccc
3661 tacccgagca ggccggagtt ggctcgcatg actccagctg aggctgcctg tgtacatttc
3721 tccagatacc ctatggctaa ttttgttata actgcacagt ggctgctgcc attttgtatt
3781 aaatatattg tgaaacaaac ctatctgggg agaagcaatc tacttgccgc tgcttcctgt
3841 ctggatccag cttgtgtcct tggagagtgg ctggcccagg tcctattcct gtcctccagc
3901 ccgttctttc atgagggaca ggaaggtaaa atcagccctt aggagagagg tctcagcctc
3961 cctttcccag atctcccagt gagttttaaa ggaagcaggg agcccagagt gctaagttct
4021 tacagccaga aggaagctta tagatttctg aaaaccgccc ctttgttttt aaaaagatca
4081 acacaatttg actttctcaa ggtcaaaacg aactagaatc cagatctgct catggcaaaa
4141 atgggggtgt tctgagaatt ccagctttgg gccgcactgt acagcagtct ggatagagtg
4201 tgatctgaga agggaatggg tctgggttgt tccacccctt ccgagttcca aaaagaggga
4261 actggttttc ttggttctca gcccagcagc acctatcctg gctcttggtc ctggcctgca
4321 gccaagtgct gttcctagcc tgaggcttga gacaggtggg gttggctcct caccaacccc
4381 agttccgtcc catcctgagg gcaagatcct gggctcatag gcagtccctt tcacttcctt
4441 gtcttgctcc ctgctatgtt ggagatgaat gtgactaaaa gggccatctt gctggcttaa
4501 tgtgtggctg gagagaccag cctggagaca atgtggcaaa atggggcgct tcatccagtc
4561 tgtctaagcc ctgtcgactt ggggaggtga tttctttcct ggttctatat gtgaagcaaa
4621 ataaatgttt taaaattaaa agcaaaaaaa acaaaatgaa ccatgaaaaa aaa
```

In some embodiments of the methods of the disclosure, the wild type human IVD/DISP2 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: transcript variant 1):

```
                                                      (SEQ ID NO: 51)
  1  maematatrl lgwrvaswrl rpplagfvsq rahsllpvdd ainglseeqr qlrqtmakfl
 61  qehlapkage idrsnefknl refwkqlgnl gvlgitapvq yggsglgyle hvlvmeeisr
121  asgavglsyg ahsnlcinql vrngneaqke kylpklisge yigalamsep nagsdvvsmk
181  lkaekkgnhy ilngnkfwit ngpdadvliv yaktdlaavp asrgitafiv ekgmpgfsts
241  kkldklgmrg sntcelifed ckipaanilg henkgvyvlm sgldlerlvl aggplglmqa
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301 vldhtipylh vreafgqkig hfqlmqgkma dmytrlmacr qyvynvakac deghctakdc

361 agvilysaec atqvaldgiq cfggngyind fpmgrflrda klyeigagts evrrlvigra

421 fnadfh
```

In some embodiments of the methods of the disclosure, the wild type human IVD/DISP2 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001159508.1, transcript variant 2):

(SEQ ID NO: 27)

```
   1 tttccgcagt taggggctgc tatttcaacg cagggagata aaaagaaaaa aacacttgct
  61 cttctacccc gctaaaaaca ctcatcctag ggagcacgcc agcatttgca gcgttcgggg
 121 cagggccact cggcctgcgg ccgttgcact ggctggaagc tggcaggcga tcacggttga
 181 ttggctcggg tgcggtccaa gggcagcaac gccttcggcg ggccgcctag ggtgattggc
 241 tgctgcagcc cacccccctag ccggtttggt gggcggcgaa gcctggattg gtggagctaa
 301 gagctggctc agtttcagcg ctggctcttc gtgcatggca gagatggcga ctgcgactcg
 361 gctgctgggg tggcgtgtgg cgagctggag gctgcggccg ccgcttgccg gcttcgtttc
 421 ccagcgggcc cactcgcttt tgcccgtgga cgatgcaatc aatgggctaa gcgaggagca
 481 gaggcaggaa ttttggaagc agctggggaa cctgggcgta ttgggcatca cagcccctgt
 541 tcagtatggc ggctccggcc tgggctacct ggagcatgtg ctggtgatgg aggagatatc
 601 ccgagcttcc ggagcagtgg ggctcagtta cggtgcccac tccaacctct gcatcaacca
 661 gcttgtacgc aatgggaatg aggcccagaa agagaagtat ctcccgaagc tgatcagtgg
 721 tgagtacatc ggagccctgg ccatgagtga gcccaatgca ggctctgatg ttgtctctat
 781 gaagctcaaa gcggaaaaga aggaaatca ctacatcctg aatggcaaca agttctggat
 841 cactaatggc cctgatgctg acgtcctgat tgtctatgcc aagacagatc tggctgctgt
 901 gccagcttct cggggcatca cagccttcat tgtggagaag ggtatgcctg gctttagcac
 961 ctctaagaag ctggacaagc tggggatgag ggctctaac acctgtgagc taatctttga
1021 agactgcaag attcctgctg ccaacatcct gggccatgag aataagggtg tctacgtgct
1081 gatgagtggg ctggacctgg agcggctggt gctggccggg ggcctcttg ggctcatgca
1141 agcggtcctg gaccacacca ttccctacct gcacgtgagg gaagcctttg gccagaagat
1201 cggccacttc cagttgatgc aggggaagat ggctgacatg tacacccgcc tcatggcgtg
1261 tcggcagtat gtctacaatg tcgccaaggc ctgcgatgag ggccattgca ctgctaagga
1321 ctgtgcaggt gtgattcttt actcagctga gtgtgccaca caggtagccc tggacggcat
1381 tcagtgtttt ggtggcaatg gctacatcaa tgactttccc atgggccgct ttcttcgaga
1441 tgccaagctg tatgagatag gggctgggac cagcgaggtg aggcggctgg tcatcggcag
1501 agccttcaat gcagactttc actagtcctg agacccttcg cccccttttc ctgcacctag
1561 tggcctttct tgggaagtag agatgtggcg gctttcccac cctgcccaca gcaggccctc
1621 ctgcccagct gctcttgtca gccctctggc ctctggatga ggttgagttc tccacaacag
1681 ctcccaagca tcatgggcct cgcagccggg cctgtgccac ggctagtgtt gtgtgattta
1741 aaatggactc agcaggaagc atattgtctg gggattgttg ggacaggttt tggtgactct
1801 gtgcccttgc tctctaactt ctgagcccac ctcccagggt aggcacctgg gggcatgcag
1861 gtgcccacct cccagggtag gcacctgggg gcatgcaggt acccacctct ttctcttggg
1921 tgaggctctg gcaaggagat ctctctgctc aagcacagca gaatcatggc ccctctccat
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1981 gaattggaac ttggtacagg ttaagtatcc ctaatcctga aatctgaaac acttgtggtt
2041 ccaagcattt tggataaggc aaattcaact ttcagtctct tttctggggg aaaaaaataa
2101 taaacctagc ctagccaggc gtggtggctc atgcttgtaa tcccagcact tcaggaggct
2161 gagatgggtg gatcacctga ggtcaggagt tcaagaccag cctggccaac atgtggaaac
2221 ctcgcctcaa ctaaaaatag aaaaaaatta gttgggcatg gtggtgggca cctgtaatcc
2281 cagctacttc aggaggctga ggcaggagaa ttacttgaac ccaggaggcg gacgttgcag
2341 tgagccgagc ttgtgccatt gcactccagc ctgggcgaca agagcaaaac tcttcaaaaa
2401 acaaaaacaa acaaaaaaac cctggccctt gtttcttcca gtttctagag gtatcagctc
2461 ctagcagctt atgaacacat atgcttgctt ggccaggcaa ggtggtgtgt gcctgtaatc
2521 ccagcacttt gggaggccaa ggcaggtgga tcacttgcag tcaggagttc aagaccagcc
2581 tgtccaacgt ggtgaaaccc catctctact aaaaatacaa aaattagcca ggggtggtgg
2641 tgcacgtctg taatcccagc tactcaggag gctgaggcag gagaatcact tgaacccggg
2701 aggtggaggt tgcaatgagc caatatgaca ccgctgcagt ccagcctggg ccatagagtg
2761 agactctgtc tcaaaaaagg aaagaaaaat aggctgggca cagtgactca tgcctgtaat
2821 cccaacactt tgggaggccg aggcaggtgg atcacgaggt caggagttca agaccagcct
2881 ggccaagatg gtaaaacctc gtctctacta aaaatacaaa aattagccag gtgtggtggc
2941 aggctcctgt aatcccagct actcaggagg ctgaggcaga gaattgcttg aacccgggag
3001 gcagagtttg cagtgagcca agatcacacc actgcactcc agcttggacg acagagcgag
3061 actctgtctc aaaaaataat aggccaggca tggtggctca acgtctgtaa tcccagcact
3121 ttgggaggcc gaggcgggca gatcacaagg tcaggagttc gagaccagcc tgacgaccaa
3181 catggtgaaa cctcgtctct actaaaaata caaaaattag ccaggcctgg tggcacgcgc
3241 ctgtaatccc agttacacag aagactgagg caggagaatc gcttgaacgc aggaggcaga
3301 ggttgcagga gctgagatcg cgccattgca ctccagcctg ggcaacagag tgagactctg
3361 tctcaaaaaa taataataaa ataaatgaac acacatgctg ctgagtccgc aggggggggca
3421 gagcagagga cagcgtgctt ttgtgtactg ttggaagact ggctcctcct gtacagcacc
3481 tctgagccct tgtgcaccgc cctgccacgg gcaccatcca gtcctggccg tgtgaccacc
3541 cacagctgac tgggcagcag gcacaggccc tacccgagca ggccggagtt ggctcgcatg
3601 actccagctg aggctgcctg tgtacatttc tccagatacc ctatggctaa ttttgttata
3661 actgcacagt ggctgctgcc attttgtatt aaatatattg tgaaacaaac ctatctgggg
3721 agaagcaatc tacttgccgc tgcttcctgt ctggatccag cttgtgtcct tggagagtgg
3781 ctggcccagg tcctattcct gtcctccagc ccgttctttc atgagggaca ggaaggtaaa
3841 atcagccctt aggagagagg tctcagcctc cctttcccag atctcccagt gagttttaaa
3901 ggaagcaggg agcccagagt gctaagttct tacagccaga aggaagctta tagatttctg
3961 aaaaccgccc ctttgttttt aaaaagatca acacaatttg actttctcaa ggtcaaaacg
4021 aactagaatc cagatctgct catggcaaaa atgggggtgt tctgagaatt ccagctttgg
4081 gccgcactgt acagcagtct ggatagagtg tgatctgaga agggaatggg tctgggttgt
4141 tccacccctt ccgagttcca aaaagaggga actggttttc ttggttctca gcccagcagc
4201 acctatcctg gctcttggtc ctggcctgca gccaagtgct gttcctagcc tgaggcttga
4261 gacaggtggg gttggctcct caccaacccc agttccgtcc catcctgagg gcaagatcct
4321 gggctcatag gcagtccctt tcacttcctt gtcttgctcc ctgctatgtt ggagatgaat
4381 gtgactaaaa gggccatctt gctggcttaa tgtgtggctg gagagaccag cctggagaca
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```
4441 atgtggcaaa atggggcgct tcatccagtc tgtctaagcc ctgtcgactt ggggaggtga
4501 tttctttcct ggttctatat gtgaagcaaa ataaatgttt taaaattaaa agcaaaaaaa
4561 acaaaatgaa ccatg
```

In some embodiments of the methods of the disclosure, the wild type human IVD/DISP2 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NP_001152980.1, transcript variant 2):

```
                                                   (SEQ ID NO: 28)
  1  maematatrl lgwrvaswrl rpplagfvsq rahsllpvdd ainglseeqr qefwkqlgnl
 61  gvlgitapvq yggsglgyle hvlvmeeisr asgavglsyg ahsnlcinql vrngneaqke
121  kylpklisge yigalamsep nagsdvvsmk lkaekkgnhy ilngnkfwit ngpdadvliv
181  yaktdlaavp asrgitafiv ekgmpgfsts kkldklgmrg sntcelifed ckipaanilg
241  henkgvyvlm sgldlerlvl aggplglmqa vldhtipylh vreafgqkig hfqlmqgkma
301  dmytrlmacr qyvynvakac deghctakdc agvilysaec atqvaldgiq cfggngyind
361  fpmgrflrda klyeigagts evrrlvigra fnadfh
```

In some embodiments of the methods of the disclosure, the wild type human DPP9 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_139159.4):

```
                                                   (SEQ ID NO: 29)
   1 caacttccgg gtcaaaggtg cctgagccgg cgggtcccct gtgtccgccg cggctgtcgt
  61 cccccgctcc cgccacttcc ggggtcgcag tcccgggcat ggagccgcga ccgtgaggcg
 121 ccgctggacc cgggacgacc tgcccagtcc ggccgccgcc ccacgtcccg gtctgtgtcc
 181 cacgcctgca gctggaatgg aggctctctg gaccctttag aaggcacccc tgccctcctg
 241 aggtcagctg agcggttaat gcggaaggtt aagaaactgc gcctggacaa ggagaacacc
 301 ggaagttgga gaagcttctc gctgaattcc gagggggctg agaggatggc caccaccggg
 361 accccaacgg ccgaccgagg cgacgcagcc gccacagatg acccggccgc ccgcttccag
 421 gtgcagaagc actcgtggga cgggctccgg agcatcatcc acggcagccg caagtactcg
 481 ggcctcattg tcaacaaggc gccccacgac ttccagtttg tgcagaagac ggatgagtct
 541 gggccccact cccaccgcct ctactacctg ggaatgccat atggcagccg agagaactcc
 601 ctcctctact ctgagattcc caagaaggtc cggaaagagg ctctgctgct cctgtcctgg
 661 aagcagatgc tggatcattt ccaggccacg ccccaccatg ggtctactc tcgggaggag
 721 gagctgctga gggagcggaa acgcctgggg gtcttcggca tcacctccta cgacttccac
 781 agcgagagtg gcctcttcct cttccaggcc agcaacagcc tcttccactg ccgcgacggc
 841 ggcaagaacg gcttcatggt gtcccctatg aaaccgctgg aaatcaagac ccagtgctca
 901 gggccccgga tggaccccaa aatctgccct gccgaccctg ccttcttctc cttcatcaat
 961 aacagcgacc tgtgggtggc caacatcgag acaggcgagg agcggcggct gaccttctgc
1021 caccaaggtt tatccaatgt cctggatgac cccaagtctg cgggtgtggc caccttcgtc
1081 atacaggaag agttcgaccg cttcactggg tactggtggt gccccacagc ctcctgggaa
1141 ggttcagagg gcctcaagac gctgcgaatc ctgtatgagg aagtcgatga gtccgaggtg
1201 gaggtcattc acgtcccctc tcctgcgcta gaagaaagga agacggactc gtatcggtac
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1261 cccaggacag gcagcaagaa tcccaagatt gccttgaaac tggctgagtt ccagactgac

1321 agccagggca agatcgtctc gacccaggag aaggagctgg tgcagccctt cagctcgctg

1381 ttcccgaagg tggagtacat cgccagggcc gggtggaccc gggatggcaa atacgcctgg

1441 gccatgttcc tggaccggcc ccagcagtgg ctccagctcg tcctcctccc cccggccctg

1501 ttcatcccga gcacagagaa tgaggagcag cggctagcct ctgccagagc tgtccccagg

1561 aatgtccagc cgtatgtggt gtacgaggag gtcaccaacg tctggatcaa tgttcatgac

1621 atcttctatc ccttccccca atcagaggga gaggacgagc tctgctttct ccgcgccaat

1681 gaatgcaaga ccggcttctg ccatttgtac aaagtcaccg ccgttttaaa atcccagggc

1741 tacgattgga gtgagccctt cagccccggg gaagatgaat ttaagtgccc cattaaggaa

1801 gagattgctc tgaccagcgg tgaatgggag gttttggcga ggcacggctc caagatctgg

1861 gtcaatgagg agaccaagct ggtgtacttc cagggcacca aggacacgcc gctggagcac

1921 cacctctacg tggtcagcta tgaggcggcc ggcgagatcg tacgcctcac cacgcccggc

1981 ttctcccata gctgctccat gagccagaac ttcgacatgt tcgtcagcca ctacagcagc

2041 gtgagcacgc cgccctgcgt gcacgtctac aagctgagcg gccccgacga cgacccccctg

2101 cacaagcagc cccgcttctg ggctagcatg atggaggcag ccagctgccc cccggattat

2161 gttcctccag agatcttcca tttccacacg cgctcggatg tgcggctcta cggcatgatc

2221 tacaagcccc acgccttgca gccagggaag aagcacccca ccgtcctctt tgtatatgga

2281 ggcccccagg tgcagctggt gaataactcc ttcaaaggca tcaagtactt gcggctcaac

2341 acactggcct ccctgggcta cgccgtggtt gtgattgacg gcaggggctc ctgtcagcga

2401 gggcttcggt tcgaaggggc cctgaaaaac caaatgggcc aggtggagat cgaggaccag

2461 gtggagggcc tgcagttcgt ggccgagaag tatggcttca tcgacctgag ccgagttgcc

2521 atccatggct ggtcctacgg gggcttcctc tcgctcatgg ggctaatcca caagccccag

2581 gtgttcaagg tggccatcgc gggtgccccg gtcaccgtct ggatggccta cgacacaggg

2641 tacactgagc gctacatgga cgtccctgag aacaaccagc acggctatga ggcgggttcc

2701 gtggccctgc acgtggagaa gctgcccaat gagcccaacc gcttgcttat cctccacggc

2761 ttcctggacg aaaacgtgca ctttttccac acaaacttcc tcgtctccca actgatccga

2821 gcagggaaac cttaccagct ccagatctac cccaacgaga gacacagtat tcgctgcccc

2881 gagtcgggcg agcactatga agtcacgttg ctgcactttc tacaggaata cctctgagcc

2941 tgcccaccgg gagccgccac atcacagcac aagtggctgc agcctccgcg gggaaccagg

3001 cgggagggac tgagtggccc gcgggcccca gtgaggcact ttgtcccgcc cagcgctggc

3061 cagccccgag gagccgctgc cttcaccgcc ccgacgcctt ttatcctttt ttaaacgctc

3121 ttgggtttta tgtccgctgc ttcttggttg ccgagacaga gagatggtgg tctcgggcca

3181 gcccctcctc tccccgcctt ctgggaggag gaggtcacac gctgatgggc actggagagg

3241 ccagaagaga ctcagaggag cgggctgcct tccgcctggg gctccctgtg acctctcagt

3301 cccctggccc ggccagccac cgtccccagc acccaagcat gcaattgcct gtcccccccg

3361 gccagcctcc ccaacttgat gtttgtgttt tgtttggggg gatatttttc ataattattt

3421 aaaagacagg ccgggcgcgg tggctcacgt ctgtaatccc agcactttgg gaggctgagg

3481 cgggcggatc acctgaggtt gggagttcaa gaccagcctg gccaacatgg ggaaaccccg

3541 tctctactaa aaatacaaaa aattagccgg gtgtggtggc gcgtgcctat aatcccagct

3601 actcgggagg ctgaggcagg agaatcgctt gaacccggga ggtggaggtt gcggtgagcc

3661 aagatcgcac cattgcactc cagcctgggc aacaagagcg aaactctgtc tcaaaataaa

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```
3721 taaaaaataa aagacagaaa gcaaggggtg cctaaatcta gacttggggt ccacaccggg

3781 cagcggggtt gcaacccagc acctggtagg ctccatttct tcccaagccc gagcagaggg

3841 tcatgcgggc cccacaggag aagcggccag ggcccgcggg gggcaccacc tgtggacagc

3901 cctcctgtcc ccaagctttc aggcaggcac tgaaacgcac cgaacttcca cgctctgctg

3961 gtcagtggcg gctgtcccct ccccagccca gccgcccagc cacatgtgtc tgcctgaccc

4021 gtacacacca ggggttccgg ggttgggagc tgaaccatcc ccacctcagg gttatatttc

4081 cctctcccct tccctccccg ccaagagctc tgccaggggc gggcaaaaaa aaaagtaaaa

4141 agaaaagaaa aaaaaaaaaa agaaacaaac cacctctaca tattatggaa agaaaatatt

4201 tttgtcgatt cttattcttt tataattatg cgtggaagaa gtagacacat taaacgattc

4261 cagttggaaa aaaaaaaaaa aaaaaa
```

In some embodiments of the methods of the disclosure, the wild type human DPP9 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_631898.3):

```
                                                        (SEQ ID NO: 30)
  1  mrkvkklrld kentgswrsf slnsegaerm attgtptadr gdaaatddpa arfqvqkhsw

 61  dglrsiihgs rkysglivnk aphdfqfvqk tdesgphshr lyylgmpygs rensllysei

121  pkkvrkeall llswkqmldh fqatphhgvy sreeellrer krlgvfgits ydfhsesglf

181  lfqasnslfh crdggkngfm vspmkpleik tqcsgprmdp kicpadpaff sfinnsdlwv

241  anietgeerr ltfchqglsn vlddpksagv atfviqeefd rftgywwcpt aswegseglk

301  tlrilyeevd esevevihvp spaleerktd syryprtgsk npkialklae fqtdsqgkiv

361  stqekelvqp fsslfpkvey iaragwtrdg kyawamfldr pqqwlqlvll ppalfipste

421  neeqrlasar avprnvqpyv vyeevtnvwi nvhdifypfp qsegedelcf lranecktgf

481  chlykvtavl ksqgydwsep fspgedefkc pikeeialts gewevlarhg skiwvneetk

541  lvyfqgtkdt plehhlyvvs yeaageivrl ttpgfshscs msqnfdmfvs hyssystppc

601  vhvyklsgpd ddplhkqprf wasmmeaasc ppdyvppeif hfhtrsdvrl ygmiykphal

661  qpgkkhptvl fvyggpqvql vnnsfkgiky lrlntlaslg yavvvidgrg scqrglrfeg

721  alknqmgqve iedqveglqf vaekygfidl srvaihgwsy ggflslmgli hkpqvfkvai

781  agapvtvwma ydtgyterym dvpennqhgy eagsvalhve klpnepnrll ilhgfldenv

841  hffhtnflvs qliragkpyq lqiypnerhs ircpesgehy evtllhflqe yl
```

In some embodiments of the methods of the disclosure, the wild type human SIGLEC14 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001098612.1):

```
                                                        (SEQ ID NO: 31)
  1 actcaccctc cggcttcctg tcggggcttt ctcagcccca ccccacgttt ggacatttgg

 61 agcatttcct tccctgacag ccggacctgg gactgggctg gggccctggc ggatggagac

121 atgctgcccc tgctgctgct gcccctgctg tggggggggt ccctgcagga gaagccagtg

181 tacgagctgc aagtgcagaa gtcggtgacg gtgcaggagg gcctgtgcgt ccttgtgccc

241 tgctccttct cttacccctg gagatcctgg tattcctctc ccccactcta cgtctactgg

301 ttccgggacg gggagatccc atactacgct gaggttgtgg ccacaaacaa cccagacaga
```

-continued

```
 361 agagtgaagc cagagaccca gggccgattc cgcctccttg gggatgtcca gaagaagaac
 421 tgctccctga gcatcggaga tgccagaatg gaggacacgg gaagctattt cttccgcgtg
 481 gagagaggaa gggatgtaaa atatagctac caacagaata agctgaactt ggaggtgaca
 541 gccctgatag agaaacccga catccacttt ctggagcctc tggagtccgg ccgcccccaca
 601 aggctgagct gcagccttcc aggatcctgt gaagcgggac cacctctcac attctcctgg
 661 acggggaatg ccctcagccc cctggacccc gagaccaccc gctcctcgga gctcaccctc
 721 acccccaggc ccgaggacca tggcaccaac ctcacctgtc aggtgaaacg ccaaggagct
 781 caggtgacca cggagagaac tgtccagctc aatgtctcct atgctccaca gaacctcgcc
 841 atcagcatct tcttcagaaa tggcacaggc acagccctgc ggatcctgag caatggcatg
 901 tcggtgccca tccaggaggg ccagtccctg ttcctcgcct gcacagttga cagcaacccc
 961 cctgcctcac tgagctggtt ccgggaggga aaagccctca atccttccca gacctcaatg
1021 tctgggaccc tggagctgcc taacatagga gctagagagg gaggggaatt cacctgccgg
1081 gttcagcatc cgctgggctc ccagcacctg tccttcatcc tttctgtgca gagaagctcc
1141 tcttcctgca tatgtgtaac tgagaaacag cagggctcct ggcccctcgt cctcaccctg
1201 atcagggggg ctctcatggg ggctggcttc ctcctcacct atggcctcac ctggatctac
1261 tataccaggt gtggaggccc ccagcagagc agggctgaga ggcctggctg agcccctccc
1321 gctcaagaca gaactgaggt gtggacactt agccctgtgg gacacatgca ggacatcact
1381 gtcagcttct ttctggaagc tcacatccca ctgactaccc ctcttttcct tcctgcccca
1441 taccccttct acttattccc ctctgcttgt gagtcttgcc ccaccacacc tgcatcccca
1501 tctgcacccc atcccctctc cacctgccct tctcttccct ctccatccac catctccagc
1561 cctgtgaagg gaatgtactt tcggtcttat acccccatta cccattaccc aaaagttacc
1621 tttttttttt tttttttttt ttgagacaga gtctcactct gttgcacagg ctggagttca
1681 gtggcacaat ctccgttcac tgcaacctcc acctctgggg ttcaagcaat tctcctgcct
1741 cagcctccct agtagctggg attacaggtg cctgccacca catccagtta attttttttt
1801 tttgtatgtt agtagagatg ggttttacc atgttggcca ggtctcgaac tcctgacctc
1861 aagcaatcca ctgcattggc ctcccaaagt gctggcatta caggtatgag ccaccgtgcc
1921 tggctgccaa aagttacctt cttaacactt gaatttctgg tctcctcagc ttccctatcc
1981 atataggcac agagaggcag catttgtttt ccagttaaaa ctctacctca ttgtgattat
2041 tatccaatac aattgttaca aaataagtaa aacttttatg aaacaataca acataactga
2101 ttttactctt taa
```

In some embodiments of the methods of the disclosure, the wild type human SIGLEC14 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001092082.1):

```
                                                      (SEQ ID NO: 32)
  1   mlpllllpll wggslqekpv yelqvqksvt vqeglcvlvp csfsypwrsw ysspplyvyw
 61   frdgeipyya evvatnnpdr rvkpetqgrf rllgdvqkkn cslsigdarm edtgsyffrv
121   ergrdvkysy qqnklnlevt aliekpdihf leplesgrpt rlscslpgsc eagppltfsw
181   tgnalspldp ettrsseltl tprpedhgtn ltcqvkrqga qvttertvql nvsyapqnla
241   isiffrngtg talrilsngm svpiqegqsl flactvdsnp paslswfreg kalnpsqtsm
301   sgtlelpnig areggeftcr vqhplgsqhl sfilsvqrss sscicvtekq qgswplvltl
361   irgalmgagf lltygltwiy ytrcggpqqs raerpg
```

In some embodiments of the methods of the disclosure, the wild type human ADM2 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_001253845.1):

```
                                                    (SEQ ID NO: 33)
   1 cgcccacgcc cggcgccccg accgcggagg actccccgag ccccgcccgc catggcccgg
  61 atcccgacgg ccgccctggg ttgcatcagc ctcctctgcc tgcagctccc tggctcgctg
 121 tcccgcagcc tgggcgggga cccgcgaccc gtcaaaccca gggagccccc agcccggagc
 181 ccttccagca gcctgcagcc caggcacccc gcaccccgac ctgtggtctg gaagcttcac
 241 cgggccctcc aggcacagag gggtgccggc ctggcccctg ttatgggtca gcctctccgg
 301 gatggtggcc gccaacactc gggcccccga agacactcgg gcccccgcag gacccaagcc
 361 cagctcctgc gagtgggctg tgtgctgggc acctgccagg tgcagaatct cagccaccgc
 421 ctgtggcaac tcatgggacc ggccggccgg caggactcag ctcctgtgga ccccagcagc
 481 ccccacagct atggctgagg tggggccggg ccacacccct gcccatccca gccagggtgc
 541 tgtgccccg tccagagctg cagctgagcc ccatctgaag cccagtccct cggagctgca
 601 gacagcaggt cctgcagcaa caatacctgc acggctttgc acacgtaaac ctaggctggt
 661 ctacacgcag tgctggtacg tcaaggagcc taaacaccct gaaattgtga cccccctgggg
 721 gacagctgcc agacacagct ggcggcagca ccagatgcta agcgcttcag agaggaggtg
 781 tctgcccaga gatgtggagc agaagctggg ccctgaacac acggggccat gtctggacga
 841 gcaggggaga gaggctgaac tggccagaag tggcccctcc gctgctggtc cagtcagact
 901 gaagcccggc cttgtgcctg ggctgttcct gctctcatgc acaaccagcc cttccacgtg
 961 cctgcctgtg ggacaggagg gggagcgtgg gatgctgtag cccccggggt tgggcaaggg
1021 aaggatggtg gccctccaga ggtcatgaag ggacctctgt ggctccagct gccaaccctg
1081 gagcccagac cgaggtggcc atggagactc cacctggatc ccctgtagga ggccagggag
1141 gggaactcag cagttcagga gccaccccaa accattctgg gacagggaca cccctttcta
1201 ccccagggca gggcagggct gggtggggca agatccccca gcccgactag acccacctca
1261 cctgaagggg gtgagaccct tgttggcagc cagacaaggg tggggctcca caggcagcac
1321 aggcgcccca ccaccaccca gtttggggac ccagtgggac caggtgcggg ggcagagggt
1381 gacttaccaa gagccaggga gggcagccca ggcccaagtg acagcaagaa caagaaccac
1441 tgccggcgtg cacagacttg gtgtgtgtcc ttccctgggg ggacggggga ctcacatgtg
1501 cctgccactg gagcctctca accgtccagc agaacacggg gttcagaaag ggctccttct
1561 gctatttagc gaacactgag catttaattt acaaatgttt gctagggtca ccctctcggc
1621 catcccacga gggtcgccat gatcacccca actctagagg ccgcagcaga gctcaggaca
1681 ttcccccaca gagcttgccc ctcagttcct acctccaagg gggagggtcc tggaagcgcc
1741 cacccaggcg ccgcccctgt gcttgctccc cgagctcagg gattgccgag tccacgtaac
1801 tgacctgtac tccacgaggc cctgtgggaa cggtccaggc tggtcctgcc ctgtggaggc
1861 ctccgtgcac tgagagatgt actaggattg cagcaaaggt ggtcagggtg atgggccgca
1921 cagcgaggca gtcaaggcca gctccctggg agaagcactg ggtcaggtga ggtctgagga
1981 cagcaggcct tccctagggg aaggagctgg gagtgccaag gccccaggtg cacaggaggc
2041 gtggctgctg agaggctgca gggtggaggg gcctcggcct cagagtcatg tgccctgtga
2101 ccactgaagg gtgtcagcag agcacacggc atgaggacag agggaggggc acggggagtg
2161 aaggaggggg ccctggggca aggctcgggg gtcaggagct cagcgtccgc tactcagccc
```

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```
2221 agccaaaacc ctcccagacg tctcctctcc tgcctgggca aagtccagct tggcaccccg

2281 tctggggcct gcctgtggtc agggccaagt gttccctcct ccaggaaagc ctttaccctc

2341 ctcatgccct gtagtcagga ggccgcctgc tgtaaccctc cgtgtcgcct cgggtgcgaa

2401 atcagaccca cctgacacca tcacgcggag gcccagcagc acctgcaccc acttccagct

2461 gctctggcca aaatctccgc tcggccaggc cccgtggctc acacctgtaa tcctagcaca

2521 ttgggaggcc aaggcaggca catcacctga gttcaggagt tcaagaccag cctggccaac

2581 atggtgaaat cccgtctcta ctaaaaacag aaaattatcc gggcgtggtg gcacatgact

2641 gtaatcccag ctactcagga ggctgaggca ggaggatcac ttgaacctgg gaggcggagg

2701 ttgcagtgag ctgagattgc gccattgcac tccagcctgg gcaacaagag caaaattctg

2761 cctcaaaaaa aaaaatagta ataatacaaa aattagctgg gcgtggtggc acatgccagt

2821 aattccatct actcgggagg ctgaggcagg agaatcgtct aagcccggga ggtggaggtt

2881 gcagtgagcc cagatggcgc tgctgcactc aagcttggat gacagagcaa gactccgttt

2941 caaaaaaaaa aaacctcctc tcttccttca caccttcctc tgaatcccac ccggtcccac

3001 ctcctgaacc tatccagaca ccttctcctg acccaggcac cacctgcttt cggggcgatg

3061 gccgtagcct cctcccaggc acctgtctgc atccctctgg ccagtgcatg ctgagcacgt

3121 gacctacccg tgttgggaca cgtgaggata cagccttgac cccaggggc tgacattcta

3181 gggggagata gaaggagaca aacgtagaag gtagaataag tgggtggtgg agtggcaggg

3241 agtgctgagt gccacaggaa gtcagacaag gaaggagagt gtggggcagg tgccgtttaa

3301 atggggggcg ctggggtctc ctcacagttg cttctcagct cagctgtgcc aggatcttgt

3361 tgagtcaggt cagctgccca cagccctctt gcctgacccc tgaagcccag aactctgatc

3421 ttcacagccc taggtatggc cccagcaccc cactgccctc tctcctgccc cagccgactg

3481 ctgttcccag acttccctgg ccacgctcca agacgccagc tctgccgcgg gcactttgtt

3541 ctcacggtgt cctccatgcc tgcagggccc atgcatggga agttgcgttg gcggcctggg

3601 tgttggcggt tccgtgcctg ctccaactct ccgtgaggcc cctctcccag agcctgacac

3661 actctgtggc cgaactctag gcaggtgccc ctgagtcctt tcctcgacga ggcctgaccc

3721 catccccatc ctcgctgggc ccgccgaccc cggtgttagc aagaatcctc taaatcagtt

3781 tatggagaat tacccaccct cgatatctga tcccattcct catctcccac ccttgatctc

3841 atcaccctgc cggcctcctg caagatcctc attgagccac tccagtgaga atccccctac

3901 cctcgaaggc cgccctaaca acttcccatc cgctgacccc tccaacgcca tcaatctcca

3961 gctgtggttg ttgaactcgg aggtgagctc ctctcaccac tctcttgaat aaagcttttc

4021 tcaccatttt aaaaaaaaaa aaaaa
```

In some embodiments of the methods of the disclosure, the wild type human ADM2 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_001240774.1):

(SEQ ID NO: 34)

```
  1  mariptaalg cisllclqlp gslsrslggd prpvkprepp arspssslqp rhpaprpvvw
 61  klhralqaqr gaglapvmgq plrdggrqhs gprrhsgprr tqaqllrvgc vlgtcqvqnl
121  shrlwqlmgp agrqdsapvd pssphsyg
```

In some embodiments of the methods of the disclosure, the wild type human TSPAN5 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_005723.3):

(SEQ ID NO: 35)

```
   1 aggcgggcgg agcgaggggt gggagggcgc gcgcgaacgg gcgggcgagc aagcgagcgg
  61 cgtctccacc agcatctgcc gcggccgcct ttgcccgaag cccggggacg aaccgacgga
 121 ccgaccgcct ggcgcacgga cgcgggcgct cgctttgtgt tcggggctag cgtcggcgag
 181 gcttgagctt gcagcgcgcg gcttccctgc tttctcgcgg ccaccccggc tccggcggcc
 241 tcggcgcgcg aggggctgga ggtgcgggag ccgctctccg ccggtcggtc cccgcgcggc
 301 tgagcccagg ccgccagcgc cgcggccccg tgcggtgtcc ctgagctcct gctccccgcc
 361 gggctgctcc gagcaacggt gcttcggagc tccaaactcg ggctgccggg gcaagtgtct
 421 tcatgaaccc agaggatgtc cgggaagcac tacaagggtc ctgaagtcag ttgttgcatc
 481 aaatacttca tatttggctt caatgtcata ttttggtttt tgggaataac atttcttgga
 541 attggactgt gggcatggaa tgaaaaagga gttctgtcca acatctcttc catcaccgat
 601 ctcggcggct ttgacccagt ttggctcttc cttgtggtgg gaggagtgat gttcattttg
 661 ggatttgcag ggtgcattgg agcgctacgg gaaaacactt tccttctcaa gtttttttct
 721 gtgttcctgg gaattatttt cttcctggag ctcactgccg gagttctagc atttgttttc
 781 aaagactgga tcaaagacca gctgtatttc tttataaaca acaacatcag agcatatcgg
 841 gatgacattg atttgcaaaa cctcatagac ttcacccagg aatattggca gtgctgtggg
 901 gcttttggag ctgatgattg gaacctaaat atttacttca attgcacaga ttccaatgca
 961 agtcgagagc gatgtggcgt tccattctcc tgctgcacta aagatcccgc agaagatgtc
1021 atcaacactc agtgtggcta tgatgccagg caaaaaccag aagttgacca gcagattgta
1081 atctacacga aaggctgtgt gccccagttt gagaagtggt tgcaggacaa tttaaccatc
1141 gttgctggta ttttcatagg cattgcattg ctgcagatat ttgggatatg cctggcccag
1201 aatttggtta gcgatatcga agctgtcagg gcgagctggt agacccccctg caaccgctgc
1261 tgcaagacac tggacagacc cagctttcgg gaccctcccg cgtgccgaac tgatcttcga
1321 gctgcatgga cctaatcaca gatgcagcct gcagtctcgc ctaatggagc tgccattagg
1381 ggagtgtaaa actgggaaat gctgctcact gacagaatta aaaaaaaaaa taaccagtat
1441 gaaagtcgtt gcgccgtgaa tctctactgt agccatgaat ttatggacag ttagatgctt
1501 accaaaaaag aaaaaaaggg agggtagggg acccagatgt acttgaatgt gcagaaaata
1561 cattcttgtc ctcatcttcc gtaattggag ggctgggaga ggcagctttg ctcttcacca
1621 caccttggac ggaccacctt ctttctgttc catggcctga aggagtgcat ctcctcaaag
1681 actcagcccc tcacctggga gggcagtggt ttgtgggcat ccctccatgt acattttagg
1741 aaacacttgc aactctcatc tgaagaagaa aacaactcat ctttgggttc agattttgtg
```

-continued

```
1801 atggtattca gcaagtcact tgggcgagca cacttggtct atcctggaaa gtctccttat
1861 aagagaagtt gtgtatttca tgtgcaccga gcaagggcat tggaagacgt catgaggctg
1921 tattttagca ggactgatcg tttttctaag tagacctgag ctttgtttat cagtgaaatt
1981 caaggagaaa atgaggttaa tgaagaggta tcagttaaat atccccttct tctcaccctg
2041 ccaaaattag cagttggatt tttggaaact ctggaatatt ctgggtcatt ttgttttgta
2101 tgtttgttgt ttttcgtctt ccaaaggtga aagctatgat acagttccac ttaaatttta
2161 gtgttttctt actcagctca agcattaatt tttgattaag tcttaatctg catgacctgt
2221 gaatctgaat ccatcatctc cctttcctgc cagcttttct acaaacattg aaatatgtta
2281 tttggtcagc acttatttcc taggttcaca gccttgggag gttgtggcat gtcctcccag
2341 tctggctggg aagagaccag ctgtaccatc caaatgcttc cctggtcttg atgatctctt
2401 ccagagtcga tctgagtggc cttttctgca ccctcccctt ctttctcttt gaatggaatt
2461 aaacccaatt tggaaacaac attgacccag tcaaaagctt ctaatggttt ctttttcttc
2521 ctccagtttt agtttgcttt tattaaaaaa agaaaatagt gcatggccat agctccttca
2581 gttctcttat tgcagactaa ccatcaggat ggtatcaaag cacaaatact ttggagggga
2641 atgcgttgaa ctggggcaag tactctgtaa cacaaagtgg gaaaccactt cctggtgctg
2701 ccgctcctgc ccccacttta ggtgggaggg acgagttttg ccctctagat tttaatccag
2761 ctggtgtcca ccggatgttg ccctcctggg gagcagatat cagtctgtgg aactctggga
2821 aaaccacagg cacatttttc ggtgcggaca gatttgccag cacataactg ggcagccagc
2881 tagaatactt tgtggaaatt aagcgaggtt ttccatttca gccccatggt gcatggtggt
2941 ggccgatgaa tgtgtcagtc tgctcagaga aaggacaaaa aggaaattat tttcaaaact
3001 gtgttcactg tttgggtgtg tgtatggctc tgcatgtgtg tgtttttgtc tctgtatagg
3061 tagaggtatt cacatcttac tccgactgta aggttgtctt acttcatctc tgccccccacc
3121 acagttgcca ttttgtaatg tccttccaac atggagaaga cacgagctct ctccagttgg
3181 catcatttgt ctttttttgtt gattgcctca ttctccagtg aactccatct ggccaattga
3241 ttcagaatca ggcaagatcc ctgccctttg gcacatccac tgaaaggcca aacagcaagt
3301 ccgagtgagt tttaaatatt aattaatcac cctttatttt ttacacttga gagtgattgt
3361 aataaaggct gtcattaata aacttggttc taccttaaaa aaaaaa
```

In some embodiments of the methods of the disclosure, the wild type human TSPAN5 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_005714.2):

```
                                                      (SEQ ID NO: 52)
  1  msgkhykgpe vsccikyfif gfnvifwflg itflgiglwa wnekgvlsni ssitdlggfd

 61  pvwlflvvgg vmfilgfagc igalrentfl lkffsvflgi iffleltagv lafvfkdwik

121  dqlyffinnn irayrddidl qnlidftqey wqccgafgad dwnlniyfnc tdsnasrerc

181  gvpfscctkd paedvintqc gydarqkpev dqqiviytkg cvpqfekwlq dnltivagif

241  igiallqifg iclaqnlvsd ieavrasw
```

In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_032294.2, transcript variant 1):

(SEQ ID NO: 53)

```
   1 ctgggcccca gcgaggcggt ggggcggggc ggggcggggc ggggcgcgca gcaggagcga
  61 gtggggccgc ccgccgggcc gcggacactg tcgcccggcg cccaggttcc caacaaggct
 121 acgcagaaga acccccttga ctgaagcaat ggaggggggt ccagctgtct gctgccagga
 181 tcctcgggca gagctggtag aacgggtggc agccatcgat gtgactcact tggaggaggc
 241 agatggtggc ccagagccta ctagaaacgg tgtggacccc ccaccacggg ccagagctgc
 301 ctctgtgatc cctggcagta cttcaagact gctcccagcc cggcctagcc tctcagccag
 361 gaagctttcc ctacaggagc ggccagcagg aagctatctg gaggcgcagg ctgggcctta
 421 tgccacgggg cctgccagcc acatctcccc ccgggcctgg cggaggccca ccatcgagtc
 481 ccaccacgtg gccatctcag atgcagagga ctgcgtgcag ctgaaccagt acaagctgca
 541 gagtgagatt ggcaagggtg cctacggtgt ggtgaggctg gcctacaacg aaagtgaaga
 601 cagacactat gcaatgaaag tcctttccaa aaagaagtta ctgaagcagt atggctttcc
 661 acgtcgccct cccccgagag ggtcccaggc tgcccaggga ggaccagcca agcagctgct
 721 gcccctggag cgggtgtacc aggagattgc catcctgaag aagctggacc acgtgaatgt
 781 ggtcaaactg atcgaggtcc tggatgaccc agctgaggac aacctctatt tggtgtttga
 841 cctcctgaga aaggggcccg tcatggaagt gccctgtgac aagcccttct cggaggagca
 901 agctcgcctc tacctgcggg acgtcatcct gggcctcgag tacttgcact gccagaagat
 961 cgtccacagg gacatcaagc catccaacct gctcctgggg gatgatgggc acgtgaagat
1021 cgccgacttt ggcgtcagca accagtttga ggggaacgac gctcagctgt ccagcacggc
1081 gggaacccca gcattcatgg cccccgaggc catttctgat tccggccaga gcttcagtgg
1141 gaaggccttg gatgtatggg ccactggcgt cacgttgtac tgctttgtct atgggaagtg
1201 cccattcatc gacgatttca tcctggccct ccacaggaag atcaagaatg agcccgtggt
1261 gtttcctgag gagccagaaa tcagcgagga gctcaaggac ctgatcctga agatgttaga
1321 caagaatccc gagacgagaa ttggggtgcc agacatcaag ttgcaccctt gggtgaccaa
1381 gaacggggag gagccccttc cttcggagga ggagcactgc agcgtggtgg aggtgacaga
1441 ggaggaggtt aagaactcag tcaggctcat ccccagctgg accacggtga tcctggtgaa
1501 gtccatgctg aggaagcgtt cctttgggaa cccgtttgag ccccaagcac ggagggaaga
1561 gcgatccatg tctgctccag gaaacctact ggtgaaagaa gggtttggtg aaggggggcaa
1621 gagcccagag ctccccggcg tccaggaaga cgaggctgca tcctgagccc ctgcatgcac
1681 ccagggccac ccggcagcac actcatcccg cgcctccaga ggcccacccc tcatgcaaca
1741 gccgcccccg caggcagggg gctggggact gcagccccac tcccgcccct cccccatcgt
1801 gctgcatgac ctccacgcac gcacgtccag ggacagactg gaatgtatgt catttggggt
1861 cttgggggca gggctcccac gaggccatcc tcctcttctt ggacctcctt ggcctgaccc
1921 attctgtggg gaaaccgggt gcccatggag cctcagaaat gccacccggc tggttggcat
1981 ggcctggggc aggaggcaga ggcaggagac caagatggca ggtggaggcc aggcttacca
2041 caacggaaga gacctcccgc tggggccggg caggcctggc tcagctgcca caggcatatg
2101 gtggagaggg gggtaccctg cccaccttgg ggtggtggca ccagagctct tgtctattca
2161 gacgctggta tgggggctcg gacccctcac tggggacagg gccagtgttg gagaattctg
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2221 attccttttt tgttgtcttt tacttttgtt tttaacctgg gggttcgggg agaggccctg
2281 cttgggaaca tctcacgagc tttcctacat cttccgtggt tcccagcaca gcccaagatt
2341 atttggcagc caagtggatg gaactaactt tcctggactg tgtttcgcat tcggcgttat
2401 ctggaaagtg gactgaacgg aatcaagctc tgagcagagg cctgaagcgg aagcaccaca
2461 tcgtccctgc ccatctcact ctctcccttg atgatgcccc tagagctgag gctggagaag
2521 acaccagggc tgactttgac cgagggccat ggacgcgaca ggcctgtggc cctgcgcatg
2581 ctgaaataac tggaacccag cctctcctcc tacaccggcc tacccatctg ggcccaagag
2641 ctgcactcac actcctacaa cgaaggacaa actgtccagg tcggagggat cacgagacac
2701 agaacctgga ggggtgtgca cgctggcagg tggcctctgc ggcaattgcc tcaccctgag
2761 gacatcagca gtcagcctgc tcagagcggg ggtgctggag cgcgtgcaga cacagctctt
2821 ccggagcagc cttcaccttc tctctgggat cagtgtccgg ctggccgacg tggcatttgc
2881 tgaccgaatg ctcatagagg ttgaccccca cagggtcacg caggactcgg acactgccct
2941 ggaaacatgg atggacaagg gcttttggcc acaggtgtgg gtgtcctgtt ggaggagggc
3001 ttgtttggag aagggaggct ggctggggga gaaacccgga tcccgctgca tctccgcgcc
3061 tgtgggtgca tgtcgcgtgc tcatctgttg cacacagctc actcgtatgt cctgcactgg
3121 tacatgcatc tgtaatacag tttctacgtc tatttaaggc taggagccga atgtgcccca
3181 ttgtcagtgg gtccacgttt ctccccggct cctctgggct aaggcagtgt ggcccgaagc
3241 ttaaaaagtt actcggtact gtttttaaga acacttttat agagttagtg gaaggcaagt
3301 taagagccaa tcactgatcc ccaagtgttt cttgagcatc tggtctgggg ggaccacttt
3361 gatcggaccc acccttggaa agctcagggg taggcccagg tgggatgctc accctgtcac
3421 tgagggtttt ggttggcatc gttgtttttg aatgtagcac aagcgatgag caaactctat
3481 aagagtgttt taaaaattaa cttcccagga agtgagttaa aaacaataaa agccctttct
3541 tgagttaaaa agaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaa
```

In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_115670.1, transcript variant 1):

```
                                                    (SEQ ID NO: 54)
  1  meggpavccq dpraelverv aaidvthlee adggpeptrn gvdppprara asvipgstsr
 61  llparpslsa rklslqerpa gsyleaqagp yatgpashis prawrrptie shhvaisdae
121  dcvqlnqykl qseigkgayg vvrlaynese drhyamkvls kkkllkqygf prrppprgsq
181  aaqggpakql lplervyqei ailkkldhvn vvklievldd paednlylvf dllrkgpvme
241  vpcdkpfsee qarlylrdvi lgleylhcqk ivhrdikpsn lllgddghvk iadfgvsnqf
301  egndaqlsst agtpafmape aisdsgqsfs gkaldvwatg vtlycfvygk cpfiddfila
361  lhrkiknepv vfpeepeise elkdlilkml dknpetrigv pdiklhpwvt kngeeplpse
421  eehcsvvevt eeevknsvrl ipswttvilv ksmlrkrsfg npfepqarre ersmsapgnl
481  lvkegfgegg kspelpgvqe deaas
```

In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_172206.1, transcript variant 2):

(SEQ ID NO: 55)

```
   1 agcagaacag agtatgcaat ttgggaagct gtggtgtggc tgcagtggag agttcccaac
  61 aaggctacgc agaagaaccc ccttgactga agcaatggag gggggtccag ctgtctgctg
 121 ccaggatcct cgggcagagc tggtagaacg ggtggcagcc atcgatgtga ctcacttgga
 181 ggaggcagat ggtggcccag agcctactag aaacggtgtg gaccccccac cacgggccag
 241 agctgcctct gtgatccctg gcagtacttc aagactgctc ccagcccggc ctagcctctc
 301 agccaggaag ctttccctac aggagcggcc agcaggaagc tatctggagg cgcaggctgg
 361 gccttatgcc acggggcctg ccagccacat ctccccccgg gcctggcgga ggcccaccat
 421 cgagtcccac cacgtggcca tctcagatgc agaggactgc gtgcagctga accagtacaa
 481 gctgcagagt gagattggca agggtgccta cggtgtggtg aggctggcct acaacgaaag
 541 tgaagacaga cactatgcaa tgaaagtcct ttccaaaaag aagttactga agcagtatgg
 601 ctttccacgt cgccctcccc cgagagggtc ccaggctgcc cagggaggac cagccaagca
 661 gctgctgccc ctggagcggg tgtaccagga gattgccatc ctgaagaagc tggaccacgt
 721 gaatgtggtc aaactgatcg aggtcctgga tgacccagct gaggacaacc tctatttggt
 781 gtttgacctc ctgagaaagg ggcccgtcat ggaagtgccc tgtgacaagc ccttctcgga
 841 ggagcaagct cgcctctacc tgcgggacgt catcctgggc ctcgagtact tgcactgcca
 901 gaagatcgtc cacagggaca tcaagccatc caacctgctc ctgggggatg atgggcacgt
 961 gaagatcgcc gactttggcg tcagcaacca gtttgagggg aacgacgctc agctgtccag
1021 cacggcggga accccagcat tcatggcccc cgaggccatt tctgattccg gccagagctt
1081 cagtgggaag gccttggatg tatgggccac tggcgtcacg ttgtactgct ttgtctatgg
1141 gaagtgccca ttcatcgacg atttcatcct ggccctccac aggaagatca agaatgagcc
1201 cgtggtgttt cctgaggagc cagaaatcag cgaggagctc aaggacctga tcctgaagat
1261 gttagacaag aatcccgaga cgagaattgg ggtgccagac atcaagttgc acccttgggt
1321 gaccaagaac ggggaggagc cccttccttc ggaggaggag cactgcagcg tggtggaggt
1381 gacagaggag gaggttaaga actcagtcag gctcatcccc agctggacca cggtgatcct
1441 ggtgaagtcc atgctgagga agcgttcctt tgggaacccg tttgagcccc aagcacggag
1501 ggaagagcga tccatgtctg ctccaggaaa cctactggtg aaagaagggt ttggtgaagg
1561 gggcaagagc ccagagctcc ccggcgtcca ggaagacgag gctgcatcct gagcccctgc
1621 atgcacccag ggccacccgg cagcacactc atcccgcgcc tccagaggcc cacccctcat
1681 gcaacagccg cccccgcagg caggggggctg ggactgcag ccccactccc gcccctcccc
1741 catcgtgctg catgacctcc acgcacgcac gtccagggac agactggaat gtatgtcatt
1801 tggggtcttg ggggcagggc tcccacgagg ccatcctcct cttcttggac ctccttggcc
1861 tgacccattc tgtggggaaa ccgggtgccc atggagcctc agaaatgcca cccggctggt
1921 tggcatggcc tggggcagga ggcagaggca ggagaccaag atggcaggtg gaggccaggc
1981 ttaccacaac ggaagagacc tcccgctggg gccgggcagg cctggctcag ctgccacagg
2041 catatggtgg agagggggggt accctgccca ccttggggtg gtggcaccag agctcttgtc
2101 tattcagacg ctggtatggg ggctcggacc cctcactggg gacagggcca gtgttggaga
2161 attctgattc cttttttgtt gtcttttact tttgttttta acctgggggt tcggggagag
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2221 gccctgcttg ggaacatctc acgagctttc ctacatcttc cgtggttccc agcacagccc
2281 aagattattt ggcagccaag tggatggaac taactttcct ggactgtgtt tcgcattcgg
2341 cgttatctgg aaagtggact gaacggaatc aagctctgag cagaggcctg aagcggaagc
2401 accacatcgt ccctgcccat ctcactctct cccttgatga tgcccctaga gctgaggctg
2461 gagaagacac cagggctgac tttgaccgag ggccatggac gcgacaggcc tgtggccctg
2521 cgcatgctga aataactgga acccagcctc tcctcctaca ccggcctacc catctgggcc
2581 caagagctgc actcacactc ctacaacgaa ggacaaactg tccaggtcgg agggatcacg
2641 agacacagaa cctggagggg tgtgcacgct ggcaggtggc ctctgcggca attgcctcac
2701 cctgaggaca tcagcagtca gcctgctcag agcgggggtg ctggagcgcg tgcagacaca
2761 gctcttccgg agcagccttc accttctctc tgggatcagt gtccggctgg ccgacgtggc
2821 atttgctgac cgaatgctca tagaggttga cccccacagg gtcacgcagg actcggacac
2881 tgccctggaa acatggatgg acaagggctt ttggccacag gtgtgggtgt cctgttggag
2941 gagggcttgt ttggagaagg gaggctggct gggggagaaa cccggatccc gctgcatctc
3001 cgcgcctgtg ggtgcatgtc gcgtgctcat ctgttgcaca cagctcactc gtatgtcctg
3061 cactggtaca tgcatctgta atacagtttc tacgtctatt taaggctagg agccgaatgt
3121 gccccattgt cagtgggtcc acgtttctcc ccggctcctc tgggctaagg cagtgtggcc
3181 cgaagcttaa aaagttactc ggtactgttt ttaagaacac ttttatagag ttagtggaag
3241 gcaagttaag agccaatcac tgatccccaa gtgtttcttg agcatctggt ctggggggac
3301 cactttgatc ggacccaccc ttggaaagct caggggtagg cccaggtggg atgctcaccc
3361 tgtcactgag ggttttggtt ggcatcgttg tttttgaatg tagcacaagc gatgagcaaa
3421 ctctataaga gtgttttaaa aattaacttc ccaggaagtg agttaaaaac aataaaagcc
3481 ctttcttgag ttaaaaagaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaa
```

In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_757343.2, transcript variant 2):

```
                                                    (SEQ ID NO: 56)
  1 mqfgklwcgc sgefptrlrr rtplteameg gpavccqdpr aelvervaai dvthleeadg

 61 gpeptrngvd pppraraasv ipgstsrllp arpslsarkl slgerpagsy leaqagpyat

121 gpashispra wrrptieshh vaisdaedcv qlnqyklqse igkgaygvvr laynesedrh

181 yamkvlskkk llkqygfprr ppprgsqaaq ggpakqllpl ervyqeiail kkldhvnvvk

241 lievlddpae dnlylvfdll rkgpvmevpc dkpfseeqar lylrdvilgl eylhcqkivh

301 rdikpsnlll gddghvkiad fgvsnqfegn daqlsstagt pafmapeais dsgqsfsgka

361 ldvwatgvtl ycfvygkcpf iddfilalhr kiknepvvfp eepeiseelk dlilkmldkn

421 petrigvpdi klhpwvtkng eeplpseeeh csvvevteee vknsvrlips wttvilvksm

481 lrkrsfgnpf epqarreers msapgnllvk egfgeggksp elpgvqedea as
```

In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_172207.2, transcript variant 3):

(SEQ ID NO: 57)

```
   1 ctgggcccca gcgaggcggt ggggcggggc ggggcggggc ggggcgcgca gcaggagcga
  61 gtggggccgc ccgccgggcc gcggacactg tcgcccggcg cccaggttcc caacaaggct
 121 acgcagaaga acccccttga ctgaagcaat ggaggggggt ccagctgtct gctgccagga
 181 tcctcgggca gagctggtag aacgggtggc agccatcgat gtgactcact tggaggaggc
 241 agatggtggc ccagagccta ctagaaacgg tgtggacccc ccaccacggg ccagagctgc
 301 ctctgtgatc cctggcagta cttcaagact gctcccagcc cggcctagcc tctcagccag
 361 gaagctttcc ctacaggagc ggccagcagg aagctatctg gaggcgcagg ctgggcctta
 421 tgccacgggg cctgccagcc acatctcccc ccgggcctgg cggaggccca ccatcgagtc
 481 ccaccacgtg gccatctcag atgcagagga ctgcgtgcag ctgaaccagt acaagctgca
 541 gagtgagatt ggcaagggtg cctacggtgt ggtgaggctg gcctacaacg aaagtgaaga
 601 cagacactat gcaatgaaag tcctttccaa aaagaagtta ctgaagcagt atggctttcc
 661 acgtcgccct cccccgagag ggtcccaggc tgcccaggga ggaccagcca agcagctgct
 721 gcccctggag cgggtgtacc aggagattgc catcctgaag aagctggacc acgtgaatgt
 781 ggtcaaactg atcgaggtcc tggatgaccc agctgaggac aacctctatt tggccctgca
 841 gaaccaggcc cagaatatcc agttagattc aacaaatatc gccaagcccc actccctgct
 901 tccctctgag cagcaagaca gtggatccac gtgggctgcg cgctcagtgt ttgacctcct
 961 gagaaagggg cccgtcatgg aagtgccctg tgacaagccc ttctcggagg agcaagctcg
1021 cctctacctg cgggacgtca tcctgggcct cgagtacttg cactgccaga agatcgtcca
1081 cagggacatc aagccatcca acctgctcct gggggatgat gggcacgtga agatcgccga
1141 ctttggcgtc agcaaccagt ttgaggggaa cgacgctcag ctgtccagca cggcgggaac
1201 cccagcattc atggcccccg aggccatttc tgattccggc cagagcttca gtgggaaggc
1261 cttggatgta tgggccactg gcgtcacgtt gtactgcttt gtctatggga agtgcccatt
1321 catcgacgat ttcatcctgg ccctccacag gaagatcaag aatgagcccg tggtgtttcc
1381 tgaggagcca gaaatcagcg aggagctcaa ggacctgatc ctgaagatgt tagacaagaa
1441 tcccgagacg agaattgggg tgccagacat caagttgcac ccttgggtga ccaagaacgg
1501 ggaggagccc cttccttcgg aggaggagca ctgcagcgtg gtggaggtga cagaggagga
1561 ggttaagaac tcagtcaggc tcatccccag ctggaccacg gtgatcctgg tgaagtccat
1621 gctgaggaag cgttcctttg ggaacccgtt tgagccccaa gcacggaggg aagagcgatc
1681 catgtctgct ccaggaaacc tactggtgta agtactggtg ggccagggac tgccgggcac
1741 tccctggagt tgggtgggga ggtctgaggc ccatcctccc actctcactg tcgttgggcc
1801 aaggccagag cctggggact tggccaggtc tcggtgttgg ccccatttgc atctctgtcc
1861 ccaaggttag tcggggctag aagggacctt ttgggcccag ctcttgcttc attcctgggg
1921 ccagcatccc tcacacacac acttccaggg atgaggagct cacgcagccc ctccatggga
1981 caggaagacc cttcttccat gcagcttgat gtcactctct cactgggtcc agcccctctg
2041 gggcttcaaa tctgtggccc cctcagccct tggcagcctg gcagaggttt gcagacaggc
2101 tgatgttggc ttcctgtagg aggctggcgg gctgtagagg aggggtgctg gcccctctgc
2161 ctggccctgg ggactgttgg ctgctctccc aagtggccca ggctgcctgc agccattgct
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2221 ggggctctgt gcccagtcag cactttgtga gtgcttgttc agtgagtaag cagggacagg

2281 ctggccggtg gaccacggga gaggaacccg cattggccga gggctcccta tggtgagcca

2341 cgcctgtggg ttcaccacct cctaggaggg tccagaaaag cagctcccca agcctgtgcg

2401 cctcgtcctc agcagatcca ccttcttcac tataataaaa gccagtctgg gatgctaaaa

2461 aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa

2521 aaaaaaaaaa aaaaa
```

In some embodiments of the methods of the disclosure, the wild type human CAMKK1 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_757344.2, transcript variant 3):

```
                                                      (SEQ ID NO: 58)
  1  meggpavccq dpraelverv aaidvthlee adggpeptrn gvdppprara asvipgstsr

 61  llparpslsa rklslqerpa gsyleaqagp yatgpashis prawrrptie shhvaisdae

121  dcvqlnqykl qseigkgayg vvrlaynese drhyamkvls kkkllkqygf prrppprgsq

181  aaqggpakql lplervyqei ailkkldhvn vvklievldd paednlylal qnqaqniqld

241  stniakphsl lpseqqdsgs twaarsvfdl lrkgpvmevp cdkpfseeqa rlylrdvilg

301  leylhcqkiv hrdikpsnll lgddghvkia dfgvsnqfeg ndaqlsstag tpafmapeai

361  sdsgqsfsgk aldvwatgvt lycfvygkcp fiddfilalh rkiknepvvf peepeiseel

421  kdlilkmldk npetrigvpd iklhpwvtkn geeplpseee hcsvvevtee evknsvrlip

481  swttvilvks mlrkrsfgnp fepqarreer smsapgnllv
```

In some embodiments of the methods of the disclosure, the wild type human MMP7 gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NM_002423.4):

```
                                                      (SEQ ID NO: 59)
  1 gaaaacacca aatcaaccat aggtccaaga acaattgtct ctggacggca gctatgcgac

 61 tcaccgtgct gtgtgctgtg tgcctgctgc ctggcagcct ggccctgccg ctgcctcagg

121 aggcgggagg catgagtgag ctacagtggg aacaggctca ggactatctc aagagatttt

181 atctctatga ctcagaaaca aaaaatgcca acagtttaga agccaaactc aaggagatgc

241 aaaaattctt tggcctacct ataactggaa tgttaaactc ccgcgtcata gaaataatgc

301 agaagcccag atgtggagtg ccagatgttg cagaatactc actatttcca aatagcccaa

361 aatggacttc caaagtggtc acctacagga tcgtatcata tactcgagac ttaccgcata

421 ttacagtgga tcgattagtg tcaaaggctt taaacatgtg ggcaaagag atccccctgc

481 atttcaggaa agttgtatgg ggaactgctg acatcatgat tggctttgcg cgaggagctc

541 atggggactc ctacccattt gatgggccag gaaacacgct ggctcatgcc tttgcgcctg

601 ggacaggtct cggaggagat gctcacttcg atgaggatga acgctggacg gatggtagca

661 gtctagggat taacttcctg tatgctgcaa ctcatgaact tggccattct ttgggtatgg

721 gacattcctc tgatcctaat gcagtgatgt atccaaccta tggaaatgga gatccccaaa

781 attttaaact ttcccaggat gatattaaag gcattcagaa actatatgga aagagaagta

841 attcaagaaa gaaatagaaa cttcaggcag aacatccatt cattcattca ttggattgta

901 tatcattgtt gcacaatcag aattgataag cactgttcct ccactccatt tagcaattat

961 gtcacccttt tttattgcag ttggtttttg aatgtctttc actcctttta aggataaact
```

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```
1021 cctttatggt gtgactgtgt cttattcatc tatacttgca gtgggtagat gtcaataaat

1081 gttacataca caaataaata aaatgtttat tccatggtaa atttaaaaaa aaaaaaaaaa

1141 aaaaaaaaaa aaa
```

In some embodiments of the methods of the disclosure, the wild type human MMP7 gene of the disclosure consists of or comprises the amino acid sequence (Genbank Accession number: NP_002414.1):

```
                                                        (SEQ ID NO: 60)
  1  mrltvlcavc llpgslalpl pqeaggmsel qweqaqdylk rfylydsetk nansleaklk

 61  emqkffglpi tgmlnsrvie imqkprcgvp dvaeyslfpn spkwtskvvt yrivsytrdl

121  phitvdrlvs kalnmwgkei plhfrkvvwg tadimigfar gahgdsypfd gpgntlahaf

181  apgtglggda hfdederwtd gsslginfly aathelghsl gmghssdpna vmyptygngd

241  pqnfklsqdd ikgiqklygk rsnsrkk
```

In some embodiments of the methods of the disclosure, the wild type human TERC gene of the disclosure consists of or comprises the nucleic acid sequence (Genbank Accession number: NR_001566.1):

```
                                                        (SEQ ID NO: 61)
  1  gggttgcgga gggtgggcct gggaggggtg gtggccattt tttgtctaac cctaactgag

 61  aagggcgtag gcgccgtgct tttgctcccc gcgcgctgtt tttctcgctg actttcagcg

121  ggcggaaaag cctcggcctg ccgccttcca ccgttcattc tagagcaaac aaaaaatgtc

181  agctgctggc ccgttcgccc ctcccgggga cctgcggcgg gtcgcctgcc cagcccccga

241  accccgcctg gaggccgcgg tcggcccggg gcttctccgg aggcacccac tgccaccgcg

301  aagagttggg ctctgtcagc cgcgggtctc tcgggggcga gggcgaggtt caggcctttc

361  aggccgcagg aagaggaacg gagcgagtcc ccgcgcgcgg cgcgattccc tgagctgtgg

421  gacgtgcacc caggactcgg ctcacacatg c
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Definitions

The following definitions are included for the purpose of understanding the present subject matter and for constructing the appended patent claims. Abbreviations used herein have their conventional meaning within the chemical and biological arts.

Unless defined otherwise, technical and scientific terms used herein have the same meaning as commonly understood by a person of ordinary skill in the art. Any methods, devices and materials similar or equivalent to those described herein can be used in the practice of this disclosure. The following definitions are provided to facilitate understanding of certain terms used frequently herein and are not meant to limit the scope of the present disclosure.

As used herein, the term "FILD" refers to fibrotic interstitial lung disease.

As used herein, the term "FIP" refers to Familial Interstitial Pneumonia.

As used herein, the term "HRCT" refers to high-resolution CT (HRCT).

As used herein, the term "ILA" refers to asymptomatic interstitial lung abnormalities.

As used herein, the term "IPF" refers to idiopathic pulmonary fibrosis.

As used herein, the term "PBMC" refers to peripheral blood mononuclear cell.

As used herein, the term "alleviate" is meant to describe a process by which the severity of a sign or symptom of a disorder is decreased. Importantly, a sign or symptom can be alleviated without being eliminated. In a preferred embodiment, the administration of pharmaceutical compositions disclosed herein leads to the elimination of a sign or symptom, however, elimination is not required. Effective dosages are expected to decrease the severity of a sign or symptom. A sign is an objective indication of a medical condition that is observable or detectable by a medical professional or lay person (e.g. family member) (for example, with respect to fibrotic pulmonary disease, signs include, but are not limited to, changes in body weight, changes in body temperature and the presence of a fibrotic lesion in one or both lungs detectable by radiography).

A symptom is an indication of disease that may be a sign but may also be exclusively observable or subjectively experienced by the subject (for example, with respect to fibrotic pulmonary disease, symptoms may include but are not limited to, a dry or hacking cough, a sore throat, a tight chest, shortness of breath, and a feeling of exhaustion or malaise).

In one aspect, the terms "co-administered" and "co-administration" as relating to a subject refer to administering to the subject a compound of the invention or salt thereof along with a compound that may also treat the disorders or diseases contemplated within the invention. In one embodiment, the co-administered compounds are administered separately, or in any kind of combination as part of a single therapeutic approach. The co-administered compound may be formulated in any kind of combinations as mixtures of solids and liquids under a variety of solid, gel, and liquid formulations, and as a solution.

As used herein, the term "composition" or "pharmaceutical composition" refers to a mixture of at least one compound useful within the disclosure with a pharmaceutically acceptable carrier. The pharmaceutical composition facilitates administration of the compound to a patient or subject. Multiple techniques of administering a compound exist in the art including, but not limited to, intravenous, oral, aerosol, parenteral, ophthalmic, nasal, pulmonary and topical administration.

A "disease" as used herein is a state of health of an animal or subject wherein the animal or subject cannot maintain homeostasis, and wherein if the disease is not ameliorated then the animal's or subject's health continues to deteriorate.

A "disorder" as used herein in an animal is a state of health in which the animal or subject is able to maintain homeostasis, but in which the animal's or subject's state of health is less favorable than it would be in the absence of the disorder. Left untreated, a disorder does not necessarily cause a further decrease in the animal's or subject's state of health.

As used herein, the terms "effective amount," "pharmaceutically effective amount" and "therapeutically effective amount" refer to a nontoxic but sufficient amount of an agent to provide the desired biological result. That result may be reduction and/or alleviation of the signs, symptoms, or causes of a disease, or any other desired alteration of a biological system. An appropriate therapeutic amount in any individual case may be determined by one of ordinary skill in the art using routine experimentation.

As used herein, the term "fibrotic lung disease" or "fibroid lung disease" or "pulmonary fibrosis" or "scarring of the lung" refers to a group of diseases characterized by the formation or development of excess fibrous connective tissue (fibrosis) in the lungs. Symptoms of pulmonary fibrosis are mainly: shortness of breath, particularly with exertion; chronic dry, hacking coughing; fatigue and weakness; chest discomfort; and loss of appetite and rapid weight loss. Pulmonary fibrosis may be a secondary effect of other diseases, most of them being classified as interstitial lung diseases, such as autoimmune disorders, viral infections or other microscopic injuries to the lung. Pulmonary fibrosis can also appear without any known cause ("idiopathic"). Idiopathic pulmonary fibrosis is a diagnosis of exclusion of a characteristic set of histologic/pathologic features known as usual interstitial pneumonia (UIP).

Diseases and conditions that may cause pulmonary fibrosis as a secondary effect include: inhalation of environmental and occupational pollutants (asbestosis, silicosis and gas exposure); hypersensitivity pneumonitis, most often resulting from inhaling dust contaminated with bacterial, fungal, or animal products; cigarette smoking; connective tissue diseases such as rheumatoid arthritis, SLE; scleroderma, sarcoidosis and Wegener's granulomatosis; infections; medications such as amiodarone, bleomycin (pingyangmycin), busulfan, methotrexate, apomorphine and nitrofurantoin; and radiation therapy to the chest.

As used herein, a "subject in need thereof" is a subject suffering from fibrotic lung disease relative to the population at large. For example, the subject is a patient who is or is about to be administered with comprising administering to the subject an effective amount of a therapeutic agent. For example, the subject is asymptomatic and is at risk of developing the fibrotic lung disease. A "subject" includes a mammal. The mammal can be e.g., any mammal, e.g., a human, primate, bird, mouse, rat, fowl, dog, cat, cow, horse, goat, camel, sheep or pig. Preferably, the mammal is a human.

As used herein, the term "pharmaceutically acceptable" refers to a material, such as a carrier or diluent, which does not abrogate the biological activity or properties of the compound, and is relatively non-toxic, i.e., the material may be administered to an individual without causing undesirable biological effects or interacting in a deleterious manner with any of the components of the composition in which it is contained.

As used herein, the term "pharmaceutically acceptable carrier" means a pharmaceutically acceptable material, composition or carrier, such as a liquid or solid filler, stabilizer, dispersing agent, suspending agent, diluent, excipient, thickening agent, solvent or encapsulating material, involved in carrying or transporting a compound useful within the invention within or to the patient such that it may perform its intended function. Typically, such constructs are carried or transported from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be "acceptable" in the sense of being compatible with the other ingredients of the formulation, including the compound useful within the invention, and not injurious to the patient. Some examples of materials that may serve as pharmaceutically acceptable carriers include: sugars, such as lactose, glucose and sucrose; starches, such as corn starch and potato starch; cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; powdered tragacanth; malt; gelatin; talc; excipients, such as cocoa butter and suppository waxes; oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; glycols, such as propylene glycol; polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; esters, such as ethyl oleate and ethyl laurate; agar; buffering agents, such as magnesium hydroxide and aluminum hydroxide; surface active agents; alginic acid; pyrogen-free water; isotonic saline; Ringer's solution; ethyl alcohol; phosphate buffer solutions; and other non-toxic compatible substances employed in pharmaceutical formulations.

Pharmaceutically acceptable carriers of the disclosure include, but are not limited to, pharmaceutically acceptable materials, compositions or carriers, such as a liquid or solid fillers, stabilizers, dispersing agents, suspending agents, diluents, excipients, thickening agents, solvents or encapsulating materials, involved in carrying or transporting a compound useful within the invention within or to the patient such that it may perform its intended function. Typically, such constructs are carried or transported from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be "acceptable" in the sense of being compatible with the other ingredients of the formulation, including the compound useful within the invention, and not injurious to the patient. Some examples of materials that may serve as pharmaceutically acceptable carriers include: sugars, such as lactose, glucose and sucrose; starches, such as corn starch and potato starch; cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; powdered tragacanth; malt; gelatin; talc; excipients, such as cocoa butter and suppository waxes; oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; glycols, such as propylene glycol; polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; esters, such as ethyl oleate and ethyl laurate; agar; buffering agents, such as magnesium hydroxide and aluminum hydroxide; surface active agents; alginic acid; pyrogen-free water; isotonic saline; Ringer's solution; ethyl alcohol; phosphate buffer solutions; and other non-toxic compatible substances employed in pharmaceutical formulations.

Suitable forms for administration include forms suitable for systemic administration, oral administration, for example by a capsule or tablet. Once formulated, the compositions of the disclosure can be administered directly to the subject.

The term "prevent," "preventing" or "prevention," as used herein, means avoiding or delaying the onset of symptoms associated with a disease or condition in a subject that has not developed such symptoms at the time the administering of an agent or compound commences.

Compounds and Compositions

In some embodiments, compounds known to be useful in treating pulmonary fibrosis are useful within the methods of the invention. Non-limiting examples of such compounds are pirfenidone (5-methyl-1-phenylpyridin-2-one, or a salt or solvate thereof) and nintedanib (methyl (3Z)-3-{[(4-{methyl[(4-methylpiperazin-1-yl)acetyl]amino}phenyl) amino] (phenyl)methylidene}-2-oxo-2,3-dihydro-1H-indole-6-carboxylate, or a salt or solvate thereof).

In some embodiments, the subject identified as having MUC5B promoter polymorphism rs35705950 is administered a compound contemplated within the disclosure. In some embodiments, the subject is a mammal. In other embodiments, the mammal is a human.

Administration/Dosage/Formulations

The regimen of administration may affect what constitutes an effective amount. The therapeutic formulations may be administered to the subject either prior to or after the onset of a disease or disorder contemplated in the invention. Further, several divided dosages, as well as staggered dosages may be administered daily or sequentially, or the dose may be continuously infused, or may be a bolus injection. Further, the dosages of the therapeutic formulations may be proportionally increased or decreased as indicated by the exigencies of the therapeutic or prophylactic situation.

Administration of the compositions of the present disclosure to a patient, preferably a mammal, more preferably a human, may be carried out using known procedures, at dosages and for periods of time effective to treat a disease or disorder contemplated in the invention. An effective amount of the therapeutic compound necessary to achieve a therapeutic effect may vary according to factors such as the state of the disease or disorder in the patient; the age, sex, and weight of the patient; and the ability of the therapeutic compound to treat a disease or disorder contemplated in the invention. Dosage regimens may be adjusted to provide the optimum therapeutic response. For example, several divided doses may be administered daily or the dose may be proportionally reduced as indicated by the exigencies of the therapeutic situation. A non-limiting example of an effective dose range for a therapeutic compound of the invention is from about 1 and 5,000 mg/kg of body weight/per day. One of ordinary skill in the art would be able to study the relevant factors and make the determination regarding the effective amount of the therapeutic compound without undue experimentation. Actual dosage levels of the active ingredients in the pharmaceutical compositions of this invention may be varied so as to obtain an amount of the active ingredient that is effective to achieve the desired therapeutic response for a particular patient, composition, and mode of administration, without being toxic to the patient.

The precise therapeutically effective amount for a human subject will depend upon the severity of the disease state, the general health of the subject, the age, weight and gender of the subject, diet, time and frequency of administration, drug combination(s), reaction sensitivities and tolerance/response to therapy. This amount can be determined by routine experimentation and is within the judgement of the clinician.

A medical doctor, e.g., physician or veterinarian, having ordinary skill in the art may readily determine and prescribe the effective amount of the pharmaceutical composition required. For example, the physician or veterinarian could start doses of the compounds of the invention employed in the pharmaceutical composition at levels lower than that required in order to achieve the desired therapeutic effect and gradually increase the dosage until the desired effect is achieved.

A suitable dose of a compound of the disclosure may be in the range of from about 0.01 mg to about 5,000 mg per day, such as from about 0.1 mg to about 1,000 mg, for example, from about 1 mg to about 500 mg, such as about 5 mg to about 250 mg per day. The dose may be administered in a single dosage or in multiple dosages, for example from 1 to 4 or more times per day. When multiple dosages are used, the amount of each dosage may be the same or different. For example, a dose of 1 mg per day may be administered as two 0.5 mg doses, with about a 12-hour interval between doses.

In some embodiments of the methods of the disclosure, the therapeutic agent comprises pirfenidone. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the effective dosage is about 2400 mg/day. In some embodiments, the effective dosage is administered according to an escalating dosage regimen. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the escalating dosage regimen comprises (a) administering to the subject about 800 mg of pirfenidone per day for a first week; (b) administering to the subject about 1600 mg of pirfenidone per day for a second week; and (c) administering to the subject about 2400 mg of pirfenidone per day for the remainder of the treatment. In some embodiments, including those embodiments wherein the therapeutic agent comprises pirfenidone, the escalating dosage regimen comprises (a) administering to the subject a capsule or tablet comprising about 250 mg of pirfenidone three times a day for a first week; (b) administering to the subject two capsules or tablets comprising about 250 mg of pirfenidone three times a day for a second week; and (c) administering to the subject three capsules or tablets comprising about 250 mg of pirfenidone three times a day for the remainder of the treatment. In some embodiments of the escalating dosage regimen, the capsule or tablet comprises 267 mg of pirfenidone.

In some embodiments of the methods of the disclosure, the therapeutic agent comprises nintedanib. In some embodiments, the effective dosage is administered orally as a capsule or a tablet. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is about 300 mg/day. In some embodiments, the effective dosage is about 150 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is about 200 mg/day. In some embodiments, the effective dosage is about 100 mg administered twice per day, wherein the daily doses are administered about 12 hours apart from one another. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the effective dosage is administered according to a modified or interrupted dosage regimen. In some embodiments, the modified or interrupted dosage regimen comprises (a) administering to the subject about 300 mg of nintedanib per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; (b) administering to the subject about 200 mg of nintedanib per day until the subject presents the control level of liver enzymes; and (c) administering to the subject about 300 mg of nintedanib per day for the remainder of the treatment; wherein the control level of liver enzymes is a level detected in the subject prior to an initiation of the treatment. In some embodiments, including those embodiments wherein the therapeutic agent comprises nintedanib, the modified or interrupted regimen comprises (a) administering to the subject a capsule or tablet comprising about 150 mg of nintedanib twice per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; (b) administering to the subject two capsules or tablets comprising about 100 mg twice per day until the subject presents an elevated level of liver enzymes compared to a control level of liver enzymes; and (c) administering to the subject a capsule or tablet comprising about 150 mg of nintedanib twice per day for the remainder of the treatment; wherein the control level of liver enzymes is a level detected in the subject prior to an initiation of the treatment.

In some embodiments, the compositions of the invention are formulated using one or more pharmaceutically acceptable excipients or carriers. In one embodiment, the pharmaceutical compositions of the invention comprise a therapeutically effective amount of a compound of the invention and a pharmaceutically acceptable carrier.

The carrier may be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and vegetable oils. The proper fluidity may be maintained, for example, by the use of a coating such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. Prevention of the action of microorganisms may be achieved by various antibacterial and antifungal agents, for example, parabens, chlorobutanol, phenol, ascorbic acid, thimerosal and the like. In many cases, it is preferable to include isotonic agents, for example, sugars, sodium chloride, or polyalcohols such as mannitol and sorbitol, in the composition.

It is also noted that the term "comprising" is intended to be open and permits but does not require the inclusion of additional elements or steps. When the term "comprising" is used herein, the terms "consisting essentially of" and "consisting" of are thus also encompassed and disclosed. Throughout the description, where compositions or combinations are described as having, including, or comprising specific components or steps, it is contemplated that compositions or combinations also consist essentially of, or consist of, the recited components. Similarly, where methods or processes are described as having, including, or comprising specific process steps, the processes also consist essentially of, or consist of, the recited processing steps.

All publications and patent documents cited herein are incorporated herein by reference as if each such publication or document was specifically and individually indicated to be incorporated herein by reference.

EXAMPLES

In order that the invention disclosed herein may be more efficiently understood, examples are provided below. It should be understood that these examples are for illustrative purposes only and are not to be construed as limiting the invention in any manner.

Example 1: Genetic Background of Asymptomatic Siblings of FIP Subjects

Asymptomatic siblings (>50 years old) of patients with established FIP underwent HRCT scan of the chest. HRCT scans were assessed for FILD by blinded thoracic radiologists; when possible, specific radiographic patterns were identified. PBMCs RNA and DNA were isolated. Genotyping for rs35705950 and microarray analysis were performed (SurePrint G3 Human Gene Expression Microarray). Data were analyzed using Partek Genomics Suite and RStudio. Four-hundred eighty-eight FIP siblings from 271 families were evaluated, 25 HRCT scans were excluded due to technically inadequacy, leaving 463 to be interpreted. Of these, 19% (n=88) met criteria for FILD. A subset of the positive FILD scans (n=58) were evaluated for specific interstitial patterns: the predominant radiographic finding was Usual Interstitial Pneumonia (UIP), documented as possible (n=37), probable (n=6), or definite (n=5) in 82.8% of these cases. DNA was available for 443 subjects (358 without and 85 with FILD). The minor allele (T) frequency (MAF) of rs35705950 was higher among those with evidence of FILD (MAF=0.29) than among those with normal appearing HRCT scans (MAF=0.21, p=0.005). The rs35705950 variant was associated with the presence of FILD (OR=1.90, 95% CI 1.10-3.30, p=0.02), and FILD was associated with age (OR=1.09, 95% CI 1.06-1.12, p=7.24×10−9), male sex (OR=1.81, 95% CI 1.04-3.16, p=0.04), and history of smoking (OR=1.94, 95% CI 1.11-3.40, p=0.02). Microarray analysis on PBMC RNA from 40 subjects with FILD and 105 unaffected siblings revealed 1,272 differentially expressed genes (FDR<0.05, fold-change>2); hierarchical clustering performed on the top 194 differentially expressed probes illustrates segregation of FILD subjects from unaffected siblings (FIG. **1**).

Example 2: Role of MUC5B in Pathogenesis of IPF

Common genetic variants play major and similar roles in the development of both familial and sporadic IPF (Table 3), indicating a similar etiology for familial and sporadic IPF. A common gain-of-function MUC5B promoter variant rs35705950 is a strong risk factor (genetic and otherwise), accounting for at least 30% of the total risk of developing IPF (10) confirmed in independent studies, including a GWAS (OR for T (minor) allele=4.51; 95% CI=3.91-5.21; P=7.21×10-95); 3) rs35705950 may be used to identify individuals with PrePF and is predictive of radiographic progression of PrePF. MUC5B promoter variant rs35705950 is present in over 50% of non-Hispanic white (NHW) patients with IPF and is associated with unique biological and clinical IPF phenotypes. PrePF can be predicted using a combination of clinical risk factors, the MUC5B promoter variant rs35705950, and a panel of biomarkers.

TABLE 1

Common IPF risk variants identified by targeted sequencing of risk loci in 3,642 IPF cases and 4,442 unaffected controls

Chrm	Common Variant	Nearest Gene	Annotation[a]	Minor Allele	MAF in cases	OR Aa vs AA (95% CI)	OR aa vs AA (95% CI)	P[b]
3q26	rs2293607	TERC	3' UTR	C	0.2999	1.30 (1.18-1.43)	1.79 (1.49-2.15)	9.11×10^{-13}
4q22	rs2609260	FAM13A	Intronic	C	0.2289	1.35 (1.22-1.50)	1.96 (1.56-2.47)	1.03×10^{-13}
5p15	rs4449583	TERT	Intronic	T	0.2641	0.68 (0.62-0.75)	0.46 (0.39-0.55)	2.67×10^{-25}
6p24	rs2076295	DSP	Intronic	G	0.5428	1.27 (1.14-1.42)	2.08 (1.83-2.37)	1.11×10^{-29}
7q22	rs6963345	ZKSCAN1	Intronic	A	0.4444	1.35 (1.22-1.50)	1.73 (1.51-1.99)	1.89×10^{-15}
10q24	rs2488000	OBFC1	Intronic	T	0.08	0.70 (0.62-0.79)[c]		7.13×10^{-9}
11p15	rs35705950	MUC5B	Promoter	T	0.3533	5.45 (4.91-6.06)	18.68 (13.34-6.17)	9.60×10^{-295}
13q34	rs1278769	AK025511	3' UTR	A	0.1996	0.77 (0.70-0.85)	0.69 (0.56-0.86)	7.48×10^{-8}
15q15	rs35700143	IVD	—	C	0.4118	0.76 (0.68-0.84)	0.63 (0.55-0.71)	3.44×10^{-12}
19p13	rs12610495	DPP9	Intronic	G	0.3398	1.22 (1.11-1.35)	1.59 (1.36-1.87)	3.11×10^{-9}

OR, odds ratio. The minor allele is defined as the minor allele in the combined case and control group.
[a]Based on SNPDOC;
[b]P value adjusted for sex;
[c]OR resulting from dominant test.

MUC5B is predicted is involved in the pathogenesis of IPF. FIG. **5** shows that MUC5B promoter variant is associated with enhanced MUC5B expression in both unaffected subjects and in patients with IPF and in IPF, MUC5B message and protein are expressed in bronchoalveolar epithelia (FIG. **6**) and honeycomb cysts. In mice, the concentration of Muc5b is directly related to the fibroproliferative response to bleomycin (FIG. **7**), Muc5b protein is expressed in the injured lung following bleomycin challenge, and enhanced production of Muc5b in mice appears to initiate endoplasmic reticulum (ER) stress in peripheral airways (FIGS. **8** and **9**). Preliminary studies, also show that mucociliary clearance is decreased in mice that overexpress Muc5b (SFTPC-Muc5b^{Tg}) and in humans with IPF (FIG. **10**).

Interstitial lung abnormalities on HRCT scans show asymptomatic relatives of patients with familial IPF and in the elderly. Similar to patients with IPF, interstitial lung abnormalities in asymptomatic subjects are associated with advanced age, cigarette smoking, reduced lung volume and decreased exercise tolerance. Moreover, the MUC5B promoter variant rs35705950 is associated with a higher prevalence of interstitial lung abnormalities on HRCT scan and is predictive of radiographic progression. Suggesting that interstitial lung abnormalities on HRCT scan are a precursor of IPF. However, interstitial lung abnormalities are not specific and include non-fibrotic and fibrotic HRCT defects, and consequently, the prevalence of interstitial lung abnormalities (>5% in the general population ≥50 years of age is orders of magnitude higher than IPF.

To address the non-specificity of interstitial lung abnormalities, a novel entity—Preclinical Pulmonary Fibrosis (PrePF) was used. PrePF is reported more frequently among smokers and in families with two or more cases of pulmonary fibrosis. In the Framingham population, data shows that PrePF is present in 1.8% of the general population ≥50 years of age (in contrast, interstitial lung abnormalities were seen in 6.7%) and that the MUC5B promoter variant rs35705950 is predictive of those with PrePF (OR=6.3 per allele [95% CI 3.1-12.7). As shown herein, among asymptomatic first-degree family members of familial interstitial pneumonia (FIP) 14% have fibrotic interstitial changes on CT scan and 35% have interstitial abnormalities on transbronchial biopsy. Moreover, in the Framingham population, it is shown that rs35705950 is predictive of radiographic progression of PrePF (OR=2.8 per allele [95% CI 1.8-4.4]) which is associated with a greater FVC decline (P=0.0001) and an increased risk of death (HR=3.7 [95% CI 1.3, 10.7]; P=0.02), indicating that in addition to having radiographic features of IPF, PrePF has similar risk factors (age, gender, smoking, and MUC5B variant) and a progressive clinical course. While the MUC5B promoter variant is predictive of PrePF, rs35705950 is present in ≈19% (minor allele frequency (MAF)=0.09) of the NHW population, however IPF occurs infrequently (<0.1%). Thus, additional biomarkers may be used in combination with rs35705950 identify PrePF within at-risk populations.

The data provided herein suggest that 1) IPF is underdiagnosed; 2) PrePF is prevalent in at-risk populations; 3) approximately 75% of the cases of PrePF are progressive; 4) radiographic progression of PrePF is associated with increased morbidity and mortality; and 5) MUC5B variant rs35705950, peripheral blood biomarkers, clinical/biological, and radiographic screening should be useful in identifying those with PrePF (FIG. **11**). While IPF takes years to develop, most patients with IPF are diagnosed in the advanced stage when little can be done to influence survival. Once the lung has undergone remodeling, the non-compliant, stiff lung matrix causes additional remodeling through activation of myofibroblasts, resulting in a feed-forward loop of lung remodeling. Earlier diagnosis of IPF detects subjects with a lower burden of fibrotic lung disease.

This disclosure provides a strategic approach to screening for early forms of IPF needs to be established (FIG. **11**). While the MUC5B promoter variant is predictive of PrePF (defined as chest HRCT consistent with probable or definite fibrosis (e.g., bilateral subpleural reticular changes, honeycombing, or traction bronchiectasis) occurring in asymptomatic subjects ≥40 years of age that emerge from at-risk populations), the MUC5B promoter variant is present in ≈ 19% of the NHW population and IPF occurs infrequently (<0.1%). To study at-risk populations (asymptomatic siblings ≥40 years of age of patients with family or sporadic IPF), identification of genetic variants and biomarkers that increase the yield of patients with PrePF are used to establish screening tools and approaches that identify early stages of IPF. This approach changes the way IPF is diagnosed and treated, and is critical to developing interventions to prevent PrePF progression to established IPF. The methods provided in this disclosure fundamentally alter the clinical approach to patients with IPF from palliative to preventive (FIG. **10**).

Example 3: Predictive Biomarker Profile for Established IPF

To address the development of a peripheral blood biomarker profile for IPF, an assay of the expression levels of

>3700 plasma proteins was performed on plasma from 70 patients with established IPF and 70 controls. After controlling for multiple comparisons and appropriate co-variables, 57 proteins were up-regulated >1.5-fold (including surfactant proteins, MMP7, and C3) in the plasma of patients with IPF and 12 were significantly down-regulated (FIG. **2**).

Example 4: Predictive Biomarker Profile for Early IPF

To evaluate a predictive biomarker profile in cases of preclinical pulmonary fibrosis (PrePF) derived from families with familial IPF (≥2 cases of IPF in a family), HRCT scans were performed on 496 asymptomatic family members ≥40 years of age previously phenotyped as unaffected from 263 families with familial IPF. PrePF, consistent with the operational definition (defined as abnormalities on chest HRCT consistent with probable or definite fibrosis (e.g., bilateral subpleural reticular changes, honeycombing, or traction bronchiectasis) occurring in asymptomatic subjects ≥40 years that emerge from at-risk populations), was present in 77 (15.5%) of 496 asymptomatic individuals from families with familial IPF. The minor allele frequency (MAF) of the MUC5B promoter variant was 0.29 in those with PrePF versus 0.21 in those without fibrosis (P=0.025). Preliminary analysis of PBMC gene expression profiles evaluated by microarrays from 38 cases of PrePF and 187 subjects without fibrosis identified 16 genes significantly differentially expressed between the two groups (p-value <0.05 and >1.5 fold change). Among genes differentially expressed in PrePF are those involved in innate immunity and inflammatory responses (SIGLEC14), antibacterial effects (ADM2), growth and motility (TSPAN5), and protein phosphorylation (CAMKK1). Moreover, PBMC gene expression appears to contribute to the ability to predict PrePF in an at-risk population (FIG. **3**).

Additionally, RNA-sequencing analysis was performed on 40 PrePF subjects and 80 subjects with a normal HRCT scan. Sequencing of the polyA-enriched libraries was prepared using Illumina TrueSEQ reagents and multiplexing 10 samples on each lane of HiSEQ4000 to obtain on average 35-40 million reads per sample. This high coverage allows for the consideration of a broad dynamic range of mRNA transcripts for biomarker selection. Platform selection of serum and plasma samples from the same subjects are used for proteomic analysis.

Example 5: Biomarker Identification

To examine for association between each biomarkers and PrePF, a multivariable logistic regression model for PrePF with biomarkers and covariates is used for inclusion and a step-wise forward selection procedure is constructed. Variables stay in the model if associated at P≤0.01 after adjustment for the variables already in the model. Protein biomarkers that are significantly associated with established IPF and the top 20 differentially expressed genes in PrePF are considered for inclusion in a multivariable model. The number of potential biomarkers allowed in the joint model is restricted to approximately 20 given the number cases of PrePF expected. Secondarily, interactions between MUC5B genotype and the other biomarkers are tested for, which allow for the possibility that different biomarker profiles are diagnostic in IPF patients with/without the MUC5B risk allele.

Example 6: Predictive Ability of Biomarkers

To test the predictive value of the combination of biomarkers associated with PrePF, the observed expression and other biomarker values from those associated with PrePF in the siblings of FIP patients is used to obtain the probability, for each sibling, having PrePF.

Following, a construct receiver operating characteristic (ROC) curves (see M. S. Pepe et al., Phases of biomarker development for early detection of cancer. Journal of the National Cancer Institute 93, 1054-1061 (2001)), is used to choose the probability threshold that maximizes the area under the ROC curve. This probability threshold is used to classify each individual as predicted to have PrePF or not, allowing calculation of the sensitivity, specificity, positive predictive value, and negative predictive value of the predictive model. The properties of the predictive model(s) in the independent set of siblings of patients with IPF are evaluated. Different aliquots are run for 10 samples for each assay at each time the assays is run in order to use those 10 samples to evaluate the need for standardization of the absolute values for each assay over time. Either the raw or standardized values, for a given model, is used to observe biomarker values in the PrePF siblings and non-PrePF siblings to obtain the probability of being in the disease group based on the model parameters developed using the FIP siblings. The thresholds identified among the FIP siblings are used to classify each individual as predicted to have PrePF or not. This categorization allows for the calculation of the sensitivity, specificity, positive predictive value, and negative predictive value of the predictive model among the siblings of independent cases of IPF to that observed in the siblings of FIP cases.

Power is calculated to detect differences between those with and without PrePF assuming 500 siblings and 10% (N=50) with PrePF. Assuming α=0.00005 (conservatively correcting for up to 1000 independent tests), we have 80% (90%) power to detect differences in protein or expression level of 0.74 (0.80) standard deviation between PrePF and unaffected siblings. These differences are larger than previously-observed protein and gene-expression levels in IPF patients and controls (see I. V. Yang et al., The peripheral blood transcriptome identifies the presence and extent of disease in idiopathic pulmonary fibrosis. PLoS One 7, e37708 (2012). With 50 PrePF and 450 unaffected, there is 90% power to bound the sensitivity of the biomarker-based classification of PrePF with a margin of error of 11% if the sensitivity is 65%, and 6.5% if the sensitivity is 95%; the margins of error for 65% and 95% sensitivity are 4.5% and 2.5%, respectively.

Example 7: MUC5B Promoter Variant r35705950 is a Risk Factor for Rheumatoid Arthritis—Interstitial Lung Disease

Methods

Study Cohorts

This study included a discovery cohort and multi-ethnic replication cohorts. The discovery cohort included patients with RA, with and without ILD (RA-noILD) as assessed by chest HRCT, and controls, from the French RA-ILD network. The multi-ethnic replication cohorts were obtained from six countries (China, Greece, Japan, Mexico, the Netherlands and United States). This included patients with RA-ILD and RA-noILD patients, and controls. All cases fulfilled the 2010 European League Against Rheumatism-American College of Rheumatology (EULAR-ACR) and/or 1987 ACR revised criteria for RA. The ILD status of patients with RA was established by chest HRCT images that were centrally reviewed by experienced readers for each participating cohort. There was one cohort, the RA-noILD cases from the USA1 cohort, which was determined by self-report. The chest HRCT ILD pattern was classified as UIP, possible UIP or inconsistent with UIP according to international criteria and all readers were blinded to the clinical and genetic data. The institutional review boards at each institution approved all protocols, and all patients provided written informed consent.

Genotyping

Genotyping of the MUC5B rs35705950 single nucleotide polymorphism (SNP) involved use of Taqman Genotyping Assays (Applied Biosystems, Foster City, CA, USA) as previously reported, by direct Sanger Sequencing or imputation from genome-wide association study data.

The additional common IPF risk variants on 3q26, 4q22, 5p15, 6p21.3, 6p24, 7q22, 10q24, 11p15.5, 13q34, 15q14-15, and 19p13 were genotyped by Taqman qPCR (Thermo Fisher Scientific, California) per the manufacturer's instructions.

Lung Tissue Analysis

In order to determine if MUC5B was expressed in RA-ILD ling tissue, we analyzed lung tissue was analyzed from nine patients with RA-ILD undergoing lung transplantation (University of California, San Francisco) compared to six unaffected controls with ILD (NHLBI Lung Tissue Research Consortium; https://ltrcpublic.com) or concordant expression of other relevant markers of pulmonary fibrosis. The tissue was formalin fixed, paraffin embedded and cut in 4 um sections. Tissue sections were deparaffinized in xylene, followed by dehydration in series of ethanol. Following citrate buffer antigen retrieval, slides were incubated overnight with primary antibodies against MUC5B (1:4000, Santa Cruz, Dallas, TX). Secondary antibody diluted 1:1000 tagged with HRP (Life Technologies) was visualized using an Aperio CS2 slide scanner (Leica, Buffalo Grove, IL).

Results

Study Cohorts

This case-control genetic study included 620 RA-ILD cases, 614 RA-noILD cases and 5448 unaffected controls. The discovery cohort included 118 RA-ILD cases, 105 RAnoILD cases and 1229 unaffected controls. The multi-ethnic replication sample included 502 RA-ILD, 509 RA-noILD cases and 4219 unaffected controls.

Characteristics of the Discovery Cohort

As compared with RA-noILD, patients with RA-ILD were more frequently male, older and more frequently smoked cigarettes (54.7% versus 36.1%) (FIG. **13**). However, after adjusting for sex, the relationship between RA-ILD and cigarette smoking was no longer statistically significant (FIG. **13**). After adjustment, RA-ILD and RA-noILD patients did not differ in rheumatoid factor (RF) and/or anti-citrullinated protein antibody (ACPA) positivity, erosive status of RA, exposure to methotrexate or the mean RA duration from diagnosis at inclusion in the cohort. Overall, 41% of patients with RA-ILD had a UIP or possible UIP HRCT pattern.

MUC5B Promoter Variant and Risk of Rheumatoid Arthritis-Associated Interstitial Lung Disease

Comparison of RA-noILD and controls revealed that none of the cohorts (discovery cohort and multi-ethnic cohorts) demonstrated a significant difference in the frequency of the MUC5B promoter variant (FIG. **14**; FIG. **16**A), suggesting a lack of association between the MUC5B promoter variant and RA. In the discovery cohort, the minor allele frequency (MAF) of the MUC5B promoter variant was 10.9% in unaffected controls and 32.6% in cases of RAILD; this variant was in Hardy-Weinberg equilibrium (HWE) in both study groups. I In the discovery population, after controlling for sex we detected a significant association between the MUC5B promoter variant and RA-ILD when compared to non-RA controls (ORadj=3.8; 95% CI, 2.8 to 5.2; P=9.7×10-17) (FIG. **14**). Similar to the discovery population, the MUC5B promoter variant was significantly over-represented among the cases of RA-ILD compared to unaffected non-RA controls in all of the multi-ethnic study case series, except in the two Asian case series (FIG. **14**). Given that the MUC5B promoter variant is under-represented in Asian populations compared to non-Hispanic whites (FIG. **14**; www.ncbi.nlm.nih.gov/projects/SNP/snp_ref.cgi?rs=35705950), a likely explanation, especially given the consistent point estimates, for the absence of a significant relationship between the MUC5B promoter variant and RA-ILD is that the analysis of the two Asian case series is likely underpowered. The relationship between the MUC5B promoter variant and RA-ILD in combined multi-ethnic study case series (ORadj=4.7; 95% CI, 3.9 to 5.8; P=1.3×10-49) (FIG. **14**) (FIG. **16**B) validated the observed association between the MUC5B promoter variant and RA-ILD in the discovery study population.n addition, the cases of RA-ILD in the study populations from Greece and USA-1 were not in HWE, suggesting (as has been observed in cases of IPF 14), that the MUC5B promoter variant and/or common variants in high or complete linkage disequilibrium with the MUC5B promoter variant should be considered as causative in these cases of RA-ILD. For the comparison with non-RA controls, the best-fitting genetic model for the three study populations (discovery population, combined multi-ethnic case series, and combined analysis) for the association of the MUC5B

MUC5B RS35705950 and Risk of Interstitial Lung Disease Among Patients with Rheumatoid Arthritis

To further investigate whether the MUC5B promoter variant rs35705950 contributes to the risk of ILD among patients with RA, we compared RA-ILD and RA-noILD patients, adjusting for sex, age at inclusion and cigarette smoking. In the discovery cohort, the MUC5B variant was associated with RA-ILD (ORadj=3.1; 95% CI, 1.6 to 6.3; $P=9.4\times10^{-4}$), and this finding was replicated in the aggregate multi-ethnic cohort (ORadj=2.9; 95% CI, 1.1 to 8.4; P=0.04) and the combined analysis (ORadj, 3.1; 95% CI, 1.8 to 5.4; $P=7.4\times10^{-5}$) (FIG. **14**; FIG. **16**C). For the comparison of RA-ILD with RA-noILD, the best-fitting genetic model for the three study cohorts (discovery population, combined multi-ethnic case series, and combined analysis) was dominant. After adjusting for covariates, no association between tobacco smoking and the risk of ILD among patients with RA was found and no interaction of tobacco smoke exposure with the MUC5B promoter variant was observed (ORadj=0.7; 95% CI, 0.3 to 1.9; P=0.51).

MUC5B RS35705950 and UIP on HRCT Scan

Limiting the RA-ILD cases to those with radiographic evidence of definite or possible UIP on HRCT scan, the association observed in the discovery cohort (ORadj=5.0; 95% CI, 2.1 to 12.3; $P=3.0\times10^{-4}$), was replicated in the combined multi-ethnic cohort (ORadj=9.2; 95% CI, 2.3 to 38.7; $P=1.8\times10^{-3}$) (FIG. **16**C), and was observed in the combined cohort analysis (ORadj=6.1; 95% CI, 2.9 to 13.1; $P=2.5\times10^{-6}$) (FIG. **16**C). In the combined analysis, the comparison of odds ratios for UIP RA-ILD vs RA-noILD (ORadj=6.1; 95% CI, 2.9 to 13.1; $P=2.5\times10^{-6}$) to non-UIP RA-ILD vs RA-noILD (ORadj=1.3; 95% CI, 0.6 to 2.8; P=0.46) was statistically significant (P=0.02), suggesting that the effect of the MUC5B promoter variant was restricted to the UIP RA-ILD sub-phenotype (FIG. **16**C). Finally, consistent with our previous findings, the MUC5B promoter variant was found to increase the risk of developing a UIP pattern among patients with RA-ILD through a dominant model in the discovery, replication and combined analysis; the odds of having a UIP and possible UIP pattern for patients with RA-ILD carrying at least one MUC5B rs35705950 T risk allele were 2.9 times greater than individuals having the GG genotype (ORadj=2.9; 95% CI, 1.7 to 4.8; P=5.1×10−5) (FIG. **15**; FIG. **16**C). After adjusting for covariates, tobacco smoking exposure did not contribute to a specific HRCT pattern for RA-ILD and no interaction with the MUC5B rs35705950 variant was detected.

Sites of MUC5B Expression in Ra-ILD

We performed immunohistochemical staining for MUC5B in nine RA-ILD lung tissue explants (5 GG and 4 GT) and 6 unaffected controls (3 GG and 3 GT). Similar to what has been reported in IPF, RA-ILD lung tissue demonstrated MUC5B in the cytoplasm of the bronchioles and in areas of microscopic honeycombing, including staining of the metaplastic epithelia lining the honeycomb cysts and the mucous within the cyst (FIG. **17**). The controls demonstrated MUC5B expression in the bronchioles only. There were no obvious differences in MUC5B expression by genotype.

Exploratory Genetic Association Study of 12 Common IPF Risk Variants in Ra-ILD

Having provided evidence for the contribution of the dominant IPF genetic risk variant, i.e. the MUC5B promoter variant, to RA-ILD, we decided to test the association of 12 additional common IPF risk variants with RA-ILD (FIG. **29**). This exploratory study included 272 RA-ILD and 242 RA-noILD patients from the France, USA-1 and Mexico case series. Taking into account the relatively small sample size and related low power of detection corresponding P-values, Odds Ratio and 95% CI for the 12 candidate variants were considered as descriptive and Bonferoni correction was therefore not applied (Table 4). Comparison between RA-ILD and RA-noILD revealed that 2 common IPF risk variants, TOLLIP rs5743890 and IVD rs2034650, were significantly associated with RA-ILD. The TOLLIP rs5743890 minor allele was associated with increased risk of RA-ILD and the IVD rs2034650 minor allele was associated with decreased risk of RA-ILD (ORadj=2.13; 95% CI, 1.13 to 4.10; P=0.02 and ORadj=0.59; 95% CI, 0.38 to 0.89; P=0.01, respectively) and the directionality of these relationships is consistent with what has been observed for IPF.16,17 No association with RA-ILD was detected for the 10 other IPF risk variants (FIG. **29**).

Example 8: MUC5B Promoter Variant is Associated with Visually and Quantitatively Detected Preclinical Pulmonary Fibrosis

Better understanding and recognition of early pulmonary fibrosis is critical because medical therapies have been shown to slow progression, not to reverse or even stabilize established fibrosis—therefore, intervention before irreversible fibrosis has become extensive has the potential to improve quality of life and decrease morbidity. While IPF affects approximately 5 million people worldwide, between 1.8 and 14% of the general population ≥50 years of age have radiologic findings of undiagnosed pulmonary fibrosis. Large cohort studies indicate that interstitial lung abnormalities, postulated to represent early pulmonary fibrosis, are associated with increased mortality, and that most of these abnormalities progress over time. Members of families with 2 or more cases of pulmonary fibrosis (FIP, Familial Interstitial Pneumonia) have been identified as an "at-risk" population. In a previous study of FIP relatives, 14% had interstitial lung abnormalities on high resolution computed tomography (HRCT), and 35% had an abnormal transbronchial biopsy indicating interstitial lung disease.

HRCT provides visualization of the lung parenchyma and plays a key role in the diagnosis of the Idiopathic Interstitial Pneumonias (TTPs), including IPF. Currently, visual diagnosis by thoracic radiologists, in conjunction with multidisciplinary clinical conference, is the gold standard for diagnosing IIPs. However, visual assessment is imprecise and hampered by inter-observer variation. Quantitative HRCT (qHRCT) evaluation provides measures of fibrosis extent that, in subjects diagnosed with IPF, correlate with degree of physiologic impairment at baseline, and may be more sensitive to subtle changes in disease status than routinely used physiological metrics. The design and utility of quantitative methods in the context of early forms of fibrotic ILD requires further study. Deep learning methods have been increasingly used in imaging to identify and classify CT patterns, and may be particularly valuable in detection of early lung fibrosis.

This study aims to: (1) examine risk factors, including two common fibrosis-associated genetic variants in MUC5B and TERT, for undiagnosed pulmonary fibrosis (PrePF) in FIP first-degree relatives; and (2) determine the utility of a deep learning, texture-based qHRCT method in the detection of early fibrosis in this cohort.

Materials and Methods

FIP Relatives Screening:

As part of a study of FIP conducted at the University of Colorado, National Jewish Health, and Vanderbilt University (COMIRB #15-1147; NJH IRB 1441a; Vanderbilt IRB #020343), non-Hispanic white (NHW) relatives of FIP patients, defined as those in families with two or more cases of pulmonary fibrosis, were contacted for enrollment. First-degree relatives without a known prior diagnosis of pulmonary fibrosis and greater than 40 years of age were offered HRCT scans of the chest and asked to undergo peripheral blood draw. Study subjects younger than 40 years of age or older than 40 years of age who reported on pre-study questionnaires to be personally affected by pulmonary fibrosis were excluded (FIG. **18**).

Visual CT Review:

HRCT scans were interpreted by study radiologists and examined for the presence of fibrotic ILD. "PrePF" was defined as the presence of "probable" or "definite" fibrotic ILD on HRCT in FIP relatives who had no known diagnosis of pulmonary fibrosis at the time of study enrollment (FIGS. **18**, **19**).

Quantitative CT:

Inspiratory HRCT series with slice thickness ≤1·25 mm and spacing ≤20·0 mm were selected for quantitative analysis. This included 212 volumetric series with thin, contiguous sections (slice thickness and spacing both <=1·25 mm) and 191 non-volumetric scans (56 with slice spacing >1·25 mm and <10 mm, 65 with slice spacing of 10 mm and 70 with slice spacing=20 mm). Scans identified as technically inadequate were omitted. In addition, 100 inspiratory volumetric HRCT of never-smoking control subjects from the COPDGene cohort were analyzed (FIG. **20**). The lungs were segmented in a semi-automatic fashion using open source software followed by manual editing, if necessary, performed by trained analysts. Examples of the categorization of different parts of CT scans are shown in FIG. **21**. Some studies were acquired with contiguous thin axial sections while others used 1 or 2 cm intervals. Also, reconstruction kernel, a parameter that affects image sharpness and noise, was not standardized.

Fibrosis quantification on CT scans was performed using a deep learning technique, with a convolutional neural network (CNN) algorithm trained with image regions of normal and abnormal lung identified by expert radiologists. Training data and an earlier algorithm version were described previously. Here, a more complex CNN architecture was employed that classifies image regions using pixel and texture features extracted by multiple convolutional layers at different scales. Classification categories included normal lung, airways, reticular abnormality, honeycombing and ground glass. An additional category, "not normal", was also included for lung regions not classified into any of the named categories. Further, pixels in the "not normal" category were split into two subcategories: "not normal" low density and "not normal" high density using the threshold value of −650 Hounsfield Units (HU). Subject level scores were computed as the percentage of total lung volume classified in each category. HRCT fibrosis score was defined as the sum of CNN classification scores for reticular abnormality, honeycombing, ground glass, and "not normal high density" (FIG. **21**).

A simpler previously described densitometric analysis of HRCTs was also performed for comparison. Percent high attenuation area (% HAA), the percentage of total lung volume with HRCT pixel intensity greater than −600 HU and less than −250 HU, has been used as a measure of interstitial lung disease on CT.

Statistical Analysis:

Analysis of the effect of specific alleles on PrePF risk was performed using minor allele frequency (MAF) for comparison of variant prevalence in the study groups; statistical significance was determined utilizing either a z-score test for proportions or a mixed effects logistic regression model when controlling for other clinical factors (age, sex, and history of smoking) and family [random effect]) in both dominant and log-additive models.

Distribution of qHRCT fibrosis scores was left skewed as was % HAA, and therefore these values were log transformed prior to analyses. Log of qHRCT fibrosis score (hereafter, "fibrosis score") and log (% HAA) were compared with visual scores using ANOVA and Tukey's honest significant difference (HSD) test. To determine the ability of qHRCT scores to predict visual diagnosis of PrePF, receiver-operating characteristic (ROC) analysis was performed. Optimal threshold for discriminating visual diagnosis of fibrotic ILD was determined with Youden's method. Five-fold cross-validation was performed to test detection accuracy, sensitivity and specificity, and consistency of optimal threshold. Linear regression was performed to test association between the MUC5B genotype and qHRCT fibrosis score and log (% HAA).

A p-value of <0.05 was considered statistically significant for differences between groups as well as for associations between individual variables and outcomes in linear and logistic regression modeling. Statistical analyses were performed using RStudio (Version 0.99.473).

Results

Study Cohort Characteristics

A total of 1,090 FIP relatives were contacted, and 523 eligible subjects were recruited and underwent HRCT screening (FIG. **18**). Of the 523 subjects, 26 were excluded due to technical inadequacy of images and one for an equivocal consensus read by study radiologists. The remaining 496 subjects from 263 families were included in the final analyses. The mean age of study subjects was 57 years (95% CI: 56.5-58), 189 (38%) were male, and 148 (29%) were either current or former smokers. The minor allele (T) frequency of the MUC5B promoter polymorphism rs35705950 was 0.22 in this cohort; 45% of the subjects in this cohort had one or two copies of the minor allele (FIG. **22**). The minor allele (C) frequency of the TERT variant rs2736100 was 0.47 in the entire cohort; 69% of the subjects in the cohort having one or two copies of the minor allele (FIG. **22**).

Prevalence of Preclinical Pulmonary Fibrosis (PrePF) in FIP Relatives

Of the 496 HRCT scans, 401 showed no CT evidence of interstitial lung disease (ILD), and 95 showed evidence of ILD, either fibrotic (27 probable and 50 definite) or non-fibrotic (n=18). Therefore, among these 496 subjects who reported being personally unaffected by pulmonary fibrosis, the PrePF prevalence was 15.5% (n=77) (FIG. **18**).

The CT patterns noted in PrePF subjects (FIG. **23**) show that possible, probable, or definite UIP pattern was the most commonly considered (n=59, 77% of all PrePF cases). NSIP was considered in 45 subjects (58% of all PrePF cases). The fibrotic changes were most commonly lower-lobe predominant and subpleural in nature, consistent with a UIP pattern (FIG. **23**). Non-fibrotic ILD scans, on the other hand, generally had more diffuse, upper-lobe predominant abnormalities.

There were 402 study subjects with HRCT scans that were technically adequate for quantitative assessment. 212 of the scans had both slice thickness and spacing <=1·25 mm (thin, contiguous); of the remaining 191 scans, 56 had slice spacing >1·25 mm and <10 mm, 65 had slice spacing=10 mm, and 70 had slice spacing=20 mm. Volumetric HRCT scans on an additional 100 COPDGene subjects were included as normal controls. Fibrosis score means were significantly different (p<0.0001) across groups defined by visual diagnosis (FIG. **24**). Comparison of means showed fibrosis score were significantly different comparing each group (all between-group comparisons p<0.01). Means of log (% HAA) scores were also significantly different across visual scoring groups (p<0.0001), and individual between-group comparisons showed log (% HAA) was significantly different in most comparisons (p<0.0001), except between the "probable" and "definite" visual scores (p=0.35).

ROC analysis showed that fibrosis score discriminates subjects with visual diagnosis of PrePF (FIG. **25**B). Average area under the curve (AUC) in five-fold cross validation was 0.85 (range 0.83-0.87) and average accuracy, sensitivity, and specificity in the test partitions were 0.83 (range 0.74-0.86), 0.74 (range 0.56-0.92), and 0.84 (range 0.76-0.89), respectively. Optimal threshold for fibrosis score ranged from 1.40-1.42, corresponding to 4.1% fibrotic area in examined lung. Utilizing a cutoff of 1.40 for fibrosis score on the entire dataset, the sensitivity was 74%, specificity was 82%, and accuracy was 81%; the negative predictive value of this test was 95%, exceeding its positive predictive value (42%) (FIG. **25**C).

Compared to the classification achieved with the CNN as described above, ROC analysis of log % HAA had lower mean AUC 0.80 (range 0.79-0.81) and average accuracy, sensitivity, and specificity of 0.67 (range 0.63-0.70), 0.82 (range 0.75-0.91) and 0.64 (range 0.62-0.70) respectively (FIG. **25**A). Optimal threshold for log % HAA ranged from 1.49-1.57. Utilizing a cutoff of 1.49 for log % HAA, the sensitivity was 88%, specificity was 55%, and accuracy was 60%; the negative predictive value of this test was 96%, exceeding its positive predictive value (26%).

Risk Factors for PrePF

Subjects with PrePF were older (mean age 65.8 years, 95% CI 63.5-68.1) than those without fibrosis (mean age 55.8, 95% CI 54.9-56.6, $p=6.36\times10^{-13}$); they were also more likely to have ever smoked (43% versus 27%, p=0.007), and to be male (48% versus 36%, p=0.05). However, there was no difference in breathlessness between the PrePF and subjects without fibrosis (mean score 0.5 versus 0.6, p=0.24, FIG. **26**). When fibrosis was defined by quantitative fibrosis score cutoff (1.4), there was a significant difference between groups in terms of mean breathlessness score (0.39 versus 0.78, p=0.003). Quantitative fibrosis score was positively associated with breathlessness score (p=0.001), even after controlling for age ($p=1.9\times10^{-9}$), male sex (p=0.7), and smoking history (p=0.8).

Screening for autoantibodies in this cohort revealed that there were no differences between PrePF and No Fibrosis subjects in terms of overall seropositivity or individual antibodies' testing in this cohort. For quantitatively defined fibrosis, there was no significant difference between groups in terms of auto-antibody testing, with similar overall seropositivity rates (11% versus 16%, p=0.30).

The MUC5B promoter polymorphism rs35705950 was associated with the visual diagnosis of PrePF (present in 40% of those without fibrosis versus 53% with PrePF; MAF 0.29 versus 0.21, respectively, p=0.03, FIG. **22**). After age 60, there was a statistically significant difference in the proportion of subjects with visually diagnosed PrePF when the cohort was stratified by MUC5B genotype (23.8% versus 39.8% prevalence, p=0.02) (FIG. **27**).

MUC5B variant carriers, regardless of their visual CT diagnosis, had significantly higher qHRCT fibrosis scores (1.3 [95% CI 1.2-1.5] versus 1.1 [95% CI 1.0-1.2], p=0.02). The association between MUC5B genotype and fibrosis score was significant even when controlling for age and male sex in linear regression (p=0.03, FIG. **28**). Age was significantly associated with fibrosis score ($p=2.17\times10^{-9}$), but male sex (p=0.63) and smoking (p=0.94) were not. To determine whether individual textural components were driving the association of the composite fibrosis score with genotype, each score component was tested individually for association with the MUC5B variant, controlling for age and sex. Quantitative scores for reticulation, honeycombing, and ground glass were significantly associated with the MUC5B variant (p=0.02, p=0.02, p=0.04, respectively), while "not normal high density" was not (p=0.18). The simpler quantitative scoring method, log % HAA, was not significantly different in MUC5B variant carriers (p=0.4).

In contrast to the MUC5B variant, the common IPF-associated TERT polymorphism (r52736100) was not significantly associated with PrePF assessed either qualitatively (MAF 0.47 in PrePF versus 0.46 in unaffected, p=0.77) or quantitatively (MAF 0.50 fibrotic versus 0.47 not fibrotic, p=0.40).

When these factors were examined individually for their contributions to risk of PrePF in our study cohort, we used a mixed effects logistic regression model to test the independent effects of age sex, smoking, and MUC5B or TERT genotypes while controlling for family. Age and the MUC5B genotype remained statistically significantly associated with PrePF (OR 1.15, 95% CI 1.09-1.22, $p=7.34\times10^{-7}$ and OR 2.18, 95% CI 1.00-4.73, p=0.05, respectively) (FIG. **22**). The common TERT polymorphism (r52736100) associated with fibrotic idiopathic interstitial pneumonia (29) was not significantly associated with PrePF (MAF was 0.45 in PrePF versus 0.45 in unaffected, p=0.88) or in a log-additive model controlling for age, sex, and smoking history (p=0.57).

Given the presence of non-fibrotic ILD (n=18, FIG. **18**) in the "No Fibrosis" cohort, secondary analyses were performed that (1) excluded non-fibrotic ILDs and (2) compared all ILD (inclusive of non-fibrotic ILD) to those without any ILD. When non-fibrotic ILDs were excluded from analyses, PrePF subjects were older ($p=4.7\times10_{-13}$), more commonly male (p=0.04), more often had a smoking history (p=0.003), and had a higher prevalence of the MUC5B promoter variant (MAF 0.29 versus 0.20, p=0.02). However, when controlling for family relatedness and the other risk factors in a mixed effects logistic regression, only age and the MUC5B promoter variant were significantly associated with PrePF with odds ratios 1.15 (95% CI 1.09-1.22, $p=9.5\times10^{-7}$) and 2.16 (95% CI 1.00-4.75, p=0.05), respectively. Another secondary analysis of the data was performed in which all subjects with CT findings of ILD (fibrotic or non-fibrotic) were compared to those without any evidence of ILD. Those with CT evidence of ILD were older (mean age 64.3 years, 95% CI 62.2-66.3) compared to those without any evidence of ILD (mean age 55.7 years, 95% CI 54.8-56.6, $p=4.1\times10_{-12}$), more likely to be male (p=0.01), more likely to have smoked (p=0.0003), and more likely to carry the MUC5B promoter variant (MAF 0.21 versus 0.30, p=0.006). When controlling for family relatedness in a mixed effects logistic regression model, age (OR 1.10, 95% CI 1.07-1.14, $p=1.21\times10^{-9}$), smoking history (OR 1.72, 95% CI 1.00-2.99, p=0.04), and the MUC5B promoter variant (OR 1.73, 95% CI 1.08-2.76, p=0·02) were significantly associated with risk of ILD.

OTHER EMBODIMENTS

It is to be understood that while the disclosure has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the disclosure, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

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gtgctgtggc ggttatggac ctgtgaactt tgcacacgcc ttctatcttc tttttcagtg  180
tccatttcag agggcatgtt ttcagatgaa accagtagaa gatggaagca gcctgtgact  240
agaatcactg cttagggtct tgctgcctag gaatcccact ctacctgcaa cagactgtga  300
aagaaccgag aaatacactg attttgaaca tagcccatac tataatgggg atgtttgtta  360
cagcagttag cattaaaaac cttggctagg cattggtcat aattgtagaa cacagcaaat  420
gaagggaaac tggaacatag aggccagtga gaactttagg gttaatgaaa aatgagggca  480
accaggataa tttggttctt kgccaaatag gaaggtgaaa ccaaaggtag actggaggtc  540
agaaaatcag tccagcacat gtgatgtttt catttagttg cctgtatgtc tgtctggtct  600
ccagctcagc ctggctcctt gaggtaagag gcagtggctg ttcacctttg catcccagca  660
cctggcatac aatagatggg atgaaatgtt caaactgagc ctaagcttca gggtgcttat  720
caaagcaggg aagatacaca agaggagatg attcaggtcc agggcaggtc aggtatctaa  780
acccagtctc ttaggaagct ggatcctccg aaccagggag aacaagctgg atatgcactg  840
gatttcccag cagtactgat ctagagactc tcatagagtc ccttttattc cttggcctag  900
ggttacaact gcttatagca tctggaaaga ctcaacacct caaaagagac tttcagtaga  960
tacagcaaat acactcatgg aattgataat taagcttcaa t                        1001

SEQ ID NO: 3            moltype = DNA  length = 1001
FEATURE                 Location/Qualifiers
source                  1..1001
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 3
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ctcaacctca tgagtagctg ggattacagg cgcccaccac cacgcctggc taatttttgt  180
atttttagta gagataggct ttcaccatgt tggccaggct ggtctcaaac tcctgacctc  240
aagtgatctg cccgccttgg cctcccacag tgctgggatt acaggtgcaa gccaccgtgc  300
ccggcatacc ttgatctttt aaaatgaagt ctgaaacatt gctacccttg tcctgagcaa  360
taagaccctt agtgtatttt agctctggcc accccccagc ctgtgtgctg ttttccctgc  420
tgacttagtt ctatctcagg catcttgaca cccccacaag ctaagcatta ttaatattgt  480
tttccgtgtt gagtgtttct ktagctttgc ccccgccctg cttttcctcc tttgttcccc  540
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ttctgtcttg ttattgctgg taaaccccag ctttacctgt gctggcctcc atggcatcta  660
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ggcggtcatc ttggcccgtg agtgtctgga gcaccacgtg gccagcgttc cttagccagt  780
gagtgacagc aacgtccgct cggcctgggt tcagcctgga aaaccccagg catgtcgggg  840
tctggtggct ccgcggtgtc gagtttgaaa tcgcgcaaac ctgcggtgtg gcgccagctc  900
tgacggtgct gcctggcggg ggagtgtctg cttcctccct tctgcttggg aaccaggaca  960
aaggatgagg ctccgagccg ttgtcgccca acaggagcat g                        1001

SEQ ID NO: 4            moltype = DNA  length = 1001
FEATURE                 Location/Qualifiers
source                  1..1001
                        mol_type = genomic DNA
                        organism = Homo sapiens
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caaagggcgc ctctctttgt ctttgtgtcg caggcggaga tggacatggt ggcctggggt  180
gtggacctgg cctcagtgga gcagcacatt aacagccacc ggggcatcca caactccatc  240
ggcgactatc gctggcagct ggacaaaatc aaagccgacc tggtacttgt ctgtgtttca  300
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cttctgaaac attgctatta tttctttccc agaagactgg aaatgtttag aaatccccact  420
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agttaaacca gaggctgagg ktgtgcaaac actcatttgc agtttgtgaa taagtctctt  540
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cagtcttata cagtagtttg tgtcattggt gaatcctttc ctaggtggta attaaaacat  660
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ttcccgcagg tctctacact tgtaaatcac taaacttttt tttttttttt ttacctgaga  840
ccatagcttc tcatcctcat ttcttcttct ggctttttgg ggcttacttt tgtccacctg  900
agcccctgac caactttctc cttcatttct ctaagaccta gggaatccta aatgatgtct  960
ttaaacttta agacaatttt ctaacacgtg agtctttaag t                      1001

SEQ ID NO: 5            moltype = DNA  length = 941
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source                  1..941
                        mol_type = genomic DNA
                        organism = Homo sapiens
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tttttttttt tttttttt tttgagatgg agtctcgccc tgttgcccag gctggagtgt  180
agtggcgaaa tctcggctca ctgcaacctc cacctccctg gttcaagcaa ttcccctgtc  240
tcagcctccc gagtagctga gattacaggt gcacaccacc acgcccggct aattttttg  300
tatttttagt agagatgggg tttcaccatg ttggccagac tggtctcgaa cttctgacct  360
caggcaatcc gcctgcctca gcctcccaaa gcgctgggat tacaggcatg agccactgcg  420
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gttcacaaag cttccttgcc tctattctct gcaaatctgt aaactcttca gctgacccaa  720
tttctctctt tagccttcag agattatttt attttatttt atttcatttc atttcatttc  780
attttgacag aatctagctc tgtcgcccag gctggagtgc agtggcacca tctttgctca  840
ctgcaacctc cccctcacag gttcaagcaa ctgtcctgcc tcagcctccc gagtagctgg  900
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SEQ ID NO: 6            moltype = DNA  length = 1001
FEATURE                 Location/Qualifiers
source                  1..1001
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 6
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gcaggatctc tggctgcact cagccctttc tgcctcccca aatgctctat atcactattt  180
gtacactgag cagagtaaag ttagagagaa ctgttttata gaatagggct ggcccccgct  240
cccctggcct acgtgatggt ccttcctggc tgccaggtac ttgtttgtat tagagacaga  300
cactccacag ggtctgttgt ggcccacagc acataggcaa tcagaggcag aaagcagagc  360
tgtttggacc cacagagggc cggctgtctg ccactgaaat gtctttccag ttggttgaga  420
agcagcagga tgctctgctg gtgatgtctg aaagtcccag gattctttgg gtctccaagg  480
agatcctagc atataccact rtcgtggttt taataaagag caaaaacact ttcagatggg  540
gagaagagtg gaacaaaagg tattcttcct gggttgaagt ctgggggaaa ggcattgaga  600
agactgggct aatggcacaa accaatgaag tactcaagtc acctgtgatg gaggccagtc  660
atccaatggt atcaactttg tatgtggcaa cacttaataa aaatctgaac aggtcttcac  720
ttgtggacac agtagacttt cttgaaaaag gacagaaaag tgagccctgt gaattttcat  780
ctcacggact gacaacaatg acttgccttt aaggacagtc actcaagatg aagatgcaac  840
aaaacccttc cagttccaag tggctgatga aaaaaaaaaa atcttaaaag catcacagaa  900
caacggagaa agagatcaga agactataac agatagtttg aattttaaaa ctcagagaaa  960
agcaactgag gaggaaatac actgcttaga aagaagaaac t                      1001

SEQ ID NO: 7            moltype = DNA  length = 601
FEATURE                 Location/Qualifiers
source                  1..601
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 7
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aatgagggac agtgacccac ccagggctgg gcctggctca gactccgtca gagccgcagg  180
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cccccccttcc cggcaggaac cggctgtgtc cctttccttc ctttatcttc tgttttcagc  300
dccttcaact gtgaagaggt gaactcttca aacacgctga gcaaacaggc ccgactccca  360
gggccgcatc cgggatgtct caatagctgt ggccttgacg tccacctcgg acccctgccc  420
cggacccagc ccagttccca atgggccctc tgcccgggga ggtgcctagt gggagggacg  480
agggcaaagt cggggccccc acttgtttgg tgtcactgtg tgccagcggc cactggcggg  540
cgaggctgtt ccagggtgga ggcggggagg gttggaccac aggcactgag cggggacaga  600
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SEQ ID NO: 8            moltype = DNA  length = 732
FEATURE                 Location/Qualifiers
source                  1..732
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 8
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ttcttagact tctcagatat tccgtttttg atagtatacc cttctgagtc taatatgtcc  180
taaagtgcga acttgtacaa ttttttttt ttttttttt ttttttttt tktgataagg  240
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tgcctcccgg gttcaagtga ttgtgatgtc tcagtctccc aagtagctgg gattacaggc  360
tcctgccacc acatgcctag ctaattgtta tactttagta gaaatggggc ttcgccgtgt  420
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gctgggatta caggcatgag ccaccgcgcc tgacccagct tcttaaatta ttctgggcca  540
ccagtaatgt gaatcatgta aattaaaata tataattaaa caaaatcata tagcgattag  600
agataatagt tgtgaaatgc ttgaaaaatc ataggcattt aataaataga agccattcca  660
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taatactcca tg                                                      732
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SEQ ID NO: 9            moltype = DNA  length = 1320
FEATURE                 Location/Qualifiers
source                  1..1320
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                        organism = Homo sapiens
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ggcaggcaga tctcttaagc ccaggagtct gagaccagcc tgggcgacat gatgaagccc  180
catctctaca aaaaatacaa aaaaattagc tgagctttat ggcaaatccc tgtaatccca  240
gttacctagg aggcccaggc aggaagatgg cttgagccca aaaggttgag gctgtagtga  300
gctgtgatca tgaacagagt gagaccctgt ttcaaaacaa aatgaaaaac aaacaaacaa  360
aaaaaccaag aaaacaagaa aacaaaaact atacaatgat gagccaaaaa gcaagatatg  420
gaagaatata tatatatata tatatatata tatagtatga gtccagctat agaaagtttg  480
aaatcaggca acctaaacaa tattgttcag ggatctatac agaggcagga agccattgag  540
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caaggagggg cacatgggct tggaatactg tcttcattga cctgcgtgtt gggtacacag  660
gagtttgtta tttttcacac tgcatatgtg catgtatata ctctcccata tataccatgc  720
atttcacaca agaacacaaa ggctgtgtgg ctctgctctg cccctttccc cttccagctc  780
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aagtgagaga gggtctttga cctcaaagat cttacagaga ggaccgtata cacaaataac  1140
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tatttgttgt gcacctatta catgccagga actgtgctgg atggtgatca tgtaaagaca  1260
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SEQ ID NO: 10           moltype = DNA  length = 1001
FEATURE                 Location/Qualifiers
source                  1..1001
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 10
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aaccaaagga catcagcact tacagggctc tcaggtcaca cacaggatgt ccgcgcccac  120
tgcagagctg caggtcccct ccagggcagt ggggagccac aagcagcgtt aggcagcggc  180
tgggaccagg accgcctgag cactcaagaa cccccactgc cccaagcact gctggcagca  240
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aatcttctgg atgtggctag actctttagc ttgagcttcc agacaggccg aggcttggtg  600
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atattgccca ggctggtctc aaactcctgg gctccagcga t                      1001
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SEQ ID NO: 11           moltype = DNA  length = 17916
FEATURE                 Location/Qualifiers
source                  1..17916
                        mol_type = genomic DNA
                        organism = Homo sapiens
SEQUENCE: 11
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ASLDGESSHK THYAVQDTKG FHTEEDIGGL HATIECEQPQ PDLYKFVGRI NVYSDLNDPV  240
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ALDTHYWTWI NHFVIWGSLL FYVVFSLLWG GVIWPFLNYQ RMYYVFIQML SSGPAWLAIV  1080
LLVTISLLPD VLKKVLCRQL WPTATERVQN GCAQPRDRDS EFTPLASLQS PGYQSTCPSA  1140
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GVPFSCCTKD PAEDVINTQC GYDARQKPEV DQQIVIYTKG CVPQFEKWLQ DNLTIVAGIF  240
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                        organism = Homo sapiens
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AAQGGPAKQL LPLERVYQEI AILKKLDHVN VVKLIEVLDD PAEDNLYLVF DLLRKGPVME  240
VPCDKPFSEE QARLYLRDVI LGLEYLHCQK IVHRDIKPSN LLLGDDGHVK IADFGVSNQF  300
EGNDAQLSST AGTPAFMAPE AISDSGQSFS GKALDVWATG VTLYCFVYGK CPFIDDFILA  360
LHRKIKNEPV VFPEEPEISE ELKDLILKML DKNPETRIGV PDIKLHPWVT KNGEEPLPSE  420
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                        organism = Homo sapiens
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```

What is claimed is:

1. A method of treating a fibrotic lung disease in an asymptomatic human subject within an at-risk population, wherein the fibrotic lung disease is familial interstitial pneumonia (FIP), pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), interstitial lung disease (ILD), an interstitial lung abnormality (ILA), an asymptomatic ILA, fibrotic interstitial lung disease (FILD), or rheumatoid arthritis-associated interstitial lung disease (RA-ILD), and

wherein the subject has a blood relative with a fibrotic lung disease selected from the group consisting of familial interstitial pneumonia (FIP), pre-clinical pulmonary disease, pulmonary fibrosis, idiopathic pulmonary fibrosis (IPF), an interstitial lung abnormality (ILA), an asymptomatic ILA, interstitial lung disease (ILD), fibrotic interstitial lung disease (FILD) and rheumatoid arthritis-associated interstitial lung disease (RA-ILD), the method comprising:

a) identifying pre-clinical pulmonary fibrosis (PrePF) in the subject by use of quantitative high resolution computed tomography (qHRCT) comprising the use of a convolutional neural network to quantify the extent of fibrosis in the subject and wherein the convolutional neural network architecture classifies image regions using pixel and texture features extracted by multiple convolutional layers at different scales;

b) determining that the subject has the T allele of the MUC5B rs35705950 polymorphism;

c) administering a therapeutic agent in an amount effective for the treatment of fibrotic lung disease to the subject, wherein the therapeutic agent:

(i) prevents the onset or development of a sign or symptom of the fibrotic lung disease;

(ii) delays the onset or development of a sign or symptom of the fibrotic lung disease when compared to the expected onset of the sign or symptom in the absence of treatment with the therapeutic agent; or

(iii) agent reduces the severity of a sign or symptom of the fibrotic lung disease when compared to the expected severity of the sign or symptom in the absence of treatment with the therapeutic agent.

2. The method of claim **1**, wherein the subject presents radiographic Usual Interstitial Pneumonia (UIP).

3. The method of claim **1**, wherein the subject is greater than 40 years in age.

4. The method of claim **1**, wherein the blood relative is a sibling.

5. The method of claim **1**, wherein the MUC5B rs35705950 polymorphism is encoded by a sequence comprising SEQ ID NO: 7.

6. The method of claim **1**, wherein the therapeutic agent comprises a N-acetylcysteine, pirfenidone, or nintedanib.

7. The method of claim **1**, wherein PrePF is identified in the subject when a fibrosis score in the subject is above a predetermined cutoff value.

* * * * *